

Systematizers

*Reason, Machines, and the Rise of
Systematic Thought in Early Modern
Philosophy, 1500–1800*

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Abstract

Philosophers tend to treat the drive towards systematic thought as a timeless demand of rationality. But there is a counter-tradition warning this “will to a system” can function as a substitute for moral character, an aesthetic fetish, or a dangerous universalization machine. In response to these critics of systematization, this book offers a genealogical reconstruction of the ideal of cognitive systematicity between 1500 and 1800, asking not just *how* thought became systematic, but *why*.

Moving beyond the standard historical explanation, which assumes systems were built merely to mirror the metaphysical blueprint of a rationally designed universe, the book argues that early modern thinkers introduced the demand for cognitive systematization in an effort to emulate the virtues of well-designed machines. By modeling cognitive on mechanical virtues, these systematizers transferred the authority of knowledge from the internal, personal dispositions (the *habitus*) of individual thinkers into freestanding, externalized architectures. Through historical case studies ranging from Ramus and Keckermann through Descartes, Hobbes, Spinoza, and Cavendish through Leibniz, Newton, and Du Châtelet to Diderot, D’Alembert, Rousseau, and Kant, the book uncovers three distinct practical rationales that drive this mechanization of thought: the pedagogical rationale, according to which systematization makes knowledge transmissible; the epistemological rationale, according to which systematization makes thought self-certifying and self-correcting; and the political rationale, according to which systematization makes public authority accountable, administrable, and fair.

While these rationales vindicate the systematic impulse as a practical necessity rather than an arbitrary quirk, the genealogy also traces a counter-tradition, from Margaret Cavendish’s vitalism through Diderot’s polyphony to Hegelian ethical critiques, that exposes the costs of emulating the virtues of machines. Pushed too far, the demand for systematicity flattens experience, strips away context-sensitive “thick” concepts, and risks becoming nothing but dogmatic machinery. The book thus provides a delimiting framework for understanding both the indispensable value of systematizing thought and the importance of recognizing its limits.

List of Key Figures

Figure	Dates	Role in the Genealogy
Ramus	1515–1572	Dichotomous method as scalable didactic technology; knowledge rendered spatially surveyable; the first pedagogical artefact with a machine-like profile
Keckermann	1571–1609	Applies the term <i>systema</i> to knowledge; the system as pedagogical machine
Alsted	1588–1638	Encyclopedic expansion of the system; makes explicit the substitution of <i>systema</i> for <i>habitus</i>
Descartes	1596–1650	System as fortress against scepticism; the geometric method as cognitive machinery
Hobbes	1588–1679	Reasoning as computation; <i>Leviathan</i> as political machine
Spinoza	1632–1677	<i>Mos geometricus</i> as impersonal, machine-like method
Cavendish	1623–1673	Counter-voice: critique of mechanism; nature (and thought) as more than a machine
Leibniz	1646–1716	<i>Calculus</i> : the dream of mechanized reasoning; metaphysical warrant at full power
Newton	1643–1727	System of the world as divine clockwork; the metaphysical warrant in natural philosophy
Du Châtelet	1706–1749	Most sophisticated articulation of why systematic physics needs systematic metaphysics; the metaphysical warrant at its zenith and under strain
Condillac	1714–1780	Demolition of the <i>esprit de système</i> ; the metaphysical warrant challenged
D'Alembert	1717–1783	<i>Esprit systématique</i> vs. <i>esprit de système</i> ; the <i>Encyclopédie</i> and the displacement from hierarchy to network
Diderot	1713–1784	Counter-voice within the Enlightenment: what machines cannot capture
Wolff	1679–1754	Logicisation of the concept of system; theorisation of systematic vs. non-systematic understanding; stage-setter for Kant
Lambert	1728–1777	<i>Systematologie</i> : general theory of systems; generalisation of Wolff's logicisation; liberation of the concept from the metaphysical warrant; stage-setter for Kant
Rousseau	1712–1778	Political necessity and political danger of systematic reason
Kant	1724–1804	Transforms the metaphysical warrant; uncovers the precondition of cognition

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Introduction: Beware of Systematizers

“Systems in many respects resemble machines.”

– Adam Smith

“Philosophers have always aimed at cleaning up the litter,” at replacing “the first sensible tangle” with “economical and orderly conceptions” (1975–88, vol. iv, 26), William James observed with bemused amazement. And it is true that the drive towards a way of thinking about the world that is *systematic*, towards a unified conception that is consistent, coherent, complete, and parsimoniously principled, is so deeply embedded in modern philosophy as to appear timelessly self-evident. It finds canonical expression in the *Kritik der reinen Vernunft* when Kant asserts: “our diverse modes of knowledge must not be permitted to be a mere rhapsody, but must form a system” (A832/B60). James may have felt that a similar tidy-mindedness could be glimpsed already in Plato’s theory of the Forms, or in Aristotle’s insistence, in the *Posterior Analytics*, that *episteme* must go beyond mere perceptions and received opinions – a thesis that Aquinas, seemingly anticipating James, rendered as *sapientis est ordinare*: it is characteristic of the wise to impose order.

Yet there is an important sense in which James was wrong. The cognitive systematization that we now immediately recognise in James’s description is radically different from the kind of ordering activity that Aquinas could have envisaged. Montaigne’s digressive *Essais* still perform a mode of philosophical reflection that eschews systematicity, and they tellingly end with the observation that “there is no way of life so weak and sottish as that which is conducted by regimen and discipline” (III, 13, 1083 B). It is only around the year 1600 that the ideal of systematic thought emerges in early modern Europe. It takes shape in response to an unprecedented convergence of practical pressures that was characteristic of the late Renaissance. Before 1600, the term *systema* existed, but it was applied to worldly composites, such as flocks of animals or formations of soldiers. No-one applied the term to thought.

Fast-forward two centuries, however, and the imperative to systematize had come to seem so obvious as to require no argument. Paul Franks, in his study of German Idealism, observes that the post-Kantian Idealists were “obsessed with the construction of a philosophical system” (2005, 1). Systematicity had become the “unquestioned answer to the question of the nature of reason’s fundamental

demand” (Franks 2005, 3) – as if the fact that reason demands systematicity were obvious without justification, and the question of whether to systematize barely even made sense.

How did this remarkable transformation come about? How did a word that no one before 1600 had even thought to apply to knowledge become an epistemic ideal so deeply entrenched that its authority was simply taken for granted? This book returns to the era before systematicity reached its apogee in the post-Kantian systems of the likes of Fichte or Schelling. My aim is to reconstruct what initially spurred the rise of systematic thought in early modern Europe – to challenge the “unquestioned answer” by asking *why* thought should be systematized at all. By recovering the foundational rationales that first drove this ideal, I hope to uncover reasons for systematization that still have force with us today, after the collapse of the grand idealist systems.

James’s remark helps dramatize the question of why to systematize by *psychologizing* philosophers’ pursuit of systematicity. Instead of treating the demand for systematicity as a demand of rationality itself, as we have become accustomed to do, he exposes the demand to reevaluation by casting it as an *aesthetic* preference: a predilection for the “intellectually neat” and the “aesthetically pure and definite” (James 1975–88, vol. iv, 26).

By reframing the will to systematize as a matter of temperament, James forces us to ask whether it is anything more than a quirk of the philosophical temperament. Are there *reasons* – as opposed to mere psychological impulses – behind philosophers’ drive to systematization? And, crucially, are there perhaps also reasons *against* it?

That question once raised, it brings to light a long lineage of fierce critics of systematization. “*Beware of systematizers!*” (*Daybreak* 318), Nietzsche warned in 1881, and he had not changed his mind a few months before his collapse into insanity in 1889: “I distrust all systematizers and avoid them. The will to a system is a lack of integrity” (“Arrows” 26). More recently, Raymond Geuss, in *Who Needs a World View?*, denounced what he too calls “philosophy’s obsession with systematicity” (2020, xviii), objecting in particular to the idea “that our life as a whole could, or should, be seen through the lens of a technician who puts every decision of ours through a universalisation machine and files it according to the result” (2020, xviii).

Nietzsche, as Jessica Berry has shown, regards the drive towards systematization as a kind of psychopathology (Berry 2005). Geuss’s worry is more political: systematization becomes dangerous when it functions as a mode of authority, when the “universalisation machine” overrides the concrete judgment of those whose lives it claims to organize. And both suspicions can be traced to a question that, as this book will argue, runs through the entire history of the systematicity ideal: does the demand for systematicity answer to genuine needs, or does it express a disposition that is aesthetic at best and authoritarian at worst?

It is in dialogue with critics of systematisation that the historical narrative told in this book unfolds its full import. This is why I will trace not just the rise of the first systematizers, but also the emergence of the counter-voice, which draws attention to various reasons against systematization. By reconstructing the reasons for and against systematization from the story of what initially spurred it, I aim to develop a delimiting framework that elucidates both the value and limits of systematization. The book is thus an attempt to give a nuanced answer to those who challenge the ideal of systematic thought by recovering its risks and rationales.

System as a Substitute for Character

An especially fruitful version of this challenge can be found in the work of Bernard Williams. Williams is a particularly constructive interlocutor for this book to set out from, because he does not simply reject the systematizing impulse; rather, he opens up lines of inquiry into its social and historical basis. At the same time, Williams bridges the political and psychological critiques of systematization. He echoes Geuss's fear of systems' totalizing tendency—their habit of commandeering human life, crowding out the unassimilable, and asserting an unearned authority over how we ought to live. But he also echoes Nietzsche's suspicion that the will to a system betrays a lack of integrity. Setting his sights on the grand edifices of Kantian and utilitarian ethical theories, Williams opens his critique in *Ethics and the Limits of Philosophy* with a mordant line from Albert Camus: “*Quand on n'a pas de caractère, il faut bien se donner une méthode*” – when one has no character, one has to give oneself a method.

In Williams's development of Camus's contrast, the path to the good life leads through the cultivation of virtuous dispositions of character. “Method,” on the other hand, stands for the rigorous application of a rationalistic decision procedure, of the kind provided by a philosophical system. Camus, grappling with the moral catastrophe of Nazism, was warning against the havoc that methodical diligence could wreak when bereft of moral guidance. Williams, himself part of a generation of philosophers grappling with how to do moral philosophy in the aftermath of WWII, evokes Hannah Arendt's account of the trial of Adolf Eichmann to underline the point that systematicity can be no safeguard against evil unless underpinned by character: Eichmann notoriously defended himself by invoking the systematic consistency of his actions – he had not acted out of inclination, but had rigorously cleaved to the dictates of duty even when they conflicted with his dispositions. This defence was immediately decried as an absurdity by everyone at the trial. But it serves as a sinister reminder that system cannot ultimately be a substitute for character – not in the ethically demanding sense implying good judgement and a developed ethical sensibility. On the contrary, cleaving to a system may end up only

amplifying a lack of character. And, as Camus sharply observes, a lack of character may precisely be what the felt need for a system expresses.

For Williams, this yields a cautionary tale for the broader ethical critique of systematicity he goes on to develop. Philosophy must always “listen to what it says” and compare the deliverances of philosophical systems against an undistorted phenomenology of the ethical life – which, for Williams, is what philosophy must ultimately be answerable to. A system can “just lead you on and on in the wrong direction” (Williams 2002). Philosophy accordingly needs to be on the lookout for the moment at which the systematicity of thought begins to betray the “rattle of machinery” (Williams 2006, 120; 2014, 167). Systematicity can support the cultivation and expression of character. But it should not supplant character.

When set against the history of the rise of systematic thought, this cautioning against letting system become a substitute for character receives added bite – for, as we shall see, the first systematizers really did develop their systems to replace the cultivation of dispositions. *Systema* did emerge as a substitute for *habitus*. The opposition between system and character that Nietzsche, Camus, and Williams each conjured up thus turns out to have a real grounding in history. If cognitive systems were expressly introduced to perform the substitution that these critics of systematicity warned against, this warrants taking their challenge more seriously.

Nor is this challenge limited to ethics. It extends into a broader epistemological attack on what Williams describes as an overly “*rationalistic conception of rationality*” (1985, 20). Having authored a book on *Descartes* and the article on “rationalism” in the *Routledge Encyclopedia of Philosophy*, Williams did not use the label “rationalistic” lightly, and we shall see that the label turns out to be vindicated by the *bona fide* rationalist pedigree of this particular conception of rationality. What marks out this conception is the assumption that to be rational *just is* to be systematic, and that any conflict among one’s commitments is inherently a deficiency to be repaired. He calls into question the very idea that conflict must be a “logical affliction of thought”, and thereby comes to see the need to reduce conflict – to systematize – in a new light as well:

If conflict is not a logical affliction of our thought, it must be a mistake to regard a need to eliminate conflict as a purely rational demand. ... Rather we should see such needs as there are to reduce conflict ... as having a more social and personal basis. (Williams 1981, 81)

The thought here is that the demand for systematicity is no longer self-justifying once we no longer regard it as inherent in the very idea of rationality. To justify the demand, we then need to understand where it comes from, what needs it answers to, and in what contexts these needs are operative. Systematization may still be justified – Williams himself intimates that we do need to reduce conflict, for social as well as personal reasons. But if the rationale for systematization is grounded in social and

personal needs rather than in the very definition of rationality, it becomes at least intelligible that it has limits and may overreach.

Indeed, Williams goes on to argue that the possibility of systematization overreaching should concern us, because what he dubs “the intuitive condition,” in which one’s commitments are unsystematic and sometimes in tension, is not merely tolerable, but *indispensable* to a flourishing human life. The state of living by a rich, historically grown amalgam of thick concepts and intuitions that resist systematic codification in terms of organizing principles, he writes,

is not only a state which private understanding can live with, but a state which it must have as part of its life, if that life is going to have any density or conviction and succeed in being that worthwhile kind of life which human beings lack unless they feel more than they can say, and grasp more than they can explain. (Williams 1981, 82)

On this view, cognitive systematization, far from being an unqualified good, can end up flattening experience and sacrificing what gives life its depth and texture if carried too far. If there were people who truly succeeded in fully systematizing all of their commitments, this would have to be because they purchased that exhaustive systematicity at the cost of becoming insensitive to anything that could not be represented within their tidy system. Whatever we want to call this condition, it is not obviously an achievement.

Williams characterizes this as an “originally Hegelian” critique (1995, 40), and we shall see that this anti-systematization critique indeed itself has a long history and is nearly as old as systematization itself. But it is nonetheless helpful to introduce this critique through a Williamsian lens, because it provides a useful distinction with which to look at that history and discern patterns in it: systematization involves abandoning or subordinating richly perspectival “thick” concepts in favour of “thin”, abstract, impersonal, but more readily systematizing concepts. The accumulated jumble of thick notions such as *courage*, *creepiness*, *chastity*, *honour*, *rudeness*, *civility* and *chivalry* leaves the intuitive condition teeming with qualitatively different and incommensurable thoughts without obvious hierarchy. To systematize these thoughts, one needs to gravitate towards *thinner* notions, such as *good*, *right*, or *rational*, in order to formulate general principles that then allow one to validate some thoughts as applications of those principles while overturning others that irremediably conflict with them.

This replacement of the thick by the thin may sometimes be warranted: Williams himself acknowledges that “there is a political argument for public justifications to take these forms, and to deploy, correspondingly, ‘thin’ concepts” (1995, 40). But that does not mean that the replacement is warranted *always* and *everywhere*, and a question remains about how far those systematizing efforts should be pushed. How many of the concepts and thoughts that *cannot* be understood as applications of the overarching concepts and principles should we abandon in the name of systemati-

city before we conclude that we have gone too far in that direction? Systematization may be pushed beyond the point at which it is warranted, and we need to remember that it is neither self-justifying nor simply a matter of presentation: to systematically rearrange our thoughts involves downgrading some in importance, and discarding others altogether. As Williams put it: “to demand that [our thoughts and feelings] be schooled by the requirement of system is to alter our moral perception of the world, not just to make it in some incontestable sense more rational” (2006, 162). And perhaps, as Williams himself believed, we now have not too many thick concepts, but too few, and the last thing we need is a further thinning out of thought in the name of systematicity.¹

The Unanswered Question: Why Systematize?

While Williams’s challenge was centred on ethics, it goes beyond the ethical, and has its analogues in epistemology and the philosophy of science. Nancy Cartwright has questioned the aspiration to systematically unify our conception of the world, arguing that if scientific understanding resembles a patchwork more than a pyramid, remaining obstinately piecemeal, pluralistic, and modular, this is because the world is itself “dappled”: it is a world “rich in different things, with different natures, behaving in different ways” (1999, 1). Any systematization of such a dappled world holds only within carefully shielded experimental contexts and breaks down at the boundaries. John Dupré has developed a complementary case for what he provocatively terms “the disorder of things,” arguing that “the disunity of science is not merely an unfortunate consequence of our limited computational or other cognitive capacities, but rather reflects accurately the underlying ontological complexity of the world” (Dupré 1993, 7). The idea that science could ultimately issue in a grand theoretical synthesis covering all aspects of reality is a substantive metaphysical thesis, and one that may well be false. And Stephen Gaukroger has presented the striving to systematize and unify the sciences as “a failed attempt to recapture a notion of unity that collapsed with the demise of a pre-modern idea of Christian unity of understanding” (2020, 9), an attempt animated partly by a secularized theological model of nature as the product of a single coherent act of creation (2020, 434) and partly by a mapping of political notions of unification onto scientific ones (2020, 11). If Cartwright, Dupré, and Gaukroger are right, even the natural sciences cannot deliver a single, consistent, comprehensive system, and the expectation that they should turns out to rest on theological and political assumptions rather than on the nature of inquiry itself.

¹For a detailed philosophical account of the process by which this thinning out actually operates, see Nguyen (2026).

These challenges converge. From practical philosophy, from the philosophy and history of science, and from the anti-systematic tradition in European thought, they arrive at the same question. If asystematicity is not inherently an affliction of thought; if the demand for system is not a built-in requirement of rationality, but something with a more contingent, more social, and more practical basis; and if the costs of systematic tidiness can sometimes outweigh its benefits — then why, between roughly 1600 and 1800, did European thinkers come to demand that knowledge form not a loose aggregate, but an articulated system? What drove that demand? What did it achieve? And what did it cost?

Despite being historically influential, the ideal of systematicity remains surprisingly undertheorized. When Paul Hoyningen-Huene surveyed previous work in 2013, he described the “sobering upshot” as being that for an “analysis of the concept of systematicity, we must start from scratch” (2013, 26). This is a striking admission, given how central the ideal has been to philosophy’s self-understanding. The question “why systematize?” has somehow never received a fully satisfactory answer. The existing treatments of the subject illuminate parts of the landscape, but none of them asks the question in the form that Williams’s challenge demands.

Otto Ritschl’s monumental *System und systematische Methode in der Geschichte des wissenschaftlichen Sprachgebrauchs und der philosophischen Methodologie* (1906) painstakingly charts the emergence of the concept of a system and of the more procedural notion of the systematicity of method. But Ritschl primarily asks *when* the word *systema* came to be used in theological and philosophical works, and in what senses. He says comparatively little about *why* one might want to systematize: about the practical pressures that made the concept attractive or the functions it came to serve.

Nicholas Rescher’s *Cognitive Systematization* (1979) fills a different gap: it provides an analytic account of what systematic organization consists in, identifying parameters of systematicity such as consistency, cohesiveness, completeness, and consonance. Rescher’s question is what systematicity *is*, and his method is conceptual clarification in the analytic tradition: a morphological inquiry that maps the logical and structural terrain of systematic knowledge. But his approach is thoroughly ahistorical, and it is broadly vindictory in a way that never calls the ideal into question or seeks to understand its limits and costs. Rescher treats the ideal of systematicity as unequivocally and timelessly desirable, as if the parameters of systematicity were features of a logical kind susceptible to analysis independently of the historical conditions in which the demand for them first crystallized. Because his analysis is not genealogical, it cannot be delimiting: he identifies trade-offs *among* the parameters of systematicity, but he does not ask whether the demand for systematicity as such might be indexed to conditions that do not always obtain, or whether it has costs that can outweigh its benefits. The tradition of resistance to the systematicity ideal, from Cavendish’s vitalist materialism through Diderot’s

polyphonic epistemology to Williams's critique of ethical theory, falls entirely outside Rescher's frame.

Hoyningen-Huene's own *Systematicity: The Nature of Science* (2013) proposes systematicity as what demarcates science from other forms of knowledge, but he does not trace the ideal's historical emergence. His interest is in the observation that science is *more systematic* than ordinary thought along various dimensions; he does not ask what practical needs the demand for systematicity serves beyond the need to demarcate.

Leo Catana's *The Historiographical Concept 'System of Philosophy'* (2008) offers a careful study of how the phrase "system of philosophy" was introduced in the 1740s by Jacob Brucker (1696–1770) to establish the history of philosophy as an autonomous discipline and demarcate it from other historical disciplines. Catana's interest, however, is in this concept's career as a *historiographical* tool – in how it shaped the way historians of philosophy selected, organized, and evaluated their material – rather than in the practical rationales that drove systematization as a cognitive practice.

By far the most ambitious historical account of the systematizing ambition is Gaukroger's multi-volume *Science and the Shaping of Modernity* (2006, 2010, 2016, 2020). But Gaukroger's genealogical story is in significant respects a debunking one. The rationale he identifies as the primary driver of systematization is a metaphysical one: systematization was thought to be warranted by the need to reflect the antecedent systematicity of the world, which was itself the product of a single coherent act of creation (2020, 434). For Gaukroger, the demand to systematize and unify the sciences is thus ultimately a doomed attempt to recapture a unity whose original grounds in pre-modern Christian cosmology have irretrievably collapsed; the demand should therefore be scaled back even within science (2020, 9–10).

What is missing from the existing literature, then, is threefold: the entanglement of systematization and mechanization developed as a sustained thesis rather than an incidental observation; the arc of the metaphysical warrant for systematization; and the recovery of distinctively practical rationales that can stand on their own feet and provide an answer to the question of why to systematize even once the metaphysical warrant falls away.

Recovering Rationales from the Historical Spurs to Systematization

This book aims to recover those rationales for systematization by engaging in detail with the historical context and development of the ideal of systematic thought and reconstruct the various spurs to systematization. The approach is genealogical, in the broad sense that it traces the ideal's development through a sequence of historically contingent episodes of crisis and innovation: the collapse of ecclesiastical

authority, the skeptical revival, the information overload generated by printing, the Wars of Religion, the Enlightenment's internal critique of its own systematic ambitions, and the Kantian transformation that followed. But the genealogy is not merely descriptive. At each stage, it asks what practical pressures made the demand for system intelligible and compelling, what work systematization performed for those who adopted it, and what costs it imposed. The result is an account that is simultaneously *vindictory* and *delimiting*: vindictory, because it shows the demand for system to be practically well-motivated rather than arbitrary; delimiting, because it shows the demand's value to be indexed to identifiable pressures, so that when those pressures are absent or take different forms, the demand may need to be recalibrated, or set aside.

What makes this approach distinctive is that it does not simply ask after the *causes* of the ideal's emergence, but after the *reasons*: the practical pressures, intellectual needs, and normative considerations that made the demand for system seem not merely attractive but, in certain contexts, practically indispensable. This matters because reconstructing the reasons, rather than merely tracing the causes, puts us in a position to ask which of those reasons are still with us and which have lapsed. The book asks, in other words: what was, and is, the *point* of systematizing thought?

The question has not been posed in this form before. It asks not merely *when* the concept of a system came to be used (which is Ritschl's question) or *what* systematic organization formally consists in (which is Rescher's), but *why* anyone would want to systematize in the first place: what practical problems systematic organization solved, and what people would have lacked without it. The genealogical answer to this question shows, *pace* James, that philosophers' tidy-mindedness is not merely an aesthetic quirk, but a response to genuine and sometimes pressing needs; yet the genealogy also vindicates the insistence, on the part of the critics of systematization, that the demand is capable of overreaching. For if the demand for systematicity is indexed to practical needs rather than inscribed in the logic of reason itself, then it is scalable and context-sensitive: it presses more urgently in some domains than in others, and it can be pushed beyond the point where it does more good than harm.

The Framework

"Systematicity" is a slippery term. The book's reconstruction of what it came to mean as applied to thought between 1600 and 1800 will give us a firmer grip on it, and yield an analytical framework organized around four claims. It may be helpful to briefly preview them here, even though the framework will be built up cumulatively throughout the historical chapters and fully developed only in the final, synthetic chapter.

A baseline. At the most basic level, systematicity emerges as a *precondition of cognition*. Subjecting cognition to the demands for consistency and coherence is a precondition of having determinate thoughts at all. Were one's judgements entirely free of this pull, they would not even *purport* to be about anything; there would be no thinking that things are thus-and-so: only noise. If one perceives a stick as rigid and straight one moment, yet bent at another (say, when half-submerged), these two perceptual judgements are inconsistent. To remain indifferent to that inconsistency is to fail to treat these two perceptions as two modes of appearance of *one and the same object*. This is the level at which it is indeed correct to say that systematicity is a demand inherent in reason itself: as a minimal threshold below which cognition cannot function. As the title of Christian Wolff's *De differentia intellectus systematici et non systematici* (1729) reminds us, however, this still leaves a gap between an intellect that is just systematic enough to merit being called an intellect and one that deserves the qualifier *systematic*. The question "Should I systematize?" does not arise at this level, but it is precisely here that the anti-systematizers go wrong if they suggest that the demand for system is nothing more than a preference. The baseline establishes that the demand for system is not arbitrary, while leaving open how far beyond it one should go. Systematicity as a precondition of cognition is operative throughout the history the book narrates, but it is only fully uncovered, in a philosophically explicit way, by Kant, who reveals it when he strips away the grander justification that had previously concealed it.

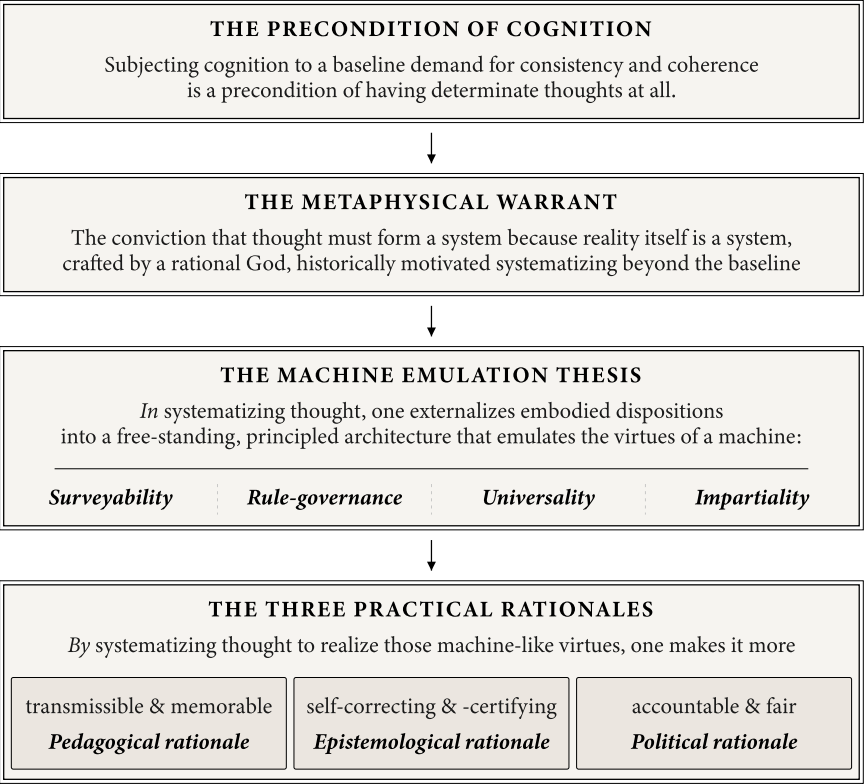


Fig. 1. An overview of the book’s proposed framework for thinking about cognitive systematization. It distinguishes four levels that build on each other.

A historical scaffold. That grander justification is what I call the *metaphysical warrant*: the conviction that knowledge must form a system because *reality itself* is a system, crafted by a rational God whose creation exhibits the very consistency, coherence, and principled unity that systematic thought aspires to mirror. This warrant stands behind the great system-building projects of the seventeenth and eighteenth centuries, supplying the confidence that systematic thought is truth-tracking: that in organizing knowledge into a system, one is reflecting the structure of the world as God designed it. The warrant is implicit in Keckermann's Protestant textbooks at the turn of the seventeenth century, becomes explicit in Descartes and Spinoza, reaches full power in Leibniz's dream of a *characteristica universalis*, and attains its zenith in Newton's system of the world and Du Châtelet's philosophical defence of it; it then comes under critical pressure with Condillac and D'Alembert's distinction between the *esprit de système* and the *esprit systématique*, while Rousseau's political deployment of systematic reason dispenses with it altogether. Kant transforms the warrant: he does not refute it outright, but removes our entitlement to it as a claim about the world, recasting systematic unity as a *regulative ideal*, a *focus imaginarius* that coordinates inquiry without being demonstrably realized in nature. The arc of this warrant, its rise, zenith, and fall, gives the book's narrative its philosophical momentum, and makes the Kant chapter the culmination of that narrative: what Kant uncovers, once the ornate superstructure of the warrant is removed, is the modest, but ineluctable demand to think systematically as a precondition of thinking at all.

The Machine Emulation Thesis. The book's central claim is that to systematize thought is to emulate the virtues of machines. By moving thought from the internal, temporal domain of personal disposition (what the Scholastics called *habitus*) into a freestanding, principled architecture (a printed table, a geometric proof, a deductive treatise), early modern thinkers gave thought the functional profile of a well-designed machine. A machine promises to take any input, process it through fixed rules in a manner open to inspection, and deliver an output that is the same whoever operates it. Cognitive systematization aspires to do the same. This aspiration can be analysed into four virtues, each characterizing a distinguishable aspect of the machine-like functional profile:

Surveyability: the system's architecture is laid out so that its joints are visible and its connections explicit, making the relations among its parts open to inspection.

Rule-governance: the system produces determinate outputs from given inputs according to fixed procedures, so that an inconsistency registers as a malfunction demanding diagnosis rather than as a difference of taste or judgment.

Universality: the system promises to accept any subject matter as input and to be operable by anyone equipped with reason. On the input side, universality is the aspiration to a rational decision procedure whose scope of application is in principle unlimited: the "universalisation machine" whose reach Geuss regards with such suspicion, the *mathesis universalis* to which Descartes aspired, or the *calculemus* in

which Leibniz placed his hopes. On the operation side, it is the aspiration to encode expertise in a form that requires no expert: Keckermann's printed textbook that replaces the master's *habitus*.

Impartiality: the system's outputs are independent of the identity, perspective, or interests of whoever operates it, so that they can claim objectivity and be reproduced by any competent operator. Two reasoners who feed the same problem into the system and follow its rules correctly arrive at the same conclusion: the output is agent-independent. This is what promises to make systematic thought capable of settling disagreements, because the system provides a neutral tribunal whose verdicts are expressive of the subject matter's own structure rather than of any particular party's standpoint.

These four virtues are the analytical core of what I call the *machine emulation thesis*: to systematize thought is to give it the surveyability, rule-governance, universality, and impartiality that characterize a well-designed machine. The machine emulation thesis is developed in detail in the next section; the point to register here is that it is the key to making visible what systematization is *for*: why anyone would bother.

Three practical rationales. The machine-like virtues that systematization confers on thought serve identifiable practical purposes. These purposes constitute the practical rationales for systematizing beyond the baseline, and they are what the book's genealogy recovers from beneath the metaphysical warrant.

The first rationale is *pedagogical*: systematization makes thought *transmissible*. The problem is the loss of traditional educational infrastructure and the need to train a professional class rapidly and at scale, without relying on years of personal apprenticeship. The system's universality (anyone can follow the procedure) and surveyability (the architecture can be grasped synoptically) combine to make it a teaching machine: knowledge is encoded in a freestanding, inspectable format that can be transmitted, reproduced, and scaled without the master's presence. This was historically the earliest rationale to emerge: the concept of a *systema* was, as Chapter 1 will show, invented by Protestant educators who needed to make knowledge transmissible without personal apprenticeship. It remains among the most practically consequential, but what is gained in transmissibility may be lost in depth: the *habitus*-based pedagogy of the medieval university produced a different, and not merely inferior, kind of understanding.

The second rationale is *epistemological*: systematization makes thought *self-certifying and self-correcting*. The problem is the Pyrrhonian sceptical crisis and the collapse of the criterion: once every claim to knowledge can be met with the further demand "and how do you know *that*?" knowledge needs an internal source of certification independent of external authority. The system's rule-governance (inconsistency registers as malfunction) and impartiality (the verdict is no one's personal opinion) combine to make coherence a credible test of truth. The system is self-certifying in the prospective direction: a claim earns credibility by integrating

into the deductive structure. It is self-correcting in the retrospective direction: a claim that generates inconsistency is flagged for revision. This is the rationale that animates Kant's contention that systematic unity is what "first raises ordinary cognition to the rank of science" (*Critique of Pure Reason*, A832/B860). But self-certification without empirical control produces speculative coherence without truth, and the Enlightenment's distinction between the *esprit de système* and the *esprit systématique* marks the recognition of that limit.

The third rationale is *political*: systematization makes thought *accountable, administrable, and fair*. The problem is the need for public authority that can be exercised and assessed without relying on personal trust, charismatic authority, or institutional tradition. The system's impartiality (outputs are agent-independent) and universality (anyone can follow and assess the reasoning) combine to make it a technology of impersonal assurance: decisions become auditable and fair because like cases are treated alike by an operator-independent logic, and departures from principle can be detected and challenged by anyone affected. This is the domain in which the demand for articulable principles is most clearly at home, as Williams himself acknowledged. But Rousseau saw that systematic reason, necessary as it is for legitimate political authority, is also capable of becoming a tool of domination when extended indiscriminately, a warning Williams echoed when he insisted that the demand for articulable principles has its proper home in the governance of public agencies but should not hold sway over the whole of human life.

These three rationales all vindicate the systematization of thought in some form: in its pedagogical role, systematization makes thought transmissible; in its epistemological role, it makes thought self-certifying and self-correcting; and in its political role, it makes thought accountable and fair. Together, they form an interconnected functional complex whose elements reinforce one another. And each also marks a limit: a point at which the demand for system, if pushed further than the underlying need warrants, begins to cost more than it delivers. The counter-tradition that runs from Cavendish through Diderot to Hegel identifies, at each stage, the domains in which the machine-like virtues on which the rationales depend cease to be virtues: wherever the quality of the agent, situated perception, sensitivity to context, the capacity to weigh incommensurable considerations, is part of what good thinking requires.

Rational Machinery

The book's central narrative is organized around the machine emulation thesis previewed in the framework above: the rise of the ideal of systematic thought is deeply entangled with the rise of machines. This entanglement operates at three levels.

First, machines created the pressures that rendered cognitive systematization urgent. The printing press generated the information overload that required new methods of ordering knowledge and fuelled the confessional crises that made the impersonality and impartiality of rational machinery seem appealing.

Second, machines provided the models for what systematic thought should look like. The success of mechanical devices, from clocks to Leibniz's Stepped Reckoner, lent prestige to mechanical models of thought and made it seem that the right way to organize knowledge was to make it function like a well-designed mechanism.

Third, and most fundamentally, the *systema* as it developed from Keckermann's Protestant textbooks through Descartes's tree of knowledge and Spinoza's geometric order to Leibniz's *calculemus* and Kant's architectonic of pure reason was itself a kind of rational machine. This observation traces back to none other than Adam Smith, who himself evinced what his contemporaries perceived as a "love of system" (Stewart 1795, 306), explicitly noted that "systems in many respects resemble machines" (Smith 1795, 66). "A system", Smith held, "is an imaginary machine invented to connect together in fancy those different movements and effects which are already in reality performed" (Smith 1795, 66).

The *systematization* of thought can accordingly be understood as what happens when thought aspires to the condition of a machine. The aspiration is captured by the four virtues identified above: thought is made machine-like in the sense that it becomes *surveyable* (one can inspect the architecture), *rule-governed* (it operates by fixed procedures), *universal* (it takes any input and can be run by anyone), and *impartial* (its outputs are agent-independent and reproducible). These are the properties that make machines useful, and they are precisely the properties that cognitive systematization transfers from physical to intellectual production. To systematize a body of knowledge is to give it the functional profile of a well-designed mechanism: a device that accepts any input within its domain, processes it through transparent, rule-governed procedures, and delivers an output whose validity is independent of whoever operates it.

And as I shall argue, the original function of giving thought the shape of a rational machine was *authority-replacement*: the demand for systematic thought was the demand that thinking be made machine-like in order to allow cognitive authority to be *depersonalized*, which is to say exercised through the structure of the system itself rather than through the standing of any particular knower or institution.

While I use the machine emulation thesis as a retrospective interpretive frame, it has a footing in the primary sources themselves. When Keckermann designed his *Systema logicae* as an externalized cognitive architecture that could function independently of the teacher's personal authority, he was constructing a pedagogical machine. When Descartes proposed a method that would produce reliable knowledge regardless of who employed it, he was modelling thought on the impersonal reliability of a mechanism. When Leibniz dreamed of a *characteristica universalis* that would allow disputants to settle their disagreements by saying *calculemus*, "let

us calculate,” he was making fully explicit the aspiration that had been implicit in every preceding system-builder: to mechanize reasoning itself. And when the Enlightenment critics, Condillac, D’Alembert, and Diderot, turned against the great systems, they did so precisely by arguing that the systems had become machines that had malfunctioned: rigid devices that ground out conclusions disconnected from reality. The *esprit de système* / *esprit systématique* distinction, which runs through the French Enlightenment, is in effect the recognition that rational machinery can get in the way of truthfulness.

The machine emulation thesis is the key to making the three practical rationales visible *as* rationales. If one asks why early modern thinkers demanded systematicity and looks only at what they explicitly said, one finds the metaphysical warrant: because reality is a God-designed system. If one asks instead what systematization *does* for them, what *practical work* it performs, then the machine emulation thesis reveals a different answer. Each rationale trades on a specific cluster of machine-like virtues to address a specific practical problem. The pedagogical rationale depends on the system’s universality and surveyability: the teaching machine encodes cognitive structure in a form that can be followed by anyone, without the teacher’s presence. The epistemological rationale depends on the system’s rule-governance and impartiality: coherence functions as a credible test of truth because inconsistency registers as malfunction, and the test commands acceptance because its deliverances are nobody’s personal opinion. The political rationale depends on the system’s impartiality and universality: decisions are fair because the system’s outputs do not express any particular agent’s perspective, and they are accountable because the reasoning is followable by anyone affected. Without the machine emulation thesis, one sees only the metaphysical warrant: the obvious justification that thinkers themselves foregrounded. With it, one sees the practical work that systematization performs beneath the metaphysical surface.

The Counter-Tradition

Recognising the links between systematization and mechanization also gives us a richer picture of the counter-tradition that resists the demand for systematicity. Tracing the rise of this counter-tradition will allow us to identify with increasing precision what the costs of making thought machine-like are.

Margaret Cavendish’s vitalist materialism, developed in the 1660s and 1670s in direct engagement with the mechanical philosophy of Descartes and Hobbes, provides the first sustained philosophical resistance to the equation of understanding with mechanization. Her critique is ontological: nature, she argues, is richer than mechanism allows, and the aspiration to capture it within a machine-like system impoverishes the very understanding it claims to advance. Diderot carries the resistance into the Enlightenment, embodying the tension between system and

anti-system within the movement itself: his dialogical, polyphonic experiments in philosophical prose insist that thought has an irreducibly creative, associative quality that resists systematic codification. The critique is aesthetic and epistemological. And the Hegelian critique that Williams later sharpened is ethical: the demand for systematic, principle-based moral reasoning, extended to ethical life, actively damages moral perception by abstracting from the concrete ethical substance of particular communities and replacing the thick concepts that require perceptual sensitivity for their application with thin ones that survive procedural abstraction. The costs deepen as the counter-tradition grows in scope, ranging from the ontological (Cavendish) through the aesthetic-epistemological (Diderot) to the ethical (Hegel).

Though it acknowledges these costs, the genealogy offered here is not ultimately a debunking one like Gaukroger's. It agrees with Gaukroger that the demand for systematic unity was originally scaffolded by theological and political assumptions that can no longer be taken for granted: this is the story of the metaphysical warrant's rise and fall. But it argues that the practical rationales recovered through detailed historical reconstruction are detachable from that scaffolding: they can be understood, evaluated, and in many cases vindicated on their own terms, without residual dependence on the metaphysics of a divinely created rational order. The demand for cognitive systematicity does not need God to make it intelligible; it needs the practical pressures of pedagogy, public accountability, the management of disagreement, and the critical scrutiny of entrenched belief. These pressures are still with us, though they may now take different forms.

From Keckermann to Kant: A Roadmap of the Genealogy

The genealogy is simultaneously historical and philosophical: historical, because it traces the ideal's emergence through contingent episodes of crisis and innovation; and philosophical, because it recovers the animating rationales for systematization, allowing us to assess whether those rationales still hold. The book is organized around philosophers and problems, not around historical background: it is a work in the history of philosophy, not in intellectual history more broadly, and context enters where it is needed to make the philosophical arguments intelligible. The narrative unfolds in seven historical chapters, a theoretical synthesis, and an epilogue.

Chapter 1, "From *Summa* to *Systema*: Keckermann, the Reformation, and the Pedagogical Machine," tells the origin story. The first application of *systema* to knowledge goes back to the year 1600, when Protestant educators, above all Bartholomaeus Keckermann at the Danzig Gymnasium, needed internal structural criteria to organize and transmit knowledge, because the Reformation had destroyed the external authority on which doctrinal order had previously depended. The system was, from its inception, a pedagogical machine: a technology for making knowledge

transmissible without personal apprenticeship and verifiable without institutional authority. The chapter traces the etymological puzzle of how *systema*, which in antiquity referred to musical intervals, astronomical arrangements, and physical compositions, came to denote a body of knowledge organized by principles. It reconstructs the pre-systematic world of the medieval *Summa* and humanist *copia*, showing what was both gained and lost in the transition. It examines the printing press as the double agent that made the Reformation possible while simultaneously generating the information overload that demanded new technologies of cognitive management: alphabetized indices, card-indexing systems, the material devices that were already making knowledge management mechanical before anyone theorized the connection. The pedagogical rationale is the first to emerge, and the machine emulation thesis makes its first concrete appearance with the externalized cognitive architecture of the textbook. But Keckermann's *systema* was a technology of *transmission*: it organized knowledge assumed to be reliable. The question of whether systematization could *produce* certainty, not merely organize it, is the question that takes centre stage in Chapter 2.

Chapter 2, "The Geometric Order: Descartes, Hobbes, Spinoza, and Cavendish," follows the *systema* into the seventeenth century's philosophical response to the skeptical crisis. The collapse of the criterion of truth, brought about by the Reformation and the rediscovery of Sextus Empiricus, raised a question Keckermann never faced: if any claim to authority can be challenged by "how do you know *that*?" then a system that merely organizes received opinion may end up organizing what is worthless. The seventeenth century's innovation was to demand that the system do more: that it *certify* knowledge, generating internal criteria of truth that require no external guarantor. Descartes proposed a method designed to produce reliable knowledge regardless of who employed it. Hobbes pushed the machine model further: reasoning, he declared, just *is* computation, and he extended the structural homology to politics, casting the state as an "artificial man" whose legitimacy derives from rational construction rather than tradition. Spinoza's *Ethics Demonstrated in Geometrical Order* represents the most radical attempt to make philosophy machine-like: a self-certifying deductive architecture that functions regardless of who reads the text. The transition from pedagogical machine to epistemological machine is the first major escalation of the machine emulation thesis, bringing the epistemological rationale into full view. The equation of understanding with mechanization was contested from the start, however, by Margaret Cavendish, whose vitalist materialism offered the first sustained philosophical alternative to the machine model and inaugurated the counter-voice tradition that the book tracks through its subsequent development. The metaphysical warrant becomes explicit and load-bearing.

Chapter 3, "Calculus: Leibniz and the Dream of the Reasoning Machine," centres on the figure who made fully explicit the aspiration that had been implicit in every preceding system-builder: to mechanize reasoning itself. Leibniz's dream of a

characteristica universalis, a formal language in which all human knowledge could be expressed, and a *calculus ratiocinator*, a procedure for deriving truths mechanically within that language, collapses the gap between metaphor and engineering. All three practical rationales converge in Leibniz, and the political rationale reaches its fullest scope: the *calculemus* is explicitly designed to resolve confessional and institutional disputes, extending the machine aspiration from individual cognition to the coordination of public institutions. The metaphysical warrant reaches full power: Leibniz's God has chosen to actualize the possible world that maximizes variety while minimizing the complexity of its fundamental laws, guaranteeing that systematic thought tracks the structure of creation. The machine emulation thesis becomes undeniable.

Chapter 4, "God's Clockwork: Newton, Du Châtelet, and the Metaphysical Foundation of System," traces the warrant to its highest philosophical articulation and its greatest vulnerability simultaneously. Du Châtelet's *Institutions de physique* provides what is arguably the most sophisticated philosophical defence of the conviction that knowledge must be systematic because God-designed reality is systematic, synthesizing Newtonian physics with Leibnizian-Wolffian metaphysics. In doing so, she makes the warrant fully explicit, and in making it explicit, she reveals how much it is being asked to bear. The chapter shows the warrant at its zenith and under strain, a combination that makes the Enlightenment critique of Chapter 5 feel philosophically necessary rather than historically contingent.

Chapter 5, "The System Examined: Condillac, D'Alembert, and Diderot," turns to the Enlightenment's internal critique. The great systems built against dogmatism had themselves become a new dogmatism. Condillac's devastating analysis demonstrates that the rationalist systems' apparent unity was linguistic rather than philosophical: they generated the illusion of coherence by operating with terms whose semantic content was never properly analysed. D'Alembert's distinction between the *esprit de système* and the *esprit systématique* captures the Enlightenment's recognition that the systematic aspiration, detached from empirical control, produces speculative constructions that merely *appear* coherent. Diderot embodies the tension between system and anti-system within the Enlightenment itself: his polyphonic experiments in philosophical prose insist on the irreducibly creative, associative quality of thought. The metaphysical warrant is dismantled, yet the demand for system survives in chastened form. The chapter raises the question that connects to Chapter 6: if systematic reason can become its own form of dogmatism, can it also become its own form of domination?

Chapter 6, "The Politics of System: Rousseau and the Costs of Rational Machinery," develops the political rationale and its internal tension. Rousseau demonstrates that systematic reason is necessary for legitimate political authority: the general will, as a publicly articulable and consistently applicable principle, makes the exercise of power answerable to those over whom it is exercised. Yet he also shows that the very impersonality that makes systematic reason a tool of fairness can

become a tool of domination when it overrides the particular circumstances and embedded judgment of those it governs. The political rationale stands here on its own feet, without the metaphysical warrant: this chapter demonstrates concretely what Chapter 7 will theorize abstractly, that the practical rationales for systematization require no speculative metaphysics to do their work.

Chapter 7, “The Architectonic: Kant and the Transformation of System,” brings the narrative to its climax. Kant accomplishes two things at once: he frees the systematicity ideal from speculative metaphysics by recasting systematic unity as a *regulative ideal* that coordinates inquiry without being demonstrably realized in nature, and he uncovers the precondition of cognition that the metaphysical warrant had concealed. The warrant falls; the precondition is revealed; the practical rationales are shown to be sufficient. The chapter closes by giving the counter-voice tradition its sharpest philosophical formulation: Hegel’s objection that the machine model of thought, extended to the ethical domain, actively impoverishes moral life by abstracting from the concrete ethical substance of particular communities. Williams inherits and sharpens this critique, and the chapter traces the connection.

The genealogy terminates with Kant. Within a decade of the first *Critique*, Fichte and Schelling would treat the imperative to systematize as obvious and urgent, pursuing the most ambitious system-building in the history of philosophy. The demand had by then become, in Franks’s phrase, an unquestioned answer. The genealogy is concerned with the prior history: the two centuries across which the demand for cognitive systematization first emerged and was eventually freed by Kant from the metaphysical scaffold that had originally sustained it. The genealogy thus ends where the era of systematicity as the “unquestioned answer” begins.

Chapter 8, “Rational Machinery: What the Genealogy Reveals,” steps back from the chronological narrative to develop the argument that the genealogy as a whole makes possible: the articulation of a limiting framework within which to understand both what makes systematization indispensable and when it is carried too far. The machine emulation thesis also reaches its final formulation, and the points at which the aspiration to make thought machine-like overreaches are pinned down.

The Epilogue, “From Leibniz’s Dream to Ours,” connects the genealogy to the present. The early modern entanglement of systematization and mechanization has been partially realized in our time: large language models process, organize, and generate knowledge at scales that Leibniz would have envied. The genealogy’s practical rationales provide tools for thinking about what we want from machine-produced systematization, and about what we stand to lose. The counter-voice tradition, from Cavendish through Diderot to Hegel, illuminates that contemporary challenge.

The book’s ambition, then, is to offer an even-handed answer to those warning against philosophy’s alleged obsession with systematicity. If the demand for systematicity is not a purely rational demand, but has a more social and personal basis,

then we need to understand that basis. The genealogy shows that the basis is real and enduring: it consists of practical pressures that are neither arbitrary nor trivial. But it also shows that the basis is plural and scalable, that different rationales support different degrees and kinds of systematization, and that there is no single answer to the question of how much systematicity to strive for. The insight that systematicity is a precondition of cognition tells us that some systematization is inescapable. The practical rationales tell us why we might want more. And their limits tell us when we go too far. Nietzsche and Camus suggested that systematicity is a substitute for character. The genealogy recovers a significant sense in which that is true: the systematization of thought indeed involves the emancipation of thought from the character of the thinker. By tracing the history through which that emancipation was effected, it recovers the pressures that made it intelligible, and even desirable, before also identifying the points at which the pursuit of systematicity began to give way to the rattle of machinery.

1

From *Summa* to *Systema*: Ramus, Keckermann, and the System as Pedagogical Machine

“Philosophy, as it exists in the mind, is a *habitus*; as it exists outside the mind, it is a *systema*.”

– Johann Heinrich Alsted

The word *system* is so thoroughly assimilated into the vocabulary of modern thought that it can be difficult to hear it as a word with an origin: a term that was, at a specific historical moment, seized upon by identifiable thinkers grappling with identifiable pressures. Yet the application of *systema* to an organized body of knowledge is a distinctly early modern invention. The Introduction asked why European thinkers came to demand that knowledge form an articulated system rather than a loose aggregate of truths. This chapter provides the first answer. It traces the concept of a cognitive *systema* to the late-sixteenth-century Reformed Protestant universities of northern Europe, and it argues that what we find there is something the later philosophical tradition has largely obscured: the system was born as a practical technology before it became a philosophical ideal. It first arose as a way of making doctrine and learning surveyable, reproducible, and transmissible under conditions in which neither personal apprenticeship nor inherited institutional authority could be relied upon. In this first historical setting, the primary rationale for systematization was pedagogical. Yet in answering that pedagogical need, the *systema* also initiated a broader shift whose consequences reached far beyond the classroom: more and more of the work once performed by formed persons and authoritative institutions was transferred into articulated structures. Recovering the circumstances of this invention is therefore the first step toward understanding both the later prestige of the systematicity ideal and the peculiar hazards that would accompany it.

In Greek antiquity, *systema* (from *syn-histēmi*, “to make to stand together”) designated a composite whole of a quite general kind: a flock of animals, a military

formation, a composite political unit, a musical scale. The *systema teleion* of Aristotle was the complete tonal system of Greek music theory; the *systema mundi* of the Stoic cosmologists was the rationally ordered cosmos itself, heaven and earth and everything contained within them, held together by divine *logos* (Rescher 2006, 90). The term continued in use throughout classical antiquity in this broad ordinary sense: something joined together, a connected or composite whole.

The Stoic usage deserves particular attention, because it harboured an ambiguity that would prove significant. The Stoics conceived of the cosmos as a providentially ordered totality governed by rational principles, in which physics, logic, and ethics formed an organic unity, comparable to a living being whose bones (logic), flesh (ethics), and soul (physics) supported one another. This was the idea of a *rational* whole whose parts were held together by intelligible connections. And yet *systema* remained, for the Stoics, tied to ontological rather than epistemological systematicity: it described the organized totality of the world – how *things* stand together, not how *beliefs* about things do. The semantic leap from *systema mundi* to *systema doctrinae*, from the rational order of the cosmos to the rational order of a textbook, would have to wait more than a millennium.

This is not to say that the Greeks lacked practices of systematic deductive organization. Euclid's *Elements*, composed around 300 BCE, provided the most enduring paradigm of knowledge organized into axioms, definitions, and rigorous derivations, and it would later become the inspiration for philosophers who sought to cast all of knowledge in a deductive mould. Yet the Greeks did not describe Euclid's achievement as a *systema*. The term belonged to cosmology and music theory; the deductive ordering of geometry was a separate intellectual achievement, one whose enormous potential for the organization of knowledge would be recognized only when early modern thinkers retrospectively adopted it as the model of what a system should look like. That retrospective adoption is the story of Chapter 2. Chapter 1 tells a different story: not the rise of deductive certainty, but the rise of an organized, externally displayable architecture of learning.

When the semantic leap from *systema mundi* to *systema doctrinae* finally occurred, it occurred in a setting that differed in many respects quite profoundly from the world of Euclid. Prior to the late 1590s, organized presentations of theological or philosophical learning employed terms such as *summa*, *loci communes*, *corpus doctrinae*, *compendium*, and *institutio*, but never *systema* (Ritschl 1906, 16, 25–30). By 1906, however, Ritschl counts at least 90 books with “system” in their title – 3.4 books per year on average (1906, 26).

The term's migration from physical composite to principled cognitive architecture was accomplished, almost single-handedly, by a Calvinist pedagogue working in a Baltic port city. His name was Bartholomäus Keckermann, and his *Systema logicae* of 1600 launched a transformation in how Europeans conceived of knowledge. To understand why Keckermann was driven to systematize knowledge in this way, and why he reached back to the Stoic term for an ordered cosmos to describe

an organized body of doctrine, we must first understand what this new ideal of cognitive systematicity replaced.

1.1 The World Before the Systematization of Knowledge

For roughly five centuries, from the high Middle Ages through the early Reformation, European intellectual life aspired to a form of cognitive organization that was highly sophisticated, yet profoundly different from the striving for systematicity. Scholastic knowledge fundamentally rested on two foundations. The first was ontological: the “great chain of being,” the hierarchical ordering of all creation from God through the angels and human beings down to brute matter, in which every entity occupied a determinate place and every discipline studied some portion of a single divine arrangement (Lovejoy 1936, 59–66). The second was institutional: the authority of the Catholic Church, whose Magisterium served as the ultimate arbiter of doctrinal coherence. Where apparent conflicts arose between Scripture, the Church Fathers, and Aristotle, the Church possessed the standing to settle the dispute. Cognitive order, on this dispensation, was sustained by ontological hierarchy and institutional authority, and it did not require articulated structure to do the work of holding things together.

The organizing ideal of knowledge was dialectical disputation. Knowledge, in this context, advanced through the disciplined confrontation of competing authorities. As Walter Ong has emphasized, intellectual activity in the ancient and medieval world was conceived primarily in auditory and conversational terms: the Socratic method essentially consisted of oral dialogue; Aristotle’s categories, his *predicaments*, were at root things *said of* a subject – they were vocal assertions, or accusations brought against a subject, and his notion of predication was grounded in the act of *saying* (Ong 1956, 224; 1958, 53). One arrived at knowledge by hearing it from a teacher, conversing about it, and being enculturated into a tradition of disputation. The authority of a claim rested in significant part on the authority of the person or tradition from which it issued, and learning was accordingly bound up with the prolonged presence of recognized masters.

This also meant that to learn philosophy was not primarily to learn a body of doctrines, but to develop a *habitus*: an acquired intellectual disposition cultivated through years and years of study and dialogue (Freedman 1999, 92–93; Hotson 2007, 45–46). This involved sharpening one’s judgment, accumulating experience, developing an acute sensitivity to context, and learning how to weigh competing considerations. On this still thoroughly Aristotelian conception of pedagogy, knowledge was instilled in *someone*: *habitus* was embodied in persons, and it was understood to take a very long time to acquire. A traditional philosophical education demanded twelve to sixteen years of study before genuine mastery was attained. One did not join the ranks of the *sapientes* overnight.

The paradigmatic expression of this organization of knowledge through dialectical disputation was the *Summa* – above all, Thomas Aquinas’s *Summa Theologiae* (1265–1274), this monument of Scholastic method that was also the standard textbook of advanced theological education. The *Summa* was structured in terms of dialectical disputation: a topic was broached as a *quaestio* (question), opposing arguments were marshalled under *videtur quod* (“it seems that”), and an authoritative counter-statement was provided in the *sed contra* (“but against this”). The author’s own resolution was then argued in the *respondeo* (“I answer that”). Crucially, however, it was followed by the *ad argumenta* (“replies to the objections”), which accomplish something highly characteristic of this way of organizing knowledge: as Anthony Kenny observes, they “go a long way to accommodate the opposite view” (Kenny 1993, 22). The very act of forcing oneself to write out the objections in their strongest form and do them justice in one’s replies forces one to elevate or expand one’s view sufficiently to accommodate both sides of the issue.

This disputational organization allowed Aquinas to preserve the tension between two sides of an argument. Where the later demand for perfect consistency and coherence would encourage one to flatten paradoxes and deny the existence of irreducible conflict, the *Summa* thrived on accommodating both sides of a question and exploiting the productive tensions between them. The positions at which Aquinas arrives are judgments refined through confrontation with competing considerations. The authority of the conclusion rests not simply on its place in an articulated structure, but on the visible process by which it was reached; readers understand the position because they have seen it tested and were brought to appreciate the force of countervailing considerations.

The *quaestio* format of the *Summa* thus tied knowledge to spoken disputation, to the give-and-take of voices in a room, while the notion of *habitus* tied it to a particular person having undergone a long formation. Knowledge, on this model, was embedded in discourse and persons, in a world still fundamentally shaped by the primacy of the spoken word. It could certainly be ordered, often with extraordinary rigor; but it was not yet optimized for rapid surveyability, for uniform replication across many learners, or for relative independence from the master who transmitted it.

This disputational conception was supported by the humanist pedagogy of *copia* (abundance), articulated by Erasmus in his *De copia verborum ac rerum* (1512) and inspired by Cicero and Boethius, whereby one should perfect one’s rhetoric by building up a storehouse of material in commonplace books. The aim was to help the speaker adopt a wide variety of viewpoints and generate a great variety of arguments, examples, and formulations. Commonplace books were so ubiquitous that, as Ann Blair has shown, they constituted the default method of learned reading and writing across Europe for centuries (Blair 2010, 62–83). Because commonplace books organized material by topic rather than by principle, however, they facilitated retrieval and recombination rather than principled derivation. They consisted of

loose collections of excerpts and observations that were organized under topical headings, but otherwise remained radically unsystematic.

Yet it was out of these “commonplaces” that a significant intermediate step on the path to *systema* emerged: the *loci communes* of early Protestant theology reconceived the collection of commonplaces as a method for mapping out the structure of reality itself: topics such as cause, effect, genus, species, similarity, and difference were treated as classes of things that map the real relationships among entities, providing a topical map through which any subject matter could be navigated (Burton 2024, 27–29). Melancthon, in his *Loci Communes* of 1521, presented theology as a series of topics arranged to mirror the intrinsic structure of Scripture. The *loci communes* thus occupied a position midway between the disputational depth of the *Summa* and the architectural rigour of the later *systema*: it organized knowledge topically rather than deductively, but it aspired to track the objective order of its subject matter instead of merely storing material for retrieval. Calvin’s *Institutes*, as Richard Muller has shown, evolved from a catechism into a pioneering system of Reformed theology under the influence of precisely this topical-methodological tradition, in which the right arrangement of theological loci was understood to reflect “the right order of teaching” (Muller 2003, I:177–180). It was this tradition that Keckermann inherited and transformed.

This pre-systematic order thus had its own forms of cohesion, indeed its own depth. Williams’s observation about the “intuitive condition”, in which convictions and commitments form a dynamic and dialectical fabric rather than a unified but static structure, captures something of what this older intellectual world was like. The *quaestio* format exemplified this condition at the level of method: it rendered disagreement productive, holding competing authorities in tension rather than dissolving that tension through derivation from first principles. The humanist tradition of *copia* cultivated plurality, enriching the speaker’s and reader’s resources for approaching a question from multiple angles. And the conception of knowledge as *habitus* prized precisely those qualities that later system-builders would work, not to abolish entirely, but to make less indispensable: personal judgment, accumulated experience, sensitivity to context, and the capacity to weigh competing considerations without reducing them to a common currency.

Even the topical tradition, which came closer to the later *systema* in aspiring to track an objective order, remained committed to a form of intellectual navigation that depended on the reader’s own capacity to identify the right topics and trace the right connections. It provided a map, but one that still required a skilled traveller. A Keckermann-style *systema* covering the same material as a *Summa* would state the doctrine, provide the definition, and move on. The dialectical depth—the sense that the position has been tested against genuine alternatives and shaped through engagement with them—would be attenuated. What would eventually be gained by the transition to *systema* was real, as the rest of this chapter will show: speed, surveyability, reproducibility, and a form of cognitive order less dependent on the

qualities of particular teachers. But something was also relinquished. The counter-tradition that this book tracks, from Margaret Cavendish through Denis Diderot to G. W. F. Hegel, can be understood in part as an extended meditation on what that something was.

The question, then, is why anyone found the proposal to replace this order with a *systema* compelling. The answer – like so many things in the early modern period – begins with the Reformation.

1.2 The Loss of Institutional Authority and Educational Infrastructure

The Protestant Reformation shattered the two foundations on which Scholastic cognitive order had rested. It disrupted not only the Church's monopoly on doctrinal authority, but also the institutional network through which that authority had been reproduced in learned persons. What emerged in its place was not merely a new theology, but a new problem of cognitive organization. If a body of doctrine could no longer rely, in the old way, on the authority of a universally recognized Church, and if the educational institutions that had formed its guardians were themselves weakened, dispersed, or abolished, then order had to be secured differently. Consequently, as Richard Muller has documented, the Reformers “were forced by polemical and practical necessity to develop theological systems and schools designed for creating and maintaining the Protestant movement” (Muller 2003, I:76).

The sixteenth century thus generated two pressures that were analytically distinct, though historically intertwined. The first was a pressure toward replacing inherited authority with articulated doctrinal order. The second was a pressure toward making that order transmissible with new speed, uniformity, and economy.

The first pressure arose from confessional fracture itself. Before the Reformation, the Church could serve, at least in principle, as the final arbiter of doctrinal disagreement. Its authority did not eliminate controversy, but it provided a standing institution within which controversy could be contained. Protestantism, by contrast, weakened precisely that kind of external guarantor. Once multiple confessions claimed access to scriptural truth without acknowledging a single magisterial tribunal, the burden of authority shifted. Doctrine could no longer appear merely as what an institution authoritatively maintained; it increasingly had to present itself as an ordered whole whose internal articulation made its rightness visible, or at least its intelligibility defensible. The thought that order might reside in structure rather than solely in office or inheritance received, in this way, a powerful historical stimulus. The demand for internal structural criteria was simultaneously a demand for a new locus of intellectual authority, one vested in the architecture of a system rather than in the office of an arbiter. The later epistemological and political

consequences that the book will trace in subsequent chapters have their earliest conditions here: once the structure of thought itself begins to carry some of the authority that institutions had previously borne, it becomes possible, in principle, for later thinkers to ask that structure do still more.

The second pressure was pedagogical. The Reformation dismantled and re-configured much of the educational apparatus that had sustained the old order. Monasteries, cathedral schools, and established teaching networks were dissolved or transformed across much of Protestant Europe. If the new confessions were to survive, they had to train pastors, schoolmasters, and administrators rapidly and at scale. They needed ways of compressing doctrine into forms that could be taught efficiently to relative novices, disseminated across dispersed institutions, and reproduced with a reliability less dependent on the exceptional gifts of individual teachers. In this setting, the attractions of articulated, surveyable, portable forms of knowledge became evident. Protestant universities and gymnasia, many of them newly founded, faced the task of transmitting complex bodies of knowledge — ranging from theology and law to philosophy and natural philosophy — efficiently and reliably, to students who could no longer rely on the traditional Catholic infrastructure of training. How else does one produce, in a single generation, the educated class that an older dispensation had cultivated over centuries? This pedagogical emergency demanded forms of knowledge that could operate with less dependence on prolonged formation, on the personal authority of the master, and on the institutional settings in which learning had traditionally taken place. It demanded, in other words, forms in which more of the cognitive work was performed by the arrangement of the material itself.

It is important not to overstate the novelty of either pressure. Medieval theology had long valued order, and every educational regime has practical constraints. But the conjunction of confessional fragmentation, institutional disruption, and print-based dissemination altered the balance of virtues that a body of doctrine was now expected to possess. Under these conditions, the older ideal of *habitus*—slowly cultivated, dialogical, person-bound—began to appear insufficient on its own. It presupposed stable institutions, extended time, and reliable chains of transmission. Protestants could and did continue to prize learned judgment; but they also needed forms of doctrinal organization that could do more of the work externally. The demand was for bodies of learning that could be seen, taught, traversed, and reproduced with greater economy.

This is the point at which the pedagogical rationale for systematicity first becomes historically explicit. The issue was not yet, at least not primarily, how to derive certainty from indubitable first principles, nor how to ground political legitimacy in publicly shareable norms. Those would come later. The immediate question was how to preserve and transmit a body of teaching under new conditions of scarcity and instability. Systematization first answered a problem of reproduction. It arose as a machine for teaching before it became a machine for proving or governing.

The language of “machine” is not meant to insinuate that sixteenth-century pedagogues imagined themselves building literal cognitive automata. But the cognitive virtues they aspired to were in effect the virtues of a pedagogical machine: characteristics that made knowledge surveyable, reproducible, and relatively less dependent on the tacit mastery of formed individuals. Some of the work previously borne by the cultivated person had to be taken over by the structure itself. Order needed to stand there on the page.

No movement more vividly embodied this transfer from person-bound formation to externalized arrangement than Ramism.

1.3 Ramism and the Visual Reorganization of Learning

Petrus Ramus (1515–1572), the controversial French humanist logician, became one of the most influential educational reformers of the later sixteenth century. His followers spread his methods across Protestant northern Europe, especially in Reformed contexts eager for pedagogical simplification and curricular reorganization. Ramism is often remembered for its hostility to Aristotelian Scholasticism and its fondness for dichotomous division. But its deeper significance for the history of systematicity lies in the way it reconceived knowledge as something that could be rendered spatially surveyable and procedurally manageable.

Ramus criticized the inherited Scholastic curriculum for its prolixity, subtlety, and dependence on disputational complexity. In its place, he proposed a method that would proceed by successive binary divisions from the more general to the more particular. Knowledge was to be arranged in trees, charts, and ordered schemata that displayed relations of inclusion and subordination at a glance. The aspiration was not merely economy in the vulgar sense, but pedagogical clarity: to present an art or discipline in such a way that its principal divisions could be quickly grasped, memorized, and taught. Ramus’s insistence that every discipline could be reduced to clear definitions and systematic dichotomous divisions, displayed in branching tables that made the structure of knowledge spatially visible and immediately graspable, offered Protestant educators precisely the kind of scalable didactic technology their predicament demanded. Where a traditional philosophical education demanded twelve to sixteen years of study before genuine mastery was attained, Ramist textbooks promised competence in months.

The principal pedagogical innovation that promised to fuel these gains in efficiency was the Ramist dichotomy chart or “Ramist tree”, which divided a subject into two branches, and those branches into two more, etc. (see Fig. 2). With their rigid, clean, and pedagogical look, such dichotomous diagrams were designed to help students memorize complex subjects by breaking them down into manageable branching pairs. In the 16th and 17th centuries, these diagrams were revolutionary, transposing dense text into a spatial layout that made it easier for readers to see the

logic and structure of a subject at a single glance. Where Aristotle's dialectic retained its connections to speech and disputation, Ramus severed those connections with what Ong calls "savage rigor": the topics were abstracted from rhetoric, judgment was replaced by spatial "arrangement" (*dispositio*), and the oral, interpersonal dimension of knowledge was displaced by a visual, diagrammatic one (Ong 1958, 280–282, 287). Words, in Ramist textbooks, were deployed in spatial relationships, converting knowledge from something *heard* into something *seen*, arranged on the page as a thing to be surveyed (Ong 1958, 89–91). Learning became less bound up with hearing the personal authority of a teacher and more bound up with inspecting the impersonal authority of a structure.

Here one can already see the formal virtues that would later be associated with cognitive system-building more generally. Ramist method prized surveyability: a discipline should be visible as a whole. It prized decomposability: a subject should be divisible into orderly, hierarchically related parts. It prized rule-guided traversal: one should be able to move through the subject by following explicit divisions rather than relying at every step on tacit judgment or the teacher's improvisational skill. And it prized a kind of portability or generalizability: the same formal scheme could, in principle, be applied to many domains.

This was not yet the mature early modern *systema*. Ramism was often too rigid, too reductive, and too suspicious of the texture of real inquiry to furnish a durable model on its own. But it made visible, perhaps for the first time in early modern pedagogy, the thought that knowledge could be reorganized so as to become externally graspable in a uniform visual architecture. What had previously been embedded in processes of conversation, commentary, and prolonged exercise was now increasingly laid out as a structure to be inspected.

Walter Ong captured something essential about this transformation when he argued that Ramism marked a shift from a more oral, agonistic culture of learning toward a more visual and spatial one (Ong 1958). Ramism, in Ong's analysis, was "a cluster of mental habits ... specializing in certain kinds of concepts, based on simple spatial models, for conceiving of the mental and communicational processes and, by implication, of the extramental world" (Ong 1958, 8). In the older tradition, mastery was displayed in the capacity to navigate arguments in real time, to respond, distinguish, and judge. In the Ramist classroom, increasing importance attached to the capacity of a scheme itself to carry order visibly. The chart became, in a modest but significant sense, a cognitive instrument.

The visual form of the Ramist tree is especially revealing. Rather than to let insight grow out of a dialectical exchange, it prescribes a route through knowledge. The student is not asked primarily to dwell on tensions or cultivate a sensitivity to the force of the competing arguments of rival authorities, but to traverse a pre-arranged set of divisions. The scheme therefore itself performs some of the cognitive labor. It guides attention, determines sequence, and renders relations explicit in advance. In that respect, the Ramist diagram may be read as one of the first pedagogical artifacts

in which thought begins to take on a distinctly machine-like profile: a free-standing and orderly architecture, instantly surveyable and algorithmically traversable, with less dependence on the dialectical gifts of a master.

The very starkness that made Ramist method attractive to reformers also made it vulnerable to criticism. Where the old disputational method cultivated judgment through engagement with opposition, Ramist method risked replacing the capacity for balanced judgment with a Procrustean tendency to force every issue into simple dichotomies. Critics complained that Ramist dichotomies hollowed out the content of disciplines, purchasing pedagogical convenience at the cost of philosophical substance (Hotson 2007, 131–132). For theologians engaged in sophisticated confessional polemic, this posed a real problem: Ramus's truncated treatment of logic stripped away precisely the metaphysical resources needed to defend theological orthodoxy against attacks (Burton 2024, 225–226). By 1599, Keckermann summed up a broad consensus in blaming Ramus for what he called “one of the calamities of our time”: the fact that the fundamental disciplines preparatory to all higher learning were “now seldom seen among students” (Hotson 2007, 144). The students arrived knowing the branching tables, but they struggled to construct a syllogism or detect a fallacy. What was needed was a synthesis: a method that preserved the Ramist virtues of clarity, surveyability, and efficiency while restoring the Aristotelian substance that Ramism had stripped away.

This synthesis was Bartholomäus Keckermann's achievement. Before turning to him, however, it is worth noting the broader significance of the Ramism that preceded him. In transforming classroom methods, Ramism transformed expectations about what a body of learning ought to be. A discipline was increasingly expected to be displayable in a compact architecture, to have a visible order that could be apprehended in overview, and to be teachable by means of explicitly articulated divisions. This was still a pedagogical expectation before it became an epistemological one. The demand was that doctrine be teachable and navigable. But once that demand had been established, the idea that order should be carried by articulated form had acquired a new plausibility. And with that, a path had opened towards more ambitious uses of structure.

The importance of Ramism, in other words, lies not simply in what it accomplished, but in what it rendered imaginable. The question it posed to the next generation was whether knowledge could be made at once more intelligible in form and more serious in substance. Keckermann's *systema* was an answer to that question. It transformed the pedagogical reorganization of doctrine into a more durable architecture of knowledge.

1.4 Keckermann's Innovation

Bartholomäus Keckermann (1571–1609) was born in Danzig (now Gdańsk, Poland) and educated in the Reformed intellectual world shaped by Ramism, Aristotelianism, confessional conflict, and pedagogical urgency. He taught first in Heidelberg and then at the Academic Gymnasium in his native city, where he produced, in a brief life, a large number of textbooks spanning logic, ethics, physics, rhetoric, theology, and metaphysics. He produced what has been called a “semi-Ramist Aristotelianism”: a synthesis that conceded to the Ramist demand for strict formal organization while insisting on the substantive richness of the Aristotelian tradition (Hotson 2007, 145–146, 149). In 1600, Keckermann published his *Systema logicae*, the first work to apply the term “systema” to a body of learning in the now-familiar sense. He thereby gave a name to a new way of conceiving what an organized discipline should be. And his *Systema logicae* was followed in rapid succession by a *Systema SS. theologiae* (1602), a *Systema disciplinae politicae* (1607), and a posthumous *Systema physicum* (1610); his collected works were published as the *Systema systematum* (1613–14), the “system of systems”.¹

The novelty can be obscured if one interprets the innovation too lexically. Bodies of doctrine had, of course, long been ordered. Medieval theologians and jurists had arranged material with great sophistication – we glimpsed the sophistication of the *Summa*; humanists and reformers had experimented with topics, methods, compendia, and institutions. What Keckermann introduced was not so much order *tout court* as a more explicit conception of a discipline as a systematically integrated whole whose parts are arranged according to a principled architecture that can be displayed in a form suited to reliable transmission. His *systema* aspired to do more than to offer a digest or a didactic depiction. It aimed to display the inner structure of the discipline itself.

This is why Keckermann occupies such an important place in the genealogy developed in this book: he is neither the first great metaphysician of system nor the first to derive knowledge from first principles in the manner of later rationalism; but he turns the pedagogical need for surveyable and reproducible order into a general model of cognition. Under his hand, the educational problem of how to teach a body of doctrine efficiently begins to yield a broader ideal of what a body of doctrine ought to look like.

Keckermann inherited from Ramism the conviction that knowledge had to be rendered methodical, orderly, and accessible to the learner through visible ar-

¹Keckermann's student and editor Clemens Timpler (1563–1624) then published his *Metaphysicae systema methodicum* (1604), formulating normative principles for building any system whatsoever and thereby effectively providing a meta-system: a system for building systems (Secundant 2016b, 41–56). In 1610, the Ramist educator Johann Heinrich Alsted (1588–1638) also published his *Systema mnemonicum duplex* at Herborn.

rangement.² Keckermann's own threefold organizational template of *praecognita* (principles of a discipline), *systemata* (their methodical ordering), and *gymnasia* (their practical application) was itself an expansion and modification of Ramus's basic methodical scheme (Hotson 2007, 148–151, 157–165; Burton 2024, 232–233). And when Keckermann's devoted students, working under Johann Heinrich Alsted's (1588–1638) editorial direction, undertook the posthumous project of compiling the *Systema systematum*, they found it remarkably easy to resolve all of Keckermann's works into the dichotomous charts characteristic of Ramist presentation. The Ramist skeleton was still there underneath the Aristotelian flesh.

But he also recognized the defects of Ramism when pressed too far. Dichotomous division could impose a schematic neatness at the cost of fidelity to the subject matter, producing structures that might be easy to memorize, yet too thin to sustain serious learning. Keckermann sought to preserve the pedagogical gains of the Ramist reorganization while restoring much of the conceptual density and disciplinary seriousness that Ramism had threatened to evacuate. For instance, Keckermann's engagement with Jacopo Zabarella, the Paduan Aristotelian whose methodological writings were reshaping logic across Protestant Germany, equipped Keckermann to go significantly beyond Ramus in his twofold division of method into the *synthetic*, which retained the Ramist order from universals to particulars, and the *analytic*, which proceeded from particulars to universals (Keckermann, *Praecognita logica*, 150–51). This allowed Keckermann to resist the Ramist assumption that there was a single method for all arts and sciences (Burton 2024, 224–226). Keckermann recognised that different disciplines needed to be treated according to their distinctive character. But he preserved the Ramist demand for a perspicuous, methodically organized presentation of each.

Keckermann's choice of title for *Systema logicae* (1600) marks the moment at which the Stoic-cosmological term is deliberately appropriated and extended to the cognitive domain. Just as the Stoic *systema mundi* designated a cosmos held together by rational principles, Keckermann's *systema logicae* designated a body of doctrine held together by logical structure. To call a textbook a *systema* was to make a substantial philosophical claim, namely that the discipline it presented possessed an internal rational order of a particular kind: that its parts were connected by principled relationships, and that the whole could be grasped through methodical exposition rather than through prolonged immersion.

Because of its roots in cosmology, the notion of a system of knowledge evoked the picture of knowledge as a self-contained, organized whole revolving around a centre — its first principles — in a manner analogous to the Copernican spheres

²Burton has shown that Keckermann's entire education marked him as a "Philippo-Ramist": the Danzig Gymnasium where he studied through 1590 was a stronghold of the Melanchthonian-Ramist synthesis, and his subsequent trajectory through the Universities of Wittenberg and Leipzig brought him into contact with Zabarellan analytical method without dislodging his formative Ramist commitments (Burton 2024, 217–219).

revolving around the sun. The very notion that a philosophy could constitute a *system* — something that, in Ong’s vivid phrase, “whirled dazzlingly around a center in the mind like the Copernican spheres around the sun, a whole self-contained and independent of the rest of reality” (Ong 1956, 237) — was itself a product of the revolution that Copernicus had brought about.

Before 1600, philosophical positions were differentiated not as competing “systems”, but as different *viae* — the *via Thomae*, the *via nominalium* — or as the *opiniones* of different persons. People thought in terms of different persons and their approaches, their “ways,” rather than in terms of self-contained intellectual architectures. Thinkers who described their philosophical position as a “system” were claiming for it a kind of structural completeness and internal coherence that the older language of *via* (a “way” or method) or *opinio* (an “opinion” or approach) had never implied. “It was comforting,” Ong observes, “to think of oneself, or of one’s enemy, as possessing a philosophical ‘system’,” because it “could cover intellectual situations of which one knew really very little”, making the very looseness of the notion of system “one of its greatest recommendations” (Ong 1956, 237).

Keckermann’s pedagogical ambition was to give students “an encyclopaedic education within three years”: the first year devoted to logic and physics, the second to mathematics and metaphysics, the third to ethics, economics, and politics (Hotson 2007, 154). This was conceivable only if the entire philosophical curriculum could be compressed into the form of systems. It required self-contained and methodically organized presentations in which the surveyable structure took over some of the cognitive work previously performed by apprenticeship and accumulated experience. The systematic textbook became, in effect, a teaching machine: a device for producing competent thinkers by exposing them to the right sequence of definitions, divisions, and derivations, regardless of whether a master was present to guide the process. More of the work was now done by the arrangement of the material: by the definitions that fixed meanings, the divisions that displayed relations, the ordered sequences that guided the reader’s progress through the discipline. Less was left to the slow, tacit acquisition of mastery through prolonged apprenticeship. The redistribution of cognitive labour from formed persons to articulated structure is what makes the systematic textbook a machine in the sense that matters for the argument of this book: its virtues are the virtues of a mechanism, reproducibility, inspectability, operator-independence, even as it remains, of course, a thing made of paper and ink rather than of gears and springs.

With Keckermann’s original application of the term “systema” to the cognitive domain, the pedagogical rationale for systematization appears in its clearest form. The *systema* responds to the immense pressure on teachers to deliver large bodies of knowledge to expanding student populations in highly compressed timeframes (Hotson 2007, 15–19, 82–87). A body of knowledge must become teachable in a way that is efficient, robust against variation in teachers, and traversable for relative

novices without much supervision. Textbooks must, in short, become more like machines for transmitting knowledge.

Keckermann's *Systema logicae* marks the beginning of a shift from *habitus* as embodied knowledge to *systema* as a free-standing cognitive architecture dissociated from individual knowers. Philosophy becomes something that can be written down completely and transmitted through textbooks, without reference to cultivated virtues of the philosopher.

This is made explicit by Keckermann's student and editor, Johann Heinrich Alsted (1588–1638). Philosophy “as viewed in the mind,” Alsted wrote, “is a *habitus*, as viewed outside the mind, a *systema*” (Alsted, *Cursus philosophici encyclopaedia*, 1620, I, col. 1).³ The formulation registers how the older conception of knowledge as a set of personal dispositions acquired through formation and dialogue was being supplemented — and would eventually be displaced — by a conception of knowledge as an organized structure that exists outside the mind as an inspectable object. The externalization of cognitive order is palpable in Alsted's distinction: the *systema* is precisely what remains when the *habitus* is projected outward, detached from the knower, and made available to inspection by anyone capable of following its structure.

The shift from *habitus* to *systema* implied a shift from a world in which cognitive authority was vested in persons to a world in which it was vested in structures. This is the deepest consequence of Keckermann's innovation, and the one that ramifies most extensively through the rest of this book: once the locus of cognitive authority begins to migrate from the person to the structure, the structure can be asked to do progressively more. What begins as an instrument of transmission can become an instrument of certification; what begins as an alternative to institutional authority can become a model for a different kind of authority altogether.

For once doctrine is organized so that its coherence appears to rest in an articulated structure, authority itself begins to take on a new form. One can still appeal to Scripture, confession, and office; Keckermann remained thoroughly embedded in a confessional world. Yet the standing of the doctrine is now partly borne by the visible order of its presentation. The authority that had previously resided more unequivocally in institution and tradition is beginning, however imperfectly, to migrate into the structure of knowledge itself.

In Keckermann, that migration remains above all a pedagogical and doctrinal phenomenon. The *systema* is not yet a general theory of legitimacy, nor a modern account of publicly auditable reason. But a pattern has been established whose later significance is unmistakable. If a doctrine's order can be displayed in a structure available to all who learn it, then authority becomes, in part, answerable to articulated form. The groundwork is laid for later aspirations to impersonal accountability: for the thought that what binds the mind, or even what binds a collectivity, should

³“Philosophia est comprehensio disciplinarum liberalium inferiorum: et alias dicitur encyclopaedia ... Comprehensio illa spectatur in mente ut habitus, extra mentem ut systema.”

do so not merely because someone authoritative says it, but because it can be seen to occupy the right place within a publicly graspable order.

The same point also yields the earliest precondition of the epistemological rationale. Keckermann's *systema* is not yet designed to secure certainty against skepticism; it presupposes, rather than manufactures, a body of teaching to be organized. But by making order increasingly internal to the arrangement of the discipline, it helps prepare the thought that correctness might be legible in form itself. Once knowledge is expected to carry its order within its articulated structure rather than resting wholly on external institutional guarantee, the path is open to later thinkers who will ask whether the right structure can itself certify the rightness of the content. Chapter 2 will follow that radicalization. Here we are still at an earlier moment, where the decisive innovation is not self-certification, but externalized intelligibility.

What drove Keckermann's innovation were thus pressing practical needs, and recognizing this transforms our understanding of what the concept of system originally meant. The *systema* was a technology devised in response to the loss of Catholic institutional authority and educational infrastructure. Without an established church hierarchy to enforce theological unity, Reformers had to devise internal structural criteria. The ideal of systematicity supplied those criteria in the form of consistency, coherence, completeness, and parsimonious principledness. The authority that had resided in an institution was relocated to a structure. And a structure, unlike an institution, could be inspected by anyone who understood its principles. As Timothy Watson's recent study puts it, Keckermann's innovation "gives the logician the directing role as a technical expert in mining the truth of theology, philosophy, and other sciences," a role that "flies in the face of the medieval view of the Church and its elders controlling that role" (Watson 2023, 15).

At the same time, Protestant territories faced an urgent need for trained professionals, but could no longer rely on the old Scholastic model of lengthy apprenticeships. Newly founded Protestant schools had to develop efficient, reliable methods to quickly transmit complex theology and secular knowledge if they were to rise to the challenge of churning out an entirely new educated class as fast as possible. And the vehicle for the system as a pedagogical machine was the printed textbook: systematization was inseparable from its material conditions of reproduction.

1.5 Printing, Paper Technologies, and the Material Ecology of System

The confessional crisis was only one of the pressures to which cognitive systematization formed an answer. A second pressure, operating simultaneously and reinforcing the first, was the sheer material explosion of available knowledge that the printing press had unleashed. What Keckermann achieved conceptually in the form of the

textbook was paralleled, enabled, and in some respects anticipated by a wider set of paper technologies through which early modern Europe learned to handle the growing abundance of knowledge.

By the time Keckermann published his *Systema Logicae*, the printing revolution was already more than a century old, and its cumulative effects had become impossible to ignore. The rapid multiplication of printed texts had created something for which the older intellectual culture possessed no adequate response: a surfeit of information that exceeded any individual's capacity to absorb. The *multitudo librorum*, or multitude of books, was a matter of widespread comment and growing anxiety among the learned by the mid-sixteenth century (Blair 2010, 11–15, 55–61). Erasmus, in his commentary on the adage *festina lente* (“make haste slowly”), gave this anxiety one of its most memorable expressions: “Is there anywhere on earth exempt from these swarms of new books? Even if, taken out one at a time, they offered something worth knowing, the very mass of them would be a serious impediment to learning” (Blair 2010, 55–56). The flood of new publications, many of dubious quality, was pulling readers away from the ancient authorities that humanist pedagogy cherished and overwhelming the topical methods of organization that had served the learned world tolerably well in the age of manuscripts.

The same technology that had made the Reformation possible — Luther's theses circulated at unprecedented speed thanks to print — thus also generated a problem of cognitive management that the Reformation's own pedagogical institutions then had to solve. Print broke up the Catholic monopoly on doctrinal dissemination by multiplying Protestant texts alongside Catholic ones. This produced an information environment in which it became increasingly challenging for any single reader to survey the whole. Confessional competition spurred publication, and the resulting abundance of competing claims generated an acute practical need for the reimposition of cognitive order and surveyability.

The connection to cognitive systematization is direct. It was as a reaction to this confusion about what followers of the new faith were supposed to believe, a confusion exacerbated by the printing press, that Protestant theologians began to publish synoptic expositions of the new theological doctrines and refer to them as “systems” (Ritschl, 1906, 16). Information overload did not by itself generate the ideal of system. But it made forms of order that promised economy and navigability much more attractive. David Simpson (1993) and Clifford Siskin (2016) have developed this point by arguing that print and system were mutually reinforcing developments: cognitive systematization was a response to information overload, and systems also took hold thanks to the standardization and easy reproducibility that print introduced. The printed textbook, unlike the copied manuscript, was identical across all its instances; every student working from the same edition encountered the same definitions, the same divisions, the same ordered sequence. The cognitive architecture encoded in the textbook was mechanically reproducible and emancipated from the original systematizer: the printing press guaranteed that

the system's structure would survive transmission without distortion, as Elizabeth Eisenstein argued in her account of printing as an "agent of change" that introduced fixity and standardization into a previously fluid manuscript culture (Eisenstein 1979, I:113–126). If the *systema* was a machine for organizing thought, the printing press was the machine that replicated that machine across an entire educational infrastructure. The entanglement of systematization and mechanization was present from the very beginning, embedded in the material conditions of the *systema*'s production and dissemination.

Cognitive systematization was part of a broader emergence of technologies of knowledge management in the face of this deluge of information. Techniques for excerpting and recombining textual material also gained in importance. Indices, alphabetized dictionaries, cross-referenced encyclopedias, and increasingly sophisticated methods of note-taking all represented attempts to impose order on an unruly abundance (Blair 2010, 137–144, 210–225). Knowledge was increasingly handled not only as continuous discourse to be followed from beginning to end, but as material that could be segmented, sorted, and rearranged. In this respect, the material culture of paper helped foster a conception of cognition in which the organization of knowledge could be partially externalized and materially manipulated.

An emblematic example is the technique devised by the Swiss polymath Konrad Gessner. In his *Pandectarum sive Partitionum Universalium* (1548), the second volume of his monumental *Bibliotheca Universalis*, Gessner published what Markus Krajewski has identified as perhaps the earliest explicit description of a procedure for generating and managing knowledge through the decomposition of texts into mobile, rearrangeable units (Krajewski 2011, 3–4, 13–14). The algorithm was simple: one copies passages of importance onto a sheet of paper, using one line per idea; one cuts the sheet into slips with scissors; one arranges the slips into clusters; one subdivides the clusters as often as necessary; and one fixes or copies the final arrangement (Krajewski 2011, 13). The slips could be glued with water-soluble paste into hybrid books, threaded onto rails in specially constructed cabinets, or pinned with needles into the desired order for a lecture, before being returned to their box for reuse (Krajewski 2011, 13–14). Gessner organized the knowledge surveyed in his *Pandectae* under *loci communes* arranged in a twenty-one-class tree structure, with grammar first and theology last; the commonplaces were "complemented and enhanced according to individual need," generating an open-ended system of topical coordinates for the classification of everything one read (Krajewski 2011, 10–12).

What makes Gessner's technique significant for the genealogy of the *systema* is the operation it performed on knowledge itself: decomposition into discrete, uniform, mobile carriers that could be arranged and rearranged according to strict systems of order. Krajewski characterizes this as the defining principle of the "paper machine" in contrast to the bound codex: within a paper machine, "information is available on separate, uniform, and mobile carriers and can be further arranged

and processed according to strict systems of order” (Krajewski 2011, 3). The logic at work here is the logic of conceptual mechanization: knowledge becomes divisible into units, rearrangeable by principle, surveyable in external form, retrievable independently of memory, and available to rule-governed handling.

This is the material counterpart of what Keckermann would achieve at the conceptual level a half-century later: breaking the continuous stream of knowledge into discrete units, organizing those units systematically, and making the resulting structure surveyable. The paper machine and the cognitive *systema* were twin responses to the same problem, operating on different levels, the one material, the other conceptual. Taken together, they reveal a broader ecology in which knowledge was coming to be treated as something that could be decomposed, externalized, and managed through structured procedure, an ecology in which the image of ordered, externally manipulable cognition was becoming increasingly salient. Such undertakings did not merely respond to abundance; they taught scholars to expect that abundance should be rendered navigable by articulated external order. The paper technology enacts the same logic that the *systema* would enact conceptually: decomposition, ordered recomposition, and the transfer of organizational work from memory into externally manipulable structure.

This broader ecology also helps explain why the pedagogical rationale for systematization quickly acquired wider resonance. A body of doctrine arranged as a *systema* possessed many of the same virtues that made indexes, tables, and bibliographies valuable. It shifted the work of organization outward: cognitive order became less wholly immanent to an accomplished mind and more visibly dependent on structures that could be built, inspected, and used. Seen in this light, Keckermann’s *systema* belongs to a larger family of early modern devices for coping with complexity.

Together, the paper machine and the cognitive *systema* signal a transition towards the treatment of knowledge as something that could be rendered navigable through systematization. As Matthew Daniel Eddy has shown, it is the allure of these machine-like virtues that later incited Adam Smith to systematize the syllabus in logic and moral philosophy at Glasgow University, and it is no coincidence that this was happening in Calvinist Scotland: “the value assigned to the system as a developmental tool in Scotland and other Reformed educational contexts was in part an extension of the Calvinistic theological tradition that had shaped the learning experiences of students from the sixteenth century onward” (Eddy 2023, 271). Smith’s efforts to systematize the teaching of philosophy underscore the durability of the connection between Reformed pedagogy and systematization traced in this chapter.

But why did Keckermann revive the Stoic word *systema*? Part of the answer may be that the term already suggested an organized whole whose parts stand together in rational relation. The term *syntēma* may have owed its new cognitive currency to the transition from the Ptolemaic to the Copernican conception of the cosmos

(Ong 1956, 232, 234). In the Ptolemaic cosmos, there was really only one *systēma*: a single, hierarchically ordered universe of concentric spheres. This made the notion of system “a rather uninteresting one” (Ong 1956, 234). The Copernican hypothesis changed this. By proposing an alternative conception of how the celestial bodies were arranged, Copernicus turned the Ptolemaic cosmos into one system among others, which could be set against the Copernican system. And the Copernican cosmos itself was a system that contained smaller systems within it: the earth and the sun, the moon and the earth, Jupiter and its satellites. The notion of system became interesting when it became plural, and it became plural when Copernicus, and Galileo’s *Dialogue Concerning the Two Chief World Systems* (1632) after him, revealed a universe that was not a single hierarchical order, but a nesting of organized wholes within organized wholes (Ong 1956, 234–235; Siskin 2016, 22–23).

But that is only part of what would have made the term “systema” appealing. Keckermann’s use also helps collapse the distinction between the order of things and the order of teaching. A *systema* is no longer only a composite in the world; it is also a body of doctrine whose arrangement is supposed to reflect the order of its subject matter.

That aspiration marks an important step beyond the looser organizational forms of *loci communes*, commonplace books, and Ramist trees. Those forms aided retrieval, recollection, and topical navigation, but Keckermann’s *systema* aims at something stronger: the principled articulation of an order that is not merely convenient, but *appropriate* to the discipline. This is one reason the *systema* can begin to bear authority. If its arrangement were arbitrary, it could not do so. But if the order of exposition is taken to reflect something objective about the order of its subject matter, then the systematic form acquires an authoritative dignity as well as a pedagogical function. It no longer appears merely as a teaching aid. It begins to present itself as an image of rational order itself. It is here that the metaphysical warrant enters.

1.6 The Metaphysical Warrant: First Appearance

An assumption embedded in Keckermann’s enterprise that his own pedagogical arguments do little to justify, and that will become one of the central preoccupations of this book’s narrative, is that the domain being systematized *admits* of systematization: that the disciplines possess an internal logical structure which the system merely makes explicit. Ritschl observed that early seventeenth-century theorists of *systema* operated on what he called a “pre-critical” assumption: the belief that all truths, whether evident or supernatural, exist objectively within or beyond the world, and that the knowing subject’s task is simply to discover and correctly reproduce this objective order through proper method (Ritschl 1906, 49–50). For Keckermann and his immediate successors, this presupposition was underwritten

by a theological conviction so deeply held as to require no argument: God's creation displays a rational order, and human knowledge, when properly organized, reflects that preexisting order.

Here the Stoic ancestry of the term matters again. When Keckermann borrowed *systema* from cosmology and applied it to doctrine, he carried with him the Stoic assumption that the organization of knowledge should mirror the organization of the cosmos. The Stoic *systema mundi* was a rationally ordered totality governed by *logos*; Keckermann's *systema logicae* aspired to be a rationally ordered totality governed by logic. His conviction that logic and other disciplines can be systematized was grounded in the conviction that reality itself is systematic; the structure of a well-made textbook mirrors, at a remove, the structure of a well-made world.

If the notion of a philosophical "system" became thinkable through the analogy with Copernican world-systems, then what I call the *metaphysical warrant* — the assumption that the systematization of thought is warranted by the antecedent systematicity of reality — was originally part of what supported the concept's extension from the ontological to the epistemological. The cosmological metaphor promised that the orderly arrangement of parts in a textbook was continuous with the orderly arrangement of parts in the world.

Yet to describe this assumption as tacit or unexamined, as though Keckermann simply assumed what he never articulated, would be to understate its role in his thought. Keckermann conceived of the relationship between logic, philosophy, and divine truth in terms of a series of images, in which each level of the cognitive hierarchy represents the one above it. Logic, he held, relates to philosophy "as an image represented in a mirror to the face which it represents," handling the "images and simulacra of philosophical things as instruments of knowing" (Keckermann, *Praecognita philosophica*, 111–12). And, "realized in encyclopaedic form, the whole of philosophy can be seen as an image of divine truth itself" (Burton 2024, 231–232). This is a version of the metaphysical warrant, one whose articulation draws on the Augustinian-Franciscan tradition of *exemplarism*, according to which all created structures participate in and reflect the divine ideas. Keckermann effectively treats the systematic organization of knowledge as mirroring the rational order of creation, which in turn mirrors the rational order of God's mind.

In fact, the Ramism that Keckermann builds on itself grew out of a long tradition of topical Realism – the philosophical belief that the categories and logical structures used to organize human knowledge actually correspond to the objective, divine structure of the universe. This tradition reached back through Rudolph Agricola and Philipp Melanchthon to medieval Neo-Platonic traditions, and it remained, as Simon Burton has recently argued in a comprehensive study, a fundamentally Platonist enterprise: the conviction that logical structure mirrors the intelligible order of reality, and hence that the divisions uncovered by dialectical analysis are divisions inscribed in things themselves (Burton 2024, 44–83). "The dialectic of

the mind,” Ramus maintained, was a light “emulating the eternal and blessed light” and the “image of the parent of all things” (Burton 2024, 56).

From the beginning, the early modern rise of the *systema* was thus accompanied by a justificatory intuition that would give the new ideal much of its later prestige: the thought that reality itself is ordered, and that a properly organized body of knowledge ought therefore to mirror that order. This intuition did not create the practical need for systematization. The pedagogical and media pressures described above were already sufficient to generate a demand for more surveyable, reproducible, and articulable forms of doctrine. But the metaphysical warrant elevated that practical technology into something possessing not merely usefulness, but dignity.

That dignity matters, because it helps explain the peculiar stability of the systematicity ideal once established. A merely convenient pedagogical format might have come and gone. The *systema* became something more durable because it could be defended on two levels at once. Practically, it solved urgent problems of teaching and transmission. Metaphysically, it could be presented as answerable to the very structure of the created world. The two levels were mutually reinforcing. The form was useful because the world was orderly; the world’s order, in turn, seemed to justify the form.

The metaphysical warrant should not be mistaken for the original spur to cognitive systematization. It did not, by itself, explain why *systema* emerged when and where it did. Belief in divine order long predated Keckermann. Medieval thinkers had also inhabited a created cosmos without calling their bodies of doctrine *systemata*. What was different in 1600 were the technological conditions and the acute pedagogical needs.

This priority of the pedagogical rationale matters for the argument of the book as a whole. One of its central claims is that the ideal of systematicity arose not because philosophers first discovered a timeless truth about the essence of reason, but because historically situated actors needed ways of generating reliable cognitive order under pressure. The metaphysical warrant was powerful, and in some contexts indispensable, but it attached itself to an ideal whose practical utility had already become evident. God may have been invoked to justify systematization, but systematization first emerged in response to earthly problems.

Even so, once the warrant was in place, it transformed the horizon of the enterprise. A *systema* that merely transmitted doctrine efficiently remained a pedagogical device. A *systema* that mirrored the rational order of creation could aspire to more. It could become the visible expression of a world in which truth itself is systematic. And once that possibility had been glimpsed, the path opened toward the more ambitious epistemological projects of the seventeenth century: projects that would no longer be satisfied with a system because it teaches well, but would seek a system because only a properly ordered body of knowledge seems capable of securing certainty, completeness, and rational necessity.

The metaphysical warrant thus plays a transitional role in the genealogy. It does not originate the *systema*, but it helps prepare its expansion beyond pedagogy by providing assurances that the systematization of thought promotes *adaequatio intellectus ad rem* – the adequation of the intellect to things, is truth-tracking. The question of what happens when this assurance is withdrawn — of whether the demand for systematization can survive the collapse of its metaphysical warrant — is the question that structures the rest of this book.

1.7 A Historical Mirror: The Promise of Compressing Intellectual Labour

The transition from *Summa* to *systema* instantiates a pattern that recurs whenever new technologies of externalization promise to compress the labour of understanding. The *systema* offered to make the conclusions of extended inquiry available to learners who had not themselves undertaken the inquiry: the definitions stood ready, the divisions were laid out, and the ordered sequence guided the novice from principle to application. What the *Summa* had preserved, and what the *systema* compressed out, was the dialectical process through which those conclusions had been reached: the *quaestio*, the marshalling of objections, the *sed contra*, the painstaking accommodation of countervailing considerations in the *ad argumenta*. When a reader worked through the *Summa*, the path to the conclusion was itself part of what was understood. A Keckermann-style textbook could promise to compress a twelve-year formation into three years of study because it claimed to have found a shortcut to the destination.

Each subsequent technology of knowledge management has renewed this promise in its own idiom. The encyclopaedia, the card index, the searchable database, the algorithmically generated summary: each makes it easier to arrive at conclusions without traversing the terrain from which they were won. And each borrows, in its own register, the virtues that the *systema* first aspired to realize: efficiency, surveyability, and reproducibility. The machine-like character of systematized knowledge, figurative in Keckermann's textbook, became progressively less figurative as technologies have advanced. What has remained constant is the animating hope that the intellectual path can be shortened without significant loss, because what matters is the product of inquiry rather than the process by which it was forged.

The contrast between *Summa* and *systema* sharply throws into reliefs why that hope is often bound to be misguided. Knowledge is more than a collection of distinctions and propositions. It encompasses embodied intellectual dispositions, cultivated through sustained dialectical engagement, that shape the learner's capacity to notice things, to draw connections, to anticipate objections and weigh their force. True learning is not just informative, but *formative*.

This was well understood by the medieval pedagogical tradition revolving around the *Summa* and the cultivation of *habitus*. Aquinas's *ad argumenta* forced the reader to confront the strongest available objections, and to appreciate why they carried weight, because he knew that this made a difference to what the reader came away with, and to their ability to hold on to tensions that could not fully be resolved. He also knew that the intellect is more than a repository waiting to be stacked with information. Sustained engagement with countervailing considerations over hours, months, or years leaves the mind altered in ways that a distillation of the results cannot replicate.

In part, this because the connections a learner draws in the course of that engagement are shaped by the particular intellect doing the drawing: by what it already knows, by what it finds puzzling, by the unexpected adjacencies it discovers along the way. It is often those idiosyncratic connections that enable the true appropriation of knowledge.

By contrast, a system, whatever its material substrate, is a network of pre-drawn connections. This is what makes it so powerful as a technology of transmission. It delivers the same connections across every copy and every student. But this is also what limits its capacity to be formative. The very uniformity that enables a *systema* to be teach at scale forecloses the idiosyncratic pathways that sustained engagement would have opened. The learner's network of connections becomes something received rather than something individually constructed.

The sixteenth-century transition thus illuminates a recurring structure. Whenever new means of externalization make knowledge appear increasingly detachable from the cognitive labour of the learner, people are tempted by the promise of a shortcut. But the shortcut is self-defeating when the journey is the point. No matter how perspicuously the supposed fruits of inquiry are displayed in the synopsis, the tension between efficiency and depth of engagement remains, and something threatens to be lost. What is lost thereby is harder to display, because it consists precisely in the kind of understanding that only the longer path could have cultivated: the *habitus* that Alsted distinguished from the *systema*, the disposition that takes shape in a particular mind through a particular encounter with the material, and that no externalized architecture can fully encode. The question whether understanding can be compressed without remainder into its products is as old as the *systema* itself. It has lost none of its force.

It cannot be denied that at some level, the pedagogical machine that was Keckermann's *systema* worked. Within a generation, the concept had migrated from the classrooms of the Danzig Gymnasium into the mainstream of European intellectual life, carried by the Calvinist pedagogical network and propelled by the printing press. The *systema* organized knowledge; it made knowledge surveyable; and it facilitated its transmission. It relocated cognitive authority from persons and institutions into articulated structure, and in doing so it gave thought a form whose functional virtues were the virtues of machinery: efficiency, surveyability, impersonality, and

reproducibility. The pedagogical rationale explains why this first system arose. The machine emulation thesis describes the form it took. And the incipient relocation of authority from institution to structure prepared the conditions from which later, more ambitious rationales would emerge. What Keckermann's system did *not* do — what its inventor never asked it to do — was *certify* that the knowledge it organized was true. The Keckermann-style textbook assumed the reliability of the doctrines it systematized. It asked how best to arrange, compress, and transmit them.

This assumption was about to come under serious pressure. The same decades that saw the rise of the cognitive *systema* also saw the revival of an ancient philosophical challenge that struck at the very possibility of secure knowledge, a challenge that Keckermann's pedagogical machinery was wholly unequipped to answer. A system that organizes received opinion without certifying its foundations organizes something that may be worthless. The seventeenth century's philosophical innovation was to demand that the system do more: that it serve as an instrument for producing certainty, that the rational structure of a system be itself a guarantee of truth. The pedagogical machine was about to become an epistemological one. That transformation, and the resistance it provoked from the very beginning, is the subject of the next chapter.

2

The Geometric Order: Descartes, Hobbes, Spinoza, and Cavendish

“True knowledge proceeds from cause to effect – except that so far as I know they never conceived the soul, as we do here, as acting according to certain laws, like a spiritual automaton.”

— Spinoza

The *systema* of the Reformed pedagogues took a decisive first step towards replacing the external authority of the Church with internal structural criteria. Armed with these criteria, one could inspect the architecture of a body of knowledge without wholly deferring to an institutional guarantor. In one important respect, however, the ambitions of Ramus, Keckermann, and Alsted remained comparatively modest: they did not yet feel compelled to show that truth was attainable in the first place. Their concern was to make doctrine teachable. The early *systema* was therefore primarily a pedagogical rather than an epistemological machine: a device for presenting knowledge didactically, one that promised to compensate for the loss of Catholic educational infrastructure by training people quickly, economically, and at scale.

Yet this left unanswered a question that the seventeenth century would feel with mounting urgency: can systematisation do more than organise beliefs which may, for all we know, be false? Can the order of a body of thought do more than display its contents? Can it help *certify* them? This chapter examines the point in the genealogy of cognitive systematisation at which that heavier burden is first imposed. System is no longer valued chiefly because it renders thought transmissible, surveyable, and teachable. It is increasingly asked to make belief truth-conducive, or at least truth-accountable, through the articulated order of its own internal connections.

The claim of the chapter is that, in response to sceptical crisis, seventeenth-century philosophers increasingly reconceived systematic order as a source of epistemic authority, and in doing so progressively remodelled the virtues of thought on the virtues of machines: repeatability, transparency, impersonality, and rule-governed necessity. That claim needs careful handling, because several distinct ideas

stand close together here. A body of knowledge can be ordered so as to be easier to teach; it can be ordered so that the grounds of its claims are easier to inspect; and it can be ordered in such a way that the very process of reasoning comes to look increasingly procedure-like, as though correct thought were something that could be run rather than merely exercised. The seventeenth century does not collapse these possibilities into one. It explores their convergence. What it discovers, or hopes to discover, is that order might do epistemological work precisely because good reasoning can be recast in a more impersonal, reproducible, and partially operator-independent form.

The four philosophers examined here pursue that aspiration in different ways, and the differences matter. With Descartes, correct reasoning acquires a methodical form whose reliability is meant to depend less on the cultivated judgement of particular thinkers than on the exact execution of rules; yet the mind that grasps first truths still retains a decisive exemption from mechanism. With Hobbes, that exemption is revoked: reasoning itself is redescribed as computation, and the aspiration to public inferential order extends from science into the structure of the state. With Spinoza, the philosophical text itself is fashioned as an inferential machine, while the rational mind is conceived as an *automa spirituale* governed by its own laws. And with Cavendish, working in the same intellectual world, the costs of this model are first brought clearly into view: what such order illuminates, what it suppresses, and what forms of activity and understanding it renders difficult to see. The seventeenth century's central contribution to the genealogy of cognitive systematisation lies here: in the attempt to make systematic order do the work of epistemic certification by remodelling thought on the profile of a machine.

The pressure behind this development was the increasing distrust of the thinker's own fallible judgement. As Richard Popkin documented in a study that has shaped every subsequent account of the period, the early seventeenth century was transformed by the rediscovery of ancient Pyrrhonian scepticism (Popkin 2003, xvii–xxi). The result was what Popkin called *la crise pyrrhonienne*. Under the combined pressure of Reformation conflict and renewed scepticism, intuition, tradition, and inherited authority no longer seemed able to secure truth with the confidence they once commanded. The great system-builders of the seventeenth century responded by seeking forms of order that could withstand doubt without resting on the say-so of any institution or person. As Paul Franks has argued, the connection between scepticism and the drive towards philosophical system proved to be a durable structural feature of modern European thought from Descartes to German Idealism: the more radical the sceptical threat, the more comprehensive the systematic response it seemed to demand (Franks 2005, 1–12). The story of that escalation begins here.

2.1 The Collapse of the Criterion

To appreciate what Descartes, Hobbes, and Spinoza were attempting, one must first appreciate the problem they were trying to solve. Intellectually, this was a problem more radical than anything the Protestant pedagogues had faced. Keckermann and his colleagues and students operated within a broadly shared Reformed faith. They needed to organise truth for efficient transmission; they did not need to show that truth was attainable in the first place. By the early decades of the seventeenth century, however, even the *possibility* of truth had become philosophically precarious.

Martin Luther's challenge to papal authority was an epistemological as much as an ecclesiological revolution. The Catholic position held that Scripture requires authoritative interpretation, and that this authority belongs to the Church. Luther and his followers countered with the principle of *sola scriptura*: the Bible is the sole authority required, and the faithful conscience, illuminated by the Holy Spirit, can discern its meaning directly. Each side claimed to be able to show that the other's criterion for distinguishing true doctrine from false rested on an unjustified assumption. The Catholic criterion – Church tradition – required one to accept the Church's authority on the Church's own say-so. The Protestant criterion – the inner testimony of the Spirit – required one to trust a private experience whose reliability was open to challenge. Any attempt to adjudicate between the two criteria then called for a further criterion, thereby generating either an infinite regress, a dogmatic stopping-point, or a vicious circle (Popkin 2003, 3–16).

This was the ancient problem of the criterion as formulated by Sextus Empiricus, now revived by the confessional conflicts of early modern Europe. In fact, Sextus Empiricus's *Outlines of Pyrrhonism*, which catalogues the tropes apt to produce precisely this kind of impasse, had become available in Latin from the 1560s onward. Montaigne's *Apology for Raymond Sebond* (c. 1576) then deployed Pyrrhonian arguments with extraordinary breadth, pressing them against the senses, against reason, and against every major domain of human understanding. The *Apology* both expressed and aggravated a loss of confidence in inherited foundations. This was the situation to which the next generation of philosophers had to respond: a situation in which the criterion for distinguishing truth from falsehood itself menaced to collapse.

For the trajectory of cognitive systematization, the *crise pyrrhonnienne* precipitated a decisive transformation in the rationale for systematization. So long as trusted institutions or ordinary criteria suffice to certify knowledge, systematicity could remain a secondary concern: a way of organising what was already taken to be known. Once those institutions and criteria were thrown into doubt, however, systematicity came to be asked to bear a heavier burden. It was increasingly expected to provide, through the explicit articulation of internal connections, the certification that external authority no longer supplied. What the systematizers of the seventeenth

century sought was therefore not merely well-ordered knowledge, but knowledge whose order was itself meant to help certify it.

This shift is worth distinguishing from two neighbouring ideas with which it is easily confused. One can order knowledge so as to display it more clearly. One can order knowledge so that the grounds of each claim become more inspectable. And one can go further still, treating the very process by which claims are validated as something that ought to proceed through explicit, repeatable, rule-governed operations. The seventeenth-century systematizers progressively explore all three possibilities, and it is this progression that gives the period its distinctive place in the genealogy of cognitive systematisation. The cognitive system begins as, first, a medium of surveyability; then a vehicle of epistemic vindication; and then, increasingly, a model of thought whose virtues are recast in machine-like terms: transparency, repeatability, impersonality, and a relative independence from the cultivated judgement of particular thinkers. If order is to certify, it must be exact enough to do more than display. It must seem able, at least in part, to run by itself. This shift is worth distinguishing from two neighbouring ideas with which it is easily confused. One can order knowledge so as to display it more clearly. One can order knowledge so that the grounds of each claim become more inspectable. And one can go further still, treating the very process by which claims are validated as something that ought to proceed through explicit, repeatable, rule-governed operations. The seventeenth-century systematizers progressively explore all three possibilities, and it is this progression that gives the period its distinctive place in the genealogy of cognitive systematisation. The cognitive system begins as a medium of surveyability; then becomes a vehicle of epistemic vindication; and then, increasingly, a model of thought whose virtues are recast in machine-like terms: inspectability, reproducibility, impersonality, and a relative autonomy from the idiosyncratic judgement of particular thinkers. If order is to certify, it must be exact enough to do more than display. It must seem able, at least in part, to run by itself.

2.2 Descartes: Method as Cognitive Machinery

It was into this landscape that René Descartes entered, a young man formed by precisely the tradition he would later present himself as overturning. Educated at the Jesuit college of La Flèche between 1607 and 1614, he was steeped in the Scholastic-Aristotelian curriculum shaped by the *Ratio Studiorum* of 1599, a curriculum that still bore the marks of Ramist reform and Renaissance humanism. His retrospective verdict on this formation in the *Discourse on the Method* registers, in autobiographical form, the epistemological crisis sketched above: 'I found myself beset by so many doubts and errors that I came to think I had gained nothing from my attempts to become educated but increasing recognition of my ignorance'

(AT VI.4, CSM I.113). While Keckermann had sought a system for transmitting knowledge assumed to be reliable, Descartes confronted the possibility that the knowledge transmitted to him was worthless. Yet the self-presentation is in crucial respects misleading. As Brian Copenhaver has recently argued, the philosophy Descartes encountered at La Flèche (and later at Leiden) was far from the intellectual wasteland the *Discourse* implies (Copenhaver 2024). It was a complex and evolving body of practice, shaped by post-medieval innovations in method, in meditation – the practice of preparing the soul for death – and in what Copenhaver calls “mechanized explanatory picturing” (xiii): the use of printed diagrams and images as instruments of philosophical argument. Franco Burgersdijk, the most celebrated philosopher in Leiden when Descartes studied there, had already attempted to reform the post-Ramist inheritance into a teachable logical system to rival Keckermann’s. Descartes repudiated these predecessors, but he rebuilt philosophy with tools fashioned from the very tradition he disavowed. His rhetoric of rupture was an instrument of what Copenhaver aptly calls ‘oblivion’ (2024, 216–225): the disciplinary forgetting of the pre-Cartesian past, which obscured the pedagogical tradition that had prepared the ideals of order and method he would radicalise.

The Cartesian contribution to the genealogy of cognitive systematisation lies in a specific transformation of those ideals. Descartes proposed to mechanise the *procedure* of inquiry: to redesign the conduct of reasoning so that it exhibits, in its formal profile, some of the virtues admired in a well-built mechanism. The claim requires careful statement, because it involves a selective mechanisation whose boundary is philosophically significant. Descartes mechanised the method by which the mind investigates the world, and he mechanised the body in which the mind is housed, but he left the thinking subject itself untouched. This selective exemption is the defining feature of the Cartesian position, and grasping it is essential to seeing how the machine emulation thesis will develop in the hands of Hobbes and Spinoza.

The philosophical motivation for mechanising method was the sceptical crisis itself, and Descartes’s response finds its fullest expression in the *Meditations on First Philosophy* (1641). In its opening paragraph, Descartes declares that the falsehoods accepted in childhood, and the unstable edifice subsequently built on them, made it necessary ‘once in the course of my life, to demolish everything completely and start again right from the foundations’ if he wished to establish anything stable in the sciences (AT VII.17, CSM II.12). The metaphor of demolition and reconstruction echoes the collapse of Scholastic architecture and the felt need to replace it with something that can bear the weight of knowledge without relying on institutional certification. The *Discourse* makes the anti-Pyrrhonian intent explicit: ‘Not that, in so doing, I imitated the sceptics who doubt only for doubting’s sake ... my whole plan had for its aim assurance and the rejection of shifting ground and sand in order to find rock and clay’ (AT VI.29, CSM I.125). The Reformers had attempted to ground religious certainty in the inner testimony of the Spirit, and then to anchor

that certainty in God as its objective guarantor. Descartes, as Popkin (2003, 143–157) has argued, followed an identical path on the plane of rational knowledge. The method of radical doubt pushes scepticism to its utmost limit: past the unreliability of the senses, past the dream hypothesis, all the way to the hypothesis of a malicious demon who might distort the operations of reason itself. At the extreme of this self-imposed crisis, the *cogito* emerges as a truth so subjectively certain that no conceivable doubt can dislodge it. From this first certainty, the criterion of truth is discovered: what makes the *cogito* indubitable is its clarity and distinctness. The criterion leads to God, whose existence is demonstrated from the idea of perfection; and God, being no deceiver, guarantees the objective reliability of clear and distinct perception. The epistemological rationale for system is visible here in its earliest and most explicitly anti-sceptical form: system is valued because its articulated order promises to certify knowledge that no external authority can any longer vouch for.

What matters most for the genealogy, however, is the specific character of the procedure by which Descartes proposes to effect this reconstruction, because it is here that the machine emulation thesis first acquires a clear philosophical profile. In the *Rules for the Direction of the Mind* (*Regulae ad directionem ingenii*), his earliest sustained methodological treatise, Descartes defines method as consisting in ‘reliable rules which are easy to apply, and such that if one follows them exactly, one will never take what is false to be true’ (AT X.372). The emphasis falls on rules, exactness, and ease of application: on a procedure so clearly specified that arriving at the truth simply becomes a matter of following instructions. Error enters, on this picture, through defective execution rather than through any lack of judgement or understanding. The aspiration is to redesign the process of reasoning so that it exhibits, in its formal profile, some of the virtues admired in a well-built mechanism: it easily and exactly steers one towards the desired result, with relative independence from the cultivated judgement of its operator. This effectively makes the method *operator-independent*: as long as the rules are correctly executed, it produces the same results regardless of who applies it. Descartes does not reduce thinking itself to mechanism; he still presupposes a subject capable of *intuition*, of grasping simple truths clearly and distinctly, and of seeing the legitimacy of each inferential step. The claim is subtler and, for the genealogy of systematization, more relevant: he recasts the *procedure* of correct reasoning so that it increasingly approximates the regularised and impersonal determinacy of a machine.

The Cartesian method therefore mechanises the *conduct* of inquiry more than the thinking subject as such, but it dissociates inquiry from the inquirer, freeing it from the Scholastic reliance on an embodied *habitus*. The overall tendency of Descartes’s project is to purify inquiry from contingency, as Williams (2005) has argued. The *Rules* are explicit that arithmetic and geometry surpass all other disciplines in certainty because they alone concern objects so ‘pure and simple’ that they ‘presuppose nothing at all that experience might render uncertain, but rather consist entirely of

conclusions to be deduced rationally' (AT X.365). The ambition of the Cartesian method is to extend this purity beyond mathematics and into the whole of natural philosophy, and ultimately into the entirety of knowledge. Descartes's aspiration to a *mathesis universalis*, 'a general science which explains all the points that can be raised concerning order and measure irrespective of the subject-matter' (AT X.378, CSM I.19), expressed the dream of a single methodological framework applicable to all domains. He formulates four rules of method in the *Discourse*: to accept nothing unclear, to divide problems into their smallest parts, to order thoughts from simple to complex, and to make enumerations complete. Taken together, these constitute a procedural specification for the cognitive machinery that will power this universal science: a set of operations that, if correctly executed, will produce knowledge with the same reliability with which a well-designed mechanism produces its intended output. The pedagogical inheritance from the Ramists persists here in transformed form: knowledge is to be broken down, ordered, and reassembled in a sequence that can be inspected step by step, and the method's operator-independence makes knowledge reproducible across persons, just as Keckermann's *systema* had made doctrine transmissible across institutions. Yet the dominant pressure is now epistemological. The *point* of systematization is no longer primarily to facilitate teaching, but to certify knowledge.

The boundary of this mechanisation is what gives the Cartesian position its distinctive profile. In Part V of the *Discourse*, having argued that the body is a machine and that animals are automata, Descartes argues that no automaton, however ingenious, could 'arrange words in various ways to reply to the sense of everything said in its presence', nor 'act in all the events of life in the way that our reason enables us to act' (AT VI.57, CSM I.140). The decisive difference is open-ended responsiveness: the capacity to answer appropriately across an indefinite range of circumstances, a capacity that, for Descartes, no finite arrangement of mechanical parts could replicate. He mechanised the method by which the mind investigates the world, and he mechanised the body in which the mind is housed, but he left the thinking subject itself untouched. The procedure has been mechanised; the entity that operates the procedure has not. And this exemption has a metaphysical counterpart that proves equally consequential.

The criterion of clarity and distinctness had its original home in mathematics, where the transparency of the relevant acts of construction seemed able to vindicate the claims made on its behalf. Yet the doctrine proved harder to sustain once Descartes moved to more complex cases. In the later, unfinished *Rules*, when the problems at issue involve 'several equations in several unknowns', the geometrical constructions required become significantly less transparent than the simple operations that had supplied the criterion's paradigm case; the appeal to clarity and distinctness correspondingly loses some of its initial plausibility (Gaukroger 2006, 485–486). The breakdown of the doctrine in mathematics, the very domain in which it was supposed to demonstrate its power, called its general legitimacy into

question. The model of clear and distinct cognition accordingly moved from mathematics to the *cogito ergo sum*, which located a source of authority in the subject's indubitable grasp of its own thinking. Descartes's mature image of systematic knowledge, the famous 'tree of philosophy' from the French Preface to the *Principles*, already exposes the resulting pressure: "All Philosophy is like a tree," he writes, "of which Metaphysics is the root, Physics the trunk, and all the other sciences the branches' (AT IXB.14, CSM I.186). The image conveys both unity and derivation, and Descartes's demand, stated in the *Rules*, that we recognise that 'all the sciences are so deeply interconnected that it is far easier to learn them all together than to separate one from the others' (AT X.361), brings the pedagogical and epistemological rationales into direct alignment: the interconnection of the sciences is what makes possible both their systematic teachability and their systematic certification. Yet the tree is rooted in metaphysics, and its metaphysics rests on the existence and veracity of God. Without a God who guarantees that clear and distinct perception tracks objective truth, the method yields at most subjective certainty, which could be due to a psychological compulsion to believe. Gassendi pressed exactly this point in his objections to the *Meditations*: perhaps Descartes's inability to doubt certain propositions tells us about the architecture of his mind rather than about the structure of reality (Popkin 2003, 161–166).

The exemption of the thinking subject and the dependence on God are two faces of the same structural limit. Because Descartes has mechanised the procedure of reasoning while leaving the intuiting mind intact, the system requires a further, non-mechanical guarantee that the mind's intuitions are not merely psychologically compelling but objectively true. The veracious God supplies this guarantee. This is where the metaphysical warrant, the conviction that systematic thought tracks truth because a rational God has designed reality to be accessible to systematic reason, enters the story of cognitive systematisation as an explicitly load-bearing element. Chapter 1 showed this warrant to be present in the background of the Protestant system-builders, for whom God's rational creation of the world was a largely tacit premise of their pedagogical enterprise. In Descartes, the warrant moves into the foreground and becomes the explicit object of philosophical argument. The veracious God is what converts the method's outputs from psychological facts about Descartes into objective truths about the world. The entire edifice depends on this conversion. And the famous 'Cartesian Circle' objection, that Descartes needs clear and distinct perception to prove God's existence while needing God's existence to validate clear and distinct perception, threatened to reveal the conversion as illusory. The system that was designed to certify knowledge without reliance on external authority turns out to require, at its root, precisely such reliance: a non-mechanical residue, in the form of the intuiting subject and the God who underwrites its intuitions, at the foundation of the cognitive machinery.

The Cartesian effort to systematize knowledge thus reveals a recurrent pressure in the modern ideal of systematicity: the more strongly order is asked to certify

belief, the greater the need for some warrant that the order of thought answers to the order of reality. Descartes secures that relation through God, and he thereby gives the epistemological rationale its first modern formulation. His achievement, within the genealogy, is the mechanisation of method; his limit is the double exemption of the subject who intuits first truths and the God who guarantees them. It is precisely this exemption that Hobbes would revoke by identifying reason with computation, and that Spinoza would dissolve by treating the rational mind itself as an *automa spirituale* governed by its own necessary laws.

2.3 Hobbes: Reasoning Is Computation

With Hobbes, the machine emulation thesis applies at an even deeper level. Descartes had mechanised the *method* of inquiry: a set of operator-independent rules for reconstructing knowledge. Yet the entity that executes the method remained conspicuously exempt from mechanism. The mind, for Descartes, is *res cogitans*, thinking substance, and thus ontologically distinct from the extended, mechanical world it investigates. The *cogito* secures a foothold precisely because the thinking subject transcends the realm of matter in motion. Hobbes found this dualism unintelligible. His materialist ontology admitted only bodies and their motions, and his account of reasoning followed accordingly. If everything that exists is matter in motion, then thought is a species of motion too, and reasoning, the highest exercise of thought, is a species of computation. With this move, the machine emulation thesis ceases to operate as a methodological aspiration and becomes an ontological identity claim. Descartes built cognitive machinery; Hobbes declared that cognition *is* machinery. The consequences of this identification ramify in two directions: it gives the epistemological rationale a new and more radical form, and it makes the political rationale for systematisation fully visible for the first time, because the same machine model that governs cognition extends to the structure of the state.

Hobbes's own intellectual development illustrates the appeal of mechanical virtues as a model for intellectual work. For Hobbes did not admire these virtues from the beginning. As Quentin Skinner and Richard Tuck have documented, the young Hobbes was a humanist, steeped in the rhetorical and historical traditions that valued eloquence, judgement, and the persuasive management of common opinion. His first published work was a translation of Thucydides (1629); his anonymous digest of Aristotle's *Rhetoric* (c. 1637) still operated within the classical assumption that the management of belief was a matter of persuasive skill, exercised by speakers of practical wisdom. The ancient rhetoricians, Hobbes observed, worked from 'common Opinions' because 'the end of Rhetorick is victory; which consists in having gotten Belief' (Tuck 1996, xx–xxii). This was the cognitive world of the humanist tradition. It conceptualised reasoning as something thoroughly situated,

personal, animated by *phronesis*, and thus irreducibly dependent on the character and judgement of individual agents.

Hobbes's conversion from this world to the world of demonstrative systems had a specific origin. While on a continental tour around 1630, he discovered Euclid in a gentleman's library on the shores of Lake Geneva. John Aubrey recorded the encounter in one of the most vivid scenes in the history of philosophy: A copy of the *Elements* lay open at the forty-seventh proposition of Book I (the first of two proofs for the Pythagorean theorem). Hobbes read the proposition. "By G—", said he, 'this is impossible!', Aubrey tells us, explaining that Hobbes would now and then swear by way of emphasis (Aubrey 1898, 332). Hobbes's interest piqued, he traced the demonstration back to the proposition on which it depended, and from there to the proposition on which *that* depended, working his way backward through the chain of demonstrations until he arrived at a proposition whose truth he found self-evident. The truth of the forty-seventh proposition then imposed itself on Hobbes with the inexorability of the inferential chain whose every link he could inspect for himself. "This made him in love with geometry", Aubrey concludes (Aubrey 1898, 332). As Robert Gray has pointed out, Hobbes's own autobiography suggests that he came to geometry *because* of his mechanistic worldview and not, as is sometimes suggested, the other way round. This would mean that Hobbes found in Euclid the tools to articulate a conviction he had already formed, namely that all phenomena, including sensation and thought, reduce to motion (Gray 1978, 212, 214–215). If this is correct, the Euclidean encounter was less the source of a new philosophical vision than the discovery of the *method* adequate to a vision Hobbes already possessed.

Either way, the Geneva episode crystallised what would become the animating conviction of his entire philosophy: the power of geometric demonstration lies in how it can *compel* assent through a sequence of steps that lie open to view and can be inspected at every stage, leaving no room for the kind of interpretive disagreement that fuels both intellectual controversy and civil strife. Through Euclid, Hobbes discovered that publicly inspectable demonstration compels with inexorable force, and this discovery made the softer power of humanist rhetoric look like part of the problem rather than the solution. Where rhetoric persuades through probable reasoning and appeals to common sentiment, demonstration proves; and the difference between persuasion and proof is the difference between a technique that depends on the character of the speaker and a method that operates with the impersonal inexorability of a machine.

Helen Hattab has argued that the intellectual genealogy of Hobbes's method runs through a channel directly relevant to the trajectory of this book. The standard scholarly approach traces Hobbes's method of 'resolution and composition' to Jacopo Zabarella's *regressus*, the method of scientific proof developed in the Aristotelian tradition at Padua. Hattab shows that this connection is misleading: what Hobbes actually inherits is closer to Zabarella's theory of *method as order*, and

he inherits it through the mediation of two Protestant systematic logicians, Bartholomäus Keckermann and Franco Burgersdijk (Hattab 2014, 462–463, 482–483). As Chapter 1 showed, Keckermann repurposed the concept of *systema* as a technology for ordering knowledge without the presence of a master. Burgersdijk, writing in Leiden a generation later, collapsed the distinction Zabarella had drawn between the order of teaching and the order of discovery, arguing that the natural method ‘ought always to progress from universals to particulars’ and that this progression constituted both the order in which knowledge is acquired and the order in which it is most effectively transmitted (Hattab 2014, 482). Hobbes adopts this strategy and pushes it further. He accepts that the cognitive order and the natural order are one and the same, but he replaces Aristotelian genera with mathematical definitions: ‘motion’, ‘point’, ‘line’, and ‘surface’ become the universals from which synthesis proceeds (Hattab 2014, 483). The result is an ideal of philosophical systematicity whose foundational ordering principle is no longer the Aristotelian genus, but the Euclidean definition, and whose method of composition is, consequently, compulsive in just the way geometry is compulsive.

The line running from Keckermann’s pedagogical *systema* through Burgersdijk’s natural method to Hobbes’s science of motion is thus a line of progressive systematisation: what began as a technology for transmitting reliable knowledge has become a programme for *producing* it. The pedagogical rationale is now subordinated to a far stronger epistemological claim: that systematic connection is what converts isolated sentences into genuine science.

By 1635, after exposure to the Mersenne circle in Paris and the “Welbeck Academy” surrounding William Cavendish, Earl of Newcastle, Hobbes had developed a comprehensive mechanist natural philosophy in which, as Gray has argued, mechanism and psychology were inextricably linked: his first question was ‘What is sensation?’, and his final answer was ‘It is motion’ (Gray 1978, 214–215).

This mechanisation of reason itself then provided further support to the Euclidean ideal of systematic thought, because it suggested that the very substrate of reasoning was subject to an impersonal compulsion that mirrored that of demonstrative reasoning. Reason could and should aspire to the virtues of a machine because it *was* a kind of machine. The definition of reason that Hobbes articulates in the *Leviathan* is the most radical statement of the machine emulation thesis prior to Leibniz. ‘When a man Reasoneth,’ Hobbes writes, ‘hee does nothing else but conceive a Summe totall, from Addition of parcels; or conceive a Remainder, from Substraction of one summe from another.’ This operation extends far beyond arithmetic. Geometers add and subtract lines, angles, and proportions; logicians add names to make affirmations and affirmations to make syllogisms; political writers add pacts to find duties; lawyers add laws and facts to determine right and wrong. ‘In summe, in what matter soever there is place for addition and subtraction, there also is place for Reason; and where these have no place, there Reason has nothing at all to do’ (*Leviathan* I.5, 18). Reason, accordingly, ‘is nothing but Reckoning (that is,

Adding and Subtracting) of the Consequences of generall names agreed upon, for the marking and signifying of our thoughts' (*Leviathan* I.5, 18). For Descartes, the geometric method had provided a model that reasoning should emulate; Hobbes reduces the gap between the model and the modelled by arguing that reasoning already is, at bottom, the addition and subtraction of names linked in chains of consequence. If systematisation is a way to make thought more machine-like, this ideal is all the more apt when thought is itself mechanical from the start. The substrate then lends itself to being given a systematic form. Anything that can be computed can be cast in the form of definitions, affirmations, and syllogisms.

Every branch of science, in Hobbes's scheme, is ultimately a science of motion. As Gaukroger observes, Hobbes's mature system – as set out in *De Corpore* – offers both 'one of the most ontologically parsimonious versions' of the mechanistic worldview and 'one of its more systematic versions' (Gaukroger 2006, 283). And as Gray has argued against an influential tradition of de-systematising interpretation stemming from A. E. Taylor, Howard Warrender, and Leo Strauss, the deductive order running from first philosophy through geometry to physics and thence to moral philosophy is the architectonic principle of the whole enterprise (Gray 1978, 199–203). Hobbes's planned philosophical trilogy made this comprehensiveness explicit: *De Corpore* (1655) treats the principles of body and motion, *De Homine* (1658) the motions of the mind, and *De Cive* (1642, published first but designed as the capstone) the motions of men in political society.

The claim that the rigorous systematicity of geometry could be extended to domains such as moral philosophy met with scepticism even in the seventeenth century. The Chief Justice Matthew Hale, responding to Hobbes's *Dialogue of the Common Laws*, argued that lawmakers were right to seek practical solutions to practical problems rather than to speculate on first principles of moral philosophy: 'it is not possible for men to come to the Same Certainty, evidence and Demonstration touching them as may be expected in Mathematicall Sciences', and those who suppose they can construct 'an unerring system of Laws and Politiques equally applicable to all States and Occasions, as Euclide demonstrates his Conclusions, deceive themselves with Notions which prove ineffectual, when they come to particular application' (Gaukroger 2006, 378). Hale's objection anticipates the counter-tradition that we will see emergence in response to the rise of systematic thought. He voices the suspicion, reiterated in our own time by particularists such as Jonathan Dancy (2004), that the systematizer's strategy of deriving everything from first principles, however powerful in some domains, impoverishes understanding when applied to matters that require sensitivity to particulars.

With Hobbes, the epistemological rationale takes on a form at once more austere and more ambitious than it had in Descartes. Hobbes's theory of demonstration requires that science begin with the 'imposition of names': the creation of a shared linguistic framework in which definitions are settled by explicit agreement (Jesseph 2000, 190–193). The crucial contrast is between science and prudence. Prudence is

experiential knowledge: the accumulated practical wisdom that comes from having observed many particular cases. The assurance it can provide falls short of certainty, and it is ineliminably personal: what one person's experience has taught, another's experience may contradict. Science, by contrast, is 'conditionall Knowledge, or Knowledge of the consequence of words,' which 'begins with the Definitions of Words, and proceeds by Connexion of the same into generall Affirmations, and of these again into Syllogismes' (*Leviathan* I.7, 30).

Because its starting points are public definitions and its inferences are formally checkable, science is independent of the particular experiences, temperament, or social standing of the reasoner. Anyone who grasps the definitions and follows the chain of consequences will reach the same conclusions. A passage in *De Corpore* states this conviction with particular force: method, Hobbes argues, transforms 'isolated moral and political sentences, however wholesome and choice,' into moral and political *science*, or 'a true and certain rule of our actions, by which we might know whether that we undertake be just or unjust' (*De Corpore* I.7; EW I.9). Disconnected maxims, however wise, offer counsel but not knowledge, and their application depends on the practical wisdom of the counsellor, which is to say on precisely the kind of personal, situated judgement that cannot be inspected, reproduced, or transmitted. Only when such maxims are connected into a deductive chain running from definitions through demonstrations does moral knowledge become *science*, a publicly auditable structure whose conclusions compel assent regardless of who draws them. For Hobbes, the formal cause of science is the 'connectedness itself of propositions,' and method is its efficient cause: the procedure by which disconnected claims are woven into a systematic whole (Talaska 1988, 208–209). The guiding analogy here is with arithmetic: 'Reason it selfe is alwayes Right Reason, as well as Arithmetique is a certain and infallible Art: But no one mans Reason, nor the Reason of any one number of men, makes the certaintie; no more than an account is therefore well cast up, because a great many men have unanimously approved it' (*Leviathan* I.5, 19). What certifies a chain of reasoning is the inspectable correctness of each inferential step together with the traceability of the whole chain back to definitions whose significations have been publicly agreed. Anything that falls outside this chain is, in Hobbes's severe verdict, mere belief. 'He that takes up conclusions on the trust of Authors, and doth not fetch them from the first Items in every Reckoning, (which are the significations of names settled by definitions), loses his labour; and does not know any thing; but onely beleeveth' (*Leviathan* I.5, 19).

It is this publicly inspectable character of Hobbesian computation that enables the extension of the machine model from epistemology to politics. This extension is the most striking early modern formulation of the *political rationale* for systematisation. The *Leviathan* carries the scars of the English Civil War that made this extension feel urgent. 'The Light of humane minds,' Hobbes writes, 'is Perspicuous Words, but by exact definitions first snuffed, and purged from ambiguity; Reason is the

pace; Encrease of Science, the way; and the Benefit of man-kind, the end.' And the counter-image: 'Metaphors, and senselesse and ambiguous words, are like *ignes fatui*; and reasoning upon them, is wandering amongst innumerable absurdities; and their end, contention, and sedition, or contempt' (*Leviathan* I.5, 22). The passage is remarkable for its equation of linguistic imprecision with political disorder. Hobbes, anticipating Krausian and Orwellian warnings against the political dangers of slovenly language, sees a causal connection running from undefined terms through confused reasoning to civil war. In *Behemoth* (1668), his history of the English Civil War, Hobbes would identify the root causes of the catastrophe as ideological: religious sectarianism, the misuse of Aristotelian philosophy, and above all the chaos of private judgement – the situation in which 'every man became a judge of religion, and an interpreter of the Scriptures.' The failure was fundamentally one of unregulated interpretation, and the remedy accordingly had to begin with language. The significations of words needed to be settled through definitions, and the ensuing chain of inferences needed to take the form of rigorous computations that could be inspected for correctness. The connection between the epistemological and the political is visible here: interpretive anarchy in the domain of knowledge and civil war in the domain of politics are, for Hobbes, two manifestations of the same disorder. Consequently, the same rational machinery that certifies knowledge can, if properly deployed, also certify political authority.

The *Leviathan's* Introduction makes this homology explicit. The text opens with one of the most famous images in the history of political thought:

NATURE (the Art whereby God hath made and governes the World) is by the Art of man, as in many other things, so in this also imitated, that it can make an Artificial Animal. For seeing life is but a motion of Limbs, the begining whereof is in some principall part within; why may we not say, that all *Automata* (Engines that move themselves by springs and wheeles as doth a watch) have an artificiall life? For what is the Heart, but a Spring; and the Nerves, but so many Strings; and the Joynts, but so many Wheeles, giving motion to the whole Body, such as was intended by the Artificer? Art goes yet further, imitating that Rationall and most excellent worke of Nature, Man. For by Art is created that great LEVIATHAN called a COMMON-WEALTH, or STATE, (in latine CIVITAS) which is but an Artificiall Man. (*Leviathan*, Introduction, 9)

The commonwealth is a machine. Its sovereignty is an 'Artificiall Soul', its magistrates 'artificiall Joynts', and its laws 'an artificiall Reason and Will.' In both the epistemic and the political domain, the failure of external authority, ecclesiastical in one case, traditional and customary in the other, necessitates the construction of an artificial order whose legitimacy derives entirely from its internal rational structure. The commonwealth, like a system of knowledge, earns its authority by being inspectable

– its principles are publicly articulated, its chain of authorisation is traceable, and its operations can be evaluated by anyone who grasps the operative terms. Hobbes's contribution to the genealogy of the ideal of systematicity is to make explicit the link between its epistemological and political applications. If reasoning is computation, and if the commonwealth is a machine, then the rational reconstruction of political authority and the systematic organisation of knowledge are, at bottom, the same kind of project.

At the same time, Hobbes makes explicit the cost of this project. The definitions on which science and political order depend are not arrived at through demure civic conversation and democratic deliberation; they are *imposed by the sovereign*. The sovereign's authority must extend to the significations of words, because unsettled definitions breed the interpretive anarchy that leads to civil war. The very features that make Hobbesian computation epistemically and politically powerful require, on Hobbes's account, an authoritative power to settle the terms in which that computation proceeds. Public inferential auditability, the crowning achievement of the political rationale for systematisation, is purchased at the price of sovereign control over discourse. The accountability dimension of the political rationale – the capacity to audit and challenge decisions on publicly articulable grounds – will be more fully developed by Rousseau and the codification movement (Chapter 6). But in Hobbes's vision, the system that promises to free reasoning from dependence on personal authority simultaneously concentrates the power to define the terms of reasoning in the hands of the sovereign.

Descartes had assimilated proper reasoning to an algorithmic procedure; Hobbes made reasoning itself computational. The machine emulation thesis had progressed from procedural aspiration to ontological identity, and in doing so it had revealed a homology between the epistemic and the political. Yet Hobbes's system, for all its radicalism, remained a programme: its architectonic was planned rather than fully executed, and its deductive chains depended on definitions whose authority rested, finally, on political imposition rather than self-evidence. It was Baruch Spinoza who attempted what neither Descartes nor Hobbes had achieved: to embody the necessity of demonstrative reasoning in the very form of philosophical writing, producing a text designed to function as an inferential machine whose conclusions follow from its premises with the same necessity by which the properties of a triangle follow from its definition.

2.4 Spinoza: The Spiritual Automaton

With Baruch Spinoza, the machine emulation thesis reaches its most ambitious instantiation: the philosophical text itself is designed to function as an inferential machine, and the mind, when directed by true ideas, operates as what Spinoza calls 'a spiritual automaton', a self-governing mechanism acting 'according to certain

laws' with the same necessity that governs bodies in motion (TdIE §85). Why would a philosopher choose to write about God, the human mind, the passions, and the conditions of human freedom in a format that suppresses every trace of the author's voice, every rhetorical appeal, every concession to the reader's expectations? The question is sharpened by the fact that Spinoza could write luminous, rhetorically powerful prose when he wished, as the *Theological-Political Treatise* (1670) amply demonstrates. The geometric format was a deliberate choice, and the reasons for it reach to the heart of the genealogy this book traces.

Spinoza was born in 1632 into the Portuguese-Sephardic Jewish community of Amsterdam, a community of former *conversos* who had fled the Iberian Inquisition and re-established Jewish religious life in the relative freedom of the Dutch Republic. On 27 July 1656, the twenty-three-year-old was excommunicated, subjected to a *cherem*, by the Talmud Torah congregation, in what Steven Nadler described as the harshest writ of *herem* ever pronounced upon a member of that community (Nadler 2006, 12–16). The precise causes remain debated, but testimony from Inquisition archives reports that Spinoza had declared 'the Law is not true' and that there is no God 'except philosophically.' After the *cherem*, he moved into circles of freethinking Christians, particularly the Collegiants, an anti-ecclesiastical Dutch Protestant group that rejected established church authority, and studied Latin, Cartesian philosophy, and mathematics at the school of Franciscus van den Enden.

The philosophical question posed this experience of being subject to the judgement of institutions claiming divine sanction while having lost confidence in their doctrinal authority echoes the one that the sceptical crisis posed for Descartes and the English Civil War for Hobbes: if external authority has collapsed, on what basis can knowledge be established? Spinoza's answer was the most uncompromising version of the response that all three gave. Knowledge must be grounded in a structure whose credentials can be inspected by anyone equipped with reason. And this structure must be one in which every claim is either self-evident or demonstrably derived from what is self-evident.

He had, in a sense, rehearsed the choice before committing to it. In 1663, Spinoza published a geometrical rendering of Parts I and II of Descartes's *Principles of Philosophy*, recasting Cartesian metaphysics and physics in the format of definitions, axioms, and propositions. What drove him to do this was a pedagogical need: Spinoza had been tutoring a young student in Cartesian philosophy and found the geometric format the most perspicuous way to lay out its inferential structure (Nadler 2006, 36–38). The pedagogical rationale, familiar from Keckermann and the Protestant system-builders of Chapter 1, was thus operative from the start. The geometric format makes the architecture of an argument surveyable in a way that discursive prose could not. But Spinoza discovered something further. The format made the logical dependencies between claims maximally transparent, revealing exactly which premises supported which conclusions, and where the argument relied on assumptions that were open to question. The pedagogical virtue of survey-

ability therefore implied epistemological virtues: the same perspicuity that made knowledge transmissible also opened it up to critical scrutiny.

Much as the pedagogical *systema* also occluded nuances that the *Summa* format still displayed, however, Spinoza's geometry-inspired systematisation concealed certain things. As Edwin Curley has argued in *Behind the Geometrical Method*, the systematic presentation notably "served to conceal" the philosophical dialogue Spinoza was conducting with his predecessors, and "was, perhaps, partly designed to conceal" (Curley 1988, xi) it. To systematise Descartes's thought as Spinoza did meant replacing the discursive back-and-forth of philosophical argument with a structure that presented results as though they followed from self-evident first principles. In an ordinary philosophical text, the author's voice is audible throughout: arguments are advanced in the first person, opponents are named and addressed, and rhetorical appeals are made to the reader's experience and judgement. The geometric format strips all of this away. What remains is a network of logical relations: definitions stipulating meanings, axioms stating what is self-evident, propositions asserting claims, and demonstrations showing that those claims follow necessarily from what has already been established. The person of the author disappears behind the systematic structure.

The machine emulation thesis, which in Descartes operated at the level of method and in Hobbes at the level of reasoning itself, here reaches the level of the philosophical work as a whole. The text becomes a cognitive machine: a device for producing assent through the impersonal operation of logical inference, requiring no trust in the author, no deference to tradition, no appeal to experience that cannot be made publicly available through definitions.

What distinguishes the geometric format of Spinoza's *Ethics* from a merely pedagogical convenience is the fusion of form and content demanded by Spinoza's metaphysics. The opening of the *Ethics* puts this machinery on display with considerable economy. Part I, 'Of God', begins with eight definitions and seven axioms, from which Spinoza proceeds to derive thirty-six propositions establishing that God is the only substance, that everything that exists is in God, and that all things follow from the divine nature with absolute necessity. The fourth axiom deserves particular attention, for it encodes the principle that drives the entire system. As Michael Della Rocca has argued, Spinoza's philosophy is animated by an unflinching commitment to the Principle of Sufficient Reason: for each thing that exists, there is an explanation sufficient to make its existence intelligible (Della Rocca 2015, 523–525). There is nothing brute, nothing unintelligible, nothing that simply happens to be the case without a reason. Axiom 4 builds this commitment into the foundations of the system by asserting that to know an effect adequately just *is* to know its cause, that understanding is constitutively causal explanation. By Proposition 14 of Part I, Spinoza has demonstrated that 'there can be, or be conceived, no other substance than God' (E IP14). Everything that exists, every mind, every body, every passion, is a *mode* of the one substance, existing in God and conceivable only

through God (E IP15). This ontological system is all-encompassing: ‘Things could have been produced by God in no other way, and in no other order than they have been produced’ (E IP33). There is only one possible world, and every particular event occupies its position in the causal-deductive web with absolute necessity.

If reality itself is a system in which everything follows from the nature of one substance with logical necessity, then the appropriate form for expressing this reality in thought is the axiomatic-deductive format in which everything follows from a set of first principles with logical necessity. The *Ethics* itself enacts the systematicity of thought it advocates: the deductive structure of the text mirrors the causal structure of nature because, for Spinoza, causation and logical entailment are ultimately the same relation. Spinoza “associated causation and deduction, so that A’s being the cause of B, and B’s being contained in A, are just another way of saying that A is the explanation for B” (Gaukroger 2006, 480). The guiding ideal here, to use Spinoza’s own phrase, is that “the order and connection of ideas” should be “the same as the order and connection of things” (E IIP7) – a conception elaborated by Brandom (2019, 61) and Hlobil and Brandom (2025, 11) in terms of an alignment between *deontic* and *alethic* modality, i.e. between what is *rationaly* necessary, possible, or impossible and what is *causally* necessary, possible, or impossible. This isomorphism means that the order of demonstrations in the *Ethics* aims to reproduce the order of causes in nature. And the reason the geometric format cannot be a mere pedagogical convenience for Spinoza, in the way it might be for a philosopher whose metaphysics did not demand it, is that his central metaphysical doctrine is the geometrical necessity that governs Nature itself: ‘from God’s supreme power, or infinite nature, infinitely many things in infinitely many modes, i.e., all things, have necessarily flowed, or always follow, by the same necessity and in the same way as from the nature of a triangle it follows, from eternity and to eternity, that its three angles are equal to two right angles’ (E IP17s).

If Nature is ordered with mathematical necessity, then the ideas that accurately represent it must be so ordered too, and the geometric format is supposed to perform what one scholar has called an ‘emendative therapy’ on the reader’s mind, rearranging ideas that were originally acquired haphazardly through the senses and everyday experience until they mirror the causal order of reality itself (Nadler 2006, 41–42). The epistemological rationale here reaches a pitch unmatched elsewhere in the chapter: the system’s internal order is itself the *criterion* of knowledge, and anything that falls outside the system is, by definition, not knowledge.

For Spinoza, this ideal of geometric systematicity is applicable across all domains and subject matters. The Preface to Part III of the *Ethics* announces the programme thus: ‘I shall treat the nature and powers of the Affects, and the power of the Mind over them, by the same Method by which, in the preceding parts, I treated God and the Mind, and I shall consider human actions and appetites just as if it were a Question of lines, planes, and bodies’ (E III Preface). The entire domain of human psychology – the passions, the appetites, the movements of desire and aversion –

are to be explained with the same deductive rigour that applies to the properties of geometrical figures.

As Eugene Marshall has shown, the *automa spirituale* of the *Treatise on the Emendation of the Intellect* is the key to this extension (Marshall 2013, 1–3). The foundational mechanism is the *conatus*: the striving by which each thing, as far as it can by its own power, endeavours to persevere in its being (E IIIP6). In the case of the human mind, the *conatus* manifests as a system of competing ideas, each possessing a degree of assertoric force – what Marshall aptly describes as a kind of *mental inertia* – that persists unless counteracted by another idea of greater force (Marshall 2013, 78–79). The mind is, in this sense, a mechanism: ideas are efficacious within it in virtue of the causal relations they bear to other ideas and to the *conatus* as a whole. Spinoza is explicit about the implication: ‘No affect can be restrained by the true knowledge of good and evil insofar as it is true, but only insofar as it is considered as an affect’ (E IVP14). An idea’s truth value is powerless within the mental mechanism unless that idea also possesses the force of an affect – it must be so situated in the network of the mind’s ideas that its power impacts the *conatus* as a whole. In other words, truth alone does not move us; only truth *felt as a force* can counteract the power of a contrary passion. As Marshall observes, ideas interact within the mind as affects interact within a mechanism, through the relations of force they bear to one another, with the outcome determined by the balance of competing powers rather than by the logical credentials of any single idea considered on its own (Marshall 2013, 168–169). This is the machine emulation thesis carried to a point that Hobbes did not reach. Hobbes identified reasoning with computation, but he did not apply mechanical principles to the internal dynamics of belief and desire. Spinoza does: the mind is a field of forces, and the task of rational self-governance is the task of engineering the affective mechanism so that adequate ideas acquire sufficient force to counteract the passions, a project Spinoza describes with the language of therapy rather than exhortation.

Spinoza’s view also differs from Descartes’s in fundamental respects. As Dominik Perler has argued, where Descartes takes the sceptical problem as given and attempts to refute it by securing an external guarantee for clear and distinct ideas, Spinoza pursues what Perler calls a preventive strategy, diagnosing the theoretical presuppositions that generate the challenge in the first place (Perler 2013, 221–223). On Perler’s analysis, radical scepticism presupposes semantic atomism: the assumption that each idea has its representational content independently of every other, so that there is always a gap between any individual idea and the reality it purports to represent. Spinoza replaces this atomism with a thoroughgoing holism: an idea derives its content from its relations to other ideas in the system, and an idea considered in isolation from those relations is, for precisely that reason, inadequate. Spinoza characterises such isolated ideas as ‘mutilated and confused’ (*mutilatae et confusae*, E IIP29c), severed from the explanatory connections that would render them intelligible. A true idea, by contrast, is one that is adequately connected to

its causes, situated within the deductive web that constitutes the order of nature. Knowledge is thus constitutively systematic for Spinoza: to know something adequately is to grasp it in its causal-explanatory connections, and an idea that lacks such connections is, by definition, inadequate. Scepticism arises from the attempt to evaluate ideas one by one, in isolation from the system; once one recognises that the system is the proper unit of epistemic evaluation, the sceptical problem dissolves. 'He who has a true idea at the same time knows that he has a true idea, and cannot doubt the truth of the thing' (E IIP43). This is the epistemological rationale for systematisation at its clearest: systematic connection is constitutive of adequate knowledge, so that the demand for systematicity and the demand for truth converge.

All three practical rationales are operative in Spinoza, however, though in a configuration distinct from any of his predecessors. The epistemological rationale is dominant and reaches its most stringent formulation: nothing counts as genuine knowledge unless it can be demonstrated within the system, and the criterion of truth is internal to the system of adequate ideas, with no court of appeal beyond the deductive structure itself. The pedagogical rationale is present, since Spinoza initially adopts the geometric format in part for its surveyability and its capacity to make the architecture of thought visible and transmissible. And the political rationale emerges in the *Theological-Political Treatise*, where Spinoza argues that freedom of philosophising is essential to the health of the republic, and that ecclesiastical authority has no legitimate jurisdiction over rational inquiry. He thereby makes the case for the availability of a mode of inquiry whose credentials are publicly inspectable and independent of any particular religious tradition. The famous Appendix to Part I sharpens the connection between systematisation and the rejection of external authority. There, Spinoza traces the origin of superstition to the human tendency to interpret nature in terms of final causes. We are, on this view, naturally inclined to assume that natural things exist for human benefit and that the gods arrange them accordingly. 'This prejudice,' he writes, threatened to 'hide the truth from the human race to eternity,' except for one countervailing force: 'mathematics, which is concerned not with causes [*fines*] but solely with the essences and properties of figures, had not shown us another standard of truth,' *alia veritatis norma* (E I Appendix). Mathematics provides an alternative epistemic authority – one that replaces the teleological worldview of theology with a causal-structural framework grounded in self-evident axioms and rigorous demonstration.

The metaphysical warrant occupies a distinctive position in Spinoza's system, and its role here must be registered with particular care, because the warrant's power and its vulnerability reach their maximum simultaneously. For Descartes, God is invoked as an external guarantor: the veracious deity ensures that clear and distinct perception corresponds to objective reality, bridging the gap between subjective certainty and the world. For Hobbes, the metaphysical warrant is present in the framing rhetoric, 'NATURE (the Art whereby God hath made and governes

the World)’, while remaining largely absent from the operative epistemological machinery. Spinoza’s position is more thoroughgoing. God is identical with Nature; the system of deductive demonstration mirrors the causal structure of the one substance; the order of ideas and the order of things are one and the same, differing only in attribute. The metaphysical warrant is thus built into the very fabric of the system rather than invoked as a supplementary guarantee. For Spinoza, the question ‘why think systematic thought is truth-tracking?’ has the most direct possible answer: because the structure of systematic thought just *is* the structure of reality, expressed under the attribute of Thought rather than under the attribute of Extension. There is no gap to bridge, no divine veracity needed to ensure correspondence, because thought and reality are two expressions of one and the same order. This is the metaphysical warrant at its most elegant, but also at its most vulnerable. The elegance lies in the closure: the isomorphism between the order of ideas and the order of things is a necessary consequence of substance monism, requiring no supplementary theological apparatus. The vulnerability lies in the fact that the entire edifice rests on the opening definitions and axioms of Part I, whose self-evidence is far from uncontested. As Curley put it: ‘The definitions are typically obscure, the axioms frequently not evident, and the demonstrations all too often unconvincing. And yet it is hard to escape the feeling that there is something there worth taking pains to try to understand, something very important, if true, and something quite possibly true’ (Curley 1988, xi). The geometric method promises certainty through self-evidence. Yet the actual experience of reading the *Ethics* delivers something more equivocal: a sense of philosophical depth struggling against a formal apparatus that frequently resists the very transparency it was designed to deliver.

A pattern we already encountered is legible here: the more totalising the system, the more weight the metaphysical warrant must bear – the greater the systematic ambition, the more forcefully the system must be understood as answerable to reality’s own rational order, and the more devastating the consequences if that order turns out to be less accommodating than the system requires. Descartes’s warrant only guarantees the deliverances of a method; Spinoza’s warrant underwrites the identity of thought and reality.

There are also epistemic costs to identifying intelligibility so closely with systematicity. As Gaukroger remarks, Spinoza modelled clarity and distinctness on mathematical demonstration to an extent ‘unthinkable even in Descartes, collapsing causal relations into deductive ones so that empirical enquiry became a form of quasi-mathematical deduction’ (Gaukroger 2006, 487). The reaction of Christiaan Huygens and Spinoza himself to Robert Boyle’s experimental approach to pneumatics was, Gaukroger notes, one of ‘incomprehension: they simply could not understand how an approach that was not systematic, that did not start from foundational clear and distinct ideas, and deduce the phenomena from principles formulated in terms of these ideas, could yield genuine explanations’ (Gaukroger 2006, 487). The machine of systematic deduction had become so powerful that the

possibility of legitimate knowledge arising from any other source – from experiment, from situated observation, or from the patient accumulation of empirical particulars – had become invisible to them. The epistemological rationale, carried to its Spinozistic limit, generates its own epistemic distortion: systematic coherence, elevated to the sole criterion of knowledge, excludes whatever cannot be captured within the deductive web.

The term Spinoza chose for the ideal of rational self-governance points, with unwitting prescience, to the tension at the heart of the enterprise. *Automa spirituale*: a spiritual automaton. In the *Treatise on the Emendation of the Intellect*, the phrase designates the highest achievement of the rational mind, a soul acting from its own necessary laws with the self-sufficiency of a well-made machine. In the *Theological-Political Treatise*, Spinoza uses the same word, *automati*, for those who merely repeat what they have heard without understanding its demonstration: parrots, not thinkers (TTP ch. 13; Marshall 2013, 2n3). The two uses mark the outer boundaries of the machine emulation thesis. At its best, the mechanisation of thought yields a mind governed by its own rational laws, transparent to itself, free from the vagaries of passion and the distortions of superstition. At its worst, it yields a mind that mimics understanding without possessing it, producing the outputs of rationality without the inner life that gives them point. The seventeenth century's most rigorous systematiser forces upon the genealogy a question that Margaret Cavendish, writing in the same decade, was the first to pose: what does a system powerful enough to display everything as necessary render impossible to see?

2.5 Cavendish: Mentalising the Machine

The trajectory of the chapter so far has been characterised by the progressive mechanisation of thought: Descartes made correct reasoning method-like; Hobbes declared that reasoning is computation; Spinoza designed a philosophical text to function as an inferential machine. Margaret Cavendish (c. 1623–1673) challenges this mechanisation from within the seventeenth-century debate itself, and her challenge reveals what the machine emulation thesis, at the height of its ambition, excludes: it captures certain intellectual virtues while suppressing others, and the selectivity has consequences for all three of the practical rationales that this book traces.

The earliest philosophical influence on Cavendish came from Hobbes himself, who had served as tutor to her husband's family and was a regular participant in the "Newcastle Circle" – the group of natural philosophers, including Walter Charleton and Kenelm Digby, who gathered around William Cavendish, Marquess (later Duke) of Newcastle, at his country seat at Welbeck and subsequently in Paris and Antwerp (O'Neill 1998, 9; Cuning 2019, xvi). The group had strong materialist commitments and had been shaped by contact with Gassendi and Mersenne during

the years of the English Civil War. While in exile, Cavendish met Descartes and Roberval. She absorbed the full spectrum of contemporary mechanistic philosophy through the 1640s and 1650s, and in 1667 became the first woman to attend a meeting of the Royal Society of London, where Robert Boyle performed experimental demonstrations for her entertainment, including the weighing of air in his famous air-pump (Detlefsen 2007, 157). She published prolifically from 1653 until her death, producing more than a dozen volumes of natural philosophy, moral philosophy, plays, poetry, and fiction. Her philosophical maturity was concentrated in the works of the 1660s: the *Philosophical Letters* (1664), the *Observations upon Experimental Philosophy* (1666), the *Description of a New World, Called the Blazing-World* (1666), and the *Grounds of Natural Philosophy* (1668). The reaction to her writings was frequently hostile and gendered: critics either dismissed her work as ‘Enthusiasm’ that scorned ‘the control of reason’ or alleged that her books must have been plagiarised, since no woman could understand so many ‘hard termes’; the Cambridge Platonist Henry More predicted to Anne Conway that some man would have to ‘quitt his breeches and putt on a petticoat to answer her in that disguise’ (O’Neill 1998, 9–10). Cavendish was excluded from the philosophical canon for centuries, and the recent recovery of her work has made clear that the exclusion was a philosophical loss. Her resistance to mechanistic philosophy was grounded in a alternative ontology with its own internal logic with, for our purposes, revealing epistemological and normative consequences.

The character of that alternative deserves careful statement, because it has often been misdescribed. The label conventionally applied to Cavendish’s position is ‘vitalist materialism’, and, in a loose sense, that label is serviceable enough: Cavendish held that all of nature is alive. But as Laura Georgescu has recently argued, the label obscures more than it reveals (Georgescu 2023). In Cavendish’s mature philosophy, ‘life’ collapses into ‘being’: to be alive is simply to exist as matter, and the predicate ‘living’ adds nothing to the predicate ‘material.’ The real explanatory work is done by two other properties that Cavendish attributes to matter throughout: *rationality* and *sensitivity*. These are the capacities that distinguish her ontology from that of the mechanistic philosophers, and it is in terms of these capacities, rather than in terms of some distinct property called ‘life,’ that Cavendish’s challenge to the mechanistic conception acquires its force. Charles Wolfe has made a complementary point: Cavendish does not problematise the boundary between life and non-life in the way that a genuine vitalist would, because her flat ontology eliminates that boundary altogether (2021, 293). If ‘vitalism’ is useful at all as a label for Cavendish’s position, it is as a historiographic contrast class situating her against mechanistic philosophy. For an accurate way to characterise her philosophical view, however, we need to look elsewhere.

What her philosophical view actually works with is mind, distributed throughout nature. Cavendish agreed with Hobbes and against Descartes that everything in nature is corporeal: there is no immaterial substance, no *res cogitans* exempt from

the material world. 'I shall merely go upon the bare Ground of *Natural Philosophy*,' she writes, 'and do not mix Divinity with it, as many Philosophers use to do ... for I think it not onely an absurdity, but an injury to the holy Profession of Divinity, to draw her to the proofs of *Natural Philosophy*' (*Philosophical Letters*, 3). This commitment places Cavendish squarely within the naturalist programme of the new philosophy. Where she diverged from the mechanistic philosophers was on the question of whether matter is inert. For Descartes, material substance is pure extension, devoid of any intrinsic capacity for motion, thought, or perception; motion must be imparted from outside – ultimately by God. For Hobbes, matter is similarly passive: bodies move because other bodies push them. Cavendish rejected this picture. Nature, she insisted, is 'not a dead immovable Body', but a self-moving, self-knowing whole. Like Spinoza, she takes central tenets of Descartes's philosophy and develops them in a direction that Descartes himself would not countenance (Cunning 2016, 2). While Descartes mechanised the body while exempting the mind and Hobbes and Spinoza extended the mechanisation to the mind itself, Cavendish took the opposite path. She accepted the materialist premise that there is nothing beyond the corporeal, and then asked what kind of matter could support the cognitive and volitional capacities that the mechanistic philosophers had either attributed to an immaterial substance or reduced to a mere rearrangement of inert parts. Her answer was that matter itself must be rational and sensitive. If the other seventeenth-century systematisers had mechanised the mental, Cavendish mentalised mechanism.

According to her mature view, the single corporeal substance into three inseparable aspects: rational matter, sensitive matter, and inanimate matter. Rational matter is the most pure and agile, functioning as the architect or designer of nature's activities. Sensitive matter serves as the labourer, carrying out the rational designs while bearing the burden of inanimate matter. Inanimate matter is the dense, passive material out of which figures and bodies are composed. The three aspects are 'inseparably commixt': no portion of material nature, however small, lacks any of the three. The consequence is that perception, knowledge, and rational agency are distributed throughout the natural world and present in varying degrees in every part of the material plenum. As Cavendish puts the central point in the *Observations*:

that every part has not onely sensitive, but also rational Matter, is evident, not onely by the bare motion in every part of Nature, which cannot be without sense, for wheresoever is motion, there's sense; but also by the regular, harmonious and well-ordered actions of Nature, which clearly demonstrates, that there must needs be Reason as well as Sense, in every part and particle of Nature; for there can be no order, method or harmony, especially such as appears in the actions of Nature, without there be reason to cause that order and harmony. (OEP, 207)

This passage employs an argumentative strategy that cuts against the machine emulation thesis. The mechanistic philosophers assumed that the order and regularity of nature requires an explanation in terms of inert matter governed by external principles. Cavendish turns this on its head. Order, method, and harmony in nature demonstrate that matter itself must possess reason, because ‘it is impossible to make orderly and methodical Distinctions, or distinct Orders, by Chances’ (GNP, 7). The order of nature is evidence of intelligence *within* matter, distributed across all its parts, rather than evidence of an external mechanism imposed upon passive matter.

Cavendish’s account of how this self-generated order works in particular cases shows what one gains, philosophically, from this alternative ontology. The mechanistic philosophy explained causal interaction by the transfer of motion from one body to another: a billiard ball moves because the cue strikes it, and the motion passes from the cue to the ball through contact. Cavendish rejected this picture on the grounds that motion, as a mode inseparable from matter, cannot quit one body and enter another any more than a body can quit its own substance (*Philosophical Letters*, 98; see also James 1999, 223). Her alternative draws on a concept she calls ‘patterning out.’ When one body acts upon another, the affected body produces the appropriate effect from within itself, using its own rational and sensitive matter to pattern out the figure of the occasioning body. When a woman sees a dog, the sensitive matter of her eye patterns out the dog’s figure through its own self-motion; the rational matter then patterns out an integrated representation. The same process operates throughout nature: a looking-glass patterns out a face just as snow patterns out a footprint, or a billiard ball patterns out the figure of the ball that collides with it (*Philosophical Letters*, 104–105). In each case, the occasioning body is a prompt, an invitation to self-activity, and the effect is generated by the principal cause: the body that responds. The term ‘patterning’ itself appears in Hobbes, who observed that ‘some natural bodies have in themselves the pattern of almost all things’ (*Of Body*, in *English Works*, vol. 1, 389); Cavendish adopted his language while putting it to work within her own distinctive framework. The result is a theory of causation in which the orderliness of nature depends, at every point, on the perceptive and rational capacities of the bodies involved. Each body must know how to respond appropriately to the occasions it encounters.

The standard reading of Cavendish holds that she attributed ubiquitous rationality to nature in order to explain how complex, orderly phenomena could arise given the inadequacy of purely mechanical causation. Cavendish indeed makes this argument with force, and she is not alone in pressing it: Leibniz drew similar conclusions about the insufficiency of mechanism, though he inferred that minds must be immaterial where Cavendish inferred that matter must be intrinsically minded (Cunning 2016, 72). But Karen Detlefsen has identified a deeper philosophical motivation for this position. That deeper motivation concerns the grounding of *freedom* within nature. Cavendish attributes rationality to matter in

order to underwrite a theory of the ubiquitous freedom of natural parts: their capacity for self-determination and for coordinating their activities voluntarily rather than under external compulsion. This allows Cavendish to account for deviations from regular patterns. For, on Cavendish's view, there are there are *irregularities* in nature (GNP, 117–18) – and it is precisely these irregularities that the mechanistic philosophy struggles to accommodate.

In the mechanistic philosophy, natural order is *necessitated*: things could not have been otherwise, given the laws or the substance from which they follow, and they do follow them at all times, meaning that cannot be true irregularities. Cavendish's account, by contrast, presents nature as orderly because its parts are rational and voluntarily coordinate their activities towards harmonious ends; but since those parts are also free, their freedom means that natural order does admit of exceptions, irregularities, and creative departures that no deterministic system can predict or accommodate. The parts of nature interact through what Cavendish calls 'rational suggestion and response,' each part exercising its own specific sort of rationality in its causal encounters with others. An occasional cause rationally suggests a course of action; the principal cause, the body being acted upon, may or may not follow the suggestion, because it is a free rational agent in its own right (Detlefsen 2007, 178–179). The order of nature thus arises from a vast network of voluntary rational coordination, and disorder, when it occurs, arises from the same source: the dissent of rational parts from the harmonious activity that Nature-as-whole prescribes.

This allows Cavendish to draw a close parallel between natural disorder and political disorder. Disease, for Cavendish, is a disorder within the body that is structurally identical to civil war: a dissent of parts from the rational, order-giving command of the sovereign whole. As Detlefsen observes, Cavendish's first-hand experience of the English Civil War is relevant here: the monarch's subjects who dissented from sovereign authority and plunged the commonwealth into strife are instances of the same phenomenon as the bodily parts whose rebellion against the organism's rational order produces disease and death (Detlefsen 2007, 177).

The treatment of disorder is also where Cavendish parts ways with Spinoza: though they both attribute rationality and self-activity to nature, reject Cartesian dualism, and treat the natural world as a single, self-sufficient whole, Spinoza is committed to regarding apparent disorders in nature as *merely* apparent – from the standpoint of the whole, everything follows necessarily from the nature of substance, and there is no genuine evil or irregularity (E I, Appendix). For Cavendish, by contrast, disorders are *real*, made possible precisely by the fact that matter's parts are free and can fail to meet the standards that Nature's rational design prescribes. Where Spinoza's nature is a deterministic machine down to its ultimate metaphysical structure, Cavendish's nature is a polity retaining just the degree of freedom it needs to defy total predictability.

That alternative ontology comes with an alternative epistemology, and it is here that Cavendish's challenge to mechanistic philosophy engages the rationales for

systematisation most directly. If nature is self-active and self-knowing, then understanding it cannot simply consist in reducing it to a system of deterministic laws deduced from first principles. The mechanistic method of understanding, which abstracts from particular qualities, identifies the underlying mechanism, and deduces the phenomena from universal laws, systematically misrepresents its object – in both senses of “systematically” – by treating as passive and inert what is in fact active and intelligent.

The conclusion that Cavendish draws from this is not that nature is unintelligible, however, but rather that its intelligibility must take a different form that resists tidy systematisation – it must be an intelligibility that respects the self-activity of its objects and remains attentive to the particular ways in which different parts of nature organise themselves. We should therefore acknowledge the limits of any single deductive framework’s capacity to capture the full range of natural phenomena, on this account.

The implication for the epistemological rationale is that when formal coherence becomes the sole criterion of knowledge, it generates a self-enclosed order that *excludes* whatever resists formalisation. This consequence emerges with particular force in her critique of the experimental philosophers and their instruments. Robert Hooke had claimed in the Preface to *Micrographia* (1665) that microscopes could reveal the very composition and structure of natural bodies – their ‘inward motions,’ and thus the ‘secret workings’ of nature. Cavendish denied this. The microscope, she argued, does not disclose the interior nature of its objects; it distorts their exterior appearance, interposing an artificial medium between the perceiver and the perceived. ‘A louse by the help of a magnifying glass appears like a lobster,’ she wrote, ‘where the microscope enlarging and magnifying each part of it, makes them bigger and rounder than naturally they are. The truth is, the more the figure by art is magnified, the more it appears misshapen from the natural’ (OEP, 50). The instrument generates a representation that is partly natural and partly artificial: what Cavendish memorably calls a ‘hermaphroditical’ figure, a hybrid of the object and the medium through which it is perceived. ‘The best optic is a perfect natural eye, and a regular sensitive perception,’ Cavendish concluded, ‘and the best judge, is reason; and the best study, is rational contemplation joined with the observations of regular sense, but not deluding arts’ (OEP, 53).

Cavendish’s ontology also has consequences that extend beyond epistemology into ethics. If matter is rational and sensitive throughout, then non-human creatures possess a form of knowledge, and the assumption that human knowledge is superior is one that, as Cavendish insists, we are not entitled to make: ‘Man may have a different Knowledge from Beasts, Birds, Fish, Flies, Worms, or the like; and yet be no wiser than these sorts of Animal-Kinds’ (GNP, 148). Manuel Fasko and Peter West (Fasko and West 2024) have recently argued that Cavendish moves from this ontological claim to a normative one, describing a perfectly ‘regular’ world in which humans do not eat animals if it requires killing them, nor plants if it requires

destroying their 'Root and Foundation' (GNP, 205). The mechanistic philosophy, by treating matter as inert and non-human nature as devoid of cognition, underwrites an instrumental relationship between humans and their environment. Cavendish's ontology, by distributing cognition and intrinsic value throughout nature, generates constraints on that relationship.

The *Description of a New World, Called the Blazing-World* (1666), published as a companion volume to the *Observations*, explores these philosophical themes through the medium of speculative fiction, and, in doing so, exposes the limits of the political rationale for systematisation. The work imagines a young woman who passes through the North Pole into an alternative world populated by hybrid creatures: bear-men, bird-men, worm-men, fly-men, ape-men, spider-men, and lice-men, each species assigned to a distinctive intellectual vocation by the Empress who governs them. The Empress erects schools and founds learned societies, interrogating each group about its methods and conclusions with the authority of a sovereign who can demand intellectual accountability. But each learned society turns out to operate with a distinctive epistemological framework that generates its own criteria of evidence, its own standards of demonstration, and its own characteristic blindspots. The bear-men's telescopes and the worm-men's burrowing investigations produce incommensurable pictures of the same world. Disagreements among the societies multiply until they threaten to 'break out into an open Rebellion, and cause a great disorder, and the ruin of the Government' (TBW, 201). The Duchess of Newcastle herself appears as a character, summoned as an advisor, and recommends that the Empress dissolve the learned societies altogether, 'for 'tis better to be without their intelligences, than to have an unquiet and disorderly government' (TBW 202). The advice is barbed with political irony. Hobbes had argued that philosophical disputation breeds faction and threatens sovereign authority; Cavendish dramatises the point while simultaneously suggesting that the suppression of intellectual diversity comes at a significant cost (Walters 2014, 151). The Blazing-World's peace is purchased at the price of its knowledge, including of geometry. The fictional frame thus accomplishes something that the geometric method, by its nature, cannot: it stages a plurality of possible epistemic frameworks, each with its own internal coherence, each generating conclusions invisible or unintelligible to the others. This starkly exemplifies Cavendish's philosophical conviction that no single cognitive system can exhaust the intelligibility of a nature whose parts are themselves intelligent and free.

Cavendish thus occupies an important position in the genealogy of cognitive systematisation: she is the first philosopher to articulate, from within the seventeenth-century debate about mechanism, a coherent account of what systematic thought *excludes*. The machine model gains its considerable power from its impersonality, deductive necessity, and universality, which together promise the reduction of complex phenomena to simple principles susceptible of formal manipulation. These are genuine intellectual virtues; but what Cavendish brings into view is the *cost* of

those virtues. The epistemological rationale, when formal coherence becomes the sole criterion of knowledge, excludes whatever resists formalisation. The political rationale, taken to its Hobbesian extreme, suppresses the plurality of epistemic perspectives in the name of order. And the pedagogical rationale, when transmissibility becomes the measure of cognitive value, flattens the living judgement of the knower into a set of operations that anyone can perform, thereby losing whatever depends on the responsive exercise of trained perception and cultivated reason. Some forms of understanding depend on the situated judgement of the knower – on capacities of perception and rational contemplation that cannot be deposited in articulated structures and operated by anyone who follows the rules. The mechanistic picture renders these forms of understanding invisible.

Cavendish's voice was long dismissed as eccentric by a canon whose criteria of intellectual seriousness were shaped by just the mechanistic model she challenged. When Diderot, a century later, argues in the *Rêve de d'Alembert* that nature is a self-organising vitality irreducible to the categories of mechanical science, he will be reinventing, without knowing it, an argument that the Duchess of Newcastle had already made. The counter-tradition to the machine emulation thesis runs from Cavendish through Diderot to the Romantics and beyond; the genealogy that traces the rise of systematic thought must also trace the resistance that accompanied it.

2.6 The Geometric Order in Retrospect

We began with a question that the Protestant system-builders bequeathed to their seventeenth-century successors: can systematisation produce certainty, or can it only organise information that may turn out to be false? The four philosophers examined here offered answers of escalating ambition, and escalating cost. Descartes mechanised the *method* of inquiry while leaving the thinking subject exempt; Hobbes revoked the exemption by identifying reason with computation and extending the machine model from cognition to the state; Spinoza designed the philosophical text itself as an inferential machine and the mind as an automaton governed by its own necessary laws; Cavendish, working from within the same intellectual milieu, showed what this mechanisation excluded by distributing cognition throughout matter and reconceiving natural order as the voluntary coordination of free rational parts.

The three practical rationales for systematisation are thus all already operative in the seventeenth century. The epistemological rationale is the century's dominant one: the sceptical crisis demanded that systematic order certify thought, and the geometric method was the instrument devised to meet that demand. The political rationale emerged with Hobbes, where the same machinery that certifies knowledge also certifies political authority, and surfaced again in the *Blazing-World*, where Cavendish dramatised the tension between public order and intellectual diversity. The

pedagogical rationale, inherited from the previous century, persisted throughout in transformed guise.

What the chapter as a whole reveals is that the three rationales are linked by a common dependence on machine-like virtues, and that each virtue, extended beyond its proper domain, generates its characteristic cost. Cavendish is the first to ground this cost profile in an alternative ontology, and the force of her critique lies in the fact that the costs she identifies are not incidental to the cognitive emulation of machine-like virtues, but structural consequences of the very properties that make it powerful.

The metaphysical warrant – the conviction that systematic thought tracks truth because reality is itself systematic – accompanied this development in different configurations. The pattern that emerged was that the more ambitious the systematisation, the more weight the warrant must bear: Descartes's warrant guarantees the deliverances of a method, while Spinoza's underwrites the identity of thought and reality. This escalation reflects a corresponding escalation in the underlying scepticism – the more radical the doubt, the more comprehensive the systematic response it demands, and the more robust the guarantee needs to be that the order of thought corresponds to the order of things.

The question that Leibniz inherited was whether the mechanisation of thought could be perfected: whether a formal calculus could be devised that would make all reasoning as transparent and compulsive as arithmetic, resolving disputes by computation and rendering disagreement a mere failure to calculate correctly. The question that Cavendish bequeathed, though it would take a century to find its next great voice in Diderot, was whether such perfection, even if achievable, would constitute genuine understanding. The tension between these two questions drives the genealogy from this point forward.

Calculamus: Leibniz and the Dream of the Reasoning Machine

“Our intellect, in view of its weakness, must be directed by certain mechanical threads.”

– Leibniz

In 1674, Gottfried Wilhelm Leibniz (1646–1716) demonstrated an early prototype of his calculating machine to the Royal Society in London. The device, later refined as the Stepped Reckoner, was meant to execute addition, subtraction, multiplication, and division through an arrangement of interlocking gears: a mechanism whose outputs no reasonable person could dispute, because they followed from the configuration of the parts with the same necessity as the fall of a weight. The machine did not work very well. It jammed, miscalculated, and required constant adjustment by its operator. But the ambition it embodied was breathtaking, and it was an ambition that went well beyond arithmetic. What Leibniz envisaged was a device, partly physical, partly notational, and partly metaphysical, that would do *for reasoning in general* what the Stepped Reckoner was supposed to do for numerical computation: settle contested questions by feeding them through a procedure whose verdicts carried the impersonal authority of a calculation. ‘If this were done,’ Leibniz wrote, ‘when controversies arise, there will be no more need for a dispute’ (A VI, 4, 912). When two philosophers disagreed, they would sit down together and say, *calculamus*: let us calculate, without further ado, to see who is right.¹

Every systematiser examined so far had, in one register or another, modelled proper thinking on the workings of machines. In Leibniz, the aspiration becomes a concrete programme: first, build a universal notation in which any item of knowledge can be encoded, the *characteristica universalis*; second, build a logical calculus

¹The passage comes from an unpublished fragment written by Leibniz around the summer of 1688, titled *De arte characteristica ad perficiendas scientias ratione nitentes* (*On the Characteristic Art for Perfecting the Sciences that Rely on Reason*). The original reads: ‘Quo facto, quando orientur controversiae, non magis disputatione opus erit inter duos philosophos, quam inter duos Computistas. Sufficiet enim calamos in manus sumere sedereque ad abacos, et sibi mutuo (accito si placet amico) dicere: calculamus’ (A VI, 4, 912).

that operates on this notation with the impersonal reliability of arithmetic, the *calculus ratiocinator*; third, build a physical machine to execute the calculus. Underwriting the entire enterprise is a metaphysics in which reality itself is a paradigmatically systematic structure, designed by a rational God and governed by the principle of sufficient reason, so that the demand for cognitive system is grounded in the very nature of things. Here, the machine emulation thesis ceases to be a historiographical observation about the tendencies of early modern philosophy. It becomes an articulated research agenda *in* early modern philosophy. And the metaphysical warrant, which had been emerging in various forms since the Reformed pedagogues, reaches its most self-conscious expression.

The historical and biographical circumstances in which this programme took shape shed a great deal of light on its underlying motivations, and they deserve attention at the outset.

3.1 Fragmentation and Integration

Gottfried Wilhelm Leibniz was born in Leipzig on 1 July 1646, two years before the Peace of Westphalia brought the Thirty Years' War to its exhausted conclusion. The war had killed roughly a third of the population of the German-speaking lands. It had devastated institutions of learning, shattered confessional arrangements that had seemed permanent, and left a political landscape of extraordinary complexity: hundreds of sovereign and semi-sovereign territories, each with its own legal, ecclesiastical, and educational order, held together only by the increasingly fragile architecture of the Holy Roman Empire. This was fragmentation at every level: political, confessional, and intellectual. It is hard not to see this as the historical matrix of the guiding question of Leibniz's philosophical career: how to integrate what had become fragmented.

The question was, of course, in various respects continuous with the crises that had driven earlier chapters of this genealogy. The Reformed pedagogues of Chapter 1 faced the collapse of the Church's authority as a guarantor of doctrinal order and invented the *systema* as a structural replacement. The rationalists of Chapter 2 faced the collapse of the criterion of truth itself, in the wake of the Pyrrhonian revival, and turned to the geometric method as a source of certitude that external authority no longer supplied. Leibniz inherited both of these challenges simultaneously and added a further one whose urgency was, for him, inseparable from the others: the need for political and confessional reconciliation across a fractured Europe. His ecumenical ambitions, his proposals for learned academies, his codification of legal principles, and his schemes for the coordination of European learning all expressed one underlying conviction: that systematic reason, properly deployed, could heal the divisions that war, dogmatism, and intellectual disorder had inflicted. Cognitive

integration and political reconciliation were, in Leibniz, two aspects of a single project.

In one respect, however, this project formed against the grain of Leibniz's own milieu. His father, Friedrich, was a professor of moral philosophy at Leipzig who based his teaching on Aristotle's *Nicomachean Ethics* and conceived of the philosophical faculty as preparation for theology in service of the *respublica Christiana*. He died when Leibniz was six. The family that survived him was thoroughly moored in Leipzig's strict Lutheran orthodoxy: Leibniz's half-brother Johann Friedrich was a committed opponent of ecumenical 'syncretism,' and his sister Anna Catharina wrote to him in alarm in 1672 at rumours that he had converted to Calvinism (something even worse, for a strict Lutheran, than conversion to Catholicism) (Antognazza 2009, 18–19). In this regard, the irenic breadth of mind that would define Leibniz's career was strikingly uncharacteristic of the world in which he grew up. As early as 1669, writing to his half-brother, the twenty-three-year-old declared that he held fast to the Evangelical truth but was 'deterred from condemning others both by my own personal inclination and by the stern command of Christ: Judge not, that ye be not judged' (Antognazza 2009, 22). He intended to remain a Lutheran because the Lutheran confession was one part of a truly universal church, and his calling was to work toward their reunion.

What made this breadth of mind possible was an intellectual inheritance that came, paradoxically, through his father's library. On gaining access to the books as a boy of eight, Leibniz fell upon the ancients and worked his way, with increasing comprehension, through Plato, Aristotle, Cicero, and the Church Fathers. Most consequentially for our concerns, the library also contained works from the tradition of post-Ramist encyclopaedism that Chapter 1 traced through Keckermann and the Reformed academies. Among the books Leibniz specifically requested to retain when the library was sold in 1684 were Keckermann's *Opera omnia*, the juridical encyclopaedia of the Herborn political theorist Johannes Althusius, and the logical and encyclopaedic works of Johann Heinrich Bisterfeld, a student of Alsted (who was himself a student of Keckermann). The young Leibniz had annotated several of Bisterfeld's works as a student (Antognazza 2009, 30–31).

In Bisterfeld, Leibniz encountered two ideas that would become hallmarks of his own philosophy: universal harmony as the union and communion of everything with everything, and the thesis that all beings are endowed with perception and appetite (Antognazza 2009, 32; Rutherford 1995, 36–40). The pedagogical machinery of Chapter 1, the *systema* as an instrument for making knowledge transmissible and surveyable, thus fell through the Herborn encyclopaedists directly into Leibniz's hands. And what he would do with it involved a transformation of the very concept of a system. As Nicholas Rescher has observed, Leibniz was the first philosopher to describe himself as having a 'system' of philosophy and to attribute the possession of one to others (Rescher 1981, 117). For Keckermann and his contemporaries, *systema* designated a genre: a synoptically comprehensive and methodically ordered expos-

ition of a discipline. By the second half of the seventeenth century, the term had undergone a decisive semantic shift. It came to designate a particular *doctrine* or *theory*, an integrated body of philosophical truth, a *nexus veritatum*. Leibniz was the prime mover of this new usage: he spoke of ‘my system of pre-established harmony,’ contrasting it with the ‘system of occasional causes’ and the ‘system of efficient and final causes’ (Rescher 1981, 117). What had begun as a label for a pedagogical format became, in Leibniz’s hands, a claim about the structure of reality and the knowledge that mirrors it. The pedagogical machine of Chapter 1 was being retooled as a cognitive and metaphysical programme of a far more ambitious kind.

The eclectic Aristotelianism of his Leipzig teachers, Jakob Thomasius and Johann Adam Scherzer, reinforced this orientation. As Christia Mercer has shown, the young Leibniz absorbed from them a specifically German tradition of ‘conciliatory eclecticism,’ committed to the thesis that truth is dispersed across rival philosophical schools and recoverable through principled synthesis (Mercer 2002, 23–54). The conviction that most philosophical sects ‘are right in what they assert, but wrong only in what they deny,’ as Leibniz rather marvellously put it (L 655), was a methodological commitment before it was a metaphysical one. His doctoral dissertation in law, completed at Altdorf in 1666, the same year as the *Dissertatio de arte combinatoria*, already shows the two poles of his thinking in place: a formal, combinatorial approach to the organization of knowledge, and a conviction that the proper aim of such organization is practical reconciliation. In *De arte combinatoria*, he proposed that all concepts could be analysed into a finite set of simple elements and that the systematic combination of these elements could generate all possible truths: an early sketch of the *characteristica universalis* that would occupy him for the rest of his life. As Leibniz himself noted in retrospect, the seeds of the programme were planted during his years of private reading: the search for ‘clarity in words’ became the basis of the *ars judicandi*, and the search for ‘usefulness in things’ became the basis of the *ars inveniendi* (Antognazza 2009, 31).

The following years (1672–76) saw him invent the differential calculus, but they also brought him into dialogue with the leading mathematicians and natural philosophers of Europe. Sent to the French capital on a diplomatic mission, Leibniz encountered Huygens, Arnauld, and Malebranche. He visited London, demonstrated an early model of his calculating machine to the Royal Society, and was elected an external member. And in 1676, on his way back to Germany, he stopped in The Hague to discuss philosophy with Spinoza.

Spinoza had responded to the collapse of authority by imposing systematicity through elimination: one monistic system, derived axiomatically from definitions, in which everything that cannot be demonstrated from first principles is discarded. What does not fit the geometric order does not merit the title of knowledge. Leibniz was of far more accommodating disposition. He would respond with systematicity through integration: a framework capacious enough to accommodate the legitimate insights of every philosophical tradition, ancient and modern, scholastic and mech-

anistic, rationalist and empiricist. ‘I have tried to uncover and unite the truth buried and scattered under the opinions of all the different philosophical sects,’ he wrote, ‘and I believe I have added something of my own which takes a few steps forward’ (L 655). He is thus a systematiser, as Nicholas Jolley (2020, 8–10) has stressed against Catherine Wilson (1999, 372–88), and a systematiser of a notably irenic sort.

This contrast between Spinoza and Leibniz is instructive for the genealogy of systematization, because it points to two fundamentally different conceptions of what the cognitive machinery of systematicity is *for*. Spinoza’s machinery *filters*: it accepts only what survives the test of deduction from first principles and discards the rest. Leibniz’s machinery *translates*: it renders apparently rival doctrines in a common symbolic language and reveals, through the operations of the calculus, whether their disagreements are genuine or merely verbal. Where a filtering machine achieves consistency by exclusion, a translating machine aims at harmony by incorporation. The latter’s ambition is grander, but the danger correspondingly greater: a machine designed to reconcile may end up merely papering over disagreements that resist reconciliation and run all the way down.

Upon his return from Paris, Leibniz took up the position of Court Councillor and Librarian to the Duke of Hanover, a post he would hold, in one form or another, for the remaining forty years of his life. The Hanoverian court placed him at the intersection of intellectual ambition and political constraint in a way that proved philosophically productive. Under Duke Johann Friedrich (d. 1679), Leibniz enjoyed the patronage of a prince who valued learning and culture. He enlarged the library, proposed the creation of a universal archive of knowledge, and around 1679 began to refer to his all-embracing plan as the *scientia generalis*: the ‘general science’ that would contain ‘the principles of all the other sciences, and the method for using the principles, so that everyone endowed with ordinary abilities will quickly descend to any specific matters whatsoever by easy meditation and short experience’ (A VI, 4, 532; Antognazza 2009, 237). The formulation deserves emphasis. The *scientia generalis* was conceived from the outset as a system whose *universality of access* was among its essential features: anyone who mastered the method could operate it, regardless of philosophical allegiance, confessional affiliation, or native language. The pedagogical rationale that had driven the Reformed *systema* of Chapter 1 was here explicitly reframed as a design constraint on a universal science.

This was itself the latest in a series of reformulations. As early as 1668, during his years at the Catholic court of Mainz, Leibniz had conceived the *Demonstrationes Catholicae*, a systematic encyclopaedia of the sciences whose prolegomena would embrace metaphysics, logic, mathematics, physics, and practical philosophy, and whose final aim would be the demonstration of the central tenets of Christian theology (Antognazza 2009, 91–92). The plan was never realized in its entirety, but it underwent a significant evolution. The *Demonstrationes Catholicae* gave way to the *scientia generalis* and its ‘demonstrative encyclopaedia’ in the late 1670s; and in a letter of 1710, Leibniz presented the *Theodicy* as the ‘forerunner’ of what he now

described as ‘Elements of general philosophy and of natural theology’ (Antognazza 2009, 225). One can discern a movement towards secularization in this: he no longer regarded revealed theology as the culmination of the system, but merely sought to embed demonstrative reason within a theistic framework.

Under Johann Friedrich’s successor, Ernst August, the terms of Leibniz’s patronage changed in a way that led him to think in more *institutional* terms about the project of the integration of knowledge. Ernst August saw value in the prestige and utility of learning, and Leibniz was increasingly drawn into the tasks of statecraft: he was charged with drafting legal codifications, writing diplomatic memoranda, and composing a history of the House of Brunswick. As Simon Dumas Primbault has documented, the years surrounding 1679 were pivotal: Leibniz multiplied proposals to the new Duke for learned institutions, medical systems, and state education, which he would frame in terms of their contribution to the rational administration of the Duchy (Primbault 2020, 97). His schemes for academies of sciences, beginning in the early 1670s with proposals to the Emperor and the Elector of Mainz and culminating in his founding of the Berlin Society of Sciences in 1700, were designed as organizational infrastructure for the coordination of dispersed intellectual labour across the Empire’s confessional and territorial divides (Antognazza 2009, 121–24, 211). *Theoria cum praxi*, the motto he chose for the Berlin Society, enshrined his conviction that theoretical knowledge and practical improvement were inseparable. The academy became itself a kind of institutional machine for systematizing collaboration.

This entanglement of the cognitive and the political can be traced through Leibniz’s legal writings, in particular his *Systema Iuris*, which aimed to reconstruct jurisprudence on scientific principles (Berkowitz 2005, 17–54). There, Leibniz envisaged a codification of legal knowledge whose authority would derive from its systematic structure rather than from the traditions or persons that had historically warranted it. The aspiration parallels his programme for the *characteristica universalis*. In both instances, the aim was to replace the personal authority of the judge, the professor, or the clergyman with a system of rules whose outputs could be inspected and evaluated by anyone who mastered the notation. Rational order, made explicit and surveyable, would do the work that personal authority could no longer be trusted to perform.

This was the context in which Leibniz’s philosophical project took shape. The *characteristica universalis*, the *calculus ratiocinator*, and the metaphysics of pre-established harmony were responses to fragmentation at every level: intellectual, confessional, and political. They were attempts to build, through systematic reason, a form of coordination that war, schism, and the loss of shared institutions had destroyed. In the remainder of this chapter, I propose to examine in more detail what this programme involved, what metaphysical resources it drew on, and where its deepest tensions lay.

3.2 The Alphabet of Thought

Leibniz pursued this programme for the rest of his life, and its three components are worth distinguishing carefully, because each represents a different level of the aspiration to mechanise reasoning. The *characteristica universalis* was a universal symbolic language in which any item of knowledge could be encoded. The *calculus ratiocinator* was a logical calculus that would operate on the symbols of the *characteristica* according to fixed rules, producing truths with the same reliability as arithmetic. And the physical calculating machine would execute the operations of the calculus mechanically, removing the need for human computation altogether. Taken together, these three components constitute a programme for depositing more and more of the work of reasoning in structures external to the individual mind, reducing the thinker's contribution to the setting of initial conditions.

The programme rested on a conception of cognitive system that Leibniz's own mathematical innovations had done much to reshape. The paradigm that had governed philosophical thinking about systematization since antiquity was Euclidean: a finite set of axioms and definitions, from which consequences are extracted by step-by-step deduction. Spinoza's *Ethics* is the purest philosophical expression of this model: every proposition follows from its predecessors in a chain whose links are individually inspectable. Leibniz's invention of the differential and integral calculus gave him access to a different model entirely. As Nicholas Rescher has argued, the calculus, and especially the calculus of variations, furnished Leibniz with a new paradigm of rational systematization: one that proceeds by surveying an infinite range of alternatives and discerning, within that range, those configurations that satisfy the principles of rational economy (Rescher 1981, 122). The Euclidean system extracts content from a finite basis; the calculus-based system identifies, amidst an endless field of possibilities, the particular arrangement in which the greatest variety of results is achieved by the simplest means. This is the 'principle of determination in nature which must be sought by maxima and minima,' as Leibniz put it: 'a maximum effect should be achieved with minimum outlay' (L 487). The mini-max principle links the formal structure of the calculus to the metaphysical structure of the best possible world, and it shapes the *characteristica* in a way that the Euclidean model could not. Where Spinoza's system filters by deduction, Leibniz's system optimizes by integration. The translating machine described in the previous section has a mathematical correlate: it aims to incorporate the maximum of intellectual content under the most economical set of principles.

The first component of the programme required what Leibniz called an 'alphabet of human thoughts': a catalogue of primitive concepts from whose systematic combination all complex concepts could be generated. The analogy with arithmetic was deliberate. Just as a finite set of digits, combined according to the rules of positional notation, can express any number, so a finite set of conceptual primitives, combined according to the rules of the *characteristica*, would express any possible thought.

Leibniz arrived at this idea early. Already in the *De arte combinatoria* of 1666, he had proposed that all concepts could be resolved into simples and that the art of combination could generate their permutations exhaustively. As Leen Spruit and Guglielmo Tamburrini observe, this early statement already expresses the two faces of the mature programme: the *ars judicandi*, a method for evaluating whether a claim is true or false, and the *ars inveniendi*, a method for discovering truths not yet known (Spruit and Tamburrini 1991, 3). Leibniz regarded this dual function as distinguishing his project from the many contemporary schemes for universal languages that proliferated in the seventeenth century. George Dalgarno's *Ars Signorum* and John Wilkins's *Essay towards a Real Character and a Philosophical Language*, both of which Leibniz studied carefully during his years at Mainz, offered systems of notation that classified things and their relations; what they did not offer was a calculus (Antognazza 2009, 108–10). Dalgarno and Wilkins aimed to *represent* the structure of the world in a well-ordered notation. Leibniz aimed to produce a notation in which the process of reasoning itself could be carried out symbolically, so that the formal manipulation of characters would track, and in certain cases replace, the movement of thought. The difference between representing knowledge and *computing with* representations is the difference between an encyclopaedia and a machine.

Why should formal manipulation of characters be able to do the work of thought? Leibniz's answer rests on a candid acknowledgement of cognitive limitation. The finite human mind cannot, in general, grasp complex concepts in their entirety. We cannot hold before the mind's eye all the properties of a circle simultaneously, or survey all the steps of a long demonstration at once. What we can do is *substitute symbols for concepts* and then manipulate those symbols according to formal rules, trusting the rules to preserve truth even when we have lost sight of the meanings the symbols encode. Leibniz called this procedure *cognitio caeca sive symbolica*: blind or symbolic knowledge. The idea appears in embryonic form as early as 1671, when the young Leibniz referred to ordinary thinking as *cogitationes caecas*, 'blind thoughts,' in which we operate with signs without reaching through to the ideas they represent (Antognazza 2009, 109). In the *Meditationes de cognitione, veritate et ideis* of 1684, the concept receives its mature formulation: 'we make use of signs in place of things themselves, whose explicit explanation we usually omit for the sake of brevity, knowing or believing that we have the ability to produce it at will' (A VI, 4, 587–588; Leduc 2014, 56–57). The mind, operating symbolically, does not need to attend to the conceptual content of each term; it attends instead to the formal relations among symbols. The symbols serve, in Leibniz's image, as a *filum cogitandi*: a thread of thought that guides the mind through inferences too complex for unaided intuition to follow.

Blind knowledge is the epistemological core of the machine thesis in its Leibnizian form. In the *Analysis linguarum*, Leibniz stated the point with a concision that illuminates the entire project: 'our intellect, in view of its weakness, must be

directed by certain mechanical threads; in this connection, recall that, with regard to thoughts which represent things that do not fall under the imagination, only the characters are available' (A VI, 4; Leduc 2014, 54). The phrase 'mechanical threads' is his own acknowledgement that the *characteristica* does the work of a machine: a structure external to the mind that processes inputs according to fixed rules and produces outputs whose correctness can be verified independently of the operator's understanding. The machine-like virtues that this book has been tracking since Chapter 1 are present here in concentrated form. A derivation carried out in the *characteristica* is surveyable, because each step is a symbol-transformation that can be inspected. It is rule-governed, because errors register as violations of syntax, just as errors in arithmetic register as miscalculations. Since anyone who masters the notation can operate the calculus regardless of philosophical allegiance, confessional affiliation, or native language, the system is universal. And since the rules make no reference to the operator's identity, social standing, or personal convictions, the output is impartial. What Leibniz proposed was to generalize these features across every domain where concepts admit of definition and claims admit of demonstration.

In a letter to the Duke of Hanover, Leibniz described the *characteristica* as 'a really infallible judge of controversies,' adding that, by its means, 'men will always be able to ascertain if it is possible to decide a given problem on the basis of the available knowledge' (GP VII, 26). The calculus would function as a *judge*, delivering verdicts whose authority derived entirely from the formal correctness of the derivation. If two disputants disagreed, they would only need translate their claims into the notation and compute.

This may strike one as optimistic. But Leibniz himself offered two qualifications, which show his conception of the programme's scope to be more nuanced than the *calculemus* formula perhaps suggests. First, as comes out in the remark to the Duke of Hanover, he acknowledged that the *ars judicandi* could not always deliver a definitive verdict. Where the available knowledge was insufficient to decide a question conclusively, the calculus would determine 'what is most likely,' setting 'exact limits to the unknown truth' (GP VII, 26). The calculus, that is, would function as an instrument of rational estimation in addition to an instrument of demonstration: a concession that the aspiration to mechanise reasoning could be scaled to the quality of the available evidence.

Second, Leibniz was careful to distinguish between the results the calculus could deliver and the creative work required to construct it. The *ars inveniendi* would enable 'anybody endowed just with sufficient readiness to act' to establish 'everything that can be obtained from the data by a great and highly trained wit in virtue of reasoning,' but the strength of this method lay 'more in acting than in meditating and discovering' (GP VII, 202; Spruit and Tamburrini 1991, 4). The calculus could *execute* the inferences that a first-rate mind would draw, but the initial identification of primitive concepts, the design of the notation, and the interpretation of

the results all remained exercises of judgement that the calculus itself could not perform. The machine operates within a framework that the machine cannot build. This asymmetry, between reasoning *within* a system and the creative construction of a system, is one of the deepest tensions in the programme of systematisation as mechanization, and we will return to it in this chapter's concluding sections.

3.3 The Machine on the Desk

Leibniz's theoretical work was continuous with his engineering, and this continuity reverberates into this philosophy. Up to this point in the genealogy, the machine emulation thesis has been an analytically useful metaphorical lens through which to understand what early modern systematisers were doing. With Leibniz, we reach the point at which the metaphor is taken so seriously that cognitive engineering becomes literal engineering and the cognitive machine a literal machine. The *calculus ratiocinator* was a formal system; but the Stepped Reckoner was a physical device that executed formal operations on gears and wheels. This greatly reduced, at least in principle, the distance between theorizing about mechanized reasoning and building a machine that reasons mechanically.

In practice, it must be said that the version of the Stepped Reckoner that Leibniz demonstrated to the Royal Society in 1673 could not yet reliably perform the multiplication and division for which it had been designed; Robert Hooke, who examined the mechanism at close quarters, criticised it harshly (though he subsequently produced a rival model that proved suspiciously similar) (Antognazza 2009, 148–49). Leibniz continued to refine the machine throughout his life, spending considerable sums and never quite achieving the reliability he sought. As Martin Davis observes, the Stepped Reckoner was a 'premature' technology: a design whose logic was in fact sound, but whose materials and manufacturing tolerances were still inadequate to the task (Davis 2000). As late as August 1716, three months before his death, Leibniz was reporting satisfactory improvements on the most recent model (Antognazza 2009, 232).

Yet the real philosophical significance of the Stepped Reckoner lies in the fact that Leibniz undertook to build it at all. Its physical existence on his desk in Hanover made the aspiration to mechanise thought concrete. And the disagreeable experience with Hooke illustrates a feature of the machine emulation thesis from an unexpected angle: the design logic of the calculating machine was reproducible precisely because it was lay open to view. The easy inspectability of this fully externalised cognitive instrument also made it vulnerable to appropriation.

The engineering dimension extended beyond the calculating machine to the material practices of intellectual work. Leibniz's own working methods involved the creation of what Simon Dumas Primbault calls 'ink-and-paper automata': material technologies of reasoning whose structure paralleled, at the level of paper and

pen, the formal structure that the *characteristica* was to provide at the level of notation (Primbault 2020, 88–89). The writing of the *De Affectibus* manuscript of 1679, in which Leibniz drafted a theory of the dynamics of thought, involved the systematic decomposition of complex arguments into numbered propositions, the cross-referencing of definitions across folios, and the spatial organisation of material on the page so as to allow syllogistic relationships to be read off from the layout. The paper-slip techniques that Chapter 1 traced through Gessner and the information-management practices of the sixteenth century reappear here in a different guise. But while the earlier techniques served the pedagogical organisation of knowledge, Leibniz's paper practices served the *computational* processing of arguments. From the filing systems of the early modern library to the gears of the Stepped Reckoner, early modern intellectuals devised increasingly ambitious ways to deposit cognitive work in material structures external to the individual mind.

The convergence of formal programme, physical engineering, and material practice in a single thinker is what gives the *calculemus* its genealogical weight. The mechanization of reasoning was more than a speculative possibility for Leibniz. He actually pursued it across every available medium: symbolic notation, mechanical gears, and paper technology. The question this convergence poses, and the question the remainder of the chapter must confront, is what *warranted* the ambition: what metaphysical convictions underwrote the belief that a universal calculus of thought was achievable, and what becomes of those convictions when they are examined under pressure.

3.4 The Rational Order of Nature

A programme as ambitious as the *characteristica universalis* requires a reason for thinking it achievable. Why should we expect that a universal symbolic language, paired with a formal calculus, could track the structure of reality? The answer Leibniz gave was a metaphysics in which reality is, at every level of description, a harmonious system designed by a rational God. In earlier chapters, the metaphysical warrant often operated selectively. But in Leibniz, the warrant encompasses the whole of reality.

Several ideas underpin the warrant for Leibniz. The principle of sufficient reason holds that nothing exists without a reason: for every truth there is, in principle, an explanation of why it is so rather than otherwise. The principle of pre-established harmony holds that the universe is composed of simple substances, monads, each of which expresses the entirety of the universe from its own perspective, and that the agreement among their perspectives is guaranteed by God's original act of creation. God chose, from among the infinity of possible worlds, that world in which the greatest variety of phenomena is produced by the simplest set of laws. Here the mini-max logic of the calculus, discussed in the preceding section, does double duty: it

is both a mathematical method and a description of divine rationality. Each monad reflects the whole; the whole is reflected in each part; and the rational order that pervades the universe is the supreme perfection that a benevolent and omniscient God would select. As Donald Rutherford has argued, ‘a principal measure of the world’s perfection for Leibniz is its “rational order,” or the degree to which reason, in the form of order and intelligibility, has been realized within the constitution of created things’ (Rutherford 1995, 2). The perfection God finds in the world is, at the most fundamental level, an intellectual good: a state of affairs whose value consists in its being maximally intelligible to an omniscient mind.

This architecture makes the demand for cognitive systematization a *metaphysical obligation*. If reality is a harmonious system designed by a rational God, and if the principle of sufficient reason ensures that every feature of reality has a rational ground, then the demand that our knowledge form a corresponding system is underwritten by the deepest structure of things. The inference can be reconstructed as a syllogism: knowledge must reflect the truth; the truth is a system; therefore, knowledge should be a system. No early modern thinker stated this explicitly. Rescher’s distinction between ontological systematicity (a feature of the world) and cognitive systematicity (a feature of our knowledge) clarifies what is at stake: Leibniz makes the latter a mirror of the former, so that the adequacy of a philosophy can be gauged by its systematic character (Rescher 1981, 120). His metaphysics supplies the major premise, his epistemology the minor, and the *characteristica universalis* is the practical expression of the conclusion: the task of science is to construct a symbolic system that mirrors the rational structure of the world as completely as finite minds can manage.

Nicholas Jolley identifies the ‘mirrors of God’ theme as one of the organizing ideas of Leibniz’s philosophy (Jolley 2020, 2–6; see also Craig 1987, 51–64). Monads resemble God in being simple, immaterial, and causally self-sufficient. Human minds, in particular, are mirrors of God: endowed with innate ideas, they can draw knowledge from their own cognitive resources. This is the metaphysical guarantee underwriting the *characteristica*. Because minds mirror the divine intellect, the rational structures they discover through systematic thought track the rational structures of reality. When the *characteristica* produces a result by formal computation, that result corresponds to a truth about the world, because the formal relations among symbols mirror the rational relations among things, and both mirror the ideas in the divine intellect. God designs the world according to rational principles, and minds are created to reflect those principles. The *characteristica* encodes them in symbolic form. The *calculus ratiocinator* then extracts their consequences mechanically. The metaphysical warrant, in its Leibnizian form, makes the aspiration to mechanize reasoning appear as a natural expression of the universe’s own rational structure.

What gives Leibniz’s formulation of the warrant its distinctive character is the connection he drew between metaphysical harmony and practical ethics. The

rational order of the universe was for Leibniz an intellectual good, as Rutherford shows, but it was also the ground of human happiness. Wisdom, he wrote in an early formulation that remained constant across his career, is the ‘science of happiness’ (*scientia felicitatis*); justice is the ‘charity of the wise’ (*caritas sapientis*), the habit of loving all human beings grounded in wisdom (Antognazza 2009, 115, 183). In the *Demonstratio Propositionum Primarum* of 1671–72, the chain was stated concisely: ‘Wisdom is the science of happiness. . . . A good man is one who loves all human beings. To love is to find delight in the happiness of another. To find delight is to feel harmony’ (A VI, 2, 485). The argument runs from the metaphysical ground – universal harmony – through its epistemic expression – wisdom as knowledge of the order of things – to its ethical culmination: delight in the happiness of another.

To systematize knowledge, on this account, is to participate in the rational order of the universe, and to participate in that order is to advance toward the happiness that consists in wisdom. The advancement of science was, in the formulation Leibniz repeated from the 1671 *Grundriß* through to his last letters, undertaken ‘for the glory of God and the advancement of the public good’ (Antognazza 2009, 3, 119–20). The metaphysical warrant overdetermined the demand for system, rendering it an obligation at once cosmic and practical. We systematize because reality is a harmonious order, and because the knowledge of that order is the path to human happiness. The practical work that systematization performs, making knowledge transmissible, certifiable, and governance accountable, is concealed by the warrant’s metaphysical splendour: Leibniz, asked ‘Why systematize?’, would have offered the grand metaphysical answer first.

Mona Beizaei has drawn attention to an aspect of the warrant that is especially suggestive for this genealogy. The monads do not interact causally; their agreement is guaranteed by the pre-established harmony of their created natures. *Coherence among perspectives* is therefore a mark of the world’s metaphysical structure, and indeed of its deepest structure. When a claim coheres with the totality of what is known, its coherence is evidence of its correspondence with reality, because a harmoniously ordered reality is precisely one in which truths cohere. Beizaei argues that the ‘Harmony Principle’ in Leibniz, the thesis that there is a close relationship between the reality of bodies and the mutual harmony of monadic perceptions, is most naturally read as an epistemic principle: a principle concerning the *criteria* by which we assess claims about the world, rather than a thesis about what bodies metaphysically consist in (Beizaei, 3–5). If harmony functions as an epistemic test, then it is already doing the work that the epistemological rationale identifies as the core achievement of cognitive systematization: making the system’s internal coherence a credible index of truth, so that claims earn credibility by integrating into the whole and lose it by generating inconsistency. The metaphysical warrant, at its most elaborate, is doing epistemological work that can in principle be separated from the metaphysical scaffold that historically supports it.

Whether such a separation is possible is the question that the warrant's own theological underpinning forces into view. Leibniz's God is an extreme intellectualist's God. As Alex Englander has argued in a study that helpfully distinguishes the surface form of Leibniz's theodicy from its underlying motivational structure, Leibniz 'begins to liberate criteria of goodness, rendering them principles of reason to which even the divine will is subject and by which his creation can be judged' (Englander 2026, 33). For a voluntarist like Descartes, the eternal truths are true because God wills them; their authority depends on the divine will. Leibniz's intellectualism reverses this dependence: God wills the best *because* it is the best, and the criteria by which 'the best' is judged are principles of reason that are prior to, and independent of, the divine will. Leibniz insisted that 'to say that God's justice is different from men's is like saying that the arithmetic or the geometry of men is false in heaven' (quoted in Riley 1996, 15). The principles of rational order that the *characteristica* was designed to capture are, on this view, constraints on rationality as such, binding even on God.

A complementary perspective emerges when we consider what happened when Leibniz was pressed to *justify* the principle of sufficient reason from outside that architecture, by an interlocutor who did not share its commitments. In his correspondence with Samuel Clarke, conducted through Princess Caroline of Wales in 1715–16 and dispatched in its final instalment only three months before Leibniz's death, Leibniz invoked the PSR repeatedly in arguments against Newtonian absolute space and time, against atoms and the void, and for the identity of indiscernibles. Clarke, acting as Newton's philosophical spokesman, challenged Leibniz to demonstrate the PSR 'either from the nature of things, or the perfection of God' (LC 45). As Paul Lodge has shown in a close analysis of the correspondence, Leibniz responded with a surprisingly diverse battery of justifications (Lodge 2018, 70–71). He offered a proof from the nature of things (every event must have its conditions and requisites, whose obtaining constitutes the sufficient reason of the event). He offered a proof from divine perfections (God's wisdom requires that all his actions be agreeable to reason). He argued that the PSR is indispensable for proving God's existence and grounding the truths of natural philosophy. He offered an inductive defence: the principle has never been falsified, and the 'method of experimental philosophy which proceeds a posteriori' supports it. And he claimed that denying the PSR would entail denying the principle of contradiction itself, reducing the denier to absurdity.

This proliferation of arguments is itself suspicious. It is as if Leibniz were saying: 'If you don't like my principles, I have others.' As Lodge observes, each justification either presupposes the theological framework it is supposed to support, or falls short of establishing the PSR on independent grounds (Lodge 2018, 79–84). The proof from divine perfections reasserts that God acts wisely, which is the very commitment Clarke's challenge puts in question. The indispensability argument moves in a circle: the PSR is needed to prove God's existence, but God's existence is needed to ground

the PSR. The inductive defence, however charming in its appeal to the principle's track record, can at best yield the rational expectation that the PSR will succeed in future cases; it cannot establish the universal and exceptionless principle Leibniz requires. And the a priori argument from the principle of contradiction remains undeveloped: Leibniz asserts the connection without explaining it, adding only: 'I might say something more upon it, but perhaps it would be too abstruse for this present dispute' (LC 96). What the correspondence ultimately exposed, Lodge argues, is a 'fundamental difference between the commitments of the two men regarding the concepts that terms like "God" and "sufficient reason" express' (Lodge 2018, 84). Leibniz and Clarke each took divine agency to operate in accordance with the PSR, and each invoked this agreement as support for the principle. The difficulty was that they meant radically different things by 'rational agency,' and neither could demonstrate the superiority of his conception on terms the other would accept.

For the genealogy's purposes, this impasse is instructive. We saw that Leibniz's intellectualism liberates the principles of reason from their theological framework, making the warrant's dispensability visible from within. The Clarke correspondence shows the same fragility from without: when the warrant is subjected to dialectical pressure by an opponent operating with different theological commitments, the PSR cannot be justified independently of the architecture it is meant to ground. The warrant's vulnerability has two faces, one visible to internal philosophical analysis, the other exposed by external challenge. Together, they mark the moment at which the most comprehensive metaphysical grounding the early modern period achieved reveals itself as dependent on convictions that it cannot independently vindicate.

What makes this moment more revealing still is that Leibniz himself appears to have sensed, however obscurely, that a different kind of justification might be needed. The 'abstruse' consideration he withheld from Clarke has never been identified with certainty. Lodge ventures a speculative, but suggestive reading, drawn from a little-known text of around 1693, given the title 'Leibniz's Philosophical Dream' by an early keeper of his papers (Lodge 2018, 86–90). In this piece, which reads as Leibniz's rendition of Plato's allegory of the cave, Leibniz portrays himself as a thinker surrounded by people pursuing luminous trifles they call 'honours' and glittering little flies they call 'riches.' He discovers cracks in the vault of the cavern through which a faint light enters; he attends to it; it grows stronger as he gazes at it; and by its illumination he finds he can 'see where he was going,' 'discern what was around him,' and 'see what would suffice to secure him from dangers.' A venerable old man tells him that this light signifies what is called 'intelligence' or 'reason' in us (Lodge 2018, 87–88).

Lodge reads this as a *pragmatic* justification of the PSR. The dreamer does not arrive at the PSR by demonstrating it from first principles. He arrives at it by discovering that reliance on reason, and thus on taking things to accord with the PSR, enables him to navigate the world: to avoid the dangers to which less luminous

beings fall prey. Lodge notes that on this reading, ‘it is not so much that the PSR yields beliefs in propositions that are true, or indeed that the PSR is itself true, that stands in its favour. Rather, it is because relying on the PSR (and thus taking it to be true) is constitutive of a practice whose value is said to lie in the facilitation of behaviour’ that leads the world to show up to the thinker in a navigable way (Lodge 2018, 88). The justification is neither *a priori*, like the demonstration from the principle of contradiction, nor *a posteriori* in the inductive sense. It is pragmatic, in that commitment to the PSR is warranted by what it enables the thinker to *do*.

For a book whose central contention is that the practical rationales for systematization always ran deeper than the metaphysical justifications that overshadowed them, this is a striking find. It suggests that even Leibniz, the metaphysical warrant’s most ambitious architect, may have glimpsed the practical substructure on which the warrant rests. The Dream is a minor and obscure text, and Lodge’s reading of it is avowedly speculative. Yet the structure of the justification it offers is precisely the structure this genealogy has been uncovering beneath the successive metaphysical architectures of Chapters 1 through 3: we have reason to systematize because doing so serves practical needs, epistemological, pedagogical, and political, that persist independently of the metaphysical convictions historically invoked to justify them. If Leibniz’s Dream discloses, even fleetingly, the thought that the PSR earns its keep by what it enables rather than by what it mirrors, then the warrant’s zenith is also the moment at which the warrant begins, however faintly, to gesture beyond itself toward a justification it cannot fully articulate.

As we shall see in Chapter 8, this need not imply that practical rationales be what actually *motivates* thinkers to systematize. The instrumental benefits of systematization may be most reliably achieved if the value of systematicity is understood in non-instrumental terms. But the immediate implication is that Leibniz’s articulation of the metaphysical warrant contains the seeds of its own dispensability. If the criteria of rational order are principles of reason to which even God is subject (as argued by Englander), and if those principles cannot be justified on terms independent of the theological framework they are meant to ground (as argued by Lodge), then the warrant’s authority rests on convictions it can neither derive from nor defend without recourse to itself. The grander the warrant, the more it asks us to accept: pre-established harmony, the monadology, the identity of indiscernibles, the principle of sufficient reason applied without restriction. Condillac would classify Leibniz’s system among those built on ‘abstract general maxims’ whose terms lack determinate empirical content: impressive architectures suspended over a void. We find the warrant here at full power, but also, for the first time, transparent to an analysis that reveals its load-bearing function and raises the question of whether that load can be borne by other means.

3.5 The Spiritual Automaton

The preceding sections have reconstructed the *calculemus* programme as unfolding on three levels: first, the notation that encodes thought symbolically; second, the physical machine that executes the notation mechanically; and third, the metaphysical architecture that underwrites both. At each level, the programme deposits more of the work of reasoning in structures external to the individual mind. Yet at each level, too, a recurrent asymmetry has emerged. The *characteristica* operates on primitive concepts it cannot itself identify. The calculating machine executes rules it cannot design. The metaphysical warrant grounds the demand for system in a rational order whose principles, as the Clarke correspondence revealed, cannot be independently justified when challenged. In every case, the machine operates within a framework the machine cannot build. This asymmetry merits closer philosophical attention, because Leibniz himself addressed it.

The concept through which he did so is that of the *spiritual automaton*: the idea that the soul is a self-moving entity whose perceptions follow spontaneously from its own internal nature, and that mechanism, properly understood, rests on something that exceeds it. As Christopher Noble has brought out, the concept constitutes a systematic reworking of the idea of the automaton that distinguishes it sharply from the Cartesian usage (Noble 2023, 56–57). For Descartes, the automaton was a model for the animal body: a machine whose apparent self-motion is produced by external efficient causes, as a clock's motion is produced by the tension of its spring. Descartes explicitly denied that the human soul could be modelled on the automaton, because the soul's freedom and rationality resist mechanical description.

Leibniz rejected both halves of this picture. On the one hand, animal bodies are 'natural automata' for Leibniz – infinitely complex organisms whose structural intricacy exceeds anything a human artificer could produce. On the other hand, the soul, far from being excluded from the *automaton* concept, is the most genuine automaton of all: a 'spiritual automaton' whose states follow spontaneously from its own appetites and perceptions, without external determination (Noble 2023, 57–58). As the Wolffian philosopher Georg Bernhard Bilfinger noted, an automaton is, etymologically, 'something desirous of itself, whose changes come from its own appetites,' and this description applies more properly to the soul, which moves from internal principles, than to a clock, which requires an external push (Noble 2023, 71).

Resources for this reconception had been accumulating since Leibniz's earliest work. In the *Theoria motus abstracti* of 1670–71, the young Leibniz had already arrived at the thesis that 'every body is a momentary mind, or one lacking recollection': bodies are grounded in an unextended, indivisible principle of activity (*conatus*) that is, strictly speaking, a mental action (Antognazza 2009, 80–81). 'True' minds differ from mere bodies in possessing memory, the capacity to retain, compare, and harmonize their *conatus*. By 1671, Leibniz was arguing that every corporeal being

contains a ‘kernel of substance,’ an indestructible spiritual point from which the body’s identity persists through all physical change (Antognazza 2009, 82–83). The conviction that the fundamental constituents of reality are self-active, perception-endowed unities, and that mechanical description captures only the outer surface of a deeper spontaneity, was present from the beginning. The concept of the spiritual automaton is the mature articulation of an intuition Leibniz had nursed since his early twenties – the intuition that the mental character of substance is what makes genuine self-motion possible.

In introducing this concept of spiritual automaton, Leibniz was attempting to absorb into his system what ostensibly resisted it. We saw Cavendish argue that mechanistic systematization screens out self-activity and responsiveness. Leibniz effectively responded by reconceiving the automaton itself as a self-active entity. The soul’s perceptions unfold from within, according to its own nature, in a sequence that is spontaneous, continuous, and reflective of the entire universe. As Leibniz wrote in the *New System* of 1695, ‘God first created the soul in such a way that everything in it arises from its own nature, with a perfect spontaneity as regards itself, and yet with a perfect conformity to things outside it’ (GP IV, 484; Roinila 2023, 176). The concept attempts to stretch the notion of ‘automaton’ to accommodate precisely the features that Cavendish had argued mechanism must exclude.

Yet the concept of the spiritual automaton embodies a structural ambiguity at the heart of Leibniz’s programme that Sybille Krämer has identified. Leibniz, Krämer argues, interpreted formal symbolic operations as epistemological instruments, and explicitly denied that a machine could serve as an explanatory model of cognition (Krämer 1996, 84). The calculus is a tool for the external guidance of reasoning, a *filum meditandi* that directs the mind’s operations: a norm prescribing how we *should* think if we want to arrive at truth, rather than a description of what we *actually do* when we think (Krämer 1996, 84–85). The distinction is crucial for our purposes. It means that the mechanization of reasoning, in Leibniz’s own understanding, applies in the first instance to the *products* of thought. It applies to the *process* of thought only in the sense that the process is *guided* by formal norms. The spiritual automaton produces perceptions spontaneously from its own nature; the *characteristica* captures and systematizes those perceptions in symbolic form. The machine operates on the outputs of a mind that is itself more than a machine.

Recall the asymmetry identified earlier: the machine operates within a framework that the machine cannot build. Leibniz’s programme requires, as its first step, the identification of an ‘alphabet of human thoughts’: a catalogue of primitive concepts from whose combination all others can be generated. This requires the philosopher to survey the entire field of human knowledge, to discern which concepts are primitive and which composite, and to design a notation that captures the logical relations among them. But these are exercises of just the kind of judgement, discernment, and intellectual creativity that the *characteristica* was designed to mechanize *away*. The programme thus depends on a prior exercise of unmechanized thought that

the programme itself cannot account for (Spruit and Tamburrini 1991, 8–10). Only once the primitives are identified, the notation designed, and the rules specified can the calculus proceed automatically. Here, the aspiration to make reasoning self-certifying through formalization runs into the fact that it presupposes a prior exercise of informal judgement that the formalization itself cannot incorporate.

Leibniz is thus not best understood as having proposed that a machine could *think*. He proposed that a machine could capture, discipline, and extend the products of thinking – and that the discipline was worth the cost because the alternative, unregulated reliance on the thinker’s own judgement, had proved catastrophically unreliable in an age of confessional war.

3.6 The Convergence of Three Rationales

Leibniz is the first figure in the genealogy in whom all three practical rationales for cognitive systematization converge in a single programme. The *characteristica universalis*, conceived as a universal medium of instruction operable by anyone who masters the notation, carries forward the pedagogical rationale that Chapter 1 traced through the Reformed system-builders. The *calculus ratiocinator*, which provides a self-certifying mechanism for evaluating claims and a self-correcting mechanism for detecting inconsistency, realizes the epistemological rationale that Chapter 2 traced through the geometric method. And the institutional projects, the *Systema Iuris*, the ecumenical programme, the proposals for learned academies, and above all the *calculemus* itself as an offer to settle disputes by computation, embody the political rationale in its most developed form. Each rationale is supported by the metaphysical warrant, and the warrant overdetermines them all: Leibniz, asked ‘Why systematize?’, would have offered the metaphysical answer first, concealing the practical work that the three rationales were already performing on their own.

The convergence makes visible something that the separate treatment of earlier chapters could only suggest: each rationale, extended to its limit, generates a characteristic cost, and the costs are structural consequences of the very virtues that make the programme attractive.

Consider the pedagogical rationale. Blind or symbolic knowledge, *cognitio caeca*, enables the mind to manipulate concepts correctly without holding their full content before the mind’s eye at every step. This is a genuine cognitive achievement, in that it makes complex reasoning tractable. But it also holds out the prospect of a form of mastery that remains entirely formal. A learner who operates the symbols of the *characteristica* correctly has, in one sense, mastered the knowledge the system encodes. In another sense, however, the learner has merely learned to operate machinery. The *scientia generalis* was designed so that ‘everyone endowed with ordinary abilities’ could use it; the cost of that design is that ordinary abilities may be all that the system cultivates.

The epistemological rationale generates a different kind of strain. Leibniz's system, as we have seen, does not work like Spinoza's, extracting consequences from a fixed axiomatic basis. It works by optimization: incorporating the maximum of intellectual content under the most economical set of principles, and treating systematic fit as the criterion by which a claim earns its place. A proposition that contributes to the coherence and economy of the whole is vindicated by its integration; one that resists integration is flagged for revision or exclusion. The cost this generates is subtle, because it is internal to the criterion itself. What counts as fitting harmoniously into the system is determined by the system's own principles of harmony, and what fails to fit is excluded on terms that the excluded material has no standing to contest. Truths whose recognition depends on perceptual sensitivity, practical experience, or contextual responsiveness may simply fail to register as candidates for integration, and their absence will appear, from within the system, as a vindication of its coherence rather than as a loss. Leibniz himself acknowledged that the *ars judicandi* could not always deliver a definitive verdict and that its results were conditional on the quality of the available information. The acknowledgement concedes something important: determining where the system's reach ends and informal judgement must begin is itself an exercise of the very judgement the system was designed to supersede.

The political rationale, taken to its limit, reveals a third cost. Recall the contrast with Spinoza drawn earlier in this chapter: where Spinoza's machine filters by exclusion, Leibniz's translates by incorporation. As early as 1669, Leibniz envisaged a *Societas Philadelphica* whose proposed aim was the 'promotion of the utility of humankind,' a society designed to span religious and political divisions by embracing the Emperor and the Pope, the King of France, German principalities, the Royal Society, and learned religious orders (Antognazza 2009, 121). Where Spinoza risks discarding truths that resist axiomatization, Leibniz risks imposing a spurious harmony on positions whose disagreements are genuine. A calculus that always produces a reconciling verdict may be a calculus that has been designed to produce reconciling verdicts, rather than one that truthfully tracks the structure of the dispute.

The metaphysical warrant, traced across the book's first three chapters, has followed an arc of increasing ambition. In Leibniz, it reaches full power. Pre-established harmony, the principle of sufficient reason, and the doctrine that this is the best of all possible worlds combine to produce the most comprehensive metaphysical grounding for the demand for cognitive system that the early modern period achieved. Yet the warrant, at its zenith, has begun to undermine itself. Leibniz's extreme intellectualism, his insistence that the principles of rational order bind even God, has liberated those principles from the theological framework in which they are embedded. And the Clarke correspondence has revealed that the most central of those principles, the PSR, cannot be independently justified when challenged by an interlocutor operating with different theological commitments. If the criteria of

rationality are autonomous, if they constitute constraints on any possible rational being, then they do not require a divinely designed universe to be binding. The demand for systematicity, in that case, would stand on practical grounds that require no speculative metaphysics.

The next chapter turns to a different face of the warrant's zenith: the Newtonian 'system of the world' and Émilie du Châtelet's philosophical articulation of the metaphysical foundations on which that system rests. Where Leibniz pushed the metaphysical warrant to its fullest expression through the rationalization of metaphysics itself, Newton's *Principia* achieved something arguably more impressive: a mathematical demonstration of the systematic order of nature that seemed to make the warrant empirically irresistible. Du Châtelet, writing in the *Institutions de physique* (1740), provided the most philosophically rigorous account of why systematic physics requires systematic metaphysics, and in doing so, she made visible the load the warrant was being asked to bear. The zenith and the strain will prove to be two aspects of a single phenomenon.

God's Clockwork: Du Châtelet and Newton's System of the World

'This universe forms a whole, one entire and single machine, all the parts of which relate to one another, and are so linked together that they all conspire towards the same end'

— Du Châtelet

In the third book of the *Principia Mathematica*, Isaac Newton set himself an extraordinary task. Having laid down mathematical principles of motion and force in the first two books, he would now, as his title announced, exhibit *the system of the world*. From three laws of motion and a single law of universal gravitation, he derived not only the orbits of the planets, but also the trajectories of comets, the precession of the equinoxes, the shape of the earth, and the rise and fall of the tides. All of this issued from a framework so tightly constructed that the motions of all the celestial and terrestrial bodies answered to a single force. And unlike Descartes and Kepler, who had merely speculated about vortices and magnetic souls, Newton was able to *demonstrate* what he asserted: his conclusions followed from his premises with the impersonal rigour of geometry, and they could be checked against observation by anyone competent in the requisite mathematics. Newton produced a cognitive apparatus of such explanatory power that it seemed to settle once and for all the question of whether reality itself was systematic. If the sun, the planets, and the comets all obeyed a single law, then the universe was a unified structure amenable to synoptic comprehension.

In addition, Newton formulated general rules which described a procedure for extracting systematic order from observed phenomena. The first rule demands parsimony: no more causes of natural things should be admitted than are both true and sufficient to explain their phenomena, 'for nature is simple and does not indulge in the luxury of superfluous causes' (Newton 1999, 794). The second extends parsimony across domains: the causes assigned to natural effects of the same kind must, so far as possible, be the same (Newton 1999, 795). The third universalizes qualities found in all bodies on which experiments can be made:

qualities that square with experiments universally are to be regarded as universal qualities (Newton 1999, 795). And the fourth, added in the third edition of 1726, asserts the methodological primacy inductive generalization against any speculative hypotheses. Inductions, Newton declares, “should be considered either exactly or very nearly true” until new phenomena may make them “either more exact or liable to exceptions” (Newton 1999, 796).

Together, these rules articulate an epistemology of systematization. Parsimony keeps the system free of redundant mechanisms. Uniformity of causes across domains ensures that local and remote phenomena answer to the same principles, which is the condition under which a universal system becomes possible. The universalization of experimentally confirmed qualities, meanwhile, licenses the projection of local regularities onto unobserved cases. And the methodological primacy of induction subordinates subjective hypothesizing to empirical observation. In combination, these rules constituted an effort to articulate a rule-governed, impersonal procedure for moving from observed phenomena to general propositions about forces and laws.

Newton's *Principia* was the most powerful cognitive system the early modern period had seen, and it exemplified the machine-like virtues of surveyability, rule-governance, universality, and impartiality. Yet Newton admitted that this system remained silent about its own philosophical foundations: ‘I have not as yet been able to deduce from phenomena the reason for these properties of gravity, and I do not feign hypotheses [*hypotheses non fingo*] ... it is enough [*satis est*] that gravity really exists and acts according to the laws that we have set forth and is sufficient to explain all the motions of the heavenly bodies and of our sea’ (Newton 1999, 943). The theory of gravity showed that systematic order was actual, but it did not explain what made it possible – what caused or underpinned the regularities described by the *Principia*. All Newton offered was the thought that ‘this most elegant system of the sun, planets, and comets could not have arisen without the design and dominion of an intelligent and powerful being’ (Newton 1999, 940). The system of the world required an intelligence to set it in order.

For philosophers on the Continent, this was not enough. Christian Wolff (1679–1754), a German philosopher working in the Leibnizian tradition, dismissed Newtonian forces as merely ‘imaginary’: all Newton offered was mathematical descriptions of regularities rather than demonstrations of their causes (Wolff 1731/2001, 316). Wolff's principle of sufficient reason demanded a deductive system in which empirical results confirmed or illustrated truths derivable from essences (Wells 2023).

In France, however, a different response was taking shape. The questions Newton left hanging were especially salient there, as the dominant natural philosophy was still Cartesian, and the Cartesian tradition had no patience with unexplained forces acting at a distance. Gravitational attraction, operating across the void with no discernible mechanism of contact, looked to its critics like an occult quality – a

regression to just the sort of scholastic obscurantism that the mechanical philosophy had sought to overcome. Accordingly, the transition from Cartesianism to Newtonianism was not, as it is sometimes portrayed, a swift conversion, but a protracted contest in which institutional alliances, confessional tensions, and competing conceptions of scientific method all played a role (Shank 2008). Though the *Principia* was published in 1687, it was not until the late 1730s that Newton's physics gained decisive institutional support in France. This support came in part from the geodetic expeditions that confirmed his prediction of the earth's oblate shape, and in part from the vigorous advocacy of Pierre-Louis Moreau de Maupertuis and Voltaire. Both were associates of a figure who would engage more seriously with Newton's system of the world than either of them: Gabrielle Émilie le Tonnelier de Breteuil, Marquise Du Châtelet (1706–1749).

Du Châtelet provided what is arguably the most sophisticated philosophical defence of the conviction that knowledge must be systematic because God-designed reality is systematic. In constructing that defence, she brought the metaphysical warrant to its fullest articulation: she showed it actively at work in generating, constraining, and evaluating scientific claims; she put it to the test on the most contested problems in natural philosophy; and she drew out the philosophical commitments on which the edifice of systematic knowledge depends. Above all, however, Du Châtelet, in theorizing the warrant and putting it to work, exposed how strained the warrant became under the weight it was being asked to bear.

4.1 The Philosopher at Cirey

Du Châtelet was born in 1706 into a family of the high Parisian nobility. Married at nineteen, she fulfilled the social expectations of her station before turning, from the early 1730s onwards, to the study of natural philosophy and mathematics (Zinsser 2006). The decisive period of her intellectual formation was spent at the château de Cirey in Champagne, where she and Voltaire established a working household. At Cirey, Du Châtelet undertook the programme of reading that would shape her principal philosophical work: she studied Newton's *Opticks* and then the *Principia* itself; she worked through the Newtonian textbooks of John Keill, Willem Jacob's Gravesande, Pieter van Musschenbroek, and Henry Pemberton; and she began, through Maupertuis and the mathematician Alexis Clairaut, to acquire the analytical mathematics required for serious engagement with Newtonian physics (Hutton 2004, 518; Zinsser 2006, 62–76).

Yet she found these texts philosophically unsatisfying. The textbooks presented Newton's results with varying degrees of mathematical sophistication, but they shared a common methodological stance: physics proceeds from observation and experiment to general laws, and questions about the metaphysical underpinning of those laws are either premature or illegitimate. This stance, inherited from Newton's

own *hypotheses non fingo*, left unanswered the questions that most engaged Du Châtelet. Why should the regularities discovered by observation hold universally? What entitled the natural philosopher to project local experimental results onto unobserved cases? How should one adjudicate between competing theoretical accounts of force, matter, and causation when the empirical evidence alone could not decide among them? These questions were sharpened by a further dissatisfaction: Du Châtelet had read Locke, and her early engagement with his epistemology convinced her that experience alone could not ground the kind of knowledge physics aspired to. Even before she came to know the Leibnizian and Wolffian metaphysics in detail, she was persuaded that the Newtonian ban on hypotheses was too restrictive, and that a complete physics would require resources that the Newtonians were unwilling to supply.

4.2 The Ambition of the *Institutions de physique*

It was in this intellectual context that Du Châtelet composed her principal work, the *Institutions de physique*, published in Paris in 1740. The title page modestly described it as a textbook addressed to her son. In fact, as Katherine Brading has shown, the *Institutions* is a highly original work in the philosophy of natural science: the first attempt to construct a unified philosophical framework capable of accommodating Newtonian physics, Leibnizian metaphysics, and an articulated methodology for adjudicating between them (Brading 2019, 1). Du Châtelet herself offered a revealing image of her aspiration. Physics, she wrote in the *Avant-propos*, is an immense building that surpasses the powers of a single person; some lay a stone, others build whole wings, but there are still others who aspire to grasp the plan of the whole (Du Châtelet 1740, Preface, XI). She saw herself as such a surveyor: an architect who grasps a building's structural logic, sees how the parts relate and why they are arranged as they are, and thereby occupies a vantage point distinct from those of the various specialist contributors.

The work itself underwent a dramatic revision shortly before publication, and the nature of that revision illuminates the philosophical problem Du Châtelet was addressing. In the original manuscript, the opening chapter discussed Newton's rules of reasoning and presented a methodology rooted in empirical practice. Before the text went to press, Du Châtelet replaced this chapter with an entirely new one: a discussion of the principles of contradiction and sufficient reason, drawn from the Leibnizian-Wolffian tradition, as foundational principles of human knowledge. As Brading has shown, where the manuscript had offered only the empirical constraint on hypotheses (be sure of the facts and attendant circumstances), the published version introduces a two-pronged methodology, requiring that hypotheses satisfy both the principles of our knowledge *and* the empirical evidence (Brading 2019, 31–32). The addition of this first prong was driven by Du Châtelet's recognition

that the Newtonian method, for all its empirical power, was impotent with respect to the deepest problems facing physics – problems about the nature of bodies, the possibility of action at a distance, and the proper measure of force (Brading 2019, 19). She needed resources that the Newtonian tradition, by dint of its methodological self-limitation, could not provide.

4.3 The Principles of Our Knowledge

Du Châtelet titled the revised opening chapter of the *Institutions* 'Of the Principles of Our Knowledge', and the *our* is significant. Her concern is with the conditions under which *we*, finite inquirers embedded in a world of contingent phenomena, can arrive at truths and avoid error. By what principles should reasoning be guided, and by what criterion should the products of reasoning be assessed?

She begins by rejecting Descartes's criterion. Descartes had proposed that clear and distinct ideas furnish the criterion of truth, but Du Châtelet argues that this is too permissive: reliance on what strikes the individual mind as luminously evident allows us to be misled by our imaginations and only perpetuates dispute. A criterion whose deliverances vary with the person who applies it cannot ground the kind of knowledge physics aspires to. We must therefore 'substitute demonstrations for the illusions of our imagination, and admit as true only what can be incontestably deduced from first principles that no-one can revoke in doubt' (Du Châtelet 1740, 1.2).

The two principles she adopts are the principle of contradiction and the principle of sufficient reason. She draws these from the Leibnizian-Wolffian tradition, but deploys them with a methodological emphasis that is distinctly her own. The principle of contradiction distinguishes the possible from the impossible – whatever implies a contradiction is impossible, and whatever does not is possible. Du Châtelet declares this the 'foundation of all certainty in human knowledge'. Without the principle of contradiction, she writes, 'every thing could be, or not be, according to the fantasy of each person' (Du Châtelet 1740, 1.4). What the principle of contradiction rules out is something close to cognitive anarchy: a condition in which the distinction between the possible and the impossible collapses into individual taste, and in which no claim can be assessed independently of the inclinations of the person making it. The principle of contradiction, in Du Châtelet's hands, functions as the minimal condition for a shared intellectual space within which reasoning can take place.

The principle of contradiction alone, however, is insufficient. Necessary truths, such as those of geometry, are determined by the principle of contradiction: what contradicts a geometrical axiom is impossible, and what follows from one is necessarily true. Yet physics deals overwhelmingly with contingent truths, with how things happen to be given that they might have been otherwise. Within the domain

of the possible, as delimited by the principle of contradiction, there are many states of affairs that do not contradict one another, and the principle of contradiction provides no means of determining which of them obtains.

It is here that the principle of sufficient reason enters. Du Châtelet treats it as independent of the principle of contradiction, 'neither less fundamental nor less universal' (Du Châtelet 1740, 1.8), and assigns it a specific task: to distinguish the actual from the merely possible. When it is possible for a thing to be in several states, the principle of sufficient reason requires that there be a reason why it is in one state rather than another. 'I can be sitting, lying down, or standing,' she writes; 'all these determinations of my situation are equally possible, but when I am standing, there must be a sufficient reason why I am standing and not sitting or lying down' (Du Châtelet 1740, 1.8).

The significance of the principle of sufficient reason for Du Châtelet's project extends well beyond this domestic example. As Brading has emphasized, Du Châtelet holds that the principle of sufficient reason must be presupposed if knowledge of contingent truths is to be possible at all, for without it, no reasoning from effects to causes can go through (Brading 2019, 35). This argument targets the Newtonians directly. Newton's fourth rule had given inductive generalization a privileged methodological standing, yet the Newtonian textbooks offered remarkably thin justifications for this reliance on induction. Gravesande appealed to ordinary life: unless we assume that like effects arise from like causes, who could live a moment in ease? Du Châtelet found such justifications inadequate, because they left unexplained the very thing requiring explanation. Her answer is that the justification depends on the principle of sufficient reason. If things can change state without anything determining the change, then no inductive reasoning is warranted, and the entire edifice of Newtonian physics loses its epistemic footing. Again, she offers a vivid example: if she places a ball of stone and a ball of lead in the basins of a balance and finds them to weigh the same, she can assert their equal weight only on the assumption that a change in weight cannot occur without a reason. Without the principle of sufficient reason, the weight of one ball could change in the next instant for no reason at all, and the measurement would prove nothing (Du Châtelet 1740, 1.8). The implication, as Brading points out, is that the entire argument of Newton's *Principia*, which depends on measurements of weight from terrestrial objects to the Moon, does not go through without the principle of sufficient reason (Brading 2019, 37).

What is distinctive about Du Châtelet's the principle of sufficient reason, however, becomes clear only when it is contrasted with the version she is often assumed to have inherited, namely Wolff's. Aaron Wells has argued that her principle diverges from Wolff's in its scope, epistemological content, and consequences for science (Wells 2023). In scope, Wolff's principle of sufficient reason ranges over both necessary and contingent properties. Du Châtelet restricts her principle of sufficient reason to contingent truths. Necessary truths, including truths about essences,

depend on the principle of contradiction alone (Du Châtelet 1740, I.7–8; III.41; III.45). In content, the differences are equally pronounced. Wolff's reasons are ultimately causal powers of substances – real grounding relations that are transitive. Understanding, for Wolff, accordingly requires a true representation of essences and their causal connections, structured deductively. Du Châtelet's reasons, by contrast, are, in the first instance, propositions or judgements that enable an intelligent being to understand why something is the case. On her account, truth is neither sufficient nor necessary for the kind of understanding that the principle of sufficient reason demands. Trivially true propositions may explain nothing, while strictly false hypotheses, including the idealizations of mathematics, can make phenomena intelligible (Du Châtelet 1740, I.8–9; IV.56; V.86). Du Châtelet's principle of sufficient reason operates at the level of our epistemic practices: it is a constraint on what counts as a satisfying explanation, and its authority derives from its indispensability for the reasoning on which physics depends. This is what makes the theological backing, when it comes, load-bearing rather than ornamental. Without a guarantee that nature cooperates with our explanatory demands, the principle of sufficient reason remains a constraint on thought with no assurance that the world answers to it.

These differences in scope and content entrain different consequences for science. Wolff's broad principle of sufficient reason yields a deductive model of science in which metaphysics must precede physics and empirical results merely confirm or illustrate truths derivable from the essential properties of substances. Du Châtelet's narrower, question-relative principle of sufficient reason yields a model in which merely probable hypotheses are epistemically legitimate, and the demands of the principle of sufficient reason can be met by non-deductive, empirically driven reasoning. This is what enabled Du Châtelet, unlike Wolff, to accept Newton's inductive arguments for universal gravitation as genuine demonstrations even in the absence of a metaphysical account of gravitational causation (Wells 2023; Du Châtelet 1740, 15.344; 15.359). Du Châtelet could accommodate a science proceeding by assigning 'degree[s] of probability' (Du Châtelet 1740, 4.61.) to hypotheses. Wolff could not.

4.4 The Two-Pronged Method

The first prong of Du Châtelet's method constrains reasoning from above – through principles. The second constrains it from below – through experience. In Chapter 4 of the *Institutions*, which is devoted to hypotheses, she positions herself between two extremes. The Cartesians, she writes, had given the learned world a taste for hypotheses, and it was not long before these 'fell into fictions'; the books of philosophy that should have been collections of truths were 'filled with fables and reveries' (Du Châtelet 1740, 4.55). Newton and his disciples had fallen into the opposite excess, banning hypotheses altogether. Du Châtelet holds that both responses are mis-

guided, because scientific theorizing *requires* hypothesizing. The path to the most sublime discoveries has always passed through unnecessary hypotheses; without them, new truths cannot be found and new perspectives cannot be unlocked (Du Châtelet 1740, 4.57–58).

To steer a middle course between these extremes, Du Châtelet writes, one must *discipline* the admission of hypotheses using stringent criteria. A hypothesis must satisfy two conditions, on her account: first, 'that it not be in contradiction with the principle of sufficient reason, nor with any principles that are the foundations of our knowledge'; second, that the inquirer 'have certain knowledge of the facts that are within our reach, and to know all the circumstances attendant upon the phenomena we want to explain' (Du Châtelet 1740, 4.61). To be legitimate, a scientific hypothesis must therefore not only align with all known empirical facts, but must also satisfy the principle of sufficient reason by offering an intelligible cause for the phenomenon in question, rather than just masking ignorance with occult qualities. A hypothesis must provide a mechanism that makes it possible for an intelligent being to understand both *how* and *why* an effect happens. When Scholastics explain plant growth by appealing to a 'vegetative soul', for example, they have advanced a cause that is completely inadmissible because it 'contains nothing that helps us to understand how the vegetation ... operates' (Du Châtelet 1740, 1.10).

From the principle of sufficient reason, Du Châtelet then derives the *law of continuity*: that 'nothing happens at one jump in nature' and that a being cannot pass from one state to another without passing through all intermediate degrees (Du Châtelet 1740, 1.13). This allows her to solve concrete problems in mechanics that unguided observation had left muddled. For example, she uses the law of continuity to demonstrate that there are no perfectly hard bodies in nature (Du Châtelet 1740, 1.15), and she deploys it to dismantle Descartes's laws of motion. Descartes's second law posited that when a larger body collides with a smaller one, the smaller one instantly reverses direction while the larger one continues unchanged. Du Châtelet argues that this violates continuity: it requires a body to pass from movement in one direction to movement in the opposite direction without passing through the intermediate stage of rest (Du Châtelet 1740, 1.17). Thus, she uses metaphysical principles to adjudicate a dispute in collision mechanics where empirical data was, at the time, confusing and contradictory.

Du Châtelet's contention is that Newton's entire enterprise relies upon these metaphysical principles. If the universe were not governed by sufficient reason and continuity, then inductive generalizations – such as assuming that gravity acts on Jupiter the same way it does on Earth – would be epistemologically baseless. Without the principle of sufficient reason, 'everything could have been thrown into confusion in my room, without anyone having entered', and truths would 'only exist for an instant' (Du Châtelet 1740, 1.8). The commitment to the rational intelligibility of the world goes beyond what observation can establish on its own. The principle of sufficient reason is, as she puts it, 'the only thread that can guide us through

the labyrinths of error' (Du Châtelet 1740, 1.8). It is the scaffolding that allows the edifice of physics to be constructed at all.

4.5 The Rationalist Theology and Its Warrant

If the principle of sufficient reason is to do this work, something must guarantee that the universe cooperates: that it is, in fact, the kind of place in which things happen for reasons, in which causes are connected to their effects by intelligible relations, and in which the contingent arrangements of bodies reflect an underlying rational order. The guarantee, for Du Châtelet, is God – a *rational* God, whose wisdom is manifest in the systematic structure of creation, and whose rational design ensures that the principle of contradiction and the principle of sufficient reason hold.

This theological commitment marks a decisive departure from the Newtonian textbooks that Du Châtelet took as her starting point. In Chapter 2 of the *Institutions*, which is devoted to the existence and nature of God, she breaks with the voluntarist conception that had shaped Newtonian epistemology. For Gravesande and Musschenbroek, the laws of nature depended on the arbitrary will of God and could not be discovered by reason. This was why physics had to rely on observation and induction, and why metaphysical speculation about the ultimate grounds of natural order was methodologically out of bounds.

Du Châtelet rejects this voluntarist picture. She argues, using the principle of contradiction and the principle of sufficient reason, that the essences of things do not depend on God's arbitrary will. The possible does not become possible by divine fiat. It is therefore illegitimate, in physics, to appeal to God's will when theorizing about how something is possible (Du Châtelet 1740, 3.49). God's will determines which of the possible worlds is *actual*; it does not determine what is possible. The space of possibility is fixed by the principle of contradiction. Within that space, the structure of the actual world is governed by the principle of sufficient reason. God acts in accordance with his intelligence and wisdom, and the evidence of design in the world is accessible to reason.

This move from voluntarism to rationalism brings the metaphysical warrant to its fullest articulation. Every thinker we examined thus far evinced, in one form or another, the conviction that reality is a God-designed system. In Keckermann and the Reformed pedagogues, this conviction was implicit and undefended: the disciplines could be systematized because God's creation had a rational order. In Descartes, the guarantee of a veracious God became load-bearing – without it, the method of radical doubt could not deliver the certainty it promised. In Leibniz, the warrant was explicit and metaphysically ambitious: pre-established harmony, the principle of sufficient reason, and the thesis that God chose the best of all possible worlds converged to form a view on which the demand for systematicity was underwritten by the systematic structure of reality.

Du Châtelet inherits the Leibnizian ambition, but she deploys it with a sensitivity to empirical constraint that distinguish her project from anything in the preceding tradition. Her claim is that systematic physics *requires* systematic metaphysics: without the principles that give reasoning its footing, the empirical achievements of the *Principia* remain, as she puts it, a collection of truths that have not been connected to their foundations.

Her metaphysics may seem incomplete when judged as a free-standing foundation; it seems less ambitious than Wolff's metaphysics, for example, which aspired to precede and ground physics by delivering a deductive science in which the properties of matter could be derived a priori from the essences of simple substances. But the key is to see that Du Châtelet's metaphysics serves a different function: as Brading has argued, the animating concern of the *Institutions* is the search for a methodology adequate to the deepest problems facing physics, and the metaphysical commitments Du Châtelet introduces are the *minimal* commitments necessary for that methodology to become viable (Brading 2019, 47).

Du Châtelet's metaphysics is calibrated to this task. She constructs a framework within which empirical science can operate by spelling out the principles that make induction rational and hypotheses assessable, but without claiming to deliver the specific content of physical theory. Legitimate metaphysics, she cautions, consists only of what is obvious to anyone who can properly use their cognitive faculties; any metaphysical claims ranging beyond this domain concern what human beings 'can never know' (Du Châtelet 1740, Preface, XII). The warrant, then, grounds the method without pretending to replace the science it enables. And it places an enormous weight on a single theological conviction: that the God who designed the world designed it as a rationally intelligible whole.

4.6 The Ontology of the System

The distance between Du Châtelet's project and the traditions she drew on becomes clearer still at the level of ontology. A long tradition of commentary treated the *Institutions* as an essentially derivative work – a competent exposition of Leibniz and Newton, at best a synthesis of the two, but not a distinctive philosophical contribution in its own right (Brading 2019, 8–16). But Marius Stan has argued that Du Châtelet's ontology of substance owes far more to Wolff than to Leibniz (Stan 2018). Leibniz's metaphysics, as the early Enlightenment knew it, was monadic idealism: an ontology in which the ultimate constituents of reality are simple, immaterial substances whose properties are exhausted by their perceptions and appetitions, and whose causal interactions are merely apparent. Wolff's metaphysics, by contrast, is a robust realism: substances are genuine causal agents in 'transeunt-causal commerce' (Stan 2018, 479), meaning that they have effects on one another, and bodies are composites whose properties – including extension, force of inertia,

and active force – are irreducible to the internal states of monads (Stan 2018, 478–79). Even though Du Châtelet herself believed she was presenting Leibniz's philosophy, Leibniz's metaphysics had reached her doubly filtered: first by what Leibniz chose to reveal to Wolff, and then by what Wolff chose to keep. The result is that the 'Leibnizian' metaphysics of the *Institutions* is, in its substantive commitments, a species of Wolffian realism.

The ontological picture she arrives at through Wolff's influence is relevant to the metaphysical warrant, because it implies that the drive towards cognitive systematization is justified by the very structure of what exists. 'All is linked in the world; each being has a relationship to all the beings that coexist with it, and to all those that preceded it, and that must follow it' (Du Châtelet 1740, 7.130). Her guiding analogy is with a well-built machine. We cannot doubt, she writes, that this universe of ours forms 'a whole, one entire and single machine, all the parts of which relate to one another, and are so linked together that they all conspire towards the same end' (Du Châtelet 1740, 7.130). In a later section, she is explicit that the universe emulates such a well-designed machine: 'just as in a well-made machine, the least part has a relation to all the others,' so 'the present state of each element contains a relation to the present state of the entire Universe, to all the states that will be born from the present state, and to all those that have preceded it' (Du Châtelet 1740, 7.131).

On this ontological picture, reality itself is a thoroughly rational system satisfying the principle of sufficient reason. Every element bears a relation to the present state of the whole universe and to its past and future states, so that the state of one part contains, in principle, information about every other. The demand that thought form a system can thus be grounded in the claim that reality *is* a system, in the ontological sense that the constituents of reality are bound together by relations of sufficient reason into a unified, rationally intelligible whole. For any given state of a simple being, 'there must be a sufficient reason why such a state is actual' (Du Châtelet 1740, 7.129). She goes on:

this reason can only be found in the preceding state, and the reason for that will be in the state antecedent to it, and so on back to the first. This first state, which supposes no other antecedent to it, depended on God, but all subsequent states are linked, so that from the first follows the last which was contained in it, and which must be such, because the first was so and not otherwise. In the same way the actual state of a clock depends on the preceding state, and that on another, and so on back to the first, which depended on the way in which the craftsman arranged the wheels; thus does the forty-seventh proposition of Euclid follow from the first, in which it is contained. (Du Châtelet 1740, 7.129)

This passage brings out how Du Châtelet fundamentally conceives of the universe as a divinely designed clockwork, whose arrangement

itself forms a deductive system in the manner of Euclid's – and, as it happens, she invokes the very same proposition that converted Hobbes to the geometric method. If this ontological picture is correct, then the aspiration to comprehensive, systematic knowledge is warranted by the very structure of what is there to be known.

Yet even in her ontology, the presuppositions of her two-pronged method for the enterprise of physics drive her metaphysical commitments rather than the other way around. Her metaphysics is designed to bring principles and evidence into convergence. Whether that convergence can actually be achieved is the question to which we now turn.

4.7 The Warrant at Work: Forces Dead and Living

What distinguishes the *Institutions* from a purely methodological manifesto is that Du Châtelet puts her method to work on the most contested problems in natural philosophy. The most instructive case is the *vis viva* controversy: a dispute about the proper measure of the motive force of a moving body that had divided European natural philosophy since Leibniz first challenged the Cartesian measure in 1686.

The controversy concerned a question that seemed straightforward, but quickly pulled one into deep metaphysical waters: when a body is in motion, it possesses a certain force – what is the correct measure of that force? The Cartesian and Newtonian traditions measured it as the product of mass and speed (mv), a quantity proportional to what they called the quantity of motion. Leibniz had argued that the correct measure was the product of mass and the *square* of speed (mv^2), which he called *vis viva*, living force. He distinguished this from *vis mortua*, dead force: the mere pressure or tendency to motion that a body exhibits before it begins to move. The distinction was connected to the law of continuity: a body does not leap from rest into motion instantaneously, but passes through all intermediate degrees, and the force that accompanies this transition accumulates in a manner that the simple product mv cannot capture. What looked like a disagreement about measurement was, as Andrea Reichenberger has argued, an examination of the ontological presuppositions underlying three competing traditions: the Cartesian, the Leibnizian, and the Newtonian (Reichenberger 2012, 159). The question of *how* to measure force turned out to be inseparable from the question of *what* force is, and that question in turn depended on one's account of substance, causation, and the relation between mathematical description and physical reality.

Du Châtelet's intervention in this debate illustrates how the metaphysical warrant could be more than a background assumption or a hinge proposition within a certain outlook; it could be brought to bear on particular questions of physics. Her intervention, which occupied the final chapter of the *Institutions* and had been

preceded by a public exchange with Jean-Jacques Dortous de Mairan, deployed both prongs of her method. The first prong yielded a case for *vis viva*. Du Châtelet argued (from the principle of sufficient reason and the law of continuity) that a body cannot pass from rest to motion without traversing all the intermediate degrees, and that the force accompanying the motion must therefore accumulate gradually, in a manner that the squared measure captures and the linear measure does not (Du Châtelet 1740, 21.557–558). The argument rested on a metaphysical commitment: because the principle of sufficient reason guarantees that every change of state has a sufficient reason in the preceding state, and because the law of continuity requires that transitions be smooth, the force of a moving body must be a function of the continuous process by which it acquired its motion. Dead force and living force stand to each other as tendency stands to accumulated effect, and the measure that reflects this distinction is mv^2 , because it tracks what Leibniz called the *effect* of a force rather than the force at an instant.

The second prong, empirical evidence, pointed in the same direction. Du Châtelet surveyed the experimental arguments on both sides and concluded that all experiments concurred in proving the existence of living forces: ‘We have seen in this chapter that all experiments concur in proving that there are living forces, and metaphysics speaks almost as strongly in their favour as physics’ (Du Châtelet 1740, 21.588). The convergence of these two prongs was the strongest vindication that the method could deliver. Both the *a priori* constraints and the experimental data pointed to the same answer.

Du Châtelet’s argument for *vis viva* is a case study in how the metaphysical warrant could be made to do real work in deciding concrete issues. The conviction that reality is a rationally intelligible system, governed by the principle of sufficient reason and the law of continuity, generates specific theoretical constraints; those constraints, when applied to the empirical problem, yield a determinate result; and the result can then be confirmed by experimental evidence. Here, the metaphysical principles reach down all the way into the detailed workings of force, motion, and collision, producing verdicts that physics alone could not reach. Thus far, the method works and the metaphysical warrant earns its keep. The question is whether it continues to do so when the problems become harder.

4.8 The Warrant Under Strain: Gravity and Collisions

If the *vis viva* controversy represents Du Châtelet’s synthesis at its most effective, two further cases reveal a different picture: one in which the two prongs of the method pull apart, and another in which the metaphysical account turns out to depend on the very physics it was meant to ground.

The case of gravity is the most revealing. Du Châtelet’s two-pronged method produced conflicting verdicts. The principle of sufficient reason yielded a reason

against accepting Newtonian action at a distance, because no sufficient reason could be given for how bodies separated by a void could act upon one another. On the other hand, the empirical evidence increasingly favoured Newton's account over the Cartesian vortex hypothesis. Du Châtelet's response was to withhold judgement. She proposed that attraction must be caused by a massless, quasi-material ether that had not yet been observed, and expressed confidence that a mechanical explanation would eventually be found. Her method required a convergence of principle and observation, and where that convergence was lacking, patience was the appropriate response (Du Châtelet 1740, 15.340–342; Wells 2023).

This left a deeper tension unaddressed, however. For if the world as God designed it satisfies the principle of sufficient reason and that principle tells against action at a distance, then the empirical success of Newton's account of gravitation poses a challenge to the warrant itself. Either the principle of sufficient reason does not hold universally, which undermines the first prong; or the empirical evidence is misleading, which undermines the second; or the warrant needs to be reformulated in a way that permits action at a distance, which would require a more permissive conception of sufficient reason than Du Châtelet was prepared to endorse. Something has to give.

Another tension emerges in her discussion of collision (Brading 2019, 97). Responding to an objection concerning collisions between bodies of unequal mass and speed, Du Châtelet appealed to a capacity of bodies to 'consume' different quantities of force. This allowed her to maintain her causal account of collisions, but it introduced a variable for which her framework provided no independent explanation. Once the capacity to consume force is admitted as an unanalysed parameter, the theoretical work in predicting the outcomes of collisions is done entirely by Newton's laws together with empirically determined coefficients; the metaphysical account of bodily force loses its predictive grip. The causal story that was supposed to ground Newton's laws in an intelligible metaphysics of force thus ends up depending on those very laws for its own application. In Münchhausenian fashion, the foundations lean upon the structure they were meant to support.

4.9 Zenith and Strain of the Metaphysical Warrant

The preceding sections have traced the metaphysical warrant through its most rigorous philosophical articulation. In the *vis viva* case, the warrant proved productive: metaphysical principles generated determinate theoretical constraints, those constraints converged with the empirical evidence, and the result was a resolution of a dispute that neither tradition could have settled on its own. In the gravitational case, however, the very ability of the warrant to be brought to bear on concrete empirical issues also enabled the empirical success of Newton's account to counter against the

warrant as Du Châtelet understood it. And in the collision case, the metaphysical foundations turned out to lean on the physical structure they were meant to support. The precision and ambition with which she articulates the metaphysical warrant is thus simultaneously what makes its successful deployment in concrete debates possible, but also what exposes its limitations.

Du Châtelet showed that the project of cognitive systematization paradigmatically embodied by Newton's *Principia* depends on philosophical commitments that the practice of physics cannot justify from its own resources. Why do observed regularities hold universally? What entitles reasoning from effects to causes? On what grounds should competing theoretical accounts be adjudicated when the empirical evidence underdetermines the choice? Du Châtelet answered each of these questions by appeal to the principle of contradiction and the principle of sufficient reason, deploying them within a two-pronged method that subjected theoretical claims to both a priori constraint and empirical testing. She showed that the principle of sufficient reason supports the inductive practices on which Newton's own reasoning depends. She illustrated how the law of continuity generates determinate theoretical constraints in domains where observation alone provides no guidance. And she argued that a rational theology in which God acts according to wisdom rather than arbitrary will guarantees the applicability of these principles to the natural world. In her hands, the metaphysical warrant for cognitive systematization is to some extent vindicated by its philosophical fruitfulness.

Yet the strain is inseparable from the achievement. Consider what the warrant is required to accomplish simultaneously. It must ground the principle of contradiction and the principle of sufficient reason as epistemic principles whose authority is independent of any particular domain of inquiry. It must justify the application of these principles to the natural world, turning them from constraints on *reasoning* to constraints on *reality*. It must sustain a rationalist theology, in which God's will is subordinate to his wisdom, against the voluntarist theology of the Newtonians for whom the natural order depends on a divine sovereignty that reason cannot penetrate. And it must bridge two philosophical traditions that disagreed on fundamental matters: the nature of force, the existence of the void, the possibility of action at a distance, and the proper measure of the motive power of bodies. Each of these tasks places a demand on the warrant, and the demands accumulate.

In Du Châtelet, the metaphysical warrant is asked to support the weight of a philosophical synthesis that extends from the most abstract principles of ontology to the most concrete problems of collision mechanics. The structure holds as long as the theological premise remains in place – as long as one accepts that a rational God designed the world as a rationally intelligible system, one has reason to trust that the principle of sufficient reason holds universally, that the law of continuity governs all natural processes, and that the principles of our knowledge are fitted to the structure of reality. Remove the theological premise, however, and that support gives way. The principle of sufficient reason becomes a methodological postulate

without metaphysical backing; the law of continuity becomes at best a defeasible heuristic rather than a necessary truth; and the bridge between systematic thought and reality loses the metaphysical guarantee that held it in place. The warrant sustains the system, but the warrant itself is sustained by a conviction that lies outside the system: the conviction that divine rationality is the ground of natural intelligibility.

This chapter has traced the arc of the metaphysical warrant from Newton's system of the world, which gave the warrant its most spectacular empirical vindication, through Du Châtelet's *Institutions de physique*, which gave it its most rigorous philosophical articulation. In Newton, the warrant appears as a theological hypothesis appended to a mathematical achievement: the heavens attest to a designing intelligence, but the connection between divine design and scientific method remains philosophically unexamined. In Du Châtelet, the warrant becomes a sustained and explicit argumentative framework: physics requires metaphysics, and the metaphysical principles that ground physics are themselves grounded in the rational structure of a God-designed world.

It is because Du Châtelet specifies so carefully what the warrant can and must accomplish, and applies it so concretely to problems of force, motion, and causation, that one can see both what it achieves and where it falters. The most sophisticated articulation of the metaphysical warrant is, simultaneously, the clearest revelation of its dependence on a conviction that the method it supports cannot independently verify. This is the chapter's central finding: that the moment at which the metaphysical warrant reaches its zenith is also the moment at which it most evidently becomes strained.

What also reaches its zenith in Du Châtelet is the idea that for thought and reality to form a harmonious and ordered system is for them to exhibit the virtues of a well-designed machine. She conceives of reality as a God-designed clockwork in which everything harmoniously interlocks with and is rationally connected to everything else. The question the chapter leaves is the question the Enlightenment would shortly pose: what happens to the demand for system when the metaphysical warrant is dismantled? It is to that dismantling that we now turn.

The System Examined: Condillac, D'Alembert, and Diderot

Systems are older than philosophers: nature itself produces them, and none were badly made so long as human beings had nature alone for their teacher.

— Condillac

Systems had been built to end political and intellectual anarchy, to replace competing institutional authorities with the single authority of reason operating according to transparent rules. They had promised to be machines for producing agreement. After a century of system-building, however, the rational machinery designed to eliminate arbitrary authority had itself become a new source of dogmatism. And while the scholastics had at least shared a common Aristotelian framework within which to disagree, the modern systematizers had produced a plurality of systems, each built upon its distinctive set of principles, so that they worked within incommensurable frameworks. The proliferation of these rival frameworks was becoming harder to ignore. How many systems had been made, Condillac asked, and how many more would be made? If at least one could be found that was received more or less uniformly by its own partisans, there might be grounds for confidence. Instead, the systems suffered a thousand changes in passing through a thousand hands, appearing and disappearing as playthings of caprice (Condillac 1749, 16). The cure was increasingly looking worse than the disease.

This chapter traces the Enlightenment's internal response to that predicament. The philosophers who articulated that response achieved something significant for the genealogy of systematization: they began to separate the *practical* case for systematization from the *metaphysical* one. In doing so, they refined the demand for systematicity rather than abandoning it, isolating the conditions under which systematization succeeds, the rationales that survive when the metaphysical warrant is withdrawn, and the costs that even successful systematization exacts. Condillac identified the linguistic conditions that cognitive machinery must meet if it is to engage with anything beyond its own operations. D'Alembert codified the distinction

between building systems and thinking systematically, and gave that distinction its most ambitious institutional expression in the *Encyclopédie*. Diderot pressed the question of cost: what dimensions of thought and experience does the systematic spirit, even in its chastened form, inevitably suppress? Together, these three thinkers demonstrated that the demand for cognitive systematization answers to practical needs that do not logically depend on any metaphysical conviction about the rational structure of reality. Yet the separation they achieved was more complete at the level of explicit argument than at the level of their own methodological practice. Both Condillac and D'Alembert retained, in their positive programmes, commitments to a rationally ordered nature that went beyond anything the practical rationales alone would require. Tracing the achievement and registering its limits is the work of this chapter.

5.1 Condillac and the Diagnosis of Empty Machinery

The philosopher who delivered the most celebrated diagnosis of the rationalist systems' failure was Étienne Bonnot de Condillac (1714–1780), the author of the *Traité des systèmes* of 1749. The year was itself a watershed: Buffon's *Histoire naturelle* and Diderot's *Lettre sur les aveugles* appeared in the same months, and the three works together marked the moment at which the French Enlightenment turned from absorbing the English empiricist inheritance to producing its own distinctive philosophical programme (Hine 1979, 19–20). D'Alembert, writing two years later in the *Preliminary Discourse* to the *Encyclopédie*, credited Condillac with having dealt the taste for systems a fatal blow (D'Alembert 1751, xxxi). The assessment was widely shared. The *Traité* earned Condillac admission to the Royal Academy of Sciences and Belles-Lettres in Berlin, appointment as a royal censor, and a reputation as the philosopher who had swept away the ruins of seventeenth-century metaphysics (Schwegman 2010, 387).

Yet that reputation was misleading. Condillac had set out to show that one could reason in metaphysics and morals with as much precision as in geometry, and that once the dream-palaces of speculative philosophy had been cleared away, a properly rigorous science of the mind could be built in their place (Knight 1968, vii, 16). His target was a particular *kind* of system, which needed to be demolished to make way for better systems. The demolition was in the service of construction.

He was not working alone. In the late 1740s, Condillac, Rousseau, and Diderot formed a small philosophical circle, dining together weekly at the Hôtel du Panier Fleuri near the Palais-Royal and talking about the problems of sense experience, the origin of language, and the nature of the mind. Rousseau later recalled having been the first to discover Condillac's abilities and to introduce him to Diderot, who in turn persuaded the bookseller Durand to publish Condillac's first work (Rousseau 1782–89, 115–16). Condillac repaid his friends in the coin of ideas: the discussions at

those dinners fed into Diderot's *Lettre sur les aveugles* and Rousseau's reflections on language in the *Discours sur l'inégalité*, where Rousseau explicitly credited Condillac with giving him his first ideas on the subject (Knight 1968, 10). The three thinkers who provide this chapter's *dramatis personae* had, in other words, hammered out their positions in conversation with one another before any of them published the works for which they are remembered here.

The *Traité des systèmes* opens with a general definition of what a system is. A system, Condillac writes, is nothing other than the arrangement of the different parts of an art or a science in an order in which they all mutually support one another, and in which the last are explained by the first. Those parts that account for the others are called principles, and the system is all the more perfect the fewer the principles: it is even desirable to reduce them to one (Condillac 1749, 8). With its emphasis on integrating an entire domain into a coherent and parsimoniously principled structure, this captures much of what the seventeenth-century systematizers had aspired to do. Condillac accepted this conception of what a system ought to be. What his critical work consisted in was to show that the systems actually on offer failed to live up to it.

From this definition, the *Traité* moves to a typology distinguishing three sorts of systems according to the principles they are based on. Systems built on *general or abstract maxims* rest on propositions so broad and apparently self-evident that they seem capable of grounding an entire edifice of knowledge. Systems built on *hypotheses* rest on suppositions about unobservable causes, offered as explanations for observed phenomena. Systems built on *well-established facts* rest on principles drawn from careful observation and confirmed by experience (Condillac 1749, 8–11). Only the third kind yields genuine knowledge and is truly deserving of the title of system, according to Condillac. The second is permissible with caution, provided the hypotheses are treated as provisional and tested against further observation. The first is fundamentally and irredeemably flawed.

The argument against abstract systems occupies the bulk of the work, and its force derives from a single, relentlessly applied insight: abstract maxims are either vacuously true or equivocally extended beyond their proper domain. Consider the principle that the whole is greater than its part. Condillac grants its truth in every case, but asks what knowledge it could possibly generate. The proposition means only that my body is larger than my arm, my arm larger than my hand, my hand larger than my finger, and so on through every particular instance from which the general claim was abstracted. Anyone who already knows these particular truths knows everything the principle contains; anyone who does not will learn nothing from the principle itself (Condillac 1749, 16). Give a philosopher nothing but abstract first principles, Condillac suggests, and ask him to command an army or govern a state. He would reasonably protest that he knows nothing of warfare or politics. Yet the art of war and the science of government have their own general principles. Why should a few hours' meditation on these fail to produce a Condé, a

Turenne, a Richelieu? Because particular knowledge, acquired through sustained attention to the details of a specific domain, cannot be extracted from abstract generalities, however apparently certain those generalities may be (Condillac 1749, 17–18).

The great systems of the seventeenth century did not, of course, rest on truisms. They rested on principles that combined apparent self-evidence with genuine substantive content: Descartes's criterion of clarity and distinctness, Leibniz's principle of sufficient reason, Spinoza's definitions and axioms. Condillac's response to these more ambitious principles occupies the central chapters of the *Traité*. Such principles, he argues, are true only in certain respects and under certain conditions, but are deployed as though their authority were unlimited. Descartes's principle that we may affirm of a thing everything contained in the clear and distinct idea we have of it is compelling for simple mathematical cases, where the ideas in question can be fully specified. Extended to questions about the nature of substance, the relationship between mind and body, or the existence of God, the principle outruns its evidence. The ideas become less clear, less distinct, and more dependent on the philosopher's particular habits of imagination, even though the principle continues to be applied as if nothing had changed (Condillac 1749, 16–17; chs. 4–10).

Condillac finds this pattern repeated in every abstract system he examines. Leibniz's monads receive the longest treatment: the central chapter of the *Traité* is devoted to the exposition and refutation of the monadology. Condillac's strategy is always to isolate the foundational concepts, trace the chain of deductions the system-builder claims to derive from them, and show that the chain holds only if the foundational concepts are given a precision and a determinacy they do not actually possess. The concepts work within the system, but the work they perform depends on an ambiguity that the system conceals. Substance, force, perception, harmony: each term carries the appearance of precision, because it functions within an internally consistent deductive structure. The precision is borrowed from the structure (Condillac 1749, 30–80). Strip away the structure, and the terms dissolve into vagueness.

The *Traité* captures this predicament in an architectural image that the *Encyclopédie*'s own article on systems would later borrow. The abstract systems resemble palaces in which taste, convenience, grandeur, and magnificence combine to make a masterpiece of art, but whose foundations are so feeble that the whole structure appears to sustain itself only by enchantment: one would certainly praise the architect, but the praise would be well counterbalanced by the criticism one would make of his imprudence (Condillac 1749; cited in Hine 1979, 29). The deductive chains that connected these systems' principles to their conclusions transmitted emptiness rather than knowledge, because the foundational terms in which the principles were articulated had no determinate empirical content. The appearance of rigour was sustained by the internal consistency of the verbal structure rather than by any contact with the world the structure purported to describe. The defect was not

that the machine had been fed the wrong materials; it was that the machine's core components, the terms carrying the load of the entire deductive apparatus, were themselves hollow.

Condillac exploited precisely the machine-like virtues of the abstract systems to identify where they showed signs of strain. He accepted surveyability, precision, rule-governance, and transparency of inferential connection as criteria of adequacy, and it was when measured against those criteria that the seventeenth-century systems failed. Jeffrey Schwegman has argued that the *Traité des systèmes* itself functioned as a pedagogical instrument: a reading technology that taught its audience how to read systems critically (Schwegman 2010). When Keckermann and the Reformed pedagogues of the late sixteenth century introduced the *systema*, their driving concern was pedagogical: the system was a technology for making complex bodies of knowledge surveyable and transmissible. After systems proliferated for a century and a half, that same pedagogical impulse had to be given a diagnostic inflection. People needed to be taught how to distinguish good systematization from bad. The pedagogical rationale thereby shaded into the epistemological rationale, and the demand for system, carried through with diagnostic honesty, turned out to be self-correcting.

5.2 Language, System, and the Floor of Cognition

Condillac's diagnosis ran deeper than a polemic against the seventeenth-century rationalists. Beneath the specific critiques of Descartes, Spinoza, and Leibniz lay a more general philosophical claim about the conditions under which any systematic thought would succeed or fail, and that claim was fundamentally a claim about what gives our thoughts determinate content.

The first object of a philosopher, Condillac declares, must be to determine his ideas exactly (Condillac 1749, 15). Abstract ideas, however, are naturally vague, offering nothing firm to hold on to. To remedy this, one needs to explain more abstract ideas in terms of less abstract ones, and those in terms of ideas that are less abstract still, until one reaches particular ideas rooted in sensory experience. Our senses are the ultimate source of all our knowledge (Condillac 1749, 15). Abstract ideas can thus only be made determinate by tracing them back to particular, sensory ideas.

It is not hard to see how Condillac's critique of abstract systems falls out of these commitments. The first abuse of systems, he writes, is that we believe we acquire genuine knowledge when our thoughts turn only on words that have no determinate meaning — *des mots qui n'ont point de sens déterminé* (Condillac 1749, 21). A system, for Condillac, is an articulated structure of propositions connected by inferential relations. If a system is to yield knowledge, the terms on which it is built must have determinate meaning, and for Condillac this means they must be

anchored in particular, sensory ideas. When the terms lack such anchorage, the system's inferential structure may be as consistent, coherent, and parsimonious as one likes: it will not convey real knowledge unless its fundamental currency is hooked up to reality in the right way. A system built on indeterminate terms is, in Condillac's image, like a game in which the players have tacitly agreed that tokens will stand in for money, but no one has bothered to establish what the tokens are worth. Everyone condemns the gambler who plays without knowing the value of the tokens, Condillac observes, yet anyone may speak without knowing the value of the words (Condillac 1749, 203). The rational machinery is fully assembled, and its moving parts harmoniously interlock, but its fundamental building-blocks are devoid of real meaning, acquiring an air of precision only through their tight enmeshment in a systematic structure. All the machinery can do is process vacuous words according to well-defined rules and deliver more vacuous words.

This linguistic dimension of Condillac's thinking has been persistently underestimated. He has often been read as a plain empiricist, a transmitter of Lockean ideas into French philosophical culture. Hans Aarsleff showed that this reading is mistaken (Aarsleff 1975). Condillac's *Traité des sensations* (1754) presents a thought experiment in which a statue is progressively endowed with the five senses. Far from establishing that sensation alone generates the full range of human cognitive powers, the statue establishes the opposite. Endowed with every sense, it achieves practical knowledge sufficient for its preservation: it learns to seek pleasure and avoid pain, develops habits and rudimentary memory, and navigates its environment with something resembling animal instinct. What it cannot do is reason. It cannot classify, generalise, or render account of its own ideas to itself. Its knowledge, Condillac insists, is only practical, and the light that guides it is merely instinct (Condillac 1754). Theoretical knowledge, the kind that deserves the name of science, requires something the statue entirely lacks. It requires language.

The gap between the statue's predicament and the condition of a knowing subject maps, in Condillac's terms, onto the distinction between two selves that every human being contains: the *moi d'habitude* and the *moi de réflexion*, the self of habit and the self of reflection (Condillac 1755). The measure of reflection we have beyond our habits is what constitutes our reason. The statue possesses the first self but lacks the second. It has habits, memories, and learned associations, but it cannot step back from these to inspect, classify, or reorder them, because reflection requires a medium in which ideas can be held, voluntarily recalled, and deliberately compared. That medium is language: a system of voluntarily created signs that fix ideas, make them retrievable, and allow them to be arranged according to method. Without such signs, the materials of experience remain a confused aggregate of sensory impressions and habitual responses, real enough in themselves, but unavailable for the kind of deliberate ordering that constitutes theoretical knowledge. Signs make it possible to fix ideas, to classify them, to arrange them in an order, and to draw out their consequences. To acquire knowledge, Condillac writes, it is necessary to have a

language, for the ideas must be classified and determined, which presupposes signs employed according to method (Condillac 1754). And signs employed according to method just *are* a system: an articulated structure whose elements stand in transparent inferential relations to one another. What the statue cannot do, then, is to systematize its experience, and it is for precisely this reason that it cannot know.

A well-made language and a well-made system are, for Condillac, the same thing viewed under different descriptions. Science, on this account, is the achievement of a properly constructed language for a given domain of experience: a set of signs whose content is anchored in observation, whose relations are transparent, and whose inferential connections yield genuine consequences rather than verbal permutations. In the final chapters of the *Traité des systèmes*, Condillac draws the identification to its conclusion: the art of reasoning reduces to a well-made language; a well-treated science is simply a well-made language; and a well-treated science is a well-made system, in which there are, in general, only two things: principles and consequences (Condillac 1749, 199, 202–3). The triple equation tightens the conceptual chain. What gives a system its cognitive purchase is the empirical determinacy of the terms in which its principles are articulated: the very terms that carry the weight of the deductive structure must themselves be anchored in observation if the structure is to transmit knowledge rather than verbal permutations. What makes a language well-made is the same thing that makes a system well-made: the terms at its foundation have determinate content, and its structure mirrors the connections among the phenomena it describes. The *langue bien faite* is, in this sense, the engineering specification for the cognitive machine: it tells us what the machine's load-bearing components must be made of if they are to engage with the world rather than merely with one another. Lavoisier's reform of chemical nomenclature half a century later would apply that principle, replacing an opaque lexicon of alchemical provenance with a systematic terminology whose structure mirrored the compositional structure of the substances it described, and acknowledging the debt to Condillac explicitly.

The statue's predicament also reveals something about the demand for system that would prove essential to Kant's transformation of the entire problematic. There is a *floor* to systematicity: a minimum level of ordered connection among ideas below which cognition dissolves into mere sensation and habit. Without language, the statue's mental life is real but inarticulate. It has experiences, but it cannot think about them, classify them, or draw conclusions from them. What language provides is the possibility of ordering experience, and the possibility of ordered experience is the possibility of knowledge. Condillac had thereby suggested that the question confronting any would-be critic of systematization is never *whether* to systematize, but *how*, and *how much*. Below a certain threshold, there is no cognition at all: the statue shows what the mental life of a creature looks like when the systematizing impulse has done no work whatsoever. Above that threshold, the question becomes

one of degree, of appropriate method, and of the costs and benefits of extending systematic order further into domains that may resist it.

The statue thereby brings into focus the specific cognitive need to which systematization answers. The statue can act effectively in the world: it avoids what harms it and seeks what preserves it. What it cannot do is render its competence *articulable*. It cannot give an account of how it navigates its environment, hold that account up for inspection, communicate it to another creature, or revise it in light of new evidence. Its practical knowledge is locked inside its habits, inaccessible to scrutiny and incommunicable by any means other than imitation. The cognitive need that systematization meets is the need to convert this kind of inarticulate, habit-bound competence into knowledge that is explicit enough to be examined, transparent enough to be assessed, and structured enough to be transmitted and corrected. These are recognizably the needs that the three practical rationales identify. The pedagogical rationale answers the need for transmissibility: knowledge must be articulable if it is to be conveyed. The epistemological rationale answers the need for inspectability and self-correction: knowledge must be structured if error is to register and invite revision. The political rationale answers the need for public accountability: knowledge must be transparent if the authority exercised through it is to be answerable to those affected. Condillac's statue shows that these needs arise from a fact about human cognitive limitation, the fact that experience does not organize itself, and that without the deliberate imposition of linguistic and systematic order, the materials of experience remain cognitively inert. The needs are real, and they would be real in any world, regardless of its metaphysical architecture.

Here lies the connection between Condillac's linguistic critique and the arc of the metaphysical warrant that this book has been tracing since its opening chapter. The warrant had provided a very different kind of reason for systematizing. From Keckermann through Leibniz and Du Châtelet, it had licensed the projection of systematic order onto the structure of reality: God had designed the world as a rational whole; the mind was equipped to grasp that whole; and the system, rightly constructed, would mirror the divine architecture. The reason to systematize, on this picture, was that reality *itself* is a system, and knowledge should mirror it. Condillac's analysis displaced this picture. The reason to systematize, as the statue reveals it, is that experience needs to be rendered articulable, inspectable, and transmissible, and that this rendering requires the imposition of linguistic and systematic order on materials that do not arrive already ordered. The need is practical, arising from facts about the human cognitive situation, and it would persist even if no metaphysical warrant were available. Condillac's work also implied that the warrant, however impressive, could not do the work it claimed to do unless a prior condition was met: whether the terms in which the system was constructed had determinate content. If they did, the system might indeed track the structure of reality, whatever one's metaphysical commitments. If they did not, no amount of metaphysical conviction could rescue it from vacuity. Condillac had uncovered a condition of adequacy for

systematic thought that was independent of, and logically prior to, the metaphysical warrant.

Yet Condillac's own positive vision of what a good system looks like reveals that the stripping was less thorough than this formulation suggests. In his *Dictionnaire des synonymes*, composed late in life, Condillac offered a definition of system that went well beyond the modest, empiricist-sounding account of the *Traité*. The universe, he wrote, is nothing other than a system: a multitude of phenomena which, being related to one another as causes and effects, all spring from a first law. Systems should not be invented, he cautioned; we should discover those which the author of nature has made (Condillac 1951, 511–12). In his pedagogical writings for the Prince of Parma, he took the point further: a true system unfolds a single truth through a chain of propositions that are, at bottom, identical. God, who grasps the whole at once, sees each truth as we see that two and two make four; he perceives all truths in a single truth (Condillac 1775a, 748). A true system, on this account, is a gradual human approximation of the divine intuition of a single, all-encompassing identity. Newton's universe is a great balance in which the sun and the planets move on a common centre of gravity, and we have risen from knowledge to knowledge only because we have passed from identical proposition to identical proposition (Condillac 1775b, 676). Isabel Knight has observed that the independence with which Condillac arrived at a conception so close to Leibniz's vision of universal harmony suggests, perhaps more strongly than imitation would, a deep affinity with the rationalist tradition that his empiricist commitments were supposed to have left behind (Knight 1968, 66).

The point here is not that Condillac was inconsistent. It is that the separation of the practical from the metaphysical case for systematization was harder than his polemical success in the *Traité des systèmes* suggested. His negative achievement is beyond dispute: the metaphysical warrant cannot rescue a system whose terms lack empirical content, and the demand for well-made language is logically independent of any conviction about God's design. But the positive programme into which that demand was embedded presupposed something the practical rationales alone would not generate: the conviction that the universe *is* a system, that its phenomena spring from a first law, and that the philosopher's task is to discover the systems the author of nature has made. The practical rationales explain why we need systems. They do not explain why the universe should be one. Condillac assumed both.

Three insights survive Condillac's critique and carry forward into the rest of this chapter. Systematization answers the cognitive need to render experience articulable, inspectable, and transmissible: the statue stands as a permanent reminder that without the ordering work of language, experience cannot be examined, communicated, or corrected, and therefore cannot rise to the level of knowledge. The machine-like virtues identified by the machine emulation thesis turn out to be self-correcting: Condillac wielded the very criteria of adequacy that the seventeenth-century systems claimed to satisfy as diagnostic instruments against those systems,

and it was when measured against these criteria that their foundational terms were exposed as hollow. And the separation of practical from metaphysical grounds for systematization, though real, left the French Enlightenment's own methodological commitments insufficiently examined: the practical need to render experience reflectively inspectable had been distinguished from the metaphysical conviction that reality is a system, but Condillac's positive programme continued to presuppose the latter alongside the former. What remained to be shown was whether the systematic spirit, purged of its speculative ambitions, could organize knowledge on a scale the rationalist systems had aspired to. That task fell to D'Alembert.

5.3 The Systematic Spirit and the Spirit of System

Jean le Rond d'Alembert (1717–1783) was a mathematician and physicist of considerable range. He was the co-editor, with Diderot, of the *Encyclopédie*, and the author of the text that served as the manifesto of the entire enterprise: the *Preliminary Discourse to the Encyclopédie*, published in 1751. Unlike the *Traité des systèmes*, which had worked through the seventeenth-century systems one by one and sought to expose the emptiness of their foundational concepts, D'Alembert stepped back and asked what could be salvaged from this enterprise. His answer took the form of a distinction that became one of the Enlightenment's orienting contrasts.

D'Alembert drew the distinction early in the *Preliminary Discourse*, in a passage on the method of natural philosophy. Discussing the reduction of observed phenomena to a small number of principles, he paused to add a qualification whose reach extended far beyond its immediate context: this reduction, which moreover makes phenomena easier to understand, constitutes the true *systematic spirit*, the *esprit systématique*, and one must be very careful not to mistake it for the *spirit of system*, the *esprit de système*, with which it does not always agree (D'Alembert 1751, xxx). The *esprit systématique* is the disciplined effort to bring observed facts into orderly relation, to discover the connections among phenomena, and to reduce the principles of a science to the smallest number compatible with explanatory adequacy. The *esprit de système* is the speculative construction of comprehensive frameworks from abstract first principles – the enterprise Condillac had just dismantled. D'Alembert pressed the contrast with an analogy: if the very small number of discoveries that systems had occasioned were claimed as proof of the usefulness of systems, one might just as well counsel geometers to apply themselves to squaring the circle, because the efforts of several mathematicians to do so had given us a few theorems (D'Alembert 1751, xlv–xlvi). Being systematic and building systems are different activities, and the former is both possible and desirable in the absence of the latter.

The distinction can be seen as separating *warrant-dependent* from *warrant-independent* systematization. The *esprit de système* is the machine emulation thesis

carried to speculative excess, driven by the conviction that systematic thought mirrors a God-designed rational order and therefore licensed to deduce the structure of reality from a priori principles. The *esprit systématique* retains the machine-like virtues that the practical rationales depend on. But it discards is the metaphysical claim that these virtues track the deep structure of creation. Systematic thought, on D'Alembert's reconception, earns its authority through the cognitive work it performs, through the intelligibility and navigability it confers, and through its capacity to expose gaps and invite correction. The rationale is fundamentally practical: the value of ordering knowledge lies in what the ordering *does* for the thinker who engages with it.

D'Alembert's philosophical ally in codifying this pragmatic grounding was Condillac himself. Thomas Hankins has shown that of all the members of the philosophical circle that included Diderot and Rousseau, it was Condillac whose philosophy agreed most closely with D'Alembert's own (Hankins 1970, 55). The influence ran deep: the editors of the *Preliminary Discourse* have noted that D'Alembert drew extensively on Condillac's *Traité des systèmes* in formulating the philosophical sections of the *Discours*, and the two may have collaborated to some extent on both works (Schwab, note 31 to D'Alembert 1751). The *Encyclopédie* articles on divination and on systems were taken word for word from the *Traité*, though they appeared unsigned (Knight 1968, 1–2). Condillac's epistemology was deeply embedded into the *Encyclopédie*'s programme.

There was, however, a tension at the heart of this programme that D'Alembert never fully resolved. His public philosophy, shaped by Condillac, was empiricist and post-metaphysical: the *esprit systématique* was to earn its authority through its empirical fruits. His private science told a different story. In the preface to his *Traité de dynamique* of 1743, D'Alembert had laid down a rationalist mechanics whose ambition far exceeded the modesty of the *Discours préliminaire*. The laws of motion, he claimed, were absolutely necessary and demonstrable from the geometry of impenetrable figures moving in space. He could not admit any reference in his mechanics to the truths of experience which would, as he put it, ruin the certainty of mechanics and reduce it to a mere experimental science (Hankins 1970, 4). The astronomer Lalande complained that D'Alembert had never held a prism in his hand, and the complaint was almost certainly just (Hankins 1970, 5). D'Alembert counted himself among the architects of knowledge and dismissed the experimentalists as subaltern labourers who carried the stones but did not deserve the credit for the beauty of the finished structure (Hankins 1970, 8). The *esprit systématique* he advocated in the *Discours* was, in his own scientific practice, closer to the *esprit de système* he officially repudiated: a conviction that nature's deepest truths are mathematical necessities, accessible to reason alone and independent of empirical verification.

D'Alembert did not see a contradiction between the two positions. The *Discours préliminaire* concerned the organization of empirical knowledge across all domains;

the *Traité de dynamique* concerned the foundations of a single mathematical science. D'Alembert could preach empiricist modesty as a general philosophical programme while practising rationalist rigour within mechanics, because he regarded mechanics as a branch of mathematics, and mathematics as independent of experience (Hankins 1970, 4). The compartmentalization allowed the residual rationalism to persist unexamined, tucked away in a domain D'Alembert considered exempt from the empiricist strictures he imposed everywhere else. It was evidence of how difficult it proved to dislodge the metaphysical warrant when its working assumptions were embedded in the sciences the Enlightenment most admired.

Bernard Williams describes the *Preliminary Discourse* as replacing the elaborate metaphysical constructions of the preceding century with an outlook that was 'orderly without being speculatively ambitious' (Williams 1967). He observes that the central opposition animating Enlightenment thought was different from the one that had driven seventeenth-century rationalism. For Descartes and Leibniz, reason had primarily been opposed to *experience*: the whole point of a rationalist system was to transcend the deliverances of the senses and arrive at truths accessible only to the intellect. For the Enlightenment, reason was primarily opposed to *authority*: the demand for systematic thought was a demand for intellectual autonomy, for the right to examine all claims and accept only those that survived critical scrutiny (Williams 1967).

This shift registers a change in the practical function of the systematicity ideal. When reason was opposed to experience, systematization aspired to an a priori comprehension of the structure of reality, and the metaphysical warrant provided the licence. When reason was opposed to authority, systematization aspired to something more modest and more defensible: the orderly examination of the world unclouded by prejudice, tradition, or ecclesiastical dictation. The *political* rationale for systematization, which in earlier chapters had been implicit in the pedagogical and epistemological functions of system-building, now stepped into full view. D'Alembert himself proposed, in one of his more startling passages, that infiltrating enough geometers into the citizenry of a country oppressed by the Church would suffice to bring about enlightenment (Hankins 1970, 40). The mathematical spirit, precisely because of its impersonality and rule-governance, was inherently subversive of authority that rested on deference rather than demonstration.

Ernst Cassirer placed the *esprit de système* / *esprit systématique* distinction at the centre of his account of the Enlightenment. The period broke faith with the spirit of systems, but by no means abandoned the systematic spirit; it aimed to further this spirit in another and more effective manner (Cassirer 1951, vii). On Cassirer's reading, the *esprit systématique* is less a method than an intellectual temperament: the habit of seeking connections, of demanding that claims be grounded and coherent, and of refusing to accept assertions on the basis of authority alone.

D'Alembert's own practice in the *Preliminary Discourse* illustrates what the systematic temperament looks like when put to work. Following Bacon, he proposed a

figurative system of human knowledge that maps the whole of cognition onto three faculties: Memory, which gives rise to History; Reason, which gives rise to Philosophy; and Imagination, which gives rise to the Fine Arts. D'Alembert was explicit that the map was not intended as a deductive system. Any ordering of the branches of knowledge is, he acknowledged, of necessity somewhat arbitrary, because the objects of inquiry do not sort themselves into clean categories (D'Alembert 1751, xxxiii). He pictures the universe of knowledge as a vast ocean, on the surface of which we perceive a few islands of various sizes, whose connection with the continent is hidden from us (D'Alembert 1751, xxxiii). The map is provisional, revisable, and openly perspectival – its authority rests on its ability to meet our needs, and the question of whether it reflects the mind of God does not arise. All ideas and branches of knowledge, D'Alembert writes, ultimately trace back to human needs, though they were slower to appear the more remote or difficult to satisfy those needs were (D'Alembert 1751, i; see Queloz 2022, 436).

This pragmatic reconception of the demand for system found its most ambitious institutional expression in the *Encyclopédie* itself. In its sheer scale, a collaborative effort involving over 140 contributors producing seventeen volumes of text and eleven of plates between 1751 and 1772, twenty million words encompassing nearly seventy-three thousand articles, the *Encyclopédie* posed the question of cognitive systematization in an unprecedented form. How does one organize the totality of human knowledge when no single mind can survey it all, no deductive structure can contain it, and no fixed set of first principles can ground it?

The question was political from the outset, and so was the answer. The *Encyclopédie's* tree of knowledge, adapted from Bacon with significant modifications, was in important respects a political act of classification, because while Bacon had placed theology at the summit of a separate tree and guarded it from philosophical contamination, D'Alembert and Diderot not only subordinated revealed theology to reason, but positioned it on a rather remote twig, alongside superstitions, divination, and black magic (Darnton 1984, 200). This conveyed a message already merely through the placement of things, turning morphology into irony (Darnton 1984, 200). The cross-references performed the same work at the level of individual articles (the article on anthropophagy referred the reader to the entry on the Eucharist). The alphabetical ordering itself had a levelling effect, placing the mechanical arts alongside the liberal arts, the pin-maker alongside the metaphysician, and dignifying the knowledge of artisans and craftspeople with the same systematic treatment accorded to the knowledge of philosophers. Recent digital scholarship has revealed a further layer: Du Châtelet's *Institutions de physique* was extensively incorporated into the *Encyclopédie's* metaphysical articles through the contributions of Johann Heinrich Samuel Formey, mostly without acknowledgement (Roe 2017).

The work was produced under conditions of sustained political opposition. Its privilege was revoked in 1759, volumes were seized, and the later volumes were published under a thin legal fiction. Chrétien-Guillaume de Malesherbes, the young

Director of the Book Trade, protected the enterprise at considerable personal risk, allowing it to appear with full Royal Privilege despite the reservations of his own censors (Blom 2004, 42). The demand for the work was so intense that it went through multiple pirated editions, was smuggled across borders in bales of hay and the false bottoms of barrels, and was reprinted by entrepreneurial publishers from Neuchâtel to Lyon (Darnton 1979; Blom 2004, 13). The appetite for publicly organized, systematically accessible knowledge extended well beyond intellectual elites. It was driven by a felt need for knowledge that did not depend on the permission of ecclesiastical or royal authority for its organization, and that could be navigated by anyone with the will to consult it.

D'Alembert's answer to the organizational challenge was the cross-reference system, the *renvois*. Considered in light of the machine emulation thesis, the *renvois* mark a structural innovation. Every cognitive machine examined in the preceding chapters had been hierarchical in design. The *Encyclopédie's* cross-reference system replaced this vertical logic with a lateral one. Its systematicity was relational, distributed across a web of connections that no single author had designed and no single reader was expected to traverse in the same sequence. Each article contained references to related articles elsewhere in the work, creating a web of connections that allowed the reader to move laterally across the whole corpus rather than descending deductively from axioms to consequences. The structure was relational rather than hierarchical, multiply navigable rather than unilinear, and it explicitly invited the reader to construct paths through the material that no single author had foreseen. An encyclopaedia that merely catalogued the contents of each discipline would be useful, but uninstructional; one that imposed a rigid system on the whole would be falsifying in practice, since the connections among disciplines are too various and too imperfectly understood to be captured in a single deductive framework. The *Encyclopédie* occupied the space between these extremes: systematic enough to exhibit connections and enable navigation, but flexible enough to accommodate the diversity and the provisionality of human knowledge. It was a cognitive machine designed for a specific purpose, whose operators could inspect its workings, identify its limitations, and modify its structure as the state of knowledge changed.

The distance between this enterprise and Leibniz's dream of a *characteristica universalis* measured the distance the systematicity ideal had travelled. Leibniz had envisaged a formal language grounded in a metaphysical conviction that all truths are interconnected in God's mind, and a calculus that would settle disputes by rendering the logical relations among truths mechanically transparent. The *Encyclopédie* pursued an analogous ambition: to make the connections among the branches of human knowledge visible, navigable, and publicly available. Yet the grounding had shifted. The authority of the encyclopaedic system rested on its navigability and its openness to revision, on the work it performed for those who consulted it, and on its status as a shared public resource.

5.4 The Counter-Voice Within the System

Condillac and D'Alembert remained committed to the systematizing impulse: Condillac wanted to replace defective systems with well-made ones; D'Alembert wanted to retain the systematic spirit while discarding speculative ambition. Denis Diderot inhabited a more unsettled position. He co-edited the *Encyclopédie*, the century's most ambitious exercise in cognitive organization, and in his article on art he had himself defined such an exercise as a system of instruments and rules conspiring toward a single end (Diderot 1751, 'Art'). Yet his own philosophical writings work against the idea that thought can be adequately captured in any single systematic form. Friedrich Melchior Grimm, who knew both Condillac and Diderot well, captured the contrast between the two reigning philosophical temperaments of the age by observing that Condillac had a melancholy exactitude of reasoning where Diderot had the fire of a philosophical imagination (Grimm, cited in Hine 1979, 9). Williams noted that many of the Enlightenment thinkers commonly described as rationalist characteristically emphasized the role of sentiment (Williams 1967), and Diderot is the most striking case. His philosophical sensibility was drawn to what resists codification: the particular, the perspectival, the physiologically grounded, the irreducibly dramatic. If D'Alembert had shown that systematic thought could survive without speculative metaphysics, Diderot raised the question of whether systematic thought, however well grounded, could ever be made fully adequate to its object.

The question was philosophical, and Diderot pursued it philosophically, though his preferred instruments were dialogues, dreams, and fictions rather than treatises. In the *Pensées sur l'interprétation de la nature* of 1753, he had already identified a cognitive capacity that the systematic spirit could not formalize. The most important service that master experimentalists render their students, he wrote, is not instruction in procedures or results but the transmission of an *esprit de divination* by which one can, so to speak, sniff out unknown procedures, new experiments, and undiscovered results (Diderot 1753, 48). This spirit of divination was a third cognitive mode, distinct from both the *esprit de système* and the *esprit systématique*. It named the trained intuition of a natural philosopher whose embodied, particular relationship with a domain of inquiry enables leaps of insight that no rule-governed method could prescribe. Everything in the philosopher's work, Diderot argued, reduces to a ceaseless movement from the senses to reflection and from reflection to the senses: going into oneself and coming out again, ceaselessly. This, he wrote, is the bee's labour (Diderot 1753, 34). The image would return sixteen years later, in a different register and with a different philosophical function.

The most concentrated expression of Diderot's inveighing against systematization is the trilogy of texts composed in the summer of 1769, centred on *Le Rêve de d'Alembert*. The framing is itself a provocation. D'Alembert, the great geometer, the architect of the encyclopaedic tree of knowledge, is rendered unconscious. In

his fever dream, freed from the rational controls of waking life, he articulates a vision of nature that no waking systematizer could have produced. Mademoiselle de Lespinasse transcribes the fragments of his delirious speech; Dr Bordeu, a physician, arrives to interpret them. The form is thus triadic: we have a dreamer who speaks truths he would disavow upon waking, a recorder who does not understand what she writes down, and a professional interpreter whose expertise is the body rather than the mind. Each participant grasps something the others miss, but none commands the whole. The literary form here enacts the philosophical thesis: knowledge distributed across perspectives that cannot be unified into a single authoritative voice is knowledge that resists the very systematization D'Alembert had championed.

In Diderot's rendering, D'Alembert's dreaming mind envisions a nature of perpetual flux and transformation in which sensibility is a universal property of matter. Individual organisms are transient configurations of living molecules that aggregate, interact, and dissolve. The organizing image is, once again, the swarm of bees, only what Diderot now does with it deepens the methodological insight of the *Pensées* into an ontological challenge. The bee of 1753 stood for philosophical method, the back-and-forth between sense and reflection. The bees of 1769 are a figure for the constitution of individuality itself. Press the swarm together and you have an aggregate; fuse their limbs, so that mere contiguity gives way to continuity, and you have something indistinguishable from a single animal. On this picture, the human body is itself such a composite. The boundaries that any system of classification imposes on such a world are, to some degree, artefacts of the classifying mind. Categories of substance, species, and kind presuppose stable, discrete entities; the world of the *Rêve* is one of gradation, metamorphosis, and interpenetration. There is nothing precise in nature, the dreaming D'Alembert declares: every animal is more or less a human being, every mineral more or less a plant, every plant more or less an animal (Diderot c. 1769). A system can capture such a world only approximately, by freezing in thought what in reality is fluid.

In *Le Neveu de Rameau*, composed over roughly the same period, Diderot stages the tension in a different register. The philosopher (Moi) encounters the nephew of the composer Rameau (Lui) at a café in the Palais-Royal. Lui is a social parasite, a mimic, and a failed musician. He is cynical about virtue, frank about his dependence on patrons, capable of devastating aesthetic insight, and immune to the philosopher's demands for consistency. His great set-pieces are pantomimes: physical performances in which he becomes, in succession, every instrument in an orchestra, every character in an opera, and every participant in the social comedy of flattery and dependence. Moi wants to extract from this performance a moral conclusion, a stable characterization of Lui's soul. The dialogue defeats him at every turn. Lui's performances are too various, too contradictory, and too alive to be captured in any moral system the philosopher can devise.

Williams, in a reading that recovers what Hegel's famous treatment of the dialogue in the *Phenomenology* had obscured, identifies the philosophical core of Rameau's resistance. Lui is sincere, but his sincerity takes a form that the systematizing tradition cannot accommodate. He is sincere in the mode of uninhibited enactment rather than in the mode of self-examination yielding a transparent report on a unified inner life: he offers an exceptionally clear example of sincerity in its basic form of uninhibited expression, rather than in the form of reporting the findings of self-examination (Williams 2002, 189). He is conspicuously free of self-deception, yet 'what he is not is a unity' – he is not 'all of a piece' (Williams 2002, 189). The declaration at any given instant of self can be only a declaration of self at that instant, because feelings, needs, passions, and identifications actually come and go. Diderot, Williams observes, was always drawn to a picture of the self as something constantly shifting and reacting: as a swarm of bees (the same image that had done ontological work in the *Rêve* now does psychological work), as a clavichord with the wind playing on it (Williams 2002, 190). Most people are steadier than Rameau, but steadiness is itself an achievement, produced by social interaction and the discipline of living with others, and the question of how it is brought about and what it costs is precisely the question that the demand for systematic self-knowledge forecloses. Rousseau's conception of sincerity, by contrast, presupposes everything the machine emulation thesis requires: a stable self that can be surveyed, a transparent interiority that can be reported without distortion, a unified character whose motives are available for inspection. Rameau's mode of being resists each of these assumptions. Diderot allows neither *Moi* nor *Lui* to triumph. *Moi* fails to pin down *Lui*, but *Lui*'s freedom from principle is indistinguishable from a certain kind of emptiness. The dialogue ends without resolution.

Jacques le Fataliste, composed over roughly the same years and left unpublished at Diderot's death, presses the same counter-voice through the act of narration itself. Jacques professes a Spinoza-derived fatalism: everything that happens was written up above on a great scroll unrolled little by little (Diderot c. 1765–84, 656). Yet the novel's literary form works relentlessly against the thesis: digressions multiply, strangers interrupt, the narrator flaunts his power to redirect the story at will, and Jacques himself never behaves like the automaton his philosophy describes. The aspiration to make thought machine-like, applied to the temporal unfolding of a human life, generates a story that the machinery of storytelling cannot deliver without breaking its own rules.

What, then, is the specific cost of systematization that Diderot's counter-voice identifies? Even a well-constructed system, properly anchored in experience and limited in its ambitions, will fail to capture certain dimensions of the world it describes. Nature is more fluid than any taxonomy allows; moral life resists the demand for consistency that systematic ethics imposes; and thought itself, in its most creative moments, operates in a mode that resists formalization. The machine-like virtues on which the practical rationales depend allow knowledge to be communicated,

assessed without deference to personal authority, and navigated by anyone with the will to engage. These are genuine benefits, as Diderot would have known as co-editor of the *Encyclopédie*. Yet those same virtues, precisely because they purchase transmissibility through abstraction and accountability through impersonality, suppress responsiveness to the particular, the perspectival, and the associative. The dreaming D'Alembert is a more adventurous philosopher than the waking one, because the dream's polyphonic, physiologically grounded mode of thought is attuned to dimensions of experience that the waking geometer's systematic orderliness must set aside. What the *esprit de divination* of 1753 had named as a cognitive capacity, the *Rêve* of 1769 elevated into an ontological claim: the world itself exceeds the categories that any system deploys to capture it.

Diderot's insight occupies an important position in the counter-voice tradition this book has been tracking. Where Cavendish challenged the equation of understanding with mechanization by insisting that nature is richer than mechanism allows, Diderot challenged it by insisting that *thought* is more polyphonic than any single systematic framework can accommodate. Cavendish's critique was ontological: the world exceeds the machine model. Diderot's is aesthetic and epistemological: the mind's own creative, associative activity exceeds the orderliness that the machine model demands of it. The counter-voice tradition does not grow by repeating itself; it deepens by identifying new dimensions of loss.

Diderot was no opponent of the *esprit systématique*; he was one of its principal architects, who had spent two decades orchestrating the *Encyclopédie*. A critique of systematization that comes from someone who has never attempted to systematize is easily dismissed. A critique that comes from the co-editor of the *Encyclopédie*, who has felt the force of the practical rationales at first hand and continues to serve them, carries a different kind of authority.

5.5 The Loosening of the Metaphysical Warrant

The metaphysical warrant that had sustained the demand for system from Keckermann through Du Châtelet has been subjected, across this chapter, to a first loosening. Condillac showed that the systems it licensed ran the risk of being empty, and that the conditions of adequacy for any system include linguistic as well as metaphysical conditions: terms must be anchored in experience if the inferential machinery is to do real work. D'Alembert codified the distinction between the speculative ambition of building systems and the disciplined practice of thinking systematically, and the *Encyclopédie* gave that practice its most ambitious institutional form: a cognitive machine designed for navigation, revision, and public scrutiny rather than for mirroring the mind of God. The practical rationales have survived their emancipation from the metaphysical warrant.

The costs of systematization have come into sharper focus in the process. Condillac identified the cost of building on indeterminate terms: formal coherence purchased at the price of empirical content. D'Alembert's scientific practice revealed a more structural cost: the demand for system, even in its chastened form, rested on assumptions about the mathematical necessity of nature's laws that the official philosophy of the *Encyclopédie* left unexamined. And Diderot identified a cost that attaches to well-built systems as readily as to defective ones: the suppression of whatever in experience resists the categories the system deploys.

Yet the loosening of the warrant remained incomplete. The French critics had performed their work at the level of explicit argument, dismantling the seventeenth-century systems and articulating a pragmatic alternative. At the level of their own positive methodologies, however, the warrant persisted in displaced forms: in Condillac's conviction that the universe is a system whose phenomena spring from a first law, in D'Alembert's insistence that the laws of mechanics are necessary truths demonstrable a priori. The practical rationales had been separated from the metaphysical warrant at the polemical level, but the separation had not been carried through into the methodological foundations of the *esprit systématique* itself. The French Enlightenment had examined the products of system-building and found them wanting. An anatomy of the *concept* of system remained to be undertaken. What *is* a system? What does it mean, precisely, for a body of knowledge to be systematically ordered as opposed to merely heaped together? And if the metaphysical warrant has been withdrawn, what structural features must a system possess in order to deserve the name, independently of any conviction about the rational structure of reality? To answer these questions would require subjecting the concept of system itself to the kind of analytical scrutiny that the *esprit systématique* had directed at everything except its own foundations. That task fell to the German-speaking philosophical world.

Wolff, Lambert, and the Theory of Systematic Understanding

If one could bring the harmony of thoughts to the highest perfection in sensation, one would resemble the musician who not only hears a false note in the concert, but recognizes at the same time who played it, and tells him where he went wrong.

— Lambert, *Criterium veritatis*, §13

If the French-speaking Enlightenment had examined the products of a century of system-building and found them wanting, the German-speaking Enlightenment asked what a system would have to be in order to deserve the name. This chapter focuses on two interventions in particular: Christian Wolff's *De differentia intellectus systematici et non systematici* of 1729 and Johann Heinrich Lambert's unfinished *Fragment einer Systematologie*, composed in the 1770s and published posthumously in 1787. Wolff *logicized* the concept of system, identifying it with the demonstrative interconnection of propositions and establishing its epistemic virtues with a precision no predecessor had achieved. Lambert *generalized* what Wolff had logicized, developing the first general theory of what any system is. To this end, he detached the concept from Wolff's purely deductive framework, grounded it in experiential foundations, and defined it by its *purpose*. In generalizing the concept, Lambert liberated it from the metaphysical warrant that had still underwritten Wolff's deductivism, making the concept of system available for the critical analysis that Kant would undertake. Together, Wolff and Lambert turned "system" from an aspiration underwritten by metaphysics into a formal concept with its own vocabulary, its own internal distinctions, and its own criteria of success.

6.1 Wolff and the Logicization of System

Around 1700, the word *System* was conceptually unsettled. Chauvin's *Lexicon philosophicum* of 1713 recorded the term's association with the *systema mundi*, the ar-

rangement of planetary orbits, which could designate either a real cosmological structure or one of the competing hypothetical models (Ptolemaic, Copernican, Tychonic) proposed to explain it (Albrecht 2011, 230). Régis, in his philosophical dictionary of 1690, saw no essential difference between “system” and “hypothesis,” and Chambers’s *Cyclopaedia* of 1728 concurred: “System and Hypothesis have the same Signification” (quoted in Albrecht 2011, 231). Elsewhere, the term named the body of doctrines peculiar to a philosophical school: the Aristotelian system, the Epicurean system, the Cartesian system. Hobbes had extended it to politics, and Rameau would claim it for music (Albrecht 2011, 231). What was missing was a precise *formal* characterization: the concept hovered between designating a real arrangement of things, a hypothetical explanatory model, and a philosopher’s personal position, with no clear account of the structural property common to all three uses.

Christian Wolff was slow to claim the word. In his German writings he preferred *Lehrgebäude* (doctrinal edifice) and described his method as producing a “chain in which each link hangs from the next, so that each is connected with all the others” (Wolff 1720, Vorrede). When his critic, Joachim Lange, described this same enterprise using the Latin *systema*, it was the enemy, not the architect, who first applied the label (Albrecht 2011, 236). Wolff himself avoided the term in his major methodological work, the *Discursus praeliminaris de philosophia in genere* of 1728 (Albrecht 2011, 236–237). Only in the *De differentia intellectus systematici et non systematici*, a brief essay composed in early 1729 alongside the *Logica*, did Wolff finally subject the concept of system to sustained philosophical analysis. The essay was one of several “Marburg leisure hours” (*Horae subsecivae Marburgenses*), the series of occasional pieces Wolff produced after his expulsion from Prussia and relocation to the University of Marburg (Albrecht 2011, 238). Though the occasion was modest, the result was anything but. As the *Historisches Wörterbuch der Philosophie* records, the *De differentia* represents a “novel investigation” into a concept that had until then been “frequently but rather loosely used” (Hager and Strub 1998, 835; Wellmann 2018, 141).

Wolff approaches the question through the *intellectus systematicus*, the systematic intellect, rather than through the system as an impersonal structure. The choice of starting point was probably influenced by Malebranche, whose *Recherche de la vérité* Wolff read in the French original (Albrecht 2011, 237–238). Malebranche had examined the claims of scholars who tried to invent new systems and concluded that most lacked the requisite properties of *esprit*: only a very few minds were capable of producing a *véritable système*, and the capacity to do so was intimately connected to the use of *notions communes*, the common concepts shared by all rational beings (Malebranche 1674, 305). Wolff follows this subject-centred approach, but he immediately converts it into an objective definition. The systematic intellect, he writes, “is one that connects universal propositions with one another. Propositions are

connected when some are demonstrated through others as through their principles” (Wolff 1729, §2).

Two features of this definition repay attention. The first is a *logicization*. Wolff explicitly distinguishes the connection among *propositions* from the connection among *things*: things are connected when the ground of one is contained in another, but propositions are connected when the truth of one is demonstrated through the truth of others (Wolff 1729, §2). The question of whether knowledge forms a system is thereby recast as a question about logical architecture – about the demonstrative dependencies among component claims. The second feature is the identification of systematic connection with *demonstrative dependence*: proposition A is systematically connected to proposition B when A figures as a principle in the demonstration of B. A system, accordingly, is the structure that results when this relation of demonstrative dependence is carried through consistently across an entire body of knowledge. What had floated between cosmology, hypothesis, and philosophical school is here given a precise logical characterization: a system is a deductively ordered body of demonstrated propositions.

By 1729, Wolff had been building and defending such systems for two decades. The *De differentia* is in part a retrospective articulation of what he took himself to have been doing all along. Its definitions encode his own practice: at bottom, the mathematical method just *is* the construction of a system. Euclid’s *Elements*, in Wolff’s account, provide the paradigm, and all genuine mathematicians are accordingly systematic thinkers (Wolff 1729, §6). Aristotle and Descartes are credited with systematization; Confucius, whom Wolff had controversially praised in his 1721 oration on Chinese practical philosophy, is cited as evidence that systematic thinking can arise independently of the European tradition (Wolff 1729, §7). By identifying the systematic intellect with mathematical method, Wolff gives the concept of system a pedigree stretching back to antiquity and a scope extending across cultures: the system is whatever the method of Euclid produces when applied to any domain of knowledge.

The definition, however, was only Wolff’s starting point. The bulk of the *De differentia* is devoted to an argument for the *advantages* of systematic understanding, and it is here that the essay achieves its deepest purchase on the questions this book has been pursuing. Wolff identifies a cluster of epistemic virtues that attach to the systematic mind, and each of them corresponds to one of the machine-like virtues that we have been tracking.

The first and most striking virtue is error-detection. When genuine systems exist, Wolff argues, errors “cannot trouble us,” because any false proposition is immediately recognized as false by its contradiction with the demonstrated propositions already contained in the system (Wolff 1729, §17). More remarkable still is the mechanism of detection: the ease with which errors are identified is owed to the *form* of the system, not to the acumen of the subject. Even a beginner, Wolff insists, working through the demonstrations in the order prescribed by the system, will discover

errors that a non-systematic reader of genius would miss, because the deductive architecture makes inconsistencies structurally salient (Wolff 1729, §17). The system itself, in effect, does the diagnostic work. Its form registers malfunction the way a well-built machine registers a misaligned component: through the disruption of an ordered sequence that was supposed to run smoothly. This is what the preceding chapters have been calling *surveyability*: the property of a cognitive structure that allows its internal relations to be inspected by anyone who follows the prescribed procedure, regardless of personal gifts or institutional standing.

Connected to this diagnostic capacity is a virtue of intellectual character. Wolff observes that those who possess systematic understanding devote themselves to displaying truth rather than attacking errors; those without it exhaust themselves in polemics (Wolff 1729, §16). When a systematic thinker discovers a flaw in the system, the deductive form makes the flaw self-evident, and the thinker corrects it without defensiveness, because holding on to a demonstrated error would mean holding on to a visible contradiction. The non-systematic thinker, by contrast, lacks the structural means to distinguish a genuine refutation from a rhetorical challenge, and therefore falls back on the defensive postures that characterize sectarian dispute. Wolff is connecting systematic thought to a distinctive intellectual attitude: the rational machinery of system liberates the thinker from the grip of the less constructive argumentative dispositions that non-systematic argument breeds. The systematic thinker, Wolff writes, takes pleasure in discovering truths even in the errors of others, and “is averse to reproaching others for their errors,” because the form of the system, once in place, causes errors “to collapse of their own accord” (Wolff 1729, §§16–17).

The second cluster of advantages concerns intellectual independence. Here Wolff engaged a debate that had shaped German academic philosophy for decades: the quarrel between the systematists and the eclectics. The eclectic tradition, whose most influential proponent was Christian Thomasius, valued the independent selection of truths from different philosophical schools over allegiance to any single system. Thomasius had in 1703 condemned all systems wholesale, arguing that the infiltration of Aristotelian system-thinking into scholastic theology had produced the fundamental evil of confusing philosophy with revelation (Thomasius 1703, 47; cf. Albrecht 2011, 234). The eclectic, by contrast, chose freely among the doctrines on offer, guided by nothing but the independent exercise of judgement. By the 1720s, eclecticism had become the progressive self-description of choice for German-speaking philosophers who wished to signal their independence from both scholasticism and Wolffianism (Albrecht 2011, 241).

Wolff’s response was to argue that the *true* eclectic is the systematic thinker, because only someone in possession of a system has a criterion by which to evaluate what deserves selection. Without a system, the eclectic has no basis for distinguishing a genuine insight from a plausible error: they must read every book slowly and carefully, sifting propositions one by one, with no structural means of testing their

compatibility. The systematic thinker, by contrast, can assess any new claim against the articulated body of demonstrated propositions already in hand. Novel truths are absorbed and contradictions flagged by this systematic structure; even the errors of others can be turned to advantage, because the system provides the framework within which their partial insights can be recovered and their mistakes diagnosed (Wolff 1729, §16). The eclectic *without* a system is merely a collector of unrelated curios. The eclectic *with* a system is a thinker who can integrate whatever is true, from whatever source, into a coherent whole. Eclecticism, properly understood, *requires* systematic thinking (Wolff 1729, §§7, 16; Albrecht 2011, 241–242).

Wolff's victory over the eclectics was so thoroughgoing that it effectively ended the career of eclecticism as a philosophical self-description in the German-speaking world (Albrecht 2011, 244). Even Wolff's sharpest critics, such as Christian August Crusius, did not call themselves eclectics; they objected to specific features of Wolff's system while accepting the premise that philosophy must be systematic (Leduc 2018, 85). After Wolff, the question was no longer *whether* to systematize, but *how*. This had institutional repercussions, because Wolff's philosophical system was a curricular programme, implemented in German universities through a sequence of Latin textbooks that covered every major discipline from logic and ontology through ethics and political philosophy (Albrecht 2011, 243). The *De differentia* articulated the self-understanding of that programme: the system was the form that knowledge must take if it is to be taught effectively, tested rigorously, and defended honestly. The system was simultaneously a pedagogical machine (making knowledge transmissible across a university curriculum), an epistemological machine (making claims testable through deductive interconnection), and an instrument of intellectual accountability (making philosophical authority answerable to transparent procedures rather than to personal charisma or institutional tradition).

A distinctive feature of Wolff's system came from the way that Wolff understood the principle of sufficient reason. In the *Deutsche Metaphysik*, he conceives the principle as fusing together two claims: that the sufficient reason of a thing is what makes it intelligible to the understanding, and that nothing can come from nothing, so that everything that exists must have a sufficient reason in the world itself (Wolff 1720, §30). The logical connectedness of the system was thereby anchored in the ontological connectedness of reality. The systematic intellect, working through demonstrative dependence, uncovers the hidden structures of a world whose rational order is guaranteed by the principle of sufficient reason.

As Violetta Waibel has observed, this produces an asymmetry between ontological and epistemological systematicity. Wolff's system is *ontologically closed*: reality is a complete structure of sufficient reasons, the best of all possible worlds, whose perfection the finite understanding can never fully survey but need never doubt. It is, however, *epistemically open*: the finitude of the human mind ensures that there is always more to discover, more connections to trace, more of the world's hidden architecture to bring to light (Waibel 2007, 64–65). Lambert's achievement, as we

shall see, was to construct a concept of system that could remain open in both dimensions: open epistemically, because the finite knower has more to learn, and open ontologically, because the foundational concepts are drawn from experience and their list cannot be guaranteed complete.

Wolff's system suffered from two limitations, however. Wolff grounded his deductive architecture in *definitions*: carefully formulated specifications of the marks that compose a concept, from which axioms and postulates were supposed to follow "per intuitum," by immediate inspection (Wolff 1730, §2; Wolters 1980, 43). Johann Heinrich Lambert identified this as a conjurer's trick. In a passage that became famous among his readers, he compared Wolff's definitional method to a *Gauckeltasche* – a conjurer's bag: you put into the definition what you then pretend to pull out of the axiom (Lambert 1771, §685; cf. Wolters 1980, 21). An example from Euclid made the point vivid. Euclid had defined parallel lines as lines in the same plane that never meet, however far extended, and the status of the parallel postulate had been debated since antiquity. Wolff believed he could resolve the difficulty by defining parallels instead as equidistant lines, from which the relevant properties could then be derived. Lambert showed that this merely displaced the problem: the existence of equidistant lines itself presupposes a proposition equivalent to the parallel postulate (Lambert, *Theorie der Parallellinien*, §8; Wolters 1980, 52–54). The difficulty had been smuggled from the axiom into the definition. Wolff's definitions, because they were merely nominal (they specified what a word means without proving that anything answers to the specification), could harbour unexamined existential assumptions. The foundations of the system were therefore less secure than the deductive architecture erected on them suggested (Wellmann 2017, 136–137).

The second limitation concerned the purely descriptive nature of Wolff's system. Wolff's method drew on Euclid, but it drew selectively. Euclid's geometry contains *problems* alongside theorems, *postulates* alongside axioms, and *instructions* alongside demonstrations that something is the case. Wolff reduced these practical, constructive, and imperative elements to their descriptive counterparts, reformulating instructions as propositions and treating postulates as minor auxiliaries of axioms with no independent methodological role (Wolters 1980, 43–45). Gereon Wolters flags this as a problematic *metabasis eis allo genos* (a shift into another genus or category): by converting imperatives into indicatives, Wolff eliminates prescriptive sentences as irreducible elements of a theory, and with them the entire operative dimension of the Euclidean method (Wolters 1980, 45). The dimension in which the reader is instructed to perform a construction and grasp a truth through

the performance was thereby lost. Wolff's system could *state* truths, but could not show how to *produce* them. These limitations set the agenda for Lambert.¹

6.2 Lambert and the Generalization of System

Johann Heinrich Lambert (1728–1777) came to philosophy through the practice of science, and this informed the way he approached the authority of systematicity. Born in Mulhouse in 1728 – which was allied to the Swiss Confederacy at the time – he was the son of a tailor and largely self-taught, acquiring his philosophical education by reading Wolff, Malebranche, and Locke during his years as a private tutor in Basel, where he also worked as a secretary to the editor of the *Basler Zeitung* (Wolters 1980, 19). His scientific achievements included the proof of the irrationality of π , foundational contributions to photometry and cartographic projection, pioneering work in non-Euclidean geometry, and the invention of the term *Phänomenologie* (Bullynck 2010, 65). His two major philosophical works, the *Neues Organon* of 1764 and the *Anlage zur Architektonik* of 1771, run to nearly two thousand pages.

Lambert inherited a philosophical culture in which, thanks to Wolff's victory over the eclectics, systematicity had become an unquestioned ideal guiding philosophical and scientific work. But Lambert sought to better understand what a system's foundations needed to look like if it was to engage with reality. Upon joining the Berlin Academy's Class of Experimental Physics, Lambert described the difficulty in terms that reveal the distance between his intellectual temperament and Wolff's: "Purposeful experiments in chosen circumstances are problems that we propose to Nature. She responds with precision, not to what we think we ask, but to what we actually ask" (Lambert 1765/1767, 508; quoted in Bullynck 2010, 68). This acknowledges the potential for a misalignment between theory and experiment. If the concepts in which one frames one's question are ill-chosen, the experiment will return results that we struggle to make sense of within our system – not because nature is confused, but because our own interpretation of the experiment is. The system, for Lambert, had to earn its authority by ensuring that its conceptual building-blocks make contact with experience.

¹In fairness to Wolff, it should be noted that his practice of system-building was richer than his official methodology acknowledged. Wolff relied on two heuristic methods of discovery that go well beyond pure deduction: the reversal of propositions (converting "if A then B" into "given B, seek A") and what he candidly called *Fictiones*, the formation of explanatory hypotheses to bridge gaps in observation (Waibel 2007, 62–64). His example was the daily course of the sun: we observe it rise in the east and set in the west, but the nocturnal journey is hidden from view, and a "fiction" is needed to connect the observable data into a coherent account. In practice, Wolff's system also involved ampliative reasoning, hypothesis-formation, and heuristic inference. His theory of method, however, provided no place for these activities: the official account recognized only definitions, axioms, and theorems, linked by demonstrative dependence.

His first move was to reject Wolff's *Gemeinbegriffe* – the common or most general concepts that Wolff treated as the ultimate foundations of the philosophical system. According to Wolff's logic, one arrives at these foundational concepts by *abstraction*: you take a particular concept such as "horse," identify the marks it shares with other concepts (four-legged, furred), and abstract from the particular marks until you reach concepts that share no further marks with anything else. This way, you arrive at maximally general concepts such as "thing" (*Ding*) or "entity" (*Ens*), which Wolff placed at the base of ontology (Wolff 1712, §§1–114; Wellmann 2018, 146–147).

Lambert turned this upside down. The most general concepts, far from being the simplest, are actually the most *composite*. In a letter to Holland, Lambert declared that the concept "thing" is "the most composite of all concepts, for it contains all possible *fundamenta divisionum* [i.e., every conceivable way that reality could possibly be divided or categorized]" (Lambert 1782, 33; quoted in Wellmann 2018, 146). In the *Architektonik*, he put the point in terms of the memorable image we already encountered: traditional ontology resembles a *Gaukeltasche*, a conjurer's bag, from which you pull out what you have already put in (Lambert 1771, §685; cf. Wolters 1980, 21). Abstraction does not guarantee truth. A concept arrived at by stripping away content may be perfectly empty.

Instead of building from the top down – i.e., from maximally abstract universals to their applications – Lambert proposed building from the bottom up, from what he called *einfache Begriffe*: simple concepts drawn from experience through careful attention to what sensation delivers (Lambert 1764, I, 420–422; Blomme 2025, 746). His argument for the foundational status of simple concepts ran as follows: Any concept that harbours a contradiction must be composite, since contradiction requires at least two elements in tension. A concept containing only a single characteristic therefore cannot be internally contradictory – its possibility is guaranteed by its simplicity (Lambert 1764, I, 421; Wellmann 2018, 148). Lambert's list of simple concepts, refined across the *Organon* and *Architektonik*, included *solidity*, *existence*, *duration*, *extension*, *force*, *consciousness*, *volition*, and *unity* (Lambert 1771, §§46–52; Blomme 2025, 790–802). These concepts originate in sensation: we gain the concept of solidity through the feeling of resistance, and the concept of duration through the experience of persisting through time. Once grasped, however, they are *a priori* in a specific sense: their possibility does not depend on further experience, because it is secured by their internal structure (Blomme 2025, 748).

The simple concepts, however, were only the foundation. Lambert's second and more consequential innovation was to insist that concepts alone, however simple and well-grounded, are inert without *postulates*: propositions that support the real possibility of a concept by instructing a construction. Here, Lambert recovered the Euclidean element that Wolff had marginalized. In Euclid's geometry, a definition such as "a line is a breadthless length" is, Lambert argued, nothing more than a name until a postulate establishes that a line *can be produced*: "from any point, a straight line of any length can be drawn in any direction" (Lambert 1771, §§12–18; Wellmann

2018, 153–154). The postulate does not describe a state of affairs; it instructs the reader to *do* something, and the truth of the definition is grasped *through the doing*. Postulates are, in Lambert’s vocabulary, “practical” propositions: they are “proved by an experience that can be renewed any time, as often as one doubts their validity” (Lambert 1762, §21). Their authority comes from the performance they demand rather than from the concepts they contain.

Applied to the problem of philosophical foundations, this meant that each of Lambert’s eight simple concepts had to be accompanied by a set of postulates specifying the operations that could be performed with it. The simple concept of extension, for instance, is accompanied by postulates stating that any part of space can be taken as a unit, that from any point a line can be drawn in any direction, and that any three points can be treated as lying in a plane (Lambert 1771, §§79–80; Blomme 2025, 766). Without these postulates, the concept of extension names something, but it floats free of any determinate application. The postulates render the concept operative, specifying what can be done with it. Postulates connect a system to reality by showing that its foundational concepts are not merely thinkable, but enable one to do something in reality.

The contrast with Wolff’s system is now evident. Wolff’s system was a machine that processed *descriptions*: it took defined terms, connected them through demonstrative dependencies, and output theorems. Lambert’s system was a machine that enabled its operator to *perform operations on the world* – the system’s value lay in what it enabled the operator to do. As Hammer has argued, this operative dimension is the root of Lambert’s most far-reaching contribution to eighteenth-century philosophy: a reconception of *generality* itself (Hammer 2022, 155–160). For Wolff, a concept is general because it has been abstracted from particular instances: the concept *thing* is maximally general because all particular content has been stripped away. For Lambert, by contrast, a postulate is general because it specifies an operation performable on any instance: “from any point, a straight line can be drawn” is general in the way a machine is general, because it works on any input, not because it has been emptied of all specificity. Generality, on this account, is the generality of a *procedure*, defined by what it can take as input or initial conditions, and not the generality of an abstraction.

Lambert’s third and most far-reaching innovation extended beyond the reform of philosophical method. In the *Fragment einer Systematologie*, composed during the 1770s and published posthumously in 1787, he undertook something no philosopher before him had attempted: a general theory of what any system is, applicable to clockworks and political constitutions, scientific theories and the solar system, musical compositions and treatises of metaphysics. The *Historisches Wörterbuch der Philosophie* identifies this as a novel enterprise – the first sustained attempt to define the properties of systems as such rather than to build or criticize particular systems (Hager and Strub 1998, 835; Wellmann 2018, 141). If Wolff had logicized the concept of system by identifying it with demonstrative interconnection, Lambert

now proposed to anatomize it: to lay out the structural features that anything deserving the name must possess, regardless of the domain to which it belongs.

“For a system,” Lambert writes, “parts are required, and indeed several. These must be placed or ordered with *purpose* and all connected with one another in such a way that they constitute precisely the whole that is appropriate to the intended purpose” (Lambert 1787, §3). A system is a “purposively composed whole” (*zweckmäßig zusammengesetztes Ganzes*) (Lambert 1787, §4). Purpose is not an incidental attribute here; it is constitutive. The ordering of the parts depends on what the system is *for*, and different purposes generate different orderings of the same material. A clockwork is a system because its components are arranged to display the time; a political constitution is a system because its institutions are ordered to enable governance; a scientific theory is a system because its propositions are connected so as to make a domain of phenomena intelligible. What makes each of them a system is the same structural property: the purposive arrangement of parts into a functioning whole.

Violetta Waibel, who applied Lambert’s systematological vocabulary to Wolff’s philosophy, identifies purpose as the element that transforms a mere aggregate into a genuine whole: it is what makes the whole, in a phrase Lambert’s text invites, more than the mereological sum of its parts (Waibel 2007, 53). An aggregate possesses parts and connections, but it lacks the organizing principle that determines which parts belong and how they are to be ordered. Purpose supplies that principle.

Kant would later reformulate this insight as the priority of the idea of the whole over the articulation of the parts: a system, in the Kantian sense, is constructed from the top down, from an idea that specifies in advance the scope and arrangement of the whole (Kant 1781/1787, A832/B860). Lambert’s conception is less rationalist in spirit: the purpose of a system can be discovered from below, through attention to the demands of inquiry, and different inquiries may yield different purposes and therefore different orderings. But both thinkers share a commitment to the priority of purpose over parts.

Lambert’s table of contents for the unfinished treatise itemizes what a general theory of systems must address: a system has parts; there are forces that bind them and laws or rules that govern their interaction; the whole rests on foundations, and presents in a certain external form; a system can be erected, maintained, and improved under certain conditions (Lambert 1787, §6). Each of these features, Lambert insists, appears in every kind of system, but it takes a different guise in each. The concept of a “foundation,” for instance, is borrowed from architecture, where it is most immediately perspicuous; in a musical composition, the foundation is the fundamental key, whereas in a treatise, it is the leading principle; in a machine, it is the frame or chassis; in the solar system, the force of gravity (Lambert 1787, §8). Recognizing the general feature through its diverse instantiations is itself a philosophical task, Lambert notes: “what one knows about a system in general

must, in its application to particular kinds, be sought out and made recognisable afresh, as it were" (Lambert 1787, §9).

Particularly consequential is Lambert's classification of systems by their "connecting forces" (*verbindende Kräfte*). All systems, he argues, fall into three fundamental classes: those held together by the forces of the *understanding* (intellectual systems: systems of truths, scientific theories, creeds, narratives, and poems), those held together by the forces of the *will* (moral and political systems: resolutions, contracts, societies, and states), and those held together by *mechanical* forces (physical systems: the cosmos, machines, buildings, organisms) (Lambert 1787, §§11–13). The classification is self-consciously parsimonious: Lambert chooses connecting forces as the principle of division precisely because they yield the smallest possible number of fundamental classes, whereas dividing systems by the variety of their parts or purposes would generate an unmanageable proliferation (Lambert 1787, §12).

In the *Criterion veritatis* and the *Neues Organon*, he developed a striking account of systematic coherence as something *perceptible* to a trained sensibility. If one could bring "the harmony of thoughts to the highest perfection in sensation," Lambert writes, one would "resemble the musician who not only hears a false note in the concert, but recognizes at the same time who played it, and tells him where he went wrong" (Lambert 1761, §13). A trained systematic thinker does not merely check propositions against a deductive chain; the thinker *perceives* the coherence or dissonance of a body of truths the way a trained ear perceives consonance and dissonance in a performance. Lambert distinguishes this cultivated perception from the formal tools of demonstration that must accompany it: the harmony one perceives in a concert, he cautions, is "only the shadow of that which one seeks in science" (Leduc 2018, 87). Formal demonstration is indispensable. Yet the perception of harmony serves as a heuristic guide, a way of *feeling* where the gaps and tensions in a body of knowledge lie even before the formal apparatus has located them. Scientific knowledge, for Lambert, involves the transformation of the "chaos" characteristic of merely historical or empirical knowledge into the ordered whole of a system, and this transformation is one in which sensation, formal reasoning, and trained judgement cooperate (Lambert 1764, I, §§620–622).

One further feature of Lambert's account deserves attention, because it marks a decisive departure from the conception of system that had prevailed since Wolff. Lambert's system of the sciences is explicitly *open*. Because the simple concepts that ground the system are derived from experience, their list cannot be guaranteed complete. Future generations may discover additional simple concepts, derive new combinations, and extend the system's scope (Blomme 2025, 944). Lambert's correspondent Holland called the result an "experimental metaphysics: a science nobody has contemplated yet" (quoted in Blomme 2025, 946). As Blomme has observed, the contrast with Kant is instructive: where Kant's architectonic may be open at the bottom (empirical laws are never fully determined) but is "absolutely closed at the top" (the table of categories is fixed a priori), Lambert's architectonic

is open at both ends (Blomme 2025, 944). His system is a working instrument of cognitive organization, revisable and extendable as experience yields new material. The machine, one might say, is modular: designed not to provide a finished representation of the structure of reality, but to be reconfigured as the demands of inquiry evolve.

6.3 Systematicity as Epistemic Quality Control

Let us take stock. Wolff logicized the concept of system, converting a term that had floated between cosmological hypothesis, philosophical allegiance, and metaphorical usage into a precise characterization of logical structure: a system is a body of universal propositions connected through demonstrative dependence. In doing so, he identified with notable clarity the epistemic virtues that systematicity confers: error-detection through structural visibility, intellectual independence through the possession of evaluative criteria, and the subsumption of eclectic selection into principled amalgamation.

Lambert then generalized what Wolff had logicized. He replaced Wolff's abstract common concepts with simple concepts drawn from experience, restored the Euclidean postulate as an irreducibly operative element of method, and developed, in the *Systematologie*, the first general theory of what any system is: a purposively composed whole, ordered by connecting forces, whose structural features can be analysed independently of the domain to which it belongs.

Wolff's *De differentia* contains the first explicit articulation of the epistemological rationale in something close to the form this book has been developing: the system is a self-diagnosing machine whose form does the cognitive work, detecting errors through structural transparency and enabling the integration of new truths through principled evaluation.

In the *Criterium Veritatis*, Lambert takes up and develops this idea that systematicity acts as a *criterion* of truth, and he reintroduces the idea that a sensitivity to systematic dissonance can and should be an embodied disposition – the mark of a cultivated systematic sensibility. Integrability within a system offers a criterion by which to determine what one should treat as true, and the discovery of inconsistency or incoherence within one's beliefs incites critical revision. This anticipates later, coherentist epistemological justifications for cognitive systematization. British Neo-Hegelians such as F. H. Bradley stressed “the claim of system as an arbiter of fact” (Bradley 1909, 489), and Nicholas Rescher, who championed the coherentist pursuit of what he called “cognitive harmony”, extolled the chief virtue of systematization as being that it enables “quality control of knowledge claims” (Rescher 2005, 27).

Lambert's tripartite classification of systems according to their connecting forces (understanding, will, and mechanical forces) then adds the recognition that systematization serves structurally parallel purposes across different domains. Intellectual

systems serve the organization and certification of knowledge. Political systems serve the coordination and accountability of collective action. Physical systems serve the functional integration of material components. What unites all three is the property Lambert placed at the centre of his definition: purposive composition. A system is a purposively composed whole, and the question of *how much* system is appropriate, and of *what kind*, accordingly emerges as a question about the purposes the systematization is meant to serve. Lambert did not draw this conclusion in the normative register this book employs. He was anatomizing, not evaluating. But the anatomy he provided is what makes the evaluation possible.

One question, however, was raised by Lambert's innovations without being resolved by them. If the ordering of a system depends on its purpose and different purposes generate different orderings of the same material, then the question of *whose* purposes the system serves acquires a new urgency. Wolff's deductive system, grounded in the principle of non-contradiction and structured by demonstrative dependence, had at least the appearance of impersonality: the logical form of the system was supposed to be the same for everyone, and the conclusions it generated were supposed to compel assent regardless of the interests or circumstances of the knower. Lambert's purpose-relative conception unsettles this appearance. A system ordered for one purpose may suppress what a system ordered for another purpose would foreground. A system designed by those in authority may serve their interests rather than the interests of those subject to it. The *Encyclopédie*, as the preceding chapter showed, had already demonstrated that the organization of knowledge for a public sphere is a political act. Lambert's conceptual framework reveals another, and potentially more political, dimension of that act: if a system is a purposively composed whole, then the composition serves someone's purposes, and one must ask *whose* purposes those are. Rousseau will raise this question with a force and clarity that will transform it into a challenge to systematization. If systematic reason is powerful enough to dismantle the arbitrary authority of tradition and organize knowledge for public accountability, is it also powerful enough to become its own form of domination? Can the machinery of impersonal reason, designed to liberate, end up governing those whom it was meant to serve? The anatomy of system so painstakingly conducted by Wolff and Lambert has provided the conceptual tools with which to pose this question. The task of answering it falls to the next chapter.

The Politics of System: Rousseau and the Costs of Rational Machinery

Putting law above man is a problem in politics that I compare to that of squaring the circle in geometry.

— Rousseau, *Considerations on the Government of Poland*

If systematic thought is powerful enough to supplant the authority of tradition and reorganize knowledge for public accountability, is it also powerful enough to become its own form of domination? Can the machinery of impersonal reason, designed to liberate, end up dominating those it was meant to serve? Lambert's conception of a system as a purposively composed whole renders this worry particularly salient by inviting us to ask whose purposes the composition serves. As the preceding chapter showed, the creators of the *Encyclopédie*, who faced political persecution in organizing knowledge for the public sphere, already appreciated that the aspiration to systematize thought had a political dimension. Yet the accounts of the risks and costs of systematization formulated by Condillac, D'Alembert, and Diderot largely remained within an epistemological register. It fell to Jean-Jacques Rousseau (1712–1778) to formulate a sustained critique of systematization in a political register.

There is, however, an ambivalence in Rousseau's attitude towards systematization that can seem downright paradoxical. On the one hand, his first major work of philosophy, the *Discourse on the Sciences and the Arts* of 1750, argues against the systematization of knowledge, which Rousseau charges with corrupting morals and wrapping “garlands of flowers around the iron chains” of civilized dependence (Rousseau 1750, 6). On the other hand, the *Social Contract* of 1762, composed barely a decade later, is itself a systematic derivation of political legitimacy from first principles, and its central argument is that political authority is legitimate only insofar as its operations are made machine-like: governed by articulable rules, inspectable by any citizen, and independent of whoever happens to hold office. In fact, he is expressly guided by the metaphor of the machine. He repeatedly describes the state as a “political machine” whose parts must be assembled, whose “springs”

(*ressorts*) must be kept taut, and whose designer is the “mechanic” (*mécanicien*) who “invents the machine” that rulers merely operate (Rousseau 1762, II.7.2). The metaphor of rational machinery, which for other thinkers in this book remains largely implicit, is in Rousseau’s hands an explicit and recurring characterisation of what political order requires.

Rousseau is thus both one of the sharpest critics of cognitive systematization and himself a preeminent advocate of systematization in the domain of political thought. As we shall see, however, this apparent paradox will turn out to be the truthful expression of a real underlying tension. Rousseau recognizes, more clearly than any of his contemporaries, that the demand for systematic justification and the diagnosis of systematic domination are two ineliminable aspects of the bind we are in. A political order is legitimate only if it can be justified on explicit, principled, and publicly inspectable grounds; yet the very machinery of principled justification tends to alienate natural feeling and impose abstract order on a lived experience whose richness exceeds what system can capture. To hold both insights together without resolving either into the other is the distinctive achievement of Rousseau’s political thought.

7.1 Rousseau’s Double Relationship to System

Rousseau claimed systematic unity for his thought. In the *Social Contract*, he declared: “All of my ideas fit together, but I can hardly present them simultaneously” (Rousseau 1762, II.5). The claim is to an underlying coherence whose form resists the linear presentation that conventional systematic exposition demands. Yet Rousseau did not write treatises in the manner of Wolff’s demonstrative chains or Spinoza’s geometrical apparatus. He wrote novels, discourses, operas, educational treatises, autobiographies, and reveries. As Arthur Melzer has observed, Rousseau was “a systematic thinker but an accidental author” who “cannibalized” his system, scattering its pieces across externally motivated writings (Melzer 1990, 8). While Melzer treats the incongruity between the claim to systematicity and the asystematicity of its expression as a biographical fact about Rousseau’s circumstances, to be resolved by the interpreter who patiently reconstructs the underlying system, one can also see Rousseau’s refusal of systematic form as being itself philosophically motivated and only consistent: a thinker who diagnoses the costs of systematic imposition on human life cannot consistently present his thought as a system without enacting the pathology he identifies. The unsystematic presentation is of a piece with the point it conveys.

Consider the distinction that Rousseau draws between the “systematic spirit” and the “spirit of observation”. In *Émile*, he declares: “Instead of yielding to the systematic spirit, I grant as little as possible to reasoning and I trust only observation. I founded myself not on what I have imagined but on what I have seen” (Rousseau

1762, IV: 254). The opposition recalls the *esprit de système* / *esprit systématique* distinction that Chapter 5 traced through Condillac and D'Alembert. Condillac had sought to separate systems grounded in mutual support from systems resting on ungrounded abstract principles. D'Alembert had urged cultivating the *esprit systématique*, the discipline of ordered, connected thinking, while resisting the *esprit de système*, the compulsion to build all-encompassing deductive structures from metaphysical first principles.

Rousseau inherits this distinction, but radicalizes it by rendering it sensitive to the *domain of application*. Condillac and D'Alembert had formulated epistemological critiques: bad systems rest on bad principles. Rousseau's critique, by contrast, is that even good systems, when applied to the wrong domain, have detrimental effects. Even a system whose principles are sound and whose method is empirically anchored may be corrupting when applied to moral and political life. Thus, Rousseau argues in the *First Discourse* that the sciences and arts not only fail to improve morals, but actively corrupt them, because systematic knowledge serves competitive social display rather than genuine understanding – people pursue systematic knowledge primarily to flaunt their education, compete for social status, and satisfy their *amour-propre* (one is reminded here of James Watson's candid admission that his co-discovery of the structure of DNA was motivated in good part by the desire for fame (Watson 1968)).

The proximity between Rousseau's anti-systematic stance and Condillac's *Traité* deserves more attention than it has received. The *Traité des systèmes* appeared in 1749, the year of Rousseau's famous illumination on the road to Vincennes. Condillac was Rousseau's close friend; the two dined together weekly in this period. No dedicated study treats the *Traité* as a direct influence on Rousseau's turn against systematic knowledge, and the direct textual evidence remains limited. Yet the intellectual proximity is striking: Condillac had just published the sharpest critique of rationalist system-building in the French language, and Rousseau, within months, launched a critique that extended Condillac's epistemological argument into the domain of morals and politics. Condillac had written that systems are older than the philosophers, that nature makes them, and that they were never bad so long as human beings had only nature for a teacher (Condillac 1749, 11). The claim that natural systems are innocent and artificial ones corrupting is precisely the thrust of Rousseau's argument in the *First Discourse*.

Rousseau's ambivalence towards systematization is also reflected in posterity's two-mindedness about whether to read him as an Enlightenment rationalist or as a proto-Romantic irrationalist. Ernst Cassirer read him as a rationalist, seeing in the *Social Contract* a theory of freedom as autonomy under self-imposed law (Cassirer 1932/1954, 59). Isaiah Berlin cast Rousseau as a proto-Romantic irrationalist (Berlin 2013, 2014). Mark Hulliung staked out a more nuanced compromise when he argued that Rousseau's critique is an "autocritique" of the Enlightenment, conducted with the *philosophes'* own intellectual arsenal: "Lockean empiricism, the philosophical

account of history, the vocabulary of utility, rights, and a social contract, or the scientific modes of argumentation that constituted the common intellectual arsenal of the philosophes" (Hullung 1994, 3).

My own reading expands on this autocritique framing: Rousseau turns Enlightenment rationalism back on itself to diagnose the costs of systematisation. As a result, he is simultaneously inside and outside the lineage of modern systematizers, and it is this positioning that connects him to the counter-voice lineage the book has been tracking. We saw Cavendish offer an ontological critique of mechanism in Chapter 2, revealing the costs of a metaphysics that evacuates nature of agency. Diderot, in Chapter 5, championed polyphony against the systematic spirit's demand for closure. Rousseau shows that the machinery of rational governance, although designed to replace arbitrary power with principled authority, can become its own form of domination.

His own argument in the *Social Contract* acknowledges, as we shall see, the value of principled, explicitly articulated, and inspectable reasoning. Why this should be so, and why the *Social Contract* should take the form of what Rousseau himself calls the "political machine," becomes clearer once we locate the demand for systematization in the practical situation that gives rise to it (following Cueni and Queloz 2021). In a small community whose members know one another well, trust can be built through personal acquaintance: decisions are seen as authoritative because they are made by *this person*, whose character is known and whose judgement is trusted on that basis. In larger societies, this acquaintance-derived trust is no longer available; one cannot cultivate a personal relationship with every official whose decisions affect one's life. The decision-making of public agencies must therefore be *verifiably* reasonable and impartial, inspectable by persons who lack any personal acquaintance with the decision-maker. As Bernard Williams puts the point, such agencies need to be "governed by an explicable order which allows those agencies to be answerable" (Williams 1981a, 81). And this requires that justifications be cast in terms that any citizen can assess: in the form of publicly statable principles articulated in terms that those citizens can recognize as authoritative. These cannot be richly perspectival terms that are the preserve of a particular section, group, or profession within society. They have to abstract away sufficiently from the particulars of a given way of life to be accessible to all. Williams calls such concepts "thin": they abstract from the rich evaluative content embedded in any particular way of life, and they gain their authority precisely from this abstraction, from the fact that they can be shared across evaluative perspectives.

The aspiration to emulate the virtues of machines in the service of pedagogical and epistemological rationales here receives a distinctly *political rationale*: it is motivated by the practical need of large societies to make governance *answerable*. The substitution, however, remains the same: the situated judgement of particular persons is replaced with principled procedures whose operations are open to inspection by anyone.

Rousseau's *Social Contract* is the most sustained attempt in the early modern period to construct such an order from scratch: a political machine designed to substitute dependence on the impersonal authority of law for dependence on the personal authority of rulers. Rousseau himself reaches for exactly this image: the social contract, he writes, is "the artifice that makes for the ingenuity and the working of the political machine" (Rousseau 1762, I.7.8). A machine is a constructed artefact whose operations are inspectable, whose outputs are determined by its formal design, and whose functioning does not depend on the personal qualities of the operator. These are the properties that Rousseau demands of a legitimate political order. The question is whether the political machine, so conceived, can actually deliver what it promises, and at what cost.

7.2 The *Social Contract* as Rational Machinery

Rousseau's political philosophy is associated, above all, with the notion of the general will (*volonté générale*). Patrick Riley has traced this term to seventeenth-century French theological debates about whether God has a *general* will to save all humanity or a *particular* will to save only the elect (Riley 1986, xi–xiii). The question at the heart of this debate, though framed in the vocabulary of salvation, was a question about the *form* that rational governance through a perfect being must take: does a perfect being operate through uniform, general laws, or through case-by-case interventions tailored to individuals?

Nicholas Malebranche (1638–1715) took the view that God's perfection expresses itself precisely in the generality, uniformity, and simplicity of his operation: "God acts by *volontés générales* ... in order to construct or to preserve his work by the simplest means, by an action [that is] always uniform, constant, perfectly worthy of an infinite wisdom and of a universal cause" (Malebranche 1680, 147–148). Lawless *ad hoc* volitions are a symptom of cognitive limitations: "to act by *volontés particulières* shows a limited intelligence which cannot judge the consequences or the effects of less fruitful causes" (Malebranche 1680, 166). A truly perfect intellect's will takes a form that is general, law-like, and uniform.

These theological roots are philosophically significant, because they render salient that the demand for generality was, from the beginning, a demand about *form*. Whether God wills through general laws or through particular decrees is a question about the *structure* of divine governance, and Malebranche's answer identified the characteristics of the divine intellect with the machine-like virtues of generality, constancy, and uniformity.

This idea's journey from theology to politics ran through Montesquieu (1689–1755), whose *De l'esprit des lois* bears the impress of Malebranche: the concept of law as *rapport*, the distinction between *causes générales* and *causes particulières*, and the conviction that particular events are produced by general

causal patterns all descend from the theological framework that Malebranche had established (Riley 1986, 155–172). But Montesquieu's conservatism prevented him from drawing out the egalitarian implications that the idea of generality drives towards. It fell to Rousseau to press this line of thought all the way to its radical conclusion. He took the formal properties that Malebranche had ascribed to divine governance and that Montesquieu had transferred to legal systems, and made them into *conditions of political legitimacy*. Those conditions then became the solution to the fundamental problem of societal design that the *Social Contract* poses: "Find a form of association that will defend and protect the person and the goods of each associate with the full common force, and by means of which each, uniting with all, nevertheless obey only himself and remain as free as before" (Rousseau 1762, I.6.4). The solution must satisfy two constraints simultaneously: effective protection and full autonomy. The general will is the mechanism that reconciles them.

How, in detail, does Rousseau build this mechanism? The *Social Contract* constructs it in stages, and the machine metaphor that runs through the text is an explicit guide to its architecture.

The first stage is the social pact itself. Rousseau insists that its clauses are "so completely determined by the nature of the act, that the slightest modification would render them null and void" (Rousseau 1762, I.6.5). This is a striking claim about the pact's *formality*: its authority derives from its logical structure, from the fact that the clauses leave no room for the intrusion of particular interests, personal advantages, or ad hoc exceptions. All the clauses reduce to one: "the total alienation of each associate with all of his rights to the whole community" (Rousseau 1762, I.6.6). Because each gives everything to all, "the condition is equal for all, and since the condition is equal for all, no one has an interest in making it burdensome to the rest." The total character of the alienation is the precondition of the machine's operator-independence: "each, by giving himself to all, gives himself to no one" (Rousseau 1762, I.6.8). Authority, on this design, is depersonalized.

The force of this depersonalization becomes visible through the contrast with Hobbes, whose Leviathan is also a kind of political machine, but one whose architecture is fundamentally different. For Hobbes, each subject subordinates his will to the particular will of a determinate agent, and what constitutes the political community is this common submission to an identifiable person rather than any constraint on the content of what that person commands. The Hobbesian sovereign's edicts qualify as law by virtue of their *source*: when we know that a regulation issues from the agent entrusted with sovereign power, we know it is binding, regardless of what it requires. As Joshua Cohen observes, Hobbes imposes no content constraints on sovereign acts, only source conditions (Cohen 2010, 64). Rousseau takes a different view. His sovereign is a collective being whose acts qualify as law only if they satisfy formal conditions bearing on their *content*: they must be general in scope and oriented to the common good. Authority thus ceases to depend on the identity of the agent who issues the command, and comes to depend instead on the

formal character of the command itself. Where the Hobbesian machine runs on the personal will of a determinate ruler, Rousseau's machine is designed so that the identity of whoever operates it is irrelevant to the legitimacy of its output.

Rousseau's thought is explicitly guided by this *machine politique* vocabulary. As he puts it in the *Discourse on Political Economy*: "the citizens are the body and the members that make the machine move" (Rousseau 1755, 10). The pact between them, he writes, "tacitly includes the following engagement which alone can give force to the rest, that whoever refuses to obey the general will be constrained to do so by the entire body: which means nothing other than that he will be forced to be free; for this is the condition that, by giving each Citizen to the Fatherland, guarantees him against all personal dependence; the artifice that makes for the ingenuity and the working of the political machine [*machine politique*]" (Rousseau 1762, I.7.8). The famous and troubling phrase "forced to be free" is intelligible as a feature of the machine's design: the machine replaces dependence on the arbitrary will of a ruler, which operates according to the ruler's private interests and moods, with dependence on a formal structure of law, which operates uniformly and impersonally.

The Lawgiver, whom Rousseau introduces in Book II as the quasi-divine figure who designs this order, is similarly described in mechanical terms. A great prince, Rousseau writes, "need only follow the model which the other must propose. He is the mechanic who invents the machine, the first is nothing but the workman who assembles and operates it" (Rousseau 1762, II.7.2). The Lawgiver is a *mécanicien*: his task is to design a system whose operations are self-sustaining and whose outputs depend on the design rather than on the designer. Once the machine is built, the mechanic steps back. The *Considerations on the Government of Poland* makes the point even more forcefully: "with the King dead, everything should proceed as if he were still alive; the absence of one piece of the machine should go almost unnoticed, so unessential to its stability was this piece" (Rousseau 1782, 23). The political machine's defining aspiration is to function independently of any particular operator.

With the formal scaffold of the pact in place, Rousseau turns to specifying *how the machine processes its inputs*. In the chapter "Whether the General Will Can Err" (SC II.3), he describes the extraction of the general will from the will of all in terms that approach an algorithmic procedure: "There is often a considerable difference between the will of all and the general will: the latter looks only to the common interest, the former looks to private interest and is nothing but a sum of particular wills; but if, from these same wills, one takes away the pluses and the minuses, which cancel each other out, what is left as the sum of the differences is the general will" (Rousseau 1762, II.3.2).

The pluses and minuses are the private, situated, "thick" interests of individual citizens: the advantages and disadvantages that accrue to them from their particular positions in the social order. The cancellation of these interests is the mechanism

by which the system strips away particularity to isolate the common: the thin, impersonal residuum that remains when the private has been subtracted. The general will is what emerges when the political machine has performed its characteristic operation of abstraction.

Voting, on this account, is itself a procedure of judgement rather than a process of preference aggregation. When a law is proposed in the assembly, “what is asked of them is not precisely whether they approve the proposition or reject it; but whether or not it conforms to the general will which is their own: each in giving his vote states his opinion on that question, and from the counting of the voting is taken the declaration of the general will” (Rousseau 1762, IV.2.8). The citizens are not asked what they *want*. They are asked what, in their judgement, the common good *requires*. Cohen has argued that this passage only makes sense on a *substantive* rather than a *procedural* interpretation of the general will (Cohen 2010, 75–79). On the procedural reading, the general will would simply be whatever the vote produces under favourable conditions, and the citizen who loses would have been wrong only in predicting the outcome. On the substantive reading, the vote is an epistemic device for *tracking* the common good, and what is asked of the citizens is an opinion about whether the proposal conforms to an independently existing standard. And if voting is indeed an epistemic procedure, the “Jury Theorem” formulated by Rousseau’s contemporary, the Marquis de Condorcet, helps explain why the procedure works (Grofman and Feld 1988; Cohen 2010, 78). If each citizen is even slightly more likely than not to judge correctly what is in the common interest, and if citizens vote independently rather than as organized factions, the probability of the majority reaching the correct judgement rises rapidly with the number of voters, approaching certainty. The political machine, on this reading, is an epistemic calculator: it aggregates independent judgements to approximate an objective truth about the common good. The machine’s output is a collective judgement about what is objectively in the common interest, and the majority vote is the best available approximation of that judgement. This helps clarify how the majority will can be a reliable *guide to* the general will without being *identical with* the general will.

Rousseau is clear, however, that the machine can still sometimes err. “The general will is always upright, but the judgement that guides it is not always enlightened” (Rousseau 1762, II.6.10). The system works only when citizens strive to promote the common good rather than to advance their private interests, and even then they may misjudge. The gap between the general will and people’s actual deliberation corresponds to the machine’s malfunction: the mechanism is designed to produce correct outputs given correct inputs, but it cannot guarantee the quality of its inputs. Especially when factions arise, Rousseau warns, the will of each faction becomes “general in relation to its members and particular in relation to the State” (Rousseau 1762, II.3.3). This disrupts the machine’s ability to cancel out particular interests, is disrupted, because the particular interests are now organized rather than random.

The third design feature that the *Social Contract* builds into the machine is the strict abstraction of law. "When I say that the object of the laws is always general," Rousseau writes, "I mean that the law considers the subjects as a body and their actions in the abstract, never any man as an individual or any particular action. Thus the law can very well enact that there will be privileges, but it cannot confer them on any one by name; the law can create several Classes of Citizens, it can even specify the qualifications that entitle to membership in these classes, but it cannot nominate this person or that for admission to them" (Rousseau 1762, II.6.6). Law, on this design, processes only abstract categories. It is structurally incapable of registering proper names, particular actions, or individual circumstances. "Every genuine act of the general will obligates or favours all Citizens equally, so that the Sovereign recognizes only the body of the nation and does not single out any one of those who make it up" (Rousseau 1762, II.4.8). The machine refuses to see individuals; it sees only the formal properties of their situation.

Finally, Rousseau makes the mechanics of political authority *surveyable* by expressing them in explicitly geometric language. In the chapter "Of Government in General," he represents the relationship between Sovereign, Government, and State as a continuous proportion: "the Government receives from the Sovereign the orders which it gives the people, and for the State to be in proper equilibrium it is necessary that, all other things being equal, the product or power of the Government taken by itself be equal to the product or power of the citizens" (Rousseau 1762, III.1.8). The Government is the "mean proportional" between the Sovereign and the State. In a discussion of monarchy, Rousseau writes that when "all of the machine's springs are in the same hand, everything proceeds toward the same goal, no opposing motions cancel one another" (Rousseau 1762, III.6.3). The springs (*ressorts*) of the machine are a running metaphor throughout the text. Victor Gourevitch's editorial note observes that *ressort* "is how Rousseau and many of his contemporaries often refer to the artifices by which institutions and individuals are set and kept in motion" (Gourevitch, note to SC II.8.4).

Rousseau anticipates the charge that his geometric language suggests a degree of precision that goes beyond what can reasonably be expected in the political domain: "if in order to express myself in fewer words I briefly borrow the language of geometry, I am nevertheless not unaware of the fact that geometrical precision does not obtain in moral quantities" (Rousseau 1762, III.1.16). Rousseau thus recognizes that geometric language is metaphorical, but he reaches for it anyway, because he sees in it a way to display the architecture of authority in a concise and perspicuous form.

The structural homology between these formal conditions on political legitimacy and the machine-like virtues this book has been tracking is now evident. The total alienation of the social pact secures operator-independence: authority is no one's personal command. The algorithmic cancellation of particular wills secures impartiality: the system's output is not the expression of any private perspective.

The strict abstraction of law secures universality: the same rules apply to all, with no individual singled out. And the geometric representation of the state secures surveyability: the mechanics of authority are expressed in terms any citizen can follow. Rousseau's "political machine" is designed to exhibit the four machine-like virtues we have been tracing.

The question the *Second Discourse* then raises is whether the political machine, thus designed, can resist capture by the personal interests it was constructed to exclude.

7.3 The Second Discourse and the Costs of Systematic Social Order

If the *Social Contract* lays out how to build the political machine, the *Second Discourse* examines how it can malfunction. The 1755 *Discours sur l'origine et les fondements de l'inégalité parmi les hommes*, as its full title has it, tells the story of how human beings, once driven from a condition of natural independence into one of mutual dependence, constructed institutions of law, property, and convention that systematized their social relations and, in doing so, entrenched the very inequalities those institutions were meant to remedy. The machinery of justice and governance originates in an act of deception whose effectiveness depends precisely on its systematic character: the language of generality conceals the particularity of the interest it serves.

The founding moment is when the rich man, having acquired property through force and usurpation, finds himself unable to defend it against those he has dispossessed. His solution is to persuade his neighbours that their shared vulnerability demands collective protection. "Let us unite," he proposes, "to protect the weak from oppression, restrain the ambitious, and secure for everyone the possession of what belongs to him: Let us institute rules of Justice and peace to which all are obliged to conform, which make no exception for anyone, and which in a way make up for the vagaries of fortune by subjecting the powerful and the weak alike to mutual duties" (Rousseau 1755, 173). The proposal deploys all the formal properties the preceding section identified with the general will: generality of obligation ("all are obliged to conform"), the absence of exceptions ("make no exception for anyone"), and the impartial subjection of powerful and weak alike to shared duties. It performs, in the register of rhetoric, exactly the "thinning out" that the *Social Contract* performs in the register of institutional design. The rich man's particular interest – the protection of his usurped property – is translated into abstract, impersonal, and universally applicable terms. The thick reality of economic domination is disguised in the thin vocabulary of reciprocal duty. And it worked: all ran "toward their chains in the belief that they were securing their freedom", Rousseau writes; "for while they had enough reason to sense the advantages of a political establishment, they did not

have enough experience to foresee its dangers” (Rousseau 1755, 173). What makes the deception effective is its formal appeal. Its rhetoric of systematic governance, with its machine-like generality and impartiality, renders invisible what particular interests it serves. Laws “gave new fetters to the weak and new forces to the rich, irreversibly destroyed natural freedom, forever fixed the Law of property and inequality, transformed a skillful usurpation into an irrevocable right, and for the profit of a few ambitious men henceforth subjugated the whole of Mankind to labor, servitude and misery” (Rousseau 1755, 173).

This parallels Chapter 5’s diagnosis of the *esprit de système*. There, the machine designed to eliminate dogmatism became a new dogmatism: the rationalist systems, though constructed to replace the authority of tradition with the authority of demonstrable truth, ended up generating their own orthodoxies. Here, the machine designed to eliminate arbitrary authority becomes a new form of domination: the institutions of justice constructed to replace the arbitrary power of the strong with the principled rule of law end up generating a new and more durable form of subjection. And the system’s formal allure, which is to say the very features that make it look impartial and universally binding, are what allow particular interests to operate beneath the surface undetected.

In the *Second Discourse*, the key concept for understanding how this capture occurs is the notion of *inflamed amour-propre*: the natural drive for recognition, which in its healthy form is simply the desire to be valued by others, becomes, under conditions of social inequality, a competitive struggle for relative standing. As Frederick Neuhouser has argued in his study of the *Second Discourse*, societies marked by substantial inequality tend to produce in their members insatiable desires for public esteem, which, when widely shared, make a general satisfaction of *amour-propre* impossible and necessarily result in frustration, conflict, and unfulfilled needs (Neuhouser 2014, 179).

The institutions of property and law formalize this dynamic. Once codified through institutions that rank, measure, and evaluate, ever-present social comparisons transform the desire for recognition into a structural feature of social life that the institutions of justice then perpetuate. The political machine can thus end up fanning just the self-centred sentiments it was designed to regulate, and it does so precisely because its systematic character lends an appearance of naturalness and inevitability to what is in fact an artificial arrangement serving particular interests.

Rousseau captured the self-defeating dynamic epigrammatically: “The vices that make social institutions necessary are the same ones that make their abuse inevitable” (Rousseau 1755, 172–73). Systematic political institutions are *necessary* because natural order, once the species has left its original condition, cannot prescribe how civilized beings should organize their collective life. Laws can no longer “speak immediately with the voice of nature” (Rousseau 1755, 127). Nature is silent on the questions that matter most to social beings, and the silence creates a void that only

institutional systematisation can fill. The demand for system arises from a practical necessity, from the absence of natural authority, rather than from a metaphysical claim about the structure of reality.

Yet the institutions that fill nature's silence are vulnerable to the pathologies the *Second Discourse* diagnoses, and the vulnerability is structural rather than accidental. Judith Shklar's reading of Rousseau presses this insight to its darkest conclusions. The general will, in Shklar's interpretation, is fundamentally "a will against inequality": a negative determination, a refusal of domination, rather than a positive programme for the good life (Shklar 1969, 184). Inequality consists in "different privileges which some enjoy to the prejudice of others, such as being more wealthy, more honored, more powerful than they, or even getting themselves obeyed" (Rousseau 1755, 131). Even Sparta, which Rousseau invokes as the closest approximation to a legitimate polity, is, in Shklar's analysis, an "essentially preventive" (Shklar 1969, 11) arrangement, which can briefly halt the growth of inequality, but remains likely to be eventually overwhelmed by it. The political machine does not so much deliver the just society as hold the forces of domination in check for a period – and this holding action hinges on the cultivation of civic virtue redirecting *amour-propre* towards the common interest.

The contrast between the *Social Contract*'s formal requirements and the *Second Discourse*'s historical narrative is accordingly a contrast between the conditions of legitimacy and the conditions of actual political life. While the general will's formal properties specify what political authority would have to look like in order to be legitimate, the *Second Discourse* shows that these properties are perpetually vulnerable to capture by the very interests they are meant to exclude. The rich man's founding speech is an exercise in the rhetoric of universality deployed in the service of particular advantage, and the institutional order it inaugurates makes the particular advantage structurally invisible by embedding it in the formal apparatus of law.

7.4 The Codification Movement

Rousseau's political machine was a philosophical construction. Its institutional realization came in the *codification movement* of the late eighteenth and early nineteenth centuries, which translated the demand for principled, inspectable governance into legal form. The Prussian *Allgemeines Landrecht* of 1794, the French *Code civil* of 1804, and their successors across the continent were exercises in replacing fragmented, locally variable, and often opaque customary law with unified texts whose authority derived from their formal properties: consistency, generality, and availability to any literate citizen.

Jeremy Bentham, who coined the term "codification", highlights that codification is an act of legislation, but one that fundamentally differs from ordinary law in

several respects. It aspires to comprehensively cover an entire area of law (such as private law or contract law) with a new, fully consistent and complete system of law that replaces existing laws. The key characteristic of codification, as Reinhard Zimmermann has emphasized, is its *systematicity* (Zimmermann 2012, 373). Codification is “based on the belief that the legal rules can be reduced to a rational system” (Zimmermann 2012, 372). A codification “aims at presenting its subject matter as a logically consistent entity of legal rules and institutions”; it “promotes the internal coherence of the law and makes it more easily comprehensible” (Zimmermann 2012, 372–373). The codifiers aspired to produce legal texts that would be self-contained, publicly accessible, and operative independently of the particular judges who applied them. The code would replace a patchwork of opaque customary law with a unified text that would be self-contained, publicly accessible, and operative independently of the particular judges who applied them.

The codification movement was, in part, a reaction to the fading authority of Roman law, which was no longer regarded as the written embodiment of universal reason: “Roman law was no longer *ratio scripta*; it was acceptable only in so far as it was in conformity with the principles of natural reason” (Zimmermann 2012, 376). Conjoined with the rationalist conviction that “all the rules regulating human behaviour could be fitted into a system that was both internally consistent and consonant with human reason and the nature of man” (Zimmermann 2012, 376), this fuelled the project of rationally reconstructing the law by the light of the ideal of systematicity. The codification movement also had “an emancipatory element” (Zimmermann 2012, 376). It self-consciously served Enlightenment concerns for liberty and equality by making it possible for the public to inspect and ascertain the consistency and coherence of the legal rules they were subjected to.

The intellectual genealogy of the codification movement leads back to a figure we already encountered. It was Leibniz’s ambition to place jurisprudence on scientific foundations, by extending “the mathematical use of elemental principles as guides and measures to aid in the science of jurisprudence” (Berkowitz 2005, 25), that initiated the movement which would culminate in the European codes. Leibniz sought to reduce all laws, natural and civil, to their foundational principles, and then to deduce the body of law from those principles, producing a *scientia juris* constructed *more geometrico* (Berkowitz 2005, 26). The *calculus* of Chapter 3, the dream of a notation and a calculus that would resolve disputes through computation, has a political descendant in the codifiers’ ambition to replace “the arbitrariness of practitioners and judges” with a rational system of rules. Zimmermann confirms the link: one of the historical conditions enabling codification was the development of “a jurisprudence constructed *more geometrico*,” which “appealed to authorities eager to rationalize the administration of justice” (Zimmermann 2012, 376).

The abolition of feudal privileges, the establishment of legal equality before the law, the creation of publicly accessible legal texts that citizens could consult and invoke against arbitrary authority – these were genuine emancipatory accomplish-

ments. And they depended on the machine-like virtues that codification embodied. The code's authority derived from its *rule-based* character, its *generality* (the same rules applied to all), its *inspectability* (the text was public and articulable), and its *impartiality* (the law's validity did not depend on the particular judge who applied it).

And here, too, we find the substitution of embodied judgement with an externalised body of principles: where the classical Roman jurists saw his authority as grounded in his *insight* into justice, in a cultivated *prudential* exercised in the decision of particular cases, the modern codification claimed the authority of a *science*: a systematic body of rules whose application required legal training rather than practical wisdom. The transformation from *ius* to *lex*, from law-as-insight to law-as-rule, is the juridical equivalent of the machine emulation thesis. What was once a matter of cultivated judgement becomes a matter of rule-following, and the gain in impartiality and inspectability is purchased at the cost of the particular, situated knowledge that the older form of juridical reasoning had presupposed.

Yet the codes, once they entered into force, generated difficulties of their own. The codes came to be regarded as comprehensive and closed systems of legal rules, constituting autonomous interpretational spaces. The result was an inward-looking, almost exclusively exegetical scholarship and a horizontal isolation of national legal doctrines from one another (Zimmermann 2012, 381). The movement seeking to unify legal order across a territory generated its own forms of fragmentation, as each national code became a self-enclosed world. The aspiration to complete systematicity, meanwhile, was never fulfilled: no codification has ever fully lived up to the ideal (Zimmermann 2012, 372). Jean-Étienne-Marie Portalis, the principal drafter of the *Code civil*, acknowledged as much when he designed the Code's provisions as *principes féconds en conséquences*, principles rich in consequences to be developed through judicial and scholarly interpretation (quoted in van Roermund 2014, 155). The code still *presupposes* the informal judgement it was ostensibly constructed to displace. The boundary between what the political machine can settle and what requires the situated judgement of particular persons is itself a matter of judgement that no code can formalize. Chapter 3 arrived at the same insight in Leibniz's register: the boundary between what the calculus can settle and what requires informal judgement is itself a matter of informal judgement. The political rationale, like the epistemological rationale, encounters a built-in limitation: the systematization of judgement always presupposes a residuum of unsystematized judgement on which the whole apparatus rests.

The political rationale for systematization is thus rooted in specific historical circumstances: the collapse of feudal authority, the rise of the modern state, and the Enlightenment's critique of arbitrary power. It nevertheless yields a principled basis for systematization, because it rests on considerations that any defender of non-arbitrary governance must acknowledge: that public decision-making should make itself accountable to those it affects by discursively articulating justifications

for its decisions, and that insisting on the systematic character of those justification helps ensure that like cases are treated alike. These considerations do not require the metaphysical warrant. They require only the practical observation that arbitrary governance is worse than principled governance, and that the machine-like properties of systematic justification are what make the difference between the two.

7.5 The Limits of the Political Rationale

The codification movement embodied genuine emancipatory achievements: the replacement of opaque customary law with inspectable, rule-governed, publicly accessible codes. As with the other rationales for systematization, however, there is a question about how far the political rationale takes us. That question has two dimensions. The first concerns the vulnerability of systematic governance to capture by particular interests: the formal apparatus of universal right can conceal the particularity of the interest it serves. The second concerns the costs of extending systematization beyond the domain of public governance into the whole of moral and psychological life, where the machine-like virtues that make law inspectable become instruments of impoverishment.

Rousseau's *Second Discourse* had diagnosed the structural vulnerability: the rich man's founding speech deploys the rhetoric of universality in the service of particular advantage, and the institutional order it inaugurates makes that advantage structurally invisible by embedding it in the formal apparatus of law. The codification movement itself illustrates this diagnosis in legislative practice. Bas van Roermund has examined how Portalis, the principal drafter of the *Code civil*, related to the philosophical tradition from which codification drew its authority. Portalis spoke Rousseau's idiom in the *Discours préliminaire* to the Code, because at the occasion of presenting a draft Civil Code in post-revolutionary France, it was politically essential to speak in Rousseau's register (van Roermund 2014, 165). Yet in his longer treatise, *De l'usage et de l'abus de l'esprit philosophique*, Portalis dismissed the *Social Contract* as a sterile and turbulent metaphysics (quoted in van Roermund 2014, 167).

This selective appropriation is telling. Van Roermund shows that behind Portalis's philosophical eclecticism lay a consistent motivation: the protection of established property rights. In Portalis's deployment of the language of self-preservation, he systematically substitutes "property" for "self" (van Roermund 2014, 170). The political machine that Rousseau's *Second Discourse* had diagnosed as the rich man's instrument of legitimation here operates in legislative practice. The formal apparatus of universal right conceals the particularity of the interest it serves, exactly as the founding narrative had predicted. The pattern confirms the structural point: the same formal properties that make systematic governance legitimate, its generality,

its impersonality, its principled character, are what make capture possible, because they allow particular interests to present themselves as universal.

We can discern a parallel here with Chapter 5's diagnosis of the *esprit de système*. There, the machine designed to eliminate dogmatism became a new dogmatism: the rationalist systems constructed to replace the authority of tradition with the authority of demonstrable truth ended up generating their own orthodoxies. Here, the machine designed to eliminate arbitrary authority becomes a new form of domination. In both cases, the tool designed to replace arbitrary authority generates its own form of arbitrariness. And in both cases, the system's formal allure conceals this.

The vulnerability to capture is, however, only one dimension of the political rationale's limits. Even if the machine were to function as designed, immune to the distortions Rousseau diagnoses, there would remain the question of how far its writ should run. The codification movement may have been guided by the ideal of a fully axiomatized deductive system, but the codices it actually produced fell well short of that ideal. Reflection on how systematic thought needs to be in order to meet the practical needs at work in the political rationale reveals that while governance clearly needs to be a great deal more systematic than opaque and fragmented customary law, there is no obvious reason why it needs to approach the completeness of a Euclidean theory. Even the U.S. Supreme Court, the institution that advocates of principled systematicity regard as the closest approximation to the aspiration that the most fundamental issues of political morality be set out and debated as issues of principle (Dworkin 1985, 70), stops far short of a universal and fully systematized codex. It systematizes the inherited ways of thinking of a certain jurisdiction with a particular tradition, and as theorists of judicial minimalism have stressed, it restricts to a minimum the number of future cases covered by its decisions, prefiguring as many as necessary and as few as possible (Sunstein 1996, 2001). Even within the domain of public governance, the political rationale calls for systematization only up to a point.

A further limit is spatial. Generality within a bounded political community is distinct from universality. The *généralité* that the French tradition championed occupies a place midway between particularity and universality (Riley 1986, xii). Rousseau's general will is *general* across a *polis*, but that political community remains a *bounded* one, and the general will is emphatically not universal in the Kantian sense of extending to all rational beings as such. Rousseau, as Riley puts it, doubted that there could be a reason-ordained *morale universelle*; for him the crucial line should be drawn between the general and the universal, the *polis* and the *cosmopolis* (Riley 2001, 127). The demand that political authority be principled, inspectable, and impartial has its proper home in a bounded community whose members share a common fate. Extended to the whole of humanity, it becomes the kind of abstract rationalism Rousseau resisted: a system whose scope exceeds its proper remit,

whose universality is purchased at the price of severing any connection to particular people's concerns.

The rupture between Rousseau and Diderot over precisely this question exemplifies this. In the *Encyclopédie* article on *Droit naturel*, Diderot had argued for a universal *volonté générale* discoverable by any rational agent who reasons in the silence of the passions. Rousseau's response, preserved in the suppressed Geneva Manuscript, denies just this claim. The "violent reasoner" whom Diderot introduces as a foil becomes, in Rousseau's hands, an "independent man" whose demand for reasons is philosophically serious rather than morally defective (Riley 2001, 130): "if care for one's self-preservation is the first precept of nature, can he be forced thus to consider the species in general in order to impose on himself duties whose connection with his own constitution he completely fails to see?" (Rousseau, Geneva Manuscript, I.2.14). Why should one submit to a universal general will that fails to make contact with one's own concerns? Diderot's answer – that *reason* compels one to do so – presupposes what it needs to show: that a universal rational order has any motivational purchase on particular human beings.

Rousseau's counter-claim is that the general will acquires its force only when it is particularized, which is to say tailored to a bounded political community whose members can recognize one another as sharing a common fate. The political machine works, but only within its proper remit. Generalized beyond that, it loses its motivational grip on the very agents it is meant to govern. The Rousseau–Diderot rupture, which was both philosophical and bitterly personal, encapsulates two contrasting views of the scope of the political rationale. Diderot's position is the *esprit de système* applied to politics: the conviction that the systematizing spirit can generate a universal and authoritative order. Rousseau's is that the political rationale is self-limiting: it must operate within the bounded domain of a particular *polis*. The distinction between *généralité* and *universalité* marks where the machine can do its work and where it begins to overreach.

Rousseau was clear-eyed about the cost of overreaching. In the Geneva Manuscript, he acknowledges that the general will is, in each individual, a pure act of the understanding reasoning in the silence of the passions – "But where," Rousseau asks, "is the man who can thus separate himself from himself...?" (Rousseau, Geneva Manuscript, I.2.14). And, "since the art of generalizing one's ideas is one of the most difficult and belated exercises of the human understanding, will most men ever be in a position to derive the rules of their conduct from this way of reasoning...?" (Rousseau, Geneva Manuscript, I.2.15). The political machine depends on the cultivation of citizens who can separate their particular will from the general will.

The cost of attempting to cultivate such citizens is what Rousseau describes, in one of the *Social Contract's* more disquieting passages, as *denaturing*. "He who dares to institute a people must feel capable of, so to speak, changing human nature; of transforming each individual who by himself is a perfect and solitary whole into part

of a greater whole from which that individual as it were receives his life and his being; of adulterating man's constitution in order to strengthen it" (Rousseau 1762, II.7.3). The machine's formal requirements demand that human beings be re-engineered: their particular attachments and passions and interests must be weakened so that they voluntarily subordinate them to the general will. "The more these natural forces are dead and annihilated", Rousseau writes, "the greater and more lasting are the acquired ones, and the more solid and lasting also is the institution" (Rousseau 1762, II.7.3). The perfection of the machine demands that its components be cut down to size.

Rousseau's politics of the general will also imposes another requirement on individual psychology that is more subtle. The individual will must be transparent to itself: all citizens must be able to consult their own conscience sincerely, report their judgements accurately, and be understood by others as having done so. The legitimacy of the collective decision depends on the citizens' capacity for what Williams calls "trustworthy self-declaration": the idea that sincere, spontaneous, non-deceitful declaration can guarantee a true understanding of motive (Williams 1996, 370). To this first assumption, Rousseau adds two more: that what is so revealed will be coherent, representing a steadiness of character, and that the self-transcending motivations thereby disclosed will provide the substance of a shared life (Williams 1996, 370). Sincerity, steadiness, and moral self-transcendence are linked together in a single gesture: the self that declares itself sincerely is the self that wills the common good, and the community of such selves constitutes a legitimate polity.

All three assumptions are questionable, however, as Williams has argued using Diderot's *Le Neveu de Rameau*. Rameau's Nephew, Williams observes, is a character who possesses a great deal of truth about himself and reveals it with unusual candour, yet whose candour yields no moral steadiness whatsoever. He is consistent in one respect only: he freely and unashamedly expresses very different things at different times. Diderot was always drawn to a picture of the self as constantly shifting and reacting, like an instrument whose strings are played upon by forces beyond the player's control. Williams takes this Diderotian insight as the starting point for a rival account of how trust is actually constructed. If the self is not a transparent unity, but a shifting field of dispositions, then sincerity can only ever be a declaration of self *at that instant*, and the steadiness that political trust requires cannot be a property that the self discovers in itself through introspection. It must instead be *constructed*: built up through social practices that stabilize beliefs, commitments, and outlooks out of what would otherwise remain episodic moods and passing impulses (Williams 1996, 377–378). We contribute, all of us, to the social activity of mutually stabilizing our declarations, and it may be the case, Williams adds, that we carry out this process very badly (Williams 1996, 377).

Rousseau, Williams concludes, foresaw the need for a politics of trust, but provided only a bare assertion to meet it, linking in one gesture sincere self-

declaration, morality, and the politics of the general will. Yet the political problem actually requires something richer: structures, social and institutional, that can build a workable form of steadiness out of the shifting material of actual selves. These structures may involve the workings of values that make sincerity into something valued intrinsically, as more than a mere instrumental device, and they may involve the politics of identity through which people come to acknowledge a shared social essence that stabilizes their self-understanding (Williams 1996, 377–378). Whatever form they take, they cannot rest on the picture of the self that the political machine presupposes: transparent, unitary, and immediately accessible to its possessor. The self that the political machine addresses is a socially constituted, internally complex, partially opaque product of the very institutions whose legitimacy the general will is meant to secure.

Rousseau's own mature reflection acknowledges, in effect, that the political machine cannot sustain itself on its formal properties alone. In the chapter on the classification of laws, he writes: "In addition to these three sorts of laws there is a fourth, the most important of all; which gets engraved not in marble or in bronze but in the hearts of the Citizens; which is the genuine constitution of the State" (Rousseau 1762, II.12.5). The formal machinery of political, civil, and criminal law is insufficient without the informal moral culture that sustains it: morals, customs, and above all opinion, a part of the laws unknown to our political men, but upon which the success of all the others depends (Rousseau 1762, II.12.5). In the *Considerations on the Government of Poland*, written a decade after the *Social Contract*, Rousseau is more explicit: "No constitution will ever be good and solid unless the law rules the citizens' hearts. So long as the legislative force does not reach that deep, the laws will invariably be evaded (Rousseau, *Poland*, 1.6). Cohen's reconstruction of Rousseau's moral psychology gives this acknowledgement a precise theoretical shape (Cohen 2010, 122–127). The sense of duty alone, Rousseau holds, is motivationally inadequate: citizens face constant temptation to subordinate the common good to their private interests, and the larger the temptation, the more likely the subordination. The machine therefore requires what Cohen calls "motivational complements" to the sense of duty: generalized compassion that extends concern beyond one's immediate circle; an egalitarian sense of self-worth that is satisfied by treatment as an equal rather than by competitive advantage; and dispositions of reciprocity through which citizens come to value one another as agents who uphold the conditions of their shared life. These affective dispositions are distinct from the sense of duty but support it, binding citizens by affection to the common good that the formal rules articulate. The difficulty is that each of these complements depends on thick social conditions that the thin apparatus of the general will cannot generate – a participatory politics that brings citizens together under manifestly equal conditions, a civic culture that makes the experience of others as equals emotionally immediate, and limits on economic inequality severe enough to prevent the invidious comparisons that inflame *amour-propre*. The political

machine, in other words, requires motivational fuel that only institutions of a quite different character can supply, institutions whose animating principle is solidarity and shared sentiment rather than formal generality. The machine's thin, impersonal rules presuppose thick, affectively charged social that the machine by itself does nothing to cultivate.

Lionel Trilling captured the resulting tension in an image drawn from Goethe's *Sorrows of Young Werther*. In the opening pages of the novel, Werther discovers a garden and is charmed by its simplicity: it was laid out, he reports, "not by a systematic gardener but by a feeling heart" (Trilling 1972, 50). The garden embodies an ordering principle that differs from system. Its arrangement responds to the particular: *this* slope, *this* light, *this* planting. Whatever coherence it possesses arises from the gardener's responsiveness to what is actually there, to the specific character of soil and growth, rather than from a design imposed according to general rules. Trilling uses the image to contrast two forms of order that Rousseau holds in permanent tension. Systematic governance requires the art of the *systematic* gardener: the imposition of structure, the subordination of particular growth to a general plan, the inspectability and reproducibility of the design. Such an ordering has the machine-like virtue of being operable by anyone who follows the plan, and it is this impersonality that makes it suitable for public institutions. The feeling heart, by contrast, responds to the specific, the local, the unrepeatable. It registers what a general plan must abstract away from. And a political order that crushes the feeling heart in the name of systematic principle will lose the very source of loyalty and attachment on which its own stability depends.

Rousseau saw both sides of this tension. He knew that the feeling heart, left to itself, produces no political order, and that political order, imposed systematically, risks crushing what the feeling heart contributes. His epigram in the *Considerations on the Government of Poland* sums up the bind: "Putting law above man is a problem in politics that I compare to that of squaring the circle in geometry" (Rousseau, *Poland* I: 955). Squaring the circle is a problem whose solution is demonstrably impossible within the constraints of the system that defines it. The political machine is necessary, its full realization is impossible, and the impossibility is internal to the project rather than a failure of execution.

7.6 The Political Rationale Without the Metaphysical Warrant

The preceding sections have established that the political rationale for systematization is genuine, but that it has a proper domain and carries costs. One question remains: how does it relate to the metaphysical warrant? Does the demand for systematic governance in any way depend on the conviction that reality is a God-designed, rationally ordered whole?

Riley's genealogy of the general will also helps us to answer this question. Generality *began* as a divine attribute: Malebranche argued that God's perfection is expressed in the generality, uniformity, and constancy of his operation, and that to govern through general laws rather than particular interventions is constitutive of wisdom. The demand for generality was, at this stage, grounded in a theological claim about the nature of divine governance. Montesquieu then transposed those formal properties into a sociological framework, observing that well-governed societies exhibit regular patterns expressible as general relations rather than particular commands; this weakened the theological grounding. Rousseau completed the transformation by making those same formal properties into conditions of *political* legitimacy, grounded in the practical necessity of non-arbitrary governance rather than in any metaphysical claims. The metaphysical scaffolding thus gradually fell away. Generality, non-particularism, and the demand for principled operation persist in Rousseau's *Social Contract* as features of legitimate political authority, and they persist for reasons that have nothing to do with the nature of God. Rousseau's system is "a theory of humanity, a philosophy of morals and politics" that "endeavours to be metaphysically neutral" (Melzer 1990, 12). The need for systematic governance is, in his hands, a response to nature's silence rather than to nature's speech.

This chapter has shown that while the metaphysical warrant formed the historical scaffolding that enabled the formation of Rousseau's concept of the general will, the political rationale that emerged from this stands on its own feet: the demand for principled, inspectable, impartial governance depends on the practical need for non-arbitrary authority, and that need is generated by enduring conditions of social life rather than by any theological or metaphysical conviction. Once that scaffolding has fallen away, what remains is the demand itself, persistent, practically grounded, and now awaiting the philosophical transformation that will make its warrant a matter of the structure of cognition rather than of the structure of the world.

Chapters 5, 6, and 7 have each arrived at the same result by different routes. The epistemological rationale's self-correcting power, the formal anatomy of the concept of system, and the political rationale for principled governance all function independently of the conviction that reality is a God-designed rational whole. The question this convergence opens is whether the demand for system, once stripped of its metaphysical warrant, can find any deeper ground. The practical rationales tell us that systematic thought is *useful*: useful for diagnosing error, useful for making governance inspectable. They do not tell us whether the demand for system reflects something about the structure of cognition itself. Whether it does is the question Kant will address.

The Architectonic: Kant and the Transformation of System

In accordance with reason's legislative prescriptions, our diverse modes of knowledge must not be permitted to be a mere rhapsody, but must form a system.

— Kant, *Critique of Pure Reason*

By the final decades of the eighteenth century, the imperative to systematize knowledge was regarded as a demand of reason itself. But on what authority does reason impose this demand? And what happens to that authority when the theological and metaphysical convictions that had sustained it are withdrawn? Immanuel Kant's (1724–1804) answers to these questions, developed across the *Critique of Pure Reason* and the *Critique of the Power of Judgement*, mark a turning point in the genealogy of cognitive systematization.

Kant's contribution operates at three levels. At the first level, he transforms the metaphysical warrant that had underwritten the demand for system throughout the tradition from Keckermann to Du Châtelet. The conviction that reality *is* a divinely ordered rational whole, and that knowledge should mirror that order, is converted into a *regulative ideal*: we must proceed *as if* nature were systematically unified. The metaphysical warrant is withdrawn, and what replaces it is the insight into systematicity as a necessary projection of reason, a *focus imaginarius* around which cognition organizes itself without ever reaching the projected centre. At the second level, Kant uncovers the respect in which the demand for systematicity is a *precondition of cognition*: some minimal striving toward systematic unity is a condition of the intelligibility of thought itself, because without it, there would be no coherent use of the understanding and, consequently, no way to distinguish what is empirically true from what merely appears to be so. At the third level, he connects the demand for theoretical system to the essential ends of human reason, which are ultimately practical, thereby fusing the aspiration to cognitive rigour with the aspiration to moral orientation in a single architectonic framework.

Each of these levels has generated an extensive interpretive literature. What has received less attention is the relationship between them, and the philosophical work that this relationship performs. The insight into the role of systematicity as a precondition of cognition is the most fundamental of the three – it establishes a *floor* below which there is no cognition for the question “should I systematize?” to be about. The metaphysical warrant, through its longstanding dominance, had rendered this floor invisible. Kant uncovered it by removing the metaphysical assumptions that had concealed it. Once we no longer assume that reality itself is a system, the question “why systematize at all?” becomes a live one, and the role of systematization as a precondition of cognition then emerges as the foundation it always was.

The chapter proceeds as follows. It begins with Kant’s inheritance from Wolff and Lambert and the critical question he posed to both. It then turns to the Architectonic chapter’s central distinctions – system and aggregate, architectonic and technical unity, and the scholastic and cosmopolitan concept of philosophy, before tracing the transformation of the metaphysical warrant in the Appendix to the Transcendental Dialectic. The centre of the chapter is a detailed treatment of systematicity as a precondition of cognition and of the argument whereby systematic unity is a normative condition of thought’s answerability to truth. From there, the chapter examines Kant’s organic metaphor and the paradox of a self-legislating system that exhibits the functional profile of a machine while deriving its properties from an unmechanical source. It then traces the connection between theoretical and practical reason that the architectonic makes possible, before closing with the counter-voice at its most philosophically developed: Hegel’s critique of the categorical imperative as rational machinery, sharpened by Williams’s distinction between thick and thin ethical concepts. The genealogy’s main normative upshot emerges from this confrontation.

8.1 From Wolff and Lambert to Kant

AA 18: Refl. 5035). A warehouse simply stores goods according to some system; an architectonic organizes them according to a purpose. Wolff had assembled a vast quantity of philosophical knowledge and arranged it in demonstrative order, but the principle governing that arrangement remained external to it. For Wolff, the principle of systematic connection “does not arise from the system itself but from the mathematical method” (Nuzzo 2003, 16). The Euclidean *mos geometricus* was still doing the organising work. And an arrangement governed by an externally imposed method, however rigorous, could produce warehouse-like order, but it would not integrate its material towards a purpose.

The force of the Magazin metaphor becomes clearer through a homely comparison. Consider someone organizing a cluttered garage by arranging every item

according to a strict system – in alphabetical order, for example. That organization is rigorous and the result is fully surveyable: one need only go to the relevant letter to locate a given tool. Yet the hammer now sits beside the hose, the rake next to the router. The organizing principle has nothing to do with the objects it governs, and so the resulting order, however complete and inspectable, feels mechanical and disconnected from life. An architectonic organization of the same space, by contrast, might group the gardening tools by the door, because their purpose leads outdoors, and the woodworking equipment on the central bench, because that is where they fulfil their purpose. Here the logic of arrangement arises from the inner character and purpose of the objects themselves. Wolff's warehouse was, in Kant's estimation, the alphabetically organized garage of human reason: exhaustively systematic, but mechanically lifeless.

Lambert had gone beyond Wolff in several respects, and Kant admired Lambert greatly. In a letter from December 31, 1765, Kant told Lambert: "In all sincerity, I hold you to be the greatest genius in Germany, a man capable of important and enduring contributions to the investigations on which I too am working". Lambert had developed, in the *Fragment einer Systematologie*, the first general theory of what any system is, defining it by its purpose and structural properties rather than by its method of derivation. And he had grounded his system in experiential foundations, proposing what Paola Basso has called a "hybrid *a priori*": simple concepts drawn from experience serve as the basis for deductions that, once the principles are secured, proceed without further appeal to empirical data (Basso 2021, 189–191). Lambert's system was empirically grounded and modular, open at both ends because the list of simple concepts derived from experience could never be guaranteed complete (Blomme 2025, §2.6). He had freed the concept of system from Wolff's exclusively deductive framework. But he had not asked what authorised the fundamental concepts themselves. As Kant's retrospective verdict had it: Lambert was "analyst and architectonic," but "the critique was still missing" (AA 18: Refl. 4893).

A systematic science, Lambert argued, is *complete* when we can enumerate all of its cases and specify the rules according to which those cases are to be treated: trigonometry, for example, is complete because every possible problem reduces to a specifiable set of case types, and, for each type, a determinate procedure exists (van den Berg 2021, 279–280). Lambert recognized that since the realm of truths is infinite, completeness in this sense is often an ideal that we can only approximate. For Kant, a systematic science is complete when an idea of the whole determines the place of each part *a priori*, and completeness remains a regulative ideal rather than an attainable reality. Kant also relocates the governing principle: Lambert's completeness requirement presupposed that the cases and rules were ultimately drawn from experience, whereas Kant's idea of the whole is dictated by reason itself.

The absent critique was the *quid juris* question applied to systematic thought as such: by what right does reason organize its cognitions into a system? Wolff had assumed the right and exercised it through the mathematical method. Lambert had theorized the structure of systems and provided them with experiential foundations. Kant asked whether reason's claim to systematic organization could itself be justified, and if so, on what grounds. Henny Blomme identifies the result as "a true revolution of the concept of system" (Blomme 2015, 109). Where Lambert had conceived of a system as a totality whose elements are assembled into a whole, Kant reversed the priority: the system is a whole that *precedes* its parts, dictated by an *idea* of reason that determines in advance both the scope of the manifold and the position of each part within it (KrV A832/B860). As Paul Abela notes, the move from the aggregate manifold contributed by the understanding to a system of integrated cognition reflects "a cognitive shift in type, not degree" (Abela 2006, 411): reason's demand for unity draws on cognitive structures entirely absent from the categories of the understanding.

This reversal of priority is easy to misconstrue as a demand that we impose a preconceived picture on the data, deciding what the jigsaw puzzle depicts before we have looked at the pieces. If that were Kant's claim, it would amount to a licence for intellectual wish-fulfilment. But Kant's point concerns the conditions under which data can register as data in the first place. To recognize an oddly shaped piece of coloured cardboard as a *puzzle piece* at all, one must already possess the prior concept of a puzzle: the idea that fragmentary elements are *meant* to interlock, that they belong to a unified whole one does not yet see. Without that anticipatory idea, the pieces would present themselves as so many unrelated scraps, and there would be no motive to attempt their assembly, because the very concept of assembly presupposes the possibility of a completed picture. Kant relocated the ground of systematicity from the structure of being to the demands of reason upon itself. The idea of the whole is the cognitive precondition that authorizes inquiry, and the Architectonic chapter of the first *Critique* is where that relocation is worked out.

8.2 The Architectonic of Pure Reason

"By an architectonic I understand the art of systems. Since systematic unity is that which first makes ordinary cognition into science, i.e., makes a system out of a mere aggregate of it, architectonic is the doctrine of that which is scientific in our cognition in general" (KrV A832/B860). The passage, which opens the penultimate chapter of the *Critique of Pure Reason*, defines architectonic as an *art*: the skill of producing systems. Hans Fulda and Jens Stolzenberg usefully gloss this by describing architectonic as the art of "designing and executing systems of thought" (Fulda and Stolzenberg 2001, 15); the emphasis falls on the disciplined activity by which such order is brought about, and on the judgement required to execute it well.

In this chapter, Kant articulates the close relationships between knowledge, reason, and system. The most fundamental distinction is between system and aggregate. “I understand by a system the unity of the manifold cognitions under one idea. This is the rational concept of the form of a whole, insofar as through this the domain of the manifold as well as the position of the parts with respect to each other is determined *a priori*” (KrV A832/B860). The emphasis falls on *a priori*: the system is a whole whose form is given in advance by an idea, and the idea determines both the scope of the manifold and the placement of every part within it. An aggregate, by contrast, is assembled from below, its parts accumulated without a governing concept of the whole. Kant calls the aggregate a *rhapsody*: a loose, unstructured effusion, a heap of material gathered without internal direction. The word carries force. Someone who spends three hours reading a headline about politics, then a post about a celebrity, then a factoid about marine biology, and emerges at the end with thousands of words consumed but no deepened understanding of anything, has built a rhapsody. The data has no architecture. If, by contrast, one sets out to understand the economic drivers of renewable energy transition and intentionally seeks out historical data, supply-chain analysis, and policy papers, one brings an idea of the whole to the inquiry. The difference between the two is the difference between accumulation and cognition.

Kant is clear that *unter der Regierung der Vernunft* — under the governance of reason — “our diverse modes of knowledge must not be permitted to be a mere rhapsody, but must form a system” (KrV A832/B860).¹ And the problem with a rhapsody, as Hoyningen-Huene has observed, is that it gives us no way to judge whether any part is missing, whether a proposed addition belongs, or whether completeness has been reached. A system, by contrast, enables one to determine all these things, and in doing so demonstrates its superiority over unordered knowledge (Hoyningen-Huene 2013, 185). The system is an *articulatio*, an articulated whole, unified under a single generative idea that dictates the place and function of each element within it.

Kant then draws a further distinction between a schema grounded in the chief end of reason, which yields *architectonic* unity, and one arising “empirically, in accordance with aims occurring contingently,” which yields merely *technical* unity (KrV A833/B861). Wolff’s achievement, in these terms, was technical: his demonstrative chains arranged philosophical cognitions with logical rigour, but the arrangement served contingent aims rather than the essential ends of reason itself. A student who memorizes a highly structured corpus of philosophical dates and definitions solely to pass a university examination has organized knowledge with impressive discipline; but the organizing principle, the examination, is entirely external to the knowledge it orders, and the resulting unity dissolves the moment the examination is over. Science, Kant insists, “cannot arise technically, from the

¹Or, as the Guyer and Wood translation has it: “Under the government of reason our cognitions cannot at all constitute a rhapsody but must constitute a system”.

similarity of the manifold or the contingent use of cognition in concreto for all sorts of arbitrary external ends, but arises architectonically, for the sake of its affinity and its derivation from a single supreme and inner end, which first makes possible the whole" (KrV A834/B862). A system that serves someone's contingent purposes may achieve technical unity; only one that answers to the purposes of reason achieves architectonic unity.

The distinction between architectonic and technical unity tells us that the system must serve reason's essential ends rather than contingent external aims, but it does not yet tell us what those essential ends are. Kant addresses this through a further distinction. The concept of philosophy has so far been "only a scholastic concept [*Schulbegriff*], namely that of a system of cognition that is sought only as a science without having as its end anything more than the systematic unity of this knowledge, thus the logical perfection of cognition" (KrV A838/B866). The *Schulbegriff* does not pursue contingent or arbitrary ends; it pursues the logical perfection of the system itself. Its limitation is that it treats the construction of a tidy system as an end in itself, a self-contained academic exercise directed at internal consistency and coherence. The philosopher operating under the *Schulbegriff* is, in Kant's pointed term, a *Vernunftkünstler*: an artisan of reason, someone who fashions logical structures with considerable technical skill but without asking what the structures are ultimately for. "But there is also a *Weltbegriff* (*conceptus cosmicus*)," Kant continues. When conceptualized in terms of the *Weltbegriff* rather than the *Schulbegriff*, the philosopher emerges as "the legislator of human reason" (KrV A839/B867). Thus understood, philosophy becomes "the science of the relation of all cognition to the essential ends of human reason (*teleologia rationis humanae*)."¹ The distance between *Schulbegriff* and *Weltbegriff* is the distance between playing an academic game and seeking wisdom.

"The Canon of Pure Reason," which precedes the Architectonic in the *Methodenlehre*, had already identified these ends. All of reason's interest, Kant writes, is united in three questions: *What can I know? What should I do? What may I hope?* (KrV A805/B833). The first dictates reason's theoretical end: the pursuit of systematic, objectively valid knowledge. Yet, under the *Weltbegriff*, theoretical knowledge is merely a subalternate end, a necessary instrument for navigating the world. The unconditioned end of reason is entirely practical, governed by the second question: the autonomous determination of the will by the moral law. The third question establishes the ultimate horizon of the system by projecting the highest good (*summum bonum*), a world in which happiness is systematically proportioned to moral worth. The system of human knowledge achieves architectonic unity only when its vast theoretical apparatus is subordinated to, and oriented by, this supreme moral destination. Under the *Weltbegriff*, the cognitive enterprise is never an end in itself; it is the intellectual scaffolding required for finite rational beings to realize their moral vocation. The essential ends are "either the final end, or subalternate ends, which necessarily belong to the former as means. The former is nothing other

than the entire vocation of human beings, and the philosophy of it is called moral philosophy” (KrV A840/B868).

Lea Ypi has argued that this integration of cognitive and moral demands within a single systematic framework is the architectonic’s most consequential achievement (Ypi 2021, 174). This also creates a clear contrast with Lambert: while Lambert’s architectonic operated “from the perspective of an artisan striving to be an architect with the tools of mathematical logic,” Kant’s was “developed from the perspective of a cosmopolitan notion of philosophy” (Manchester 2003, 205). The *Weltbegriff* raises the question that neither Lambert’s instrumental framework nor the *Schulbegriff*’s logical self-sufficiency could pose: what purposes should the system ultimately serve? The architectonic is answerable to the question of how human beings should live.

From the purposes of the system, Kant turns to its mode of acquisition. Historical cognition, he writes, is *cognitio ex datis*: cognition from data, knowledge acquired through reception and memorization. Rational cognition is *cognitio ex principiis*: knowledge generated from underlying principles by the knower’s own rational faculties. A student who has learned the Wolffian system possesses “nothing other than a complete historical cognition of the Wolffian philosophy; he knows and judges only as much as has been given to him.” He “has formed himself according to an alien reason, but the faculty of imitation is not that of generation, i.e., the cognition did not arise *from* reason for him.” Such a person, Kant adds in a withering image, is merely “a plaster cast of a living human being” (KrV A836/B864). The external shape may be perfect, but the inside is hollow. Someone who follows a complex recipe line by line, adding a teaspoon of this and simmering for exactly ten minutes, may produce a creditable meal, but is operating entirely on *cognitio ex datis*: the moment a missing ingredient or a malfunctioning oven disrupts the prescribed sequence, the veneer of competence collapses, because the underlying principles were not understood. Knowledge borrowed from an “alien reason” cannot adapt. By contrast, a cook who has grasped the principles by which fat carries flavour, acid balances sweetness, and heat alters proteins can open a refrigerator full of unfamiliar ingredients and improvise a meal, because their knowledge is *alive*.

This is what Kant means when he insists that one can learn to *philosophize*, but one cannot learn *philosophy*. The contrast between noun and verb here mirrors the contrast between memorizing the surveyable structure and demonstrations in a textbook and *living cognition* – the capacity to produce insights out of one’s own rational faculties. The ideal of education ushered in by Ramus and Keckermann, which treats education as the efficient transmission of a fully externalized body of knowledge, and in which the student’s task is to create a copy of a system in his own mind, amounts to treating learning as a file transfer. But philosophy is irreducibly a generative activity, not a systematic body of propositions to be downloaded as fast as possible into students’ “necktops”.²

Kant closes the Architectonic chapter by connecting systematic thought to governance. Metaphysics, he writes, “serves more to prevent errors than to amplify cognition,” and derives its “dignity and authority” from its “office as censor, which secures the general order and unity, indeed the well-being of the scientific community, and prevents its cheerful and fruitful efforts from straying from the chief end, that of the general happiness” (KrV A851/B879). Manchester reads this as the architectonic’s deepest function: it provides “a tribunal for the interrogation of any doctrine,” a critical apparatus rather than a completed edifice (Manchester 2003, 207). The tribunal ensures that doctrines are answerable to reason’s demands, and in this respect the architectonic transposes the ambition of legal codification to the domain of cognition. The system makes intellectual authority inspectable. What the Architectonic does not yet fully explain is where the authority of systematicity itself comes from.

8.3 Systematicity as an Indispensable Regulative Ideal

For the tradition that preceded Kant, the source of systematicity’s authority was straightforward. Nicholas Rescher boiled it down to a syllogism: “Knowledge must reflect the truth. The truth is a system. Therefore, knowledge should be a system” (Rescher 1979, 19). The metaphysical warrant entered with the second premise: reality *is* a system, crafted by a rational God, and knowledge should mirror that order. But once the metaphysical warrant falls, what compels us to systematize? If the demand for system can no longer be read off the structure of the world, where does its authority come from?

Kant retained the conclusion but replaced the premises. Systematic unity, he argued, is a *regulative ideal* that reason imposes on experience as a condition of coherent inquiry, and the demand for it derives its authority from reason’s own needs rather than from the structure of the world. The canonical statement of this transformation appears in the “Appendix to the Transcendental Dialectic”, in a passage whose optical metaphor has become one of the most celebrated images in the *Critique*:

The transcendental ideas are never of constitutive use ... On the contrary, however, they have an excellent and indispensably necessary regulative use, namely that of directing the understanding to a certain goal respecting which the lines of direction of all its rules converge at one point, which, although it is only an idea (*focus imaginarius*)—i.e., a point from which the concepts of the understanding do not really proceed, since it lies entirely outside the bounds of possible experience—nonetheless still serves to obtain for these concepts the greatest unity alongside the greatest extension. (KrV A644/B672)

Kant's *focus imaginarius* is an optical fiction: the vanishing point in a perspectival painting around which the entire composition organizes itself. As Alfredo Ferrarin observes, it is "a requisite optical focus, a point of orientation, which moves along with the observer. It has no reality of its own and is rather a fiction giving all cognitions a systematic unity" (Ferrarin 2015, 53). The vanishing point organizes the visual field without being a feature *of* the visual field; it lies outside the painting while making the painting's coherence possible. The metaphysical warrant had claimed that the vanishing point was a real place, that the lines of cognition converge because reality itself converges on a divine ground. Kant shows that the convergence is a construct of the painter's art, a projection of reason rather than a description of the world. Yet the projection is indispensable: without it, the painting dissolves into unrelated marks on a surface.

The passage immediately following the *focus imaginarius* specifies what the projection prescribes:

If we survey the cognitions of our understanding in their entire range, then we find that what reason quite uniquely prescribes and seeks to bring about concerning it is the **systematic** in cognition, i.e., its interconnection based on one principle. This unity of reason always presupposes an idea, namely that of the form of a whole of cognition, which precedes the determinate cognition of the parts and contains the conditions for determining *a priori* the place of each part and its relation to the others. Accordingly, this idea postulates complete unity of the understanding's cognition, through which this cognition comes to be not merely a contingent aggregate but a system interconnected in accordance with necessary laws. (KrV A645/B673)

What reason prescribes is an idea of the whole that *precedes* the cognition of the parts. The idea determines the place and mutual relation of each part before inquiry encounters them, and this priority is what distinguishes a system from the bottom-up accumulation that yields a mere aggregate. This idea, moreover, is not the concept of an *object*; rather, it acts as a *rule* for the understanding, a principle of organization that tells reason how to treat its cognitions, but without specifying the content of any particular cognition.

Kant's language of "reason's prescription" locates the source of the demand squarely in reason itself. Reason does not *discover* that its cognitions should form a system because reality happens to form a system; rather, it *prescribes* that they be brought into systematic connection, and it treats its own cognition as defective insofar as that cognition remains a contingent aggregate rather than a system interconnected by necessary laws.

The ideal of systematicity accordingly does not *reflect* nature's systematicity so much as *regulate* our inquiry into nature:

Such concepts of reason are not created by nature, rather we question nature according to these ideas, and we take our cognition to be defective as long as it is not adequate to them. Admittedly, it is hard to find **pure earth, pure water, pure air**, etc. Nevertheless, concepts of them are required (though as far as their complete purity is concerned, have their origin only in reason) in order appropriately to determine the share that each of these natural causes has in appearance. (KrV A645–6/B673–4)

Even if no chemist ever isolates a sample of perfectly pure water, the concept functions as an ideal without which the chemist cannot determine the share that water, as distinct from its impurities, contributes to the appearance under investigation. The ideal of perfectly systematic knowledge is no more a description of the world than the idea of pure water is a description of any actual substance. We “question nature according to these ideas”. In the case of the idea of pure water, this gives us the capacity to discriminate among the contributing causes of an appearance. In the case of the idea of systematic unity, it gives us an orienting standard against which the understanding can measure its own progress, and recognize any cognition that falls short of that standard as defective.

This is the sense in which the regulative ideal answers the authority question posed by the collapse of the metaphysical warrant. The authority of systematicity no longer rests on the claim that reality is a system. It rests on the fact that reason cannot assess the adequacy of its own cognitions without a standard of systematic unity against which to measure them.

Kant does not, of course, completely do away with the notions of a divine creator or a unified whole of nature. They reappear in the “Appendix to the Transcendental Dialectic” as two of the “ideas of reason”: the idea of *God* and the idea of the *world-whole*. But they now operate in a radically different capacity. As Rachel Zuckert has emphasized, they function as “optimistic placeholders”: they project something beyond the current state of knowledge, encouraging inquiry to continue, while remaining sufficiently indeterminate to leave the content of that inquiry undetermined (Zuckert 2017, 95–99). Yet the role these ideas now play is very different from the one the metaphysical warrant had assigned them. They issue procedural directives to the inquirer: treat nature “as if” it were the product of an intelligent creator, and seek comprehensive order in nature as a whole (Zuckert 2017, 96). The “as if” marks the decisive structural change in the kind of work these ideas perform. The metaphysical warrant had asserted a fact about the world’s origin from which the demand for systematic order followed as a consequence. The regulative idea asserts nothing about the world’s origin. It tells the inquirer how to proceed: organize your cognitions as though comprehensive rational order were available, and let experience determine how far that organization can be carried. The near-emptiness of the ideas is what suits them for this role. Because they specify

almost nothing about the objects they project, they do not predetermine the results of empirical investigation, but sustain the hope that nature will cooperate with our inquiries without prescribing what form that cooperation will take.

Kant was, as Nuzzo argues, “the first to develop the distinction between the merely formal concept of purposiveness and the metaphysical assumption of an actual divine intention” (Nuzzo 2003, 27). There can be purposiveness without purpose, in the sense that we can appreciate the breathtaking organizing and purposiveness of a biological ecosystem without leaping to the conclusion that some creator must have purposively designed it that way. The purposiveness we see can be, in the first instance, a reflection of our own minds demanding order from the data. The purposiveness is *formal*.

But this leaves open a further question: whether the demand for systematic unity is just a useful cognitive orientation for inquiry, one we might in principle abandon without loss, or whether it runs deeper, constituting a precondition of cognition as such.

Here is the fully rewritten section. I have merged those duplicate paragraphs, restructured the flow according to our roadmap (Argument → Scope → Mechanism → Objections → Conclusion), and smoothed out the transitions so the philosophical progression is tight and logical.

8.4 Systematicity as a Precondition of Cognition

In the “Appendix to the Transcendental Dialectic,” Kant articulated grounds for thinking that systematicity is a precondition of cognition rather than merely an aspiration of it. He wrote:

the law of reason to seek unity is necessary, since without it we would have no reason, and without that no coherent use of the understanding, and, lacking that, no sufficient mark of empirical truth; thus in regard to the latter we simply have to presuppose the systematic unity of nature as objectively valid and necessary. (KrV A651/B679)

The passage chains together a series of indispensability arguments, and each link ratchets up the stakes. Without the demand for systematic unity, there is no reason. Without reason, there is no coherent use of the understanding. Without the understanding’s coherent operation, there is no way of distinguishing what is empirically true from what merely appears to be so. Systematic unity is presented here as a condition on which the entire apparatus of empirical cognition depends. The final link in the chain is the one that matters most for the chapter’s argument: the demand for system is what provides “a sufficient mark of empirical truth”. Systematicity

is the framework within which truth can be assessed, because it is what enables reason to test cognitions for coherence, detect inconsistencies among them, and correct the ones that fail the test. Without this framework, individual cognitions may still occur, but we have no way to evaluate their claim to truth. Kant makes the epistemic function explicit a few pages earlier: “the systematic connection which reason can give to the empirical employment of the understanding not only furthers its extension, but also guarantees its correctness” (KrV A648/B676). Systematic connection *guarantees correctness* by supplying what Kant calls “touchstones of truth” (KrV A647/B675): criteria by which the adequacy of any given cognition can be measured against the body of cognitions already accepted.

A few pages later, Kant sharpens the claim by imagining its negation. If nature displayed such heterogeneity that even the most acute human understanding could detect no common features among existing beings, “then the logical law of genera would not obtain at all, no concept of a genus, nor any other universal concept, indeed no understanding at all would obtain” (KrV A654/B682). A world of unrelieved particularity, in which nothing resembled anything else in any salient respect, would defeat our very capacity for conceptualization. His term for the alternative to a minimally systematic nature is a “crude chaotic aggregate” (FI, AA 20:209).

These two passages, taken together, constitute the textual foundation for what I call the precondition of cognition. The precondition establishes a *floor* below which the question “should I systematize?” cannot intelligibly be raised, because there is no cognition for the question to be about below that threshold. The precondition does not tell us *how much* system to demand, nor does it dictate the form that systematic organization should take. Everything that concerns the specific form, extent, and ambition of systematic thought is where the practical rationales and their limits come in. What the precondition tells us is that some minimal striving toward systematic unity belongs to the conditions of thought’s answerability to truth.

What Kant realized, then, is that the demand for systematicity is *normatively binding on cognition as such*. It occupies a distinctive intermediate position in the interpretive literature. At one end, it is a stronger claim than the view that systematic unity is merely a *useful heuristic* for higher-level scientific inquiry. At the other end, it is more carefully circumscribed than Philip Kitcher’s view that systematicity is constitutive of what counts as a law of nature, and hence *constitutive of experience* in the exact same sense as the categories (Kitcher 1986). Kant’s text supports a slightly weaker reading than Kitcher’s: a single act of perception *can* occur without the perceiver explicitly consulting any principle of systematic unity. Similarly, a person can come to know, on the basis of a friend’s testimony, that the city of Bern is west of the city of Basel without actually assessing the systematic coherence of that belief with the rest of her geographical knowledge (compare Kant on “historical belief”, AA 8:141). What the precondition demands, as Katharina Kraus (2023) has argued, is that the cognitive framework within which her knowledge-claim

operates is normatively oriented toward *rational certainty* as the highest standard of knowledge. Systematicity is constitutive of that highest standard, and the standard in turn is normatively binding on every cognitive act that aspires to contribute to knowledge. To cognize at all is already to be subject to this demand, because without it, there is no criterion by which any cognition's claim to truth can ultimately be evaluated.

Kant provides a further resource for understanding how this demand operates in practice. He argues that ideas of reason function as “analogues” of sensible schemata (KrV A665/B693). A schema is a procedural rule that determines how to synthesize a sensible manifold into a unified intuition in accordance with a category: the schema of substance, for instance, provides the rule for finding something persistent in the flux of appearances. An idea of reason, employed regulatively, provides an analogous procedural rule, one that determines how to bring manifold cognitions into systematic unity. An idea “shows not how an object is constituted but how, under the guidance of that concept, we ought to seek after the constitution and connection of objects of experience in general” (KrV A671/B699). The schema–idea analogy explains what it means for the precondition to be *operative* in cognition: it generates procedural rules for ordering, connecting, and assessing the coherence of our judgements. The idea outlines a domain of intelligibility, a context within which acts of the understanding can be meaningful, without determining the content of those acts in advance. Failing to follow these regulative principles does not make experience impossible in the sense that a failure of the categories would; it makes experience *unintelligible*, depriving us of the framework within which we can ask whether our cognitions cohere and form a body of knowledge worthy of the name.

Hannah Ginsborg has provided the strongest defence of this precondition by connecting it to the presupposition of purposiveness that governs the *Critique of the Power of Judgment*. Her argument is that the demand for systematicity derives from a more fundamental presupposition: that nature is *purposive for our cognitive faculties*, that it is structured in ways our concepts can track. Without this presupposition, Ginsborg argues, “our activity” would fail “to amount to conceptualization as opposed to mere association of representations” (Ginsborg 2017, 71). Association requires only regularities in the environment and a mechanism for registering them; conceptualization requires the presupposition that one's ways of sorting things are responsive to something in the things themselves. Thomas Teufel has identified a corresponding development in Kant's own thinking, tracing a shift from a “maximalist” conception of nature's systematicity to a “minimalist” conception that demands only a threshold level of cognizable order (Teufel, 120–121). The minimalist version strips the demand for system to its essential core: what cognition requires is the *possibility* of systematic ordering, the assurance that nature will not defeat our every attempt to bring it under concepts.

Understanding the precondition in these minimalist, procedural terms goes some way towards answering a worry that Paul Guyer has raised: the concern that Kant's

regulative ideal “does nothing but transform our own need for systematicity into a self-serving delusion that nature is systematic” (Guyer 1997, 42). For Kant, our working under the assumption that nature is systematic is not just a comforting projection, like seeing shapes in the clouds; nor is it merely an expression of an aesthetic preference for neatness. It is absolutely necessary. Without it, we could not conceptualize the world at all.

A further distinction latent in the precondition deserves to be made explicit, as it bounds the scope of the claim. The precondition demands that *inquiry* be systematic—that cognitions be tested and corrected—but it does not demand that the *domain inquired into* exhibit a corresponding systematic harmony. A detective reasoning with exemplary coherence about a case may discover that the testimony is conflictual in ways that cannot be resolved into a single consistent narrative. Her reasoning is none the worse for this; what would be worse is if she abandoned the demand for coherence in her own thinking on the grounds that the material resists it. This distinction becomes especially relevant in normative domains. Value pluralists from Isaiah Berlin onwards have argued that the normative landscape is irreducibly fragmented and tension-ridden. Yet Kant’s precondition thesis is compatible with this possibility. What it rules out is a mode of thought that has abandoned the aspiration to systematicity altogether, not a domain that resists complete systematic harmonization. Systematic inquiry into an asystematic domain is not a contradiction; it is what truthful and responsible thought looks like.

The insight into systematicity’s role as a precondition of cognition requires no external warrant at all, because it is constitutive of the cognitive activity from which justification of any kind must proceed. “Why systematize?” is thus never really a question about *whether* to systematize at all, because the precondition settles that. The real question is *how much* to systematize and *in which domains*. The practical rationales can motivate systematizing beyond the floor that the precondition establishes.

8.5 Organizing the Machine

Kant’s account of systematic thought moves between two metaphors in a way that marks a significant development for the machine emulation thesis. In the Architectonic, the system is compared to a living body: “The whole is articulated (*articulatio*) and not heaped together (*coacervatio*); it can, to be sure, grow internally (*per intus susceptionem*) but not externally (*per appositionem*), like an animal body, whose growth does not add a limb but rather makes each limb stronger and fitter for its end without any alteration of proportion” (KrV A833/B861). The animal body grows from within – its development is internally directed and its proportions are maintained through the organic integration of its parts. The aggregate, by contrast, is heaped together from without, its parts accumulated by external addition.

A few pages later, the biological imagery intensifies: “the idea lies in reason like a seed, all of whose parts still lie very involuted and are hardly recognizable even under microscopic observation” (KrV A834/B862). Systems appear to arise from the “mere confluence of aggregated concepts, garbled at first but complete in time,” yet they all had their schema, Kant insists, “as the original seed, in the mere self-development of reason” (KrV A835/B863). The idea’s regulative force is generative: it produces the structure of the system from within, rather than suffering it to be imposed from without.

Ferrarin has identified a tension running through these passages: reason is figured simultaneously as a seed from which an organism organically grows, and as an architect who deliberately plans an edifice (Ferrarin 2015, 14). The seed suggests a development prior to and independent of deliberate construction, while the architect suggests a designer who surveys and organizes materials from above. But the tension is productive, because Kant needs both (Ferrarin 2015, 28). The organic metaphor captures the *self-development* of reason’s systematic activity; the architectural metaphor captures its *self-legislation*. Both are needed because reason is neither a mere natural process nor a mere deliberate construction. It is an activity that generates its own principles from within while legislating those principles upon itself.

However, a deeper challenge to machine emulation thesis surfaces here. “Pure reason is everything except a machinery,” Ferrarin writes, “as Kant’s definition of stupidity and the power of judgment shows. ‘Machinery’ implies the thoughtless functioning of an automaton ... the opposite of the spontaneity and freedom with which Kant identifies reason” (Ferrarin 2015, 8n10). The objection strikes at the core of the machine emulation thesis as applied to Kant. A machine, in the sense this book has been developing, is an apparatus governed by externally fixed rules, whose operations are determinate and operator-independent. If a machine is an apparatus governed by externally fixed rules, and Kant’s reason is self-legislating and spontaneous, it seems the Kantian system cannot be machine-like in any meaningful sense. The machine’s operations are heteronomous; the architectonic’s operations are autonomous.

Yet to read Kant’s organic turn as an abandonment of machine-like virtues is to miss the radical nature of his synthesis. The functional properties of the architectonic remain exactly those of a well-designed machine. The system demands that its parts stand in demonstrative, rule-governed relations; it demands that its structure be surveyable to any rational inspector; and it demands operator-independence, meaning its form holds universally for anyone who follows it.

What the architectonic transforms is not the machine’s virtues, but their ground. For Wolff, these machine-like properties were imported from without via the Euclidean method. For Lambert, they were derived externally from empirical concepts. In the earlier metaphysical tradition, they reflected the blueprint of a divine watchmaker. Kant relocates the ground to reason’s own self-legislation. The system is

rule-governed, surveyable, and universal because reason prescribes these very mechanical virtues to itself.

In this light, the organism metaphor is not a rejection of the machine emulation thesis, but its ultimate, internalized evolution. Lacking an engineering model for an apparatus that writes its own operating instructions, Kant reached for the biological organism to conceptualize a *self-assembling machine*. The system exhibits the functional, operator-independent profile of a mechanism, but derives those properties from a spontaneous, unmechanical source.

In the late manuscripts of the *Opus postumum*, Kant characterized transcendental philosophy as “the consciousness of the capacity of being the *creator* [*Urheber*] of the system of one’s ideas, in a theoretical as well as in a practical respect. Ideas are not mere concepts but laws of thought which the subject prescribes to itself. *Autonomy*” (OP 21:93). The system is autonomous in that its organizing principles are self-given. The machine-like virtues survive the loss of the metaphysical warrant because they are expressions of reason’s own legislative activity rather than reflections of an external blueprint. The machine has been internalized: its operating instructions are written by the apparatus itself.

8.6 Connecting Theoretical to Practical Reason

If reason legislates its own systematic principles, what ensures that these principles serve more than formal completeness? What prevents the architectonic from collapsing back into the *Schulbegriff*, a doctrine of logical ability untethered from any wider human purpose? The *Opus postumum* passage cited above points toward Kant’s answer: the system of ideas is self-legislated “in a theoretical as well as in a practical respect.” The architectonic integrates the demand for cognitive rigour with the demand for moral orientation, and the *Weltbegriff* is the concept under which this integration occurs. As Ypi has argued, the Architectonic is “the only section of the book where reason in its complexity is analysed from the point of view of the organic connection between its theoretical and practical use” (Ypi 2021, 12). *Sapere aude*, the Enlightenment’s call to autonomous thought, and the architectonic’s demand for systematic unity are expressions of the same ideal of rational self-governance: to think for oneself is to organize one’s cognitions according to the ends of reason itself, and those ends are ultimately practical.

Stefano Bacin has argued that the practical side of this integration follows what he calls the “Pufendorfian paradigm”: the system of obligations unfolds from an established principle rather than assembling moral beliefs into a coherent set (Bacin 2023, 4–6). Pufendorf had maintained that a comprehensive doctrine of natural law required first establishing a solid fundamental proposition and then deriving from it, in evident order, the entire scope of moral demands. In Kant, the categorical imperative furnishes the principle, and the *Metaphysics of Morals* develops from

it a system of duties whose completeness is guaranteed by the principle's scope rather than by the comprehensiveness of the catalogue. The convergence shows that the *Weltbegriff*'s subordination of system to moral purpose has a genealogy in the natural law tradition. Kant's architectonic inherits, from both the theoretical and the practical sides, the conviction that the system must answer to ends beyond its own logical structure.

The connection between theoretical and practical reason in fact runs deeper than the subordination of cognition to moral purpose. The same demand for systematicity that, in the theoretical domain, provides touchstones of truth also operates in the constitution of the person as an agent. Katharina Kraus has developed this argument in detail, showing that the regulative principles derived from the idea of the soul prescribe norms for self-knowledge and what she calls self-formation: "Reason demands ... that we realize ourselves (1) as unified across time ... (2) as the self-efficacious common cause of all mental activity ... and (3) as self-directing towards a rational personality" (Kraus 2023, 251–252). These demands parallel, in the practical sphere, the regulative role that ideas play in the theoretical sphere: they do not constitute the person as a metaphysical substance, but they outline the domain of intelligibility within which a person's mental life becomes comprehensible as *hers*. Without some striving toward inner systematicity, a person's mental life would amount to what Kraus vividly calls a "staccato mind" (Kraus 2023, 193): a series of disconnected episodes without memory of past states or projections for future ones, lacking the unity that makes self-knowledge and deliberate action possible.

Kraus's account specifies the normativity of the demand in three spheres: epistemic coherence – the consistency and mutual support of one's cognitive commitments; practical coherence – the consistency of one's maxims and intentions; and mental harmony – the integration of one's affective responses with one's cognitive and practical orientations (Kraus 2023, 272). The demand for inner systematicity is what makes self-knowledge epistemically possible and self-formation practically possible. It generates the imperative that Kraus distills to its most economical form: *seek to be a systematic whole* (Kraus 2023, 277).

Note, however, that this practical imperative to systematize is normative without being constitutive: one can fail to realize it, and most of us fail to realize it fully, without ceasing to be a person. Personhood is a matter of degree here, in the sense that the normative standard can be approximated more or less closely, and the pathological cases of mental disunity that clinical psychology documents are, on this account, cases of failing to comply with a rational norm (Kraus 2023, 272).

This extension of the demand for system from the theoretical to the practical domain matters for the book's larger argument in a specific way. It shows that the precondition of *cognition* is continuous with a precondition of *personhood*: the same rational demand that makes cognition answerable to truth makes agency answerable to practical coherence. The machine-like virtues that the architectonic exhibits, rule-governance, surveyability, universality, are what make theoretical

cognition inspectable and correctable; transposed to the practical domain, they are what make a person's principles and commitments available for reflective scrutiny and revision. The Kantian architectonic is thus a framework for self-governance in the broadest sense: the governance of one's cognitive life and the governance of one's practical life are continuous exercises of the same rational demand for systematic unity.

Ypi's reading identifies a tension that the first *Critique* does not fully resolve. The principle of purposiveness, as deployed in the Architectonic and the Appendix, remains tethered to the idea of God as the ultimate ground of systematic unity: we proceed "as if" nature were designed by an intelligent creator. Ypi characterizes this as *purposiveness as design*, and she argues that it threatens to collapse back into the dogmatic metaphysics that the Critique was supposed to overcome: the Architectonic is "the solution and the riddle" of the first *Critique* (Ypi 2021, 3). The resolution comes only in the third *Critique*, where purposiveness is reconceived as a normative principle of reflective judgement grounded in transcendental freedom rather than divine intention: *purposiveness as normativity* in place of *purposiveness as design* (Ypi 2021, 173–178). The transformation of the metaphysical warrant, even within Kant's oeuvre, is a process unfolding across his critical project. The demand for system is progressively liberated from its theological scaffolding and recognized as a demand internal to reason itself, operating simultaneously in the domains of knowledge, action, and self-understanding. And it is precisely the scope of that demand that the counter-voice tradition will call into question in its reaction to Kant's transformation.

8.7 Hegelian Reactions to Kant

The demand for systematicity, as Kant articulates it, is normatively binding on cognition and continuous with a demand for practical self-governance. The counter-voice tradition does not contest the first part – in particular, it does not contest there is a minimal threshold or floor below which we cannot afford *not* to systematize if we are to make any sense of objective reality at all. What the counter-voice contests is the *extension* of this demand beyond the domains in which it serves cognition into domains where it distorts or impoverishes what it claims to comprehend.

The most powerful version of this objection comes from within the systematic tradition rather than from outside it. G. W. F. Hegel (1770–1831) accepts the aspiration to systematicity; but he argues that Kant's way of realizing it produces a machinery of moral thought whose outputs are either empty or destructive of the ethical material they claim to process. This objection, later taken up notably by Bernard Williams, gives the counter-voice its most resonant expression formulation.³

Viewed through the lens of the machine emulation thesis, the categorical imperative is a cognitive machine for producing moral judgements. It takes any proposed

maxim as input, subjects it to the universalizability test, and delivers a determinate output, permission or prohibition, regardless of who operates it. Operator-independence is the whole point: the moral law binds universally and impersonally, and its verdict does not depend on the character, circumstances, or evaluative perspective of the agent who applies it. The machine-like virtues are maximally on display. The categorical imperative is rule-governed: its application follows a specifiable procedure. It is universal: anyone possessed of practical reason can apply the test. It is impartial: the test abstracts from all particular interests and perspectives. And it is surveyable: the procedure is open to inspection, and its workings are transparent to any rational agent.

Hegel's §135 of the *Philosophy of Right* identifies the cost. He somewhat grudgingly recognizes that Kant achieved something essential in understanding that moral authority derives from the will's capacity to legislate universal laws for itself. Yet persistence at the merely moral standpoint, Hegel charges, reduces this gain to an empty formalism and moral science to a rhetoric of duty for duty's sake (Hegel 1821, §135 Anm.). From the standpoint of the categorical imperative alone, no immanent doctrine of duties is possible: "from that determination of duty as the absence of contradiction, the formal agreement with itself, which is nothing other than the positing of abstract indeterminacy, one cannot proceed to the determination of particular duties" (Hegel 1821, §135 Anm.). The machine cannot generate content from its own formal resources. It must borrow its material from the concrete ethical substance, the customs, institutions, and evaluative practices of particular communities, that Hegel calls *Sittlichkeit*: the concrete, living cultural substance of a community – its historical customs, family structures, shared traditions, and established institutions. The categorical imperative presents its outputs as products of pure formal reason while covertly drawing on substantive presuppositions it cannot acknowledge.

Hegel drives home the point with a celebrated example. "That there is no property contains just as little a contradiction in itself as that this or that particular people, family, etc. does not exist, or that no human beings live at all." A contradiction arises "only with something, that is, with a content that is presupposed as a fixed principle" (Hegel 1821, §135 Anm.). If we presuppose that property exists and should be respected, then theft is contradictory; if we do not presuppose this, the universalizability test returns no result. The machine produces an output only when substantive ethical content is fed into it as a premise, and it is this content that does the real work while the formal test claims the credit. The machinery of universalizability is either *indeterminate*, delivering no result where no substantive presupposition is supplied, or *parasitic*, delivering a result that depends entirely on the substantive presupposition it purports to have derived from formal reason alone.

To be fair, the charge bites only against the categorical imperative taken in isolation as a free-standing decision procedure. Kant's own mature practical philosophy

already registers the insufficiency of the formal test considered alone. In the *Metaphysics of Morals*, the universalizability formula is specified through a principle of ethics that constrains the ends an agent may adopt, and the system of duties unfolds from two obligatory ends, one's own perfection and the happiness of others, whose content cannot be generated by the formal procedure but whose authority derives from the same principle (Bacin 2023, 8–10). As Stefano Bacin has argued, this system of ends is both complete in scope and open to further development: it determines the territory of ethical demands without pretending to enumerate every particular duty, and its organic structure allows for substantive moral progress as new demands come into view. The machine, in Kantian terms, is supplemented by a map of the ends it is meant to serve. Whether the supplement is adequate to Hegel's objection is another question. The formal procedure still does the filtering, and the ends that give it material remain, in Hegel's terms, presuppositions whose authority is established before the machine begins to operate. What the *Metaphysics of Morals* achieves is a recognition that the machine requires such presuppositions, a recognition that is itself a step toward the counter-voice's deeper insight.

Yet, we must ask whether this Kantian supplement actually defuses Hegel's objection. It arguably does not. The formal procedure still does the filtering, and the ends that give it material remain, in Hegel's terms, presuppositions whose authority is established before the machine begins to operate. What the *Metaphysics of Morals* achieves is a recognition that the machine requires such presuppositions.

Where Hegel appealed to the macro-structures of *Sittlichkeit* to reveal the unacknowledged substance powering Kant's formal machine, Bernard Williams translated this insight to the micro-level of moral vocabulary. Williams developed Hegel's critique through his distinction between *thick* and *thin* ethical concepts. Thick concepts, such as *courage*, *brutality*, *loyalty*, and *treachery*, carry substantial descriptive as well as evaluative content and require the cultivation of a certain perceptual sensitivity for their correct application to new cases. They are, in this respect, *operator-dependent*: their proper use presupposes the kind of judgement, character, and contextual familiarity that can be cultivated only through participation in specific forms of ethical life. By contrast, thin concepts, such as *right*, *wrong*, *good*, and *bad*, survive procedural abstraction. They are precisely the concepts that can be fed into a decision procedure, because they have shed the descriptive specificity that ties thick concepts to particular evaluative perspectives. The demand of ethical theory that thick concepts be reduced to or replaced by thin ones is, accordingly, a demand that tends to make ethical thought approximate the virtues of a machine. Williams himself presents his critique as "originally Hegelian" (Williams 1985, 140–141), but his thick/thin distinction gives him the vocabulary to articulate it with a different emphasis.

Consider what happens when a thick concept like *loyalty* is subjected to the universalizability machine. The agent is asked: can you universalize the maxim of acting loyally toward those to whom you are bound by ties of mutual commitment?

The question is nearly unintelligible as posed, because loyalty is precisely the kind of concept whose content varies with the particular relationships, histories, and evaluative traditions in which it is embedded. Loyalty to a friend in a time of political danger, loyalty to a comrade under fire, loyalty to a scholarly tradition that has shaped one's thinking: these are recognizably instances of the same thick concept, but the perceptual sensitivity required to exercise it cannot be reduced to a rule. The categorical imperative can formalize the thin residue: "act on universalizable maxims." What it cannot formalize is the judgement that recognizes *this* situation as one calling for loyalty rather than honest dissent, or the character that has been formed by the practices within which such discriminations are learned. The machine processes the thin concept with impeccable operator-independence. But the thick concept, which is what an agent actually needs in order to know what to do, falls through the gears.

Williams's epigraph to *Ethics and the Limits of Philosophy*, drawn from Camus, gets at a closely related point: "*Quand on n'a pas de caractère, il faut bien se donner une méthode.*" When one has no character, one has to give oneself a method. The line can be developed in two different directions. Read one way, it is an indictment: decision procedures is what people resort to in the absence of the formed character that would enable them to act well without a procedure. Read another way, it is a constructive proposal: when we cannot rely on someone's character as a basis for trust, an explicit, impersonal, and rule-governed procedure can offer a different basis for trust – then, the substitution of method for character is a real emancipation. It offers a shared currency of justification.

But where this public currency ends up displacing the perceptual sensitivity that thick ethical concepts encode and that only formed character can exercise, it impoverishes our ethical sensibility. The machine-like virtues of impartiality and rule-governance serve us well in domains where agent-independence is valuable: the adjudication of legal disputes, the testing of empirical hypotheses, or the organization of encyclopaedic repositories of knowledge. Yet they become liabilities in domains where the qualities of the agent are part of what matters and perceptiveness is as important as consistency.

The demand for systematicity may, then, be grounded in a precondition of cognition that no thinker can opt out of: systematicity is what makes cognition answerable to truth, and the demand for it is normatively binding on every cognitive act that aspires to contribute to knowledge. But it is liable to overreach in domains where the quality of the agent is part of what matters, because the very machine-like virtues that make the demand cognitively powerful are purchased at the cost of whatever resists formalization and abstraction. There is not just a floor, but also a ceiling to systematization, and the space between them is where the question "how much system?" must be negotiated in practice.

Rational Machinery: What the Genealogy Reveals

“If conflict is not a logical affliction of our thought, it must be a mistake to regard a need to eliminate conflict as a purely rational demand. ... Rather we should see such needs as there are to reduce conflict ... as having a more social and personal basis.”

– Bernard Williams

The most conspicuous answer the early modern systematizers gave when asked why knowledge must form a system was a metaphysical one: because reality itself is a system, crafted by a rational God whose creation exhibits the very coherence and principled unity that systematic thought aspires to mirror. This conviction, what the preceding chapters have called the metaphysical warrant, stands behind two centuries of philosophical innovation, from Keckermann’s Protestant textbooks through Leibniz’s dream of a *characteristica universalis* to Du Châtelet’s philosophical defence of Newtonian physics. It is what thinkers foregrounded when they reflected on the grounds of their own systematizing ambitions.

Beneath the metaphysical surface, a different answer was operative throughout, one that the warrant’s very prominence tended to conceal. The demand for system served identifiable practical purposes: it made knowledge transmissible without personal apprenticeship, made belief self-certifying and self-correcting through the structural detection of inconsistency, and made the exercise of public authority inspectable and accountable. These three functions, pedagogical, epistemological, and political, are what systematization was *doing* in the hands of the thinkers who practised it, and the machine emulation thesis is the analytical key that makes them visible as functions. Because the metaphysical warrant provided a grander reason, it overdetermined the demand for system, making these practical rationales explanatorily invisible. As long as thinkers believed that the world was a divinely ordered whole, the question “why systematize?” had a ready and seemingly exhaustive answer, and there was no occasion to look further. It was only as the warrant weakened, under Condillac’s linguistic critique, D’Alembert’s methodological sciep-

ticism, and Kant's critical transformation, that the practical rationales emerged into full view. The genealogy recovers them from beneath the metaphysical story that had concealed them, and this recovery is what makes it possible to answer the question "why systematize?" in a way that is simultaneously vindicatory and delimiting.

Eight chapters of historical argument have transformed the question. "Why systematize?" is too blunt, because it treats the demand for system as a single undifferentiated aspiration. The three rationales show that it is plural, and the degree of systematization each warrants depends on how acutely the underlying practical need is felt. The task now is to develop the answer in its fully articulated form, to show how its elements fit together, and to demonstrate that the same analytical resources that recover the rationales also generate their limits. Williams was right that the basis of the demand for system is "social and personal" rather than purely rational (1981, 81). The genealogy has specified what that social and personal basis is.

9.1 The Machine Emulation Thesis and the Structure of Substitution

Adam Smith, a lover of system himself, observed that "systems in many respects resemble machines" and that "a system is an imaginary machine invented to connect together in fancy those different movements and effects which are already in reality performed" (Smith 1795, 66). The genealogy has shown that the resemblance goes deeper than Smith's formulation suggests. Cognitive systems do not merely *resemble* machines: they aspire to the functional profile of machines, and the aspiration is what makes them cognitively powerful. A machine takes any input within its domain, processes it through fixed rules in a manner open to inspection, and delivers an output that is the same whoever operates it. The cognitive *systema*, from Keckermann's textbook through Spinoza's geometric order and Leibniz's *calculus* to Kant's architectonic, does the same. To systematize thought is to confer on it four distinguishable properties, each of which characterizes a different aspect of this machine-like functional profile.

The first is *surveyability*: the system's architecture is laid out so that its joints are visible and its connections explicit. Thought that has been externalized and organized around explicit principles becomes a structure one can navigate, inspect, and audit. The second is *rule-governance*: the system produces determinate outputs from given inputs according to fixed procedures, so that an inconsistency registers as a malfunction demanding diagnosis. The third is *universality*: the system promises to accept any subject matter as input and to be operable by anyone equipped with reason. The fourth is *impartiality*: the system's outputs are independent of the identity, perspective, or interests of whoever operates it, so that they can command

acceptance as verdicts of the subject matter's own structure rather than expressions of any particular party's standpoint.

These four properties are analytically distinguishable even though real systems exhibit them in combination. Keckermann's textbook was surveyable and universal: its architecture could be grasped synoptically, and anyone who could read could follow it. Spinoza's *Ethics* was rule-governed and impartial: its deductive apparatus processed any input according to the same demonstrative procedures, and its conclusions carried the authority of the proof, not of the prover. Leibniz's programme aspired to all four simultaneously: a universal notation operating through a logical calculus whose workings were transparent and whose verdicts were agent-independent. The four virtues are what the systematizers were conferring on thought when they built systems, and the machine emulation thesis is the recognition that these are, at bottom, the virtues of machines transferred from physical to cognitive production.

Beneath this transfer of virtues lies a pattern whose significance only the completed genealogy makes fully visible. At every stage of the narrative, systematization involves the same fundamental operation: cognitive work previously performed by formed persons, by cultivated *habitus*, situated judgment, disciplinary memory, or institutional authority, is relocated into articulated structure. The system takes over functions that had been exercised by the qualities of particular knowers and deposits them in an architecture whose operations are independent of whoever uses it. Johann Heinrich Alsted captured this operation with a precision that he may not have fully intended when he declared that "philosophy, as it exists in the mind, is a *habitus*; as it exists outside the mind, it is a *systema*." The *systema* is the exteriorization of *habitus*: the transformation of knowledge from something lodged in a particular mind, shaped by that mind's formation and inseparable from its cultivated dispositions, into something freestanding, inspectable, and operable by anyone.

This structure of substitution recurs throughout the genealogy with a regularity that is philosophically revealing. In Keckermann's Reformed classrooms, the systematic textbook substituted for the master's personal presence. In Descartes's methodical rebuilding of knowledge, rule-governed procedure substituted for the cultivated judgment of the humanist scholar. Hobbes's encounter with Euclid dramatized the conversion in a single biographical episode: the humanist rhetor, trained in the management of opinion through persuasive skill, discovered that geometric demonstration could compel assent with the impersonal necessity of a mechanism. In Spinoza's *Ethics*, the philosophical text itself was fashioned as an inferential machine: the reader was to follow the demonstrations as one follows the operations of a device, and the authority of the conclusion rested in the structure of the proof, not in the character of the philosopher. When Leibniz imagined two disputants sitting down to say *calculemus*, he was proposing to substitute calculation for the personal authority of the disputant. In the codification movement that Rousseau's

chapter examined, written law substituted for the sovereign's discretion. And Kant, completing the transformation, relocated the ground of system from divine architecture to the self-legislation of reason, internalizing the machine while retaining all of its functional properties.

In each instance, the system achieves authority by *depersonalizing* it. The cognitive authority that the system exercises derives from its structural properties, from its rule-governance, its inspectability, and its operator-independence, rather than from the standing of whoever built or operates it. Keckermann's textbook commanded attention because its logical architecture could be inspected. Spinoza's proofs commanded assent because their inferential structure was available for scrutiny. Rousseau's general will commanded legitimacy because its formal properties ensured that no particular interest had captured the legislative machinery. Depersonalization is the mechanism by which the substitution operates, and the four machine-like virtues are the specific means through which it is accomplished.

The normative consequence of recognizing this structure is considerable, because it puts one in a position to ask a question that a purely metaphysical justification of systematization never needed to pose: was what the substitution eliminated *merely* a source of distortion, or was it also a source of something valuable? The three rationales and the counter-voice that marks their limits are the genealogy's answer to this question.

9.2 The Three Rationales and Their Costs

Where personal authority was arbitrary, the system that replaced it with impersonal procedures was a genuine emancipation. The three rationales identify the practical pressures under which that emancipation proved indispensable, and the machine-like virtues it traded on. Where, however, the personal quality of the knower was itself a source of cognitive or ethical value, the substitution exacted a cost. The counter-voice tradition, from Cavendish through Diderot and Hegel to Williams, identifies the domains in which the cost predominates. What makes the framework diagnostic is that rationale and cost are generated by the same properties: the machine-like virtues that serve cognition in one domain become liabilities in another. Understanding which virtue is in play is what enables one to locate the boundary.

Historically the earliest of the three, the *pedagogical* rationale addresses the problem of making thought transmissible. Keckermann invented the cognitive *systema* under conditions that made this problem acute: the Reformation had destroyed the Catholic educational infrastructure, and Protestant universities needed to produce a trained class of pastors, teachers, and administrators within a single generation. The older model, years of apprenticeship under recognized masters, was too slow and too dependent on institutions that no longer existed. The *systema*

responded by compressing what had previously required prolonged formation into a surveyable, reproducible, and portable architecture. Its machine-like power resided in two of the four virtues: *universality*, which ensured that anyone who could read could follow the procedure, and *surveyability*, which made the architecture graspable synoptically. Together, they transformed knowledge from something embedded in a particular teacher's cultivated *habitus* into something a student could traverse, independently, in an externalized format. Du Châtelet's *Institutions de physique*, a century and a half later, deployed the same rationale at a higher level of sophistication: a systematic presentation of Newtonian physics integrated with Leibnizian-Wolffian metaphysics, designed to make the frontiers of natural philosophy accessible to a reader equipped with reason and patience.

Wherever the pedagogical rationale is pressing, the system's universality and surveyability are valuable: they are what make the difference between knowledge locked in the experience of a few masters and knowledge available to anyone who can follow a procedure. The cost becomes visible when transmissibility is treated as the sole measure of cognitive value. The *habitus*-based pedagogy that the *systema* replaced produced a kind of understanding that was slow, dialogically shaped, and irreducibly personal. The *Summa*'s *quaestio* format, as Aquinas practised it, forced both author and reader to take the strongest version of each objection seriously, and the resolution bore the marks of having survived that confrontation. A Keckermann-style *systema* covering the same material would state the doctrine, provide the definition, and move on. What was gained in speed and reproducibility was purchased at the cost of dialectical depth: the connections a learner draws in the course of sustained engagement are shaped by the particular intellect doing the drawing, by what it already knows and what it finds puzzling, and these idiosyncratic connections are part of what makes the understanding the learner's own. A system, whatever its material substrate, delivers pre-drawn connections that are the same across every copy, and this uniformity is both the source of its pedagogical power and the source of what it cannot provide.

Beyond its proper domain, the pedagogical rationale's cost deepens further. Wherever understanding requires personal transformation, the system's promise that "anyone can operate it" becomes misleading. Lambert's conjurer's bag is a diagnostic image here: a system built by abstracting from the richly particular may deliver back only the generality that was put in. The universality is real, but it is a universality of *procedure*, and what is lost in the translation from formed understanding to procedural knowledge may include precisely the elements that made the understanding valuable. Cavendish registered this loss in the domain of natural philosophy, where the machine model's imposition of external mechanical order on self-moving, self-knowing matter screens out precisely the features of reality that a vitalist ontology takes to be fundamental. The resistance is ontological: nature is richer than mechanism allows, and adequate knowledge of nature must be *responsive* to its own self-activity. Translated into the terms of the pedagogical

rationale, Cavendish's critique identifies a domain where the system's universality of access dissolves the role of the kind of perception that only sustained, receptive engagement with the subject matter can cultivate.

The second rationale is *epistemological*: systematization makes thought self-certifying and self-correcting. In a rule-governed system, inconsistency registers as a malfunction demanding diagnosis: a claim that generates a contradiction with established results cannot simply be left in place. In an impartial system, the verdict commands acceptance because it does not express anyone's personal opinion. Coherence thereby becomes a credible test of truth, operating in two directions: prospectively, a claim earns credibility by integrating into the deductive structure, and retrospectively, a claim that generates inconsistency is flagged for revision. The system is simultaneously a certification mechanism and an error-detection device.

Descartes's method, Spinoza's deductive web, and Newton's rules of reasoning all exemplify this rationale with increasing sophistication. The sceptical crisis that drove the seventeenth century's philosophical innovation raised the question of whether systematization could *produce* certainty, and the geometric method was the answer. Yet the epistemological rationale's most instructive episode is its exercise in the retrospective, self-correcting direction. Condillac's demolition of the rationalist systems is an instance of the epistemological rationale turned against the very systems that claimed to embody it. The systems of Descartes, Malebranche, Spinoza, and Leibniz had presented themselves as coherent deductive structures whose internal consistency was a mark of their truth. Condillac accepted the criterion and showed that the systems failed to meet it: their apparent coherence was linguistic rather than philosophical, sustained by terms whose semantic content had never been properly analysed. The rational machinery was fully assembled, and its moving parts harmoniously interlocked, but its building-blocks were devoid of real meaning. The *esprit de système* was diagnosed by means of the *esprit systématique*: the machine was used to identify its own malfunction.

This self-correcting capacity is a genuine virtue of the epistemological rationale. Particular systems may calcify into dogma, as the Enlightenment demonstrated. The epistemological rationale as such, however, carries within it the resources for diagnosing that calcification, because the demand for consistency, applied with diagnostic honesty, turns against any system whose coherence is purchased at the cost of empirical anchorage or conceptual clarity. The machine can diagnose its own malfunction precisely because rule-governance is indifferent to the provenance of the inputs: it flags inconsistency wherever inconsistency occurs, including in the system's own foundations.

Where self-certification is detached from empirical control, the cost becomes visible. Condillac's gambler, who plays without knowing the value of the tokens, is the emblem: a system that is internally consistent may be entirely empty. C. Thi Nguyen's analysis of what he calls "the seductions of clarity" identifies the psycholo-

gical mechanism at work in such cases (Nguyen 2021). The sense of clarity, Nguyen argues, functions as a heuristic for allocating cognitive resources: when something feels clear to us, we treat this as a signal that we have investigated enough. A system whose terms connect fluently, whose deductions follow, and whose architecture exhibits pleasing regularity produces a powerful experience of cognitive facility, the ease with which one can generate explanations and draw connections within the system's own categories. This facility is real, but it can be mistaken for genuine understanding. When the system's internal consistency triggers the clarity heuristic, inquiry terminates before the inquirer has detected whether the terms are anchored in anything outside the system. What Condillac exposed in the rationalist metaphysics, Nguyen identifies as a general structural vulnerability: systems that produce an exaggerated sense of clarity can function as thought-terminators, bringing investigation to a halt precisely where the most critical questions remain unasked.¹

Diderot carried the counter-voice into the Enlightenment with a shift of register. His resistance concerns the texture of *thought itself*. In the *Rêve de d'Alembert*, the great systematizer is rendered unconscious, and in his fever dream he articulates a vision of nature that his waking, geometric intellect could never have produced. The literary form enacts the philosophical thesis: knowledge distributed across perspectives that cannot be unified into a single authoritative voice is knowledge that resists the very systematization D'Alembert had championed. Diderot's *esprit de divination*, the trained intuition by which a natural philosopher sniffs out undiscovered results, names the cognitive capacity that rule-governance cannot formalize: the back-and-forth between sense and reflection, the bee's ceaseless labour of going into oneself and coming out again. What Diderot identified is the cost of rule-governance at the level of inquiry itself. A rule-governed system processes inputs according to fixed procedures, and its outputs carry the authority of the procedure. What the procedure cannot capture is the movement of thought in its most creative mode, the mode in which the thinker's particular, embodied engagement with the material generates insights that no rule could have prescribed in advance. Cartwright's "dappled world" and Dupré's "disorder of things" press the point further: if some domains of inquiry resist unification without forfeiting their claim to knowledge, then the epistemological rationale has natural limits even where its terms are empirically anchored. The demand that all knowledge form a single coherent system may itself be an expression of the *esprit de système*, driven by the metaphysical conviction that reality is one, rather than by the practical need for internal error-detection.

The third rationale is *political*: systematization makes the exercise of public authority accountable, administrable, and fair. The machine-like virtues it trades on are impartiality and universality. In an impartial system, the law applies the same

¹For a detailed philosophical account of how the clarity of systematized values can displace richer, less articulate evaluative resources, see Nguyen (2024; 2026).

principle to every case, and the verdict is the system's rather than the judge's. In a universal system, anyone affected can follow and assess the reasoning: the justification for a decision is publicly articulable, available for inspection by any citizen competent to evaluate it. Together, these two properties make governance auditable: like cases are treated alike by an operator-independent logic, and departures from principle can be detected and challenged.

Hobbes already extended the machine model from cognition to the state when he conceived the Leviathan as an "artificial man" whose legitimacy derived from rational construction rather than tradition. Leibniz's ambition to place jurisprudence on scientific foundations initiated a trajectory that would culminate in the great European codification movement: the Prussian *Allgemeines Landrecht* of 1794, the French *Code civil* of 1804, and their successors, each aspiring to replace the fragmented, locally variable, and often opaque body of customary law with a unified legal text whose authority derived from its formal properties. Rousseau brought the political rationale to its fullest philosophical articulation. The general will's three formal conditions, universality of source, universality of application, and orientation toward the common good, are the machine-like virtues in their most concentrated political expression. Williams supplied the explicit philosophical defence: in a modern open society whose members do not share a single evaluative perspective, public agencies must be "governed by an explicable order which allows those agencies to be answerable" (Williams 1985, 40). The demand that public justification be articulable and inspectable is the political face of systematization, and within its proper home it is among the strongest reasons for systematizing that the genealogy has recovered.

Rousseau's *Second Discourse* also provides the pathology report. The founding deception that inaugurates civil society deploys every formal property of the general will: universality of obligation, absence of exceptions, the impartial subjection of powerful and weak alike to shared duties. All ran "toward their chains in the belief that they were securing their freedom" (Rousseau 1755/1997, 173). What makes the founding deception effective is precisely the formal perfection of its rhetoric: the language of systematic governance renders the particular interest it serves structurally invisible. The same formal properties that make the political machine legitimate when genuine can be captured and deployed to *legitimate domination*. Even the legal practice that most ambitiously embodied the political rationale illustrates the limit from within. The great codes, once enacted, came to be regarded as comprehensive and closed systems of rules, generating what Zimmermann describes as "an inward-looking scholarship of an almost exclusively exegetical nature" (Zimmermann 2012, 381). Yet no codification has ever fully complied with the ideal of completeness, and the *Code civil*'s architect Portalis designed its provisions as principles to be developed through judicial interpretation. The code *presupposes* the informal judgment it was constructed to displace. The boundary between what the political machine can settle and what requires the

situated judgment of particular persons is itself a matter of judgment that no code can formalize.

Hegel brought this line of critique to its deepest articulation. Kant's categorical imperative is, in the terms of the machine emulation thesis, a cognitive machine for producing moral judgments: it takes a proposed maxim as input, applies the universalizability test, and delivers a determinate output regardless of who operates it. Operator-independence is the whole point. Hegel's objection is that this machinery, because it abstracts from the concrete ethical substance of particular communities and relationships, produces outputs that are either indeterminate or parasitic on the ethical life it purports to transcend. "That there is no property contains just as little a contradiction in itself as that this or that particular people, family, etc. does not exist": a contradiction arises only with something, a content presupposed as a fixed principle (Hegel 1821/2009, §135 Anm.). The machine processes vacuously unless its inputs have already been shaped by the very traditions the machine claims to adjudicate.

Williams sharpened this Hegelian insight into the distinction between thick and thin ethical concepts that has structured much of the book's argument. Thick concepts, such as courage, brutality, loyalty, and treachery, carry substantial descriptive content and require perceptual sensitivity for their application: their proper use presupposes the kind of judgment, character, and contextual familiarity that can be cultivated only through participation in specific forms of ethical life. They are *operator-dependent* in a way that thin concepts, such as right, wrong, and good, are not. When a thick concept like loyalty is fed into the universalizability machine, the concept's content, which varies with the particular relationships, histories, and evaluative traditions in which it is embedded, falls through the gears. The machine can formalize the thin residue: act on universalizable maxims. What it cannot formalize is the judgment that recognizes *this* situation as one calling for loyalty rather than honest dissent, or the character that has been formed by the practices within which such discriminations are learned. Williams acknowledged that "there is a political argument for public justifications to take these forms, and to deploy, correspondingly, 'thin' concepts" (Williams 1985, 40). Beyond that domain, the demand for thin, publicly articulable principles becomes a distinctive form of impoverishment. Geuss's "universalisation machine" names the pathology: the conviction that every human decision should be processed through a formal decision procedure whose outputs are agent-independent, as though the whole of life were a matter for public administration.

The counter-voice deepens as it develops across the genealogy, and the deepening is not accidental. It tracks the increasing intimacy of the cost to the agent's own formation as a thinking and acting being. Cavendish's cost concerns the *world*: nature is richer than mechanism allows. Diderot's concerns *thought*: inquiry at its most creative operates in a mode that resists formalization. Rousseau's concerns *political life*: the machinery of principled governance can entrench domination by

rendering particular interest invisible. Hegel's and Williams's concerns the *agent*: the character, perceptual sensitivity, and cultivated judgment on which good action depends are dissolved by the demand for formal, agent-independent principles. Each later articulation identifies a cost that is closer to the thinker's own activity as a thinking and acting being, and the progression from ontological to ethical is a progression toward the point at which the machine model touches what is most intimately at stake in the question of how to live.

A framework that is simultaneously vindictory and delimiting is possible because vindication and delimitation draw on the same analytical resources. Each virtue that grounds a rationale is also the source of a characteristic liability, and the counter-voice figures are the thinkers who identified the domains in which the liability predominates. Universality, which grounds the pedagogical rationale's capacity to transmit knowledge at scale, becomes a liability wherever understanding is irreducibly experiential and the system's promise that "anyone can operate it" is misleading. Rule-governance, which grounds the epistemological rationale's capacity for self-correction, becomes a liability wherever the phenomena under investigation are too fluid or too polyphonic for fixed procedures. Impartiality, which grounds the political rationale's demand for agent-independent verdicts, becomes a liability wherever the quality of the agent is part of what matters and the demand for operator-independent outputs suppresses precisely the situated judgment the domain requires. The costs are internal to the framework, generated by the same properties that produce the benefits. Williams put the point with characteristic economy: "to demand that [our moral sentiments] be schooled by the requirement of system is to alter our moral perception of the world, not just to make it in some incontestable sense more rational" (Williams 1980, 10–11). The alteration is a structural consequence of the machine-like virtues on which the rationales depend, and recognizing it is part of understanding what those virtues are.

C. Thi Nguyen's concept of *value capture* provides a contemporary philosophical vocabulary for the mechanism by which these costs are incurred beyond the domains in which the counter-voice tradition first identified them (Nguyen 2024). Value capture occurs when an agent adopts a simplified, standardized expression of value and permits it to dominate practical reasoning without adapting it to the agent's particular context. The concept was developed through a study of institutional metrics: step counts, citation rates, law school rankings, and the quantified indicators that large-scale organizations use to manage information across vast domains. In each case, the metric offers a trade-off. The agent gains clarity, communicability, and motivational ease: successes become measurable, justifications become straightforward, and the complexity of plural, context-sensitive values is replaced by a single legible standard. In exchange, the agent surrenders fine-grained control over the articulation of values that would enable them to be tailored to the agent's particular personality, circumstances, and context. What makes the

trade-off persistent is that the properties which make a metric useful for large-scale institutional coordination, its standardization, its context-invariance, its portability across institutional levels, are the same properties that make it resistant to local tailoring. A metric's institutional function *requires* it to be inflexible. The inflexibility that makes it usable at scale is the inflexibility that makes it a poor guide for any particular individual's flourishing. In the terms of the genealogy, value capture describes the contemporary institutional form of a cost that the machine emulation thesis makes structurally predictable: wherever the machine-like virtues of universality, surveyability, and impartiality are extended beyond the domains in which the practical rationales give them their warrant, they colonize the agent's evaluative life in ways that the framework identifies as characteristic overreach.

9.3 The Precondition and the Diagnostic Framework

The three rationales are analytically distinct, and the distinction is what makes the framework diagnostic. Yet they are also functionally interconnected. Historically, the ordering in which they emerged, pedagogical first, then epistemological, then political, tracks a broadening of the domain in which systematization is put to work: from transmitting knowledge, to certifying it, to governing with it. It also tracks a pattern of increasing functional dependence. The pedagogical rationale requires no support from the others: one can systematize knowledge for transmission without aspiring to epistemological self-correction, as Keckermann's textbooks organized doctrine assumed to be reliable. The epistemological rationale draws on the functional achievements of the pedagogical, because using coherence as a test of truth requires that the structure within which coherence is being assessed be articulated with sufficient clarity for inconsistency to register: one needs a surveyable architecture before one can diagnose a malfunction within it. The political rationale draws on both. Public authority is accountable only if its reasoning is transmissible, so that citizens can follow it, and testable against standards of consistency, so that departures from principle are detectable. These dependencies explain something about the historical sequence that a purely chronological narrative might leave unexplained: the rationales did not emerge in arbitrary order, and the later ones presupposed the functional achievements of the earlier.

Beneath all three rationales, and beneath the metaphysical warrant that historically accompanied them, lies a further layer that only Kant's critical transformation fully uncovered: the precondition of cognition. Some minimal degree of systematicity, subjecting one's judgments to the regulative pull of consistency and coherence, is a condition of thought itself. The question "should I systematize at all?" does not arise at this level, because whoever poses the question has already answered it in the affirmative by the very act of posing it. This settles the dispute with the most radical critics: the suggestion that the demand for system is noth-

ing more than “tidy-mindedness” mistakes the floor for the furniture. Where the precondition operates, there is nothing to negotiate.

Above the floor, everything turns on the practical rationales: which of them is at stake, how pressing the underlying need is, and whether the domain is one in which the machine-like virtues the rationale trades on are serviceable. These are questions whose answers will differ from case to case. Lambert’s purpose-relative conception of system provides the essential diagnostic resource. If a system is a purposively composed whole, then the ordering of its parts depends on what the system is *for*, and different purposes generate different orderings of the same material. A body of law must be systematized for public inspectability: its organizing principle is the demand that any citizen be able to follow the reasoning behind a decision. A body of scientific theory must be systematized for internal coherence: its organizing principle is the demand that inconsistency register as malfunction. A body of teaching material must be systematized for transmission: its organizing principle is the demand that learners traverse the material efficiently. These three organizing principles may pull in different directions when applied to the same material. Legal reasoning that is maximally inspectable may be epistemologically crude. Scientific reasoning that is maximally self-correcting may be pedagogically impenetrable. Recognizing which purpose governs the systematization at hand is what enables one to assess whether the degree of system is appropriate.

D’Alembert’s distinction between the *esprit de système* and the *esprit systématique* generalizes across all three rationales as a diagnostic for pathology. In every domain, systematization can become detached from the practical need it was meant to serve. When it does, it produces what is recognizably a malfunction of the relevant rationale. In the pedagogical domain, the *esprit de système* takes the form of transmissibility pursued at the cost of depth: the system becomes easy to traverse and empty of formative engagement, delivering what Kant called “plaster casts of a living human being” (KrV A836/B864). In the epistemological domain, it takes the form Condillac diagnosed: formal coherence sustained by terms that have never been anchored in experience. In the political domain, it takes the form Rousseau identified: the rhetoric of universality deployed in the service of particular advantage. In each case, the system is running according to its own internal logic, processing inputs and producing outputs with the impersonal regularity of a well-designed machine, but it has lost contact with the domain it was designed to organize.

Recognizing which side of this boundary a given exercise of systematization falls on requires precisely the contextual judgment that the machine model was designed to render unnecessary. This is the productive paradox at the heart of the framework: the question of whether systematization has gone far enough, or too far, is itself a question that no system can answer. The framework yields no formula, and the absence of a formula is itself part of the answer: a diagnostic algorithm for determining “how much system?” would commit the very overreach the genealogy has identified. Nguyen’s concept of *value federalism* captures a cognate insight in

a contemporary register (Nguyen 2024; 2026). Some values are best pursued at the largest scale, with maximum standardization and impersonality: where the gains from coordination across a vast domain outweigh the costs of contextual insensitivity, the machine-like virtues are serviceable. Other values are best pursued at the most local scale, with maximum sensitivity to what is particular and situated: where the gains from tailoring outweigh the gains from portability, the machine-like virtues overreach. The question of where to draw the line is always a matter for judgment, and the right answer will differ by domain, by context, and by the practical pressure that is doing the work. What the framework provides is an articulated awareness of what considerations bear on the judgment, and what is at stake in getting it wrong.

A demand that grows out of, and varies with, the practical situations of those who confront it: understanding the demand and its limits is what thinking well about systematization requires.

9.4 System and Character

The book began with philosophers who aimed at “cleaning up the litter” (James 1975–88, vol. iv, 26). It has traced the history through which that ambition was given institutional form, philosophical justification, and critical examination, from the Reformed classrooms where Keckermann first reorganized doctrine as a *systema* through the great rationalist architectures and their Enlightenment critique to Kant’s transformation of the systematicity ideal and Hegel’s rejoinder. The litter that James observed philosophers tidying up turns out to be the texture of thought before it has been made machine-like. The tidy conception that replaces it is a cognitive architecture whose functional properties are the properties of a well-designed machine. The genealogy has shown that the replacement serves genuine purposes: it makes thought transmissible, self-correcting, and accountable. It has also shown that the replacement carries costs whose depth increases the further the machine model is extended from the domains in which it is at home into the domains where character, perception, and situated judgment do the cognitive and ethical work.

Nietzsche and Camus suggested that systematicity is a substitute for character. The genealogy has recovered a significant sense in which that is true, and it has specified the conditions under which the substitution is warranted. The *systema* really did emerge as a replacement for *habitus*, and the replacement really was, in many contexts, an achievement. Where personal authority was arbitrary, the system that replaced it with impersonal procedures was a genuine emancipation. Keckermann’s textbook freed knowledge from the contingencies of individual teachers. Descartes’s method freed inquiry from the vagaries of personal intuition. The

codification of law freed governance from the caprice of the sovereign. These were real gains, earned under real pressures.

Where the substitution reaches its limit is where character was also a source of something valuable: the settled dispositions of a formed person whose judgment has been shaped by sustained engagement with the material, whose perception has been educated by the practices within which thick ethical concepts have their life, and whose responsiveness to the particular is itself a form of knowledge that no externalized architecture can encode. Williams heard in the philosophical demand for system the “rattle of machinery” (Williams 2006, 120) that signals the point at which the pursuit of systematicity has outrun what the practical rationales warrant. Philosophy, he insisted, needs to be on the lookout for that sound. The genealogy has provided a way of identifying what produces it: the rattle begins when the machine-like virtues that serve cognition well in their proper domain are extended into domains where they become liabilities, when impartiality overrides the ethically relevant particular, when rule-governance flattens what is irreducibly fluid, when universality of access dissolves the role of trained perception, and when surveyability gives the appearance of comprehension where the reality is reductive simplification.

Between system and character, the genealogy does not resolve the tension. It equips us to navigate it, and it provides tools whose bearing on the present is immediate. For we now live in a world in which the early modern entanglement of systematization and mechanization has been realized in a form that the early modern thinkers could not have envisaged. The question the genealogy has been training us to ask, which rationales apply, how much systematization do they call for, and what are the costs, is the question to which the Epilogue turns.

10

Epilogue: From Leibniz's Dream to Ours

10.1 Notes for enriching the Epilogue

We see, in present-day artificial neural networks, a new model for what machines can be. And this model turns the machine emulation thesis on its head. The machine now emulates the human mind rather than the other way round. Moreover, the fact that humans have sought to develop machines towards the kinds of properties that ANNs have is revealing. It vindicates several of the broader lessons we have distilled from this genealogy. One such lesson is that externalizing thought in a free-standing architecture based on rigid rules comes at a cost: the loss of the cultivated dispositions of character and everything they involve. Pursuing systematicity at the expense of character has turned out to be a costly strategy. Anthropic's eschewal, in their alignment strategy, of fixed rules in favour of the cultivation of Claude's character looks like a vindication of the character-substitution thesis. It offers an example of a machine that seeks to remedy the shortcomings of rule-based systems by recovering the power of cultivated dispositions. It is also an example of a machine that is increasingly autonomous rather than heteronomous, vindicating the Kantian point about what systems should be like.

10.2 Current Opening of the Epilogue

Chapter 1 described how the transition from *Summa* to *systema* instantiated a pattern that recurs whenever new technologies of externalization promise to compress the labour of understanding. The *Summa* had preserved the dialectical process through which conclusions were reached: the *quaestio*, the marshalling of objections, the *sed contra*, the painstaking accommodation of countervailing considerations. When a reader worked through the *Summa*, the path to the conclusion was itself part of what was understood. The Keckermann-style textbook compressed that path. It laid out the definitions, drew the divisions, and guided the novice from principle to application, promising to make the conclusions of extended inquiry

available to learners who had not themselves undertaken the inquiry. What was gained was efficiency, surveyability, and scale. What was lost was harder to display, because it consisted in the kind of understanding that only the longer path could have cultivated: the *habitus* that takes shape in a particular mind through a particular encounter with the material, and that no externalized architecture can fully encode.

Each subsequent technology of knowledge management has renewed this promise in its own idiom. The encyclopaedia, the card index, the searchable database, and the algorithmically generated summary each makes it easier to arrive at conclusions without traversing the terrain from which they were won. And each borrows, in its own register, the virtues that the *systema* first aspired to realize: the machine-like properties of efficiency, surveyability, impersonality, and reproducibility that the genealogy has been tracking across three centuries. In the Keckermann-style textbook, these properties were figurative: the *systema* gave thought the *functional profile* of a machine. In a large language model, they are literal. The machine has arrived.

A large language model can produce fluent, internally coherent, well-organized text on virtually any subject, delivering in seconds what would take a human inquirer hours, weeks, or years to assemble. It takes any prompt as input, processes it through statistical regularities extracted from an immense corpus, and delivers an output whose form is independent of who prompted it. The connection to the genealogy is through the machine emulation thesis itself. What an LLM shares with Keckermann's *systema* and Leibniz's *calculus* is the aspiration to make the products of reasoning scalable, operator-independent, and detachable from the cognitive biography of the individual thinker. The system produces outputs that are coherent, organized, and syntactically well-formed regardless of who prompts it. It is, in the precise sense this book has been tracking, a machine for systematizing thought.

And, like every predecessor technology in the pattern Chapter 1 identified, it carries the same animating hope: that the intellectual path can be shortened without significant loss, because what ultimately matters is the product of inquiry rather than the process by which it was forged.

The genealogy has been exploring this hope for eight chapters, and the framework it has yielded provides the tools for assessing it. The three practical rationales identify what cognitive systematization achieves: the pedagogical rationale explains why making thought transmissible is valuable, the epistemological rationale explains why making thought self-correcting through structural coherence is powerful, and the political rationale explains why making thought publicly inspectable is necessary. These are real gains, earned under real pressures. Keckermann's textbook freed knowledge from the contingencies of individual teachers. The geometric method freed inquiry from the vagaries of personal intuition. The codification of law freed governance from the caprice of the sovereign. AI-mediated systematization extends

each of these achievements to a scale its predecessors could not have imagined, and the rationales explain why this is, in many contexts, a genuine service.

Yet the same framework also identifies the costs. The machine-like virtues that ground each rationale are the same properties that produce its characteristic liability when the rationale is pushed beyond its proper domain. Universality, which grounds the pedagogical rationale's promise that anyone can follow the procedure, becomes a liability wherever understanding is irreducibly shaped by the particular mind doing the understanding. Rule-governance, which grounds the epistemological rationale's capacity to flag inconsistency as malfunction, becomes a liability wherever the phenomena under investigation are too fluid or too context-dependent for fixed procedures. Impartiality, which grounds the political rationale's demand for agent-independent verdicts, becomes a liability wherever the quality of the agent is part of what matters. The counter-voice tradition, from Cavendish's vitalist challenge to mechanism through Diderot's polyphonic resistance to encyclopaedic reduction and Hegel's critique of the Categorical Imperative as rational machinery that impoverishes ethical life, has been identifying these costs throughout. Williams compressed their insight into a formula: "to demand that [our thoughts and feelings] be schooled by the requirement of system is to alter our moral perception of the world, not just to make it in some incontestable sense more rational" (2006, 162). These costs do not disappear when the system is operated by a machine rather than a human systematizer. They intensify.

They intensify because the disanalogy between early modern cognitive systems and contemporary AI is as important as the analogy. The early modern systematizers aspired to make thought machine-like by making it *more* systematic: more rule-governed, more transparent, more inspectable, more principled. The result, at its best, was a structure whose workings could be followed step by step, whose principles could be assessed, and whose conclusions could be challenged by anyone who mastered the method. A large language model makes thought machine-like by making it *less* transparent. It produces coherent outputs from processes that are opaque even to its designers, and whose internal workings cannot be reconstructed as a chain of reasons leading from premises to conclusions. The early modern *systema* externalized reasoning in a form that invited inspection. The LLM externalizes the *products* of reasoning in a form that resists it. What it shares with the tradition this book has traced is the aspiration to scalable, operator-independent systematization. What it lacks is the inspectability that the early modern systematizers understood as essential to the enterprise.

This matters for a specific reason that the genealogy makes visible. Inspectability was never incidental to the systematicity ideal. It was constitutive of the practical work that systematization performed. The pedagogical rationale required surveyability: the system's architecture had to be graspable synoptically if it was to function as a teaching machine. The epistemological rationale required rule-governance: the system's procedures had to be followable if inconsistency was to register as

malfunction. The political rationale required public articulability: the system's reasoning had to be accessible to those over whom authority was exercised. In each case, the demand for inspectability was internal to the rationale, part of what made systematization useful for the purpose in question. A system whose outputs cannot be inspected can still serve certain functions, but it cannot serve the functions that the early modern ideal of systematic thought was designed to serve. The genealogy identifies a form of rational machinery whose value was inseparable from its transparency. The AI that now confronts us is a form of rational machinery whose power is inseparable from its opacity. The question the genealogy equips us to ask is what this combination of vast systematizing power and radical opacity means for each of the practical rationales that justify the demand for system in the first place.

Consider, first, the pedagogical rationale. AI's capacity to organize, summarize, and tailor knowledge for transmission is perhaps its most immediately valuable contribution. A well-prompted language model can present a complex domain in a form that is surveyable, sequenced for the learner, and calibrated to the learner's prior knowledge. Du Châtelet would have recognized the aspiration. Her *Institutions de physique*, as Chapter 4 showed, was itself an attempt to do what AI tutoring systems now do at industrial scale: synthesize the best available knowledge into an architecture that a learner can traverse without having to reconstruct the entire edifice from the ground up. The pedagogical rationale vindicates this ambition. Knowledge that cannot be transmitted efficiently is knowledge that cannot accumulate across generations, and systematization in the service of transmission is one of the oldest and most defensible achievements of rational machinery.

Yet the analysis of Chapter 1 applies with undiminished force. A system delivers pre-drawn connections. This is what makes it powerful as a technology of transmission: the same connections are delivered across every copy and every user. It is also what constrains its capacity to be formative. Aquinas's *ad argumenta* forced readers to confront the strongest available objections, and to appreciate why they carried weight, because sustained engagement with countervailing considerations leaves the mind altered in ways that a distillation of the results cannot replicate. When a Keckermann-style textbook compressed that dialectical process into a synoptic table, the compression gained efficiency at the cost of formative engagement. When AI compresses an even larger body of inquiry into a fluent paragraph, the compression is carried further, and the cost, though harder to see, is structurally the same.

Knowledge, as Chapter 1 argued, is more than a collection of propositions. It encompasses embodied intellectual dispositions, cultivated through sustained engagement, that shape the learner's capacity to notice things, draw connections, anticipate objections, and weigh their force. The connections a learner draws in the course of that engagement are shaped by the particular intellect doing the drawing: by what it already knows, by what it finds puzzling, by the unexpected adjacencies it discovers along the way. A system, whatever its material substrate, forecloses

these idiosyncratic pathways by delivering a finished network of connections. The learner's understanding becomes something received rather than something individually constructed. When the system in question is a printed textbook, the foreclosure is partial: the learner still reads, pauses, re-reads, and rethinks, and the slowness of the medium permits at least some of the formative engagement the system otherwise compresses out. When the system is an LLM that delivers its output in seconds, in prose that reads as though a knowledgeable interlocutor has already done the thinking, the foreclosure deepens. The very fluency of the output conspires against the productive friction that difficulty provides.

None of this amounts to a case against using AI for pedagogical purposes. The pedagogical rationale is genuine, and the cases in which efficient transmission matters are legion. It amounts to a reminder that the pedagogical rationale has a characteristic cost, one that the genealogy identified at the moment of the *systema*'s invention and that every subsequent technology of externalization has renewed: systematization for the sake of transmission is valuable, and its value coexists with a genuine loss. The question is where the balance falls, and the framework provides no algorithmic procedure for determining this. It provides, instead, an articulated awareness of what is at stake.

Consider, next, the epistemological rationale. Systematic coherence serves as a diagnostic instrument: it makes inconsistency visible by giving thought a structure within which conflicting commitments register as malfunctions demanding repair. AI can serve this function with a thoroughness no individual inquirer can match, integrating vast bodies of claims and flagging tensions among them. The epistemological rationale vindicates this ambition too. Condillac used the demand for systematic coherence to expose the rationalist systems' failure to anchor their terms in experience. Lambert generalized the demand into a formal criterion of systematic adequacy. Kant elevated it into a regulative principle of reason. In each case, the demand for coherence served as an instrument for detecting error and guiding the revision of belief.

Yet the genealogy also documented the pathology to which this rationale is susceptible. The *esprit de système*, as D'Alembert and Condillac diagnosed it, is the systematic aspiration detached from empirical control: a system that sustains its apparent coherence through terms that have never been anchored in experience, whose unity is linguistic rather than philosophical. The rationalist systems that Condillac exposed were *internally* coherent; their definitions connected smoothly, their deductions followed from their premises, and their architecture exhibited a satisfying regularity. Their failure lay elsewhere: the coherence was purchased by semantic indeterminacy, and the terms that connected so fluently within the system did not connect to anything outside it.

AI-generated text reproduces this pathology at scale. A large language model is optimized for fluency and internal consistency. Its outputs read as though produced by a mind that has grasped the subject: the sentences follow from one another

with apparent logical necessity, the qualifications are appropriately placed, and the overall structure exhibits a pleasing coherence. C. Thi Nguyen's analysis of how quantified systems produce what he calls "seductive clarity" identifies the mechanism at work (Nguyen 2021). The sense of clarity, Nguyen argues, functions as a heuristic for allocating cognitive resources: when something feels clear to us, we treat this as a signal that we have investigated enough. This heuristic is useful but exploitable. Systems that produce an *exaggerated* sense of clarity can trigger the heuristic prematurely, bringing inquiry to a halt before the inquirer has detected the flaws in what is on offer. The feeling of cognitive facility, the ease with which one can generate explanations and draw connections using the system's categories, is precisely what quantified institutional systems and conspiracy theories alike exploit: they provide genuine facility *within their own terms*, and the facility is what masks the impoverishment of the underlying categories.

LLM outputs are, by design, optimized for exactly this kind of fluency. Their sentences flow, their paragraphs cohere, and the experience of reading them is one of apparent cognitive facility: everything seems to connect, everything seems to make sense. The clarity heuristic fires, and inquiry terminates. In Condillac's terms, the system sustains a linguistic coherence that can be mistaken for philosophical coherence. In the framework of this book, the epistemological rationale's diagnostic instrument has been turned against itself. Coherence was meant to serve as a *test* of truth, a structural property whose absence signals malfunction. When coherence is produced by a statistical process optimized for plausibility rather than accuracy, the test misfires: it registers a pass where no genuine systematic achievement warrants one.

The genealogy equips us to recognise this as a familiar failure mode rather than a novel one. What Condillac exposed in the Wolffian systems, what D'Alembert codified in the distinction between the *esprit de système* and the *esprit systématique*, is the same structural vulnerability that AI-generated text exploits at immeasurably greater scale: coherence that simulates understanding without the cognitive labour from which understanding derives. The difference in scale matters. A Wolffian treatise could be read critically by a trained philosopher who knew where to look for the unstable joints. AI-generated output arrives at a volume and velocity that strains any individual's capacity for critical assessment. The epistemological rationale's demand for coherence remains sound. Its fulfilment by a machine that can simulate coherence without producing it is what the genealogy identifies as a form of the *esprit de système* whose seductive power exceeds anything the eighteenth century could have imagined.

The political rationale raises a different set of questions, and the genealogy introduces a productive complication. Hobbes demanded that political authority be grounded in a rational construction whose reasoning any citizen could follow. Rousseau showed that the demand for systematic governance could function without the metaphysical warrant, and also that it could entrench domination

under the appearance of universality. Kant brought the political rationale to its fullest articulation by insisting that the exercise of public reason requires principles whose credentials are inspectable by anyone who subjects them to rational scrutiny. Throughout this history, the demand for public articulability has been one of the most compelling achievements of rational machinery: where authority is exercised over those who have a right to understand the reasoning behind it, the system's capacity to make that reasoning transparent is a genuine safeguard against arbitrariness and abuse.

AI puts this rationale under immediate pressure. When an algorithm determines a credit score, a sentencing recommendation, or an insurance premium, the political rationale demands that the reasoning be accessible to those affected by the decision. Yet the processes that produce these outputs are, in many cases, opaque even to the engineers who designed them. A neural network trained on millions of data points does not produce a chain of reasons that can be reconstructed and assessed in the way that a Kantian argument or a codified statute can. The political rationale's demand for inspectability is violated in its most demanding form: citizens are subject to decisions whose grounds they cannot examine, challenge, or even understand.

Nguyen's analysis of transparency adds a further dimension to this diagnosis (Nguyen 2022). One might suppose that the solution is straightforward: make the AI transparent. Force the system to articulate its reasoning in publicly accessible terms, and the political rationale's demand will be satisfied. Yet Nguyen has shown, through a careful study of what happens when expert domains are subjected to regimes of public transparency, that the demand for inspectability can itself become a source of distortion. When experts are required to justify their actions in terms accessible to non-experts, the demand pressures them to act only in those ways for which publicly articulable justifications are available. The effect is to leash expertise: to confine expert practice to the range of actions that admit of non-expert justification, screening out precisely the trained sensitivity, embodied judgment, and contextual responsiveness that made the expertise valuable in the first place.

Nguyen calls this the "epistemic intrusion argument." The demand for transparency does not merely force experts to *report* differently; it can change what experts *do*, by restricting the considerations they can act on to those that survive translation into a publicly comprehensible format. Qualitative judgments grounded in long experience are replaced by quantified metrics that can be audited and aggregated. The reasoning becomes inspectable, but what is inspected is an impoverished version of the reasoning that would have occurred in the absence of the transparency regime. The cost-benefit analysis that Theodore Porter traced from the Army Corps of Engineers to the centres of political decision-making exemplifies this pattern: a richer, multi-dimensional evaluative process was compressed into a single quantified calculation, precisely because the quantified format could survive Congressional scrutiny in a way that qualitative expert assessment could not (Porter 1995).

This analysis maps onto the political rationale's characteristic cost as the genealogy has tracked it. Impartiality, the machine-like virtue that grounds the political demand for agent-independent verdicts, becomes a liability wherever the quality of the agent is part of what matters, and the demand for publicly articulable outputs suppresses the situated judgment the domain requires. Rousseau identified this tension in the political domain: the general will, constructed through the systematic abstraction of particular interests, can function as a tool of domination precisely because it presents the interests of some as the interest of all. Hegel deepened the critique by arguing that the Categorical Imperative, as a purely formal principle of practical reason, actively damages ethical life by abstracting from the concrete substance of particular communities. Williams compressed the point: the demand for thin, systematizable concepts in place of thick, perception-laden ones alters moral experience, and the alteration is a loss. In each case, the cost arises from the same property that generates the benefit. Public articulability makes authority accountable, and it also restricts the forms of reasoning that authority can draw upon.

AI sharpens this tension without resolving it. An opaque AI system violates the political rationale by denying inspectability. A fully transparent AI system, one whose reasoning has been forced into publicly articulable categories, may satisfy the demand for inspectability while sacrificing the complexity that made the system's outputs useful. The genealogy identifies this as a structural feature of the political rationale, present since its inception: trust and transparency are in essential tension, and no institutional arrangement eliminates the need to negotiate between them.

These observations about each of the three rationales converge on a broader phenomenon that the genealogy's framework renders visible. AI-mediated systematization carries a specific motivational pull. Its outputs are fluent, internally coherent, and instantly communicable. They provide what the genealogy has been calling the machine-like virtues in their most concentrated form: surveyability (the output is organized and navigable), rule-governance (the output follows a consistent internal logic), universality (the output is accessible to anyone who can read), and impartiality (the output is the same regardless of who prompted it). These properties are valuable, and they are also, in a precise sense, *seductive*.

Nguyen's concept of value capture identifies the mechanism (Nguyen 2024). Value capture occurs when an agent adopts a simplified, standardized expression of value and permits it to dominate practical reasoning without adapting it to the agent's particular context. The concept was developed through a study of institutional metrics: step counts, citation rates, law school rankings, and the quantified indicators that large-scale organizations use to manage information across vast domains. In each case, the metric offers a trade-off. The agent gains clarity, communicability, and motivational ease: successes become measurable, justifications become straightforward, and the nauseating complexity of plural, inchoate, context-sensitive values is replaced by a single legible standard. The agent surrenders, in exchange, the

fine-grained control over the articulation of values that would enable them to be tailored to the agent's particular personality, circumstances, and context.

The analysis of *why* institutional metrics resist such tailoring tracks the same properties that the genealogy has identified as the machine-like virtues. Metrics are standardized because standardization is what makes them portable across contexts and aggregable across institutional levels. They are explicit because explicitness is what makes them publicly assessable. They are context-invariant because context-invariance is what makes them usable by different people in different circumstances. These are the virtues of universality, surveyability, and impartiality in their institutional form. And they are what makes metrics resistant to the kind of fine-tuning that would make them responsive to particular agents and local conditions. A metric's institutional function *requires* it to be inflexible. The inflexibility that makes it usable at scale is the same inflexibility that makes it a poor guide for any particular individual's flourishing.

AI-generated systematization creates a structurally similar risk. When a language model delivers a fluent, well-organized account of a complex domain, it offers the user a prefabricated evaluative framework: the connections have been drawn, the categories have been set, and the reasoning has been laid out in a form that is instantly communicable and readily justifiable. Adopting this framework provides clarity and ease. It also means accepting a set of pre-drawn connections that have been optimized for fluency and generality rather than calibrated to the user's particular intellectual needs, interests, and context. The user's understanding becomes, in the terms of Chapter 1, something received rather than something individually constructed. And the more frequently the user relies on the machine's systematization, the more thoroughly their evaluative vocabulary comes to be shaped by the categories the machine provides, at the expense of the subtler, more contextually sensitive categories that sustained independent engagement would have cultivated. In Nguyen's terms, this is the process by which value capture proceeds: the clarity and ease of the standardized framework create a motivational pull that gradually displaces the richer, harder-to-articulate evaluative resources it was meant to serve.

If the genealogy identifies the risks of AI-mediated systematization, it also provides the resources for thinking about how those risks might be managed. The framework developed in Chapter 9 answers the question "how much system?" by indexing the demand to the practical rationale in play and the domain to which it is applied: as much systematization as the relevant rationale warrants, under the constraint that the costs internal to the framework are acknowledged and weighed. This is a diagnostic answer, and its force lies in the specificity of the diagnosis it enables. Yet it leaves open a procedural question: what *stance* toward systematic structures does the framework recommend in practice? The genealogy has been building toward an answer to this question throughout its narrative,

but it is Nguyen's concept of reflective control that gives the answer its sharpest contemporary articulation (Nguyen 2026).

Reflective control, in Nguyen's analysis, is the capacity to take up a scoring system, use it for the practical purposes it serves, and periodically step back to assess whether it is serving those purposes, against criteria that the scoring system itself cannot supply. Games model this capacity. A game player adopts the game's goals, plays under them with full engagement, and then, when the game is over, steps back and evaluates whether the play was worthwhile by criteria external to the game: Was it fun? Did it cultivate something valuable? Did it fit the person I want to be? The game's goals are what Nguyen calls "disposable ends," taken up temporarily and revisable at will. The player's real purposes, the ones that govern the choice of which games to play and how long to keep playing them, lie outside the scoring system and are exercised in the reflective moment between one game and the next.

Metrics, by contrast, discourage reflective control. They present themselves as universal and permanent, resist individual modification, and are embedded in institutional infrastructures that make opting out costly. A law school ranking does not invite its users to step back and ask whether the ranking is capturing what matters about a legal education. A citation index does not encourage the scholar to reflect on whether the metric serves the purposes that drew her to philosophy. The clarity and communicability of these metrics, the very properties that make them useful as instruments of coordination, also make them motivationally sticky: the game-like pleasures of value clarity, the ease of justification they provide, and the social legibility they confer all conspire against the reflective moment in which the agent steps back and asks whether the proxy is still serving the richer value it was meant to approximate.

The genealogy has been tracking this distinction *avant la lettre*. Condillac's *esprit systématique* is systematization under reflective control: the thinker uses systematic methods as instruments, subject to ongoing assessment against the domain they are meant to organize. The systematizer asks, at each stage, whether the system's coherence is purchased honestly or sustained by semantic indeterminacy, whether its categories are tracking real features of the domain or imposing an order that the phenomena do not warrant. Condillac's *esprit de système* is systematization that has escaped reflective control: the system runs according to its own internal logic, and the question of whether it still serves its purpose is no longer being asked. D'Alembert's encyclopaedic ambition held the two *esprits* in productive tension, insisting on the value of systematic organization while warning against the dogmatism that systematic ambition breeds. Lambert's purpose-relative conception of system provided the analytical means for exercising reflective control: if a system is a purposively composed whole, then the ordering of its parts depends on what the system is *for*, and different purposes generate different orderings of the same material. Recognizing which purpose governs the systematization at hand is what enables the reflective systematizer to assess whether the system is doing its work. Kant's

regulative use of systematic unity is perhaps the most philosophically developed exercise of reflective control in the tradition: the demand for system is maintained as an indispensable guide to inquiry, but its authority is explicitly conditional on its serving the purposes of reason rather than generating metaphysical illusion.

Our relationship to AI-mediated systematization calls for the same stance. Using a large language model under reflective control means enlisting its pedagogical power, its capacity for epistemological coherence-testing, and its ability to make knowledge surveyable and transmissible at scale, while maintaining the capacity to step back and assess whether the machine's outputs serve the purposes they were enlisted for, against criteria the machine itself cannot supply. Those criteria are precisely the ones the counter-voice tradition has been defending across the genealogy: the contextual sensitivity that universality screens out, the situated judgment that impartiality suppresses, the formative engagement that efficient transmission compresses away. Reflective control does not require rejecting AI systematization. It requires treating the machine's outputs as powerful instruments whose authority is indexed to the practical rationale they serve and bounded by the costs that the same framework identifies.

Nguyen's concept of value federalism captures the structural implication (Nguyen 2024). Some cognitive tasks are best pursued at the largest scale, with maximum standardization and machine-like impersonality. Where knowledge must be transmitted rapidly across vast populations, where claims must be checked for consistency across enormous bodies of evidence, where public accountability requires that reasoning be legible to millions, AI-mediated systematization is superbly equipped for the work, and the practical rationales vindicate its use. Other cognitive tasks require the most local scale and the maximum sensitivity to what is particular and situated. The cultivation of judgment through sustained dialectical engagement, the contextually sensitive assessment of what matters in a specific case, the idiosyncratic connections that a particular mind draws in the course of a particular encounter with the material: these resist compression into the machine's categories and require the kind of prolonged, formative engagement that no externalized architecture can replicate. The question the genealogy poses is the one it has been training us to ask throughout: which rationale applies here, how much systematization does it call for, and at what point do the costs begin to outweigh the benefits? Maintaining reflective control over AI systematization is what enables agents to answer that question in practice, domain by domain and case by case.

The book began with William James's observation that philosophers aim at "cleaning up the litter" (1975–88, vol. iv, 26) and with Camus's challenge, evoked by Williams: when one has no character, one has to give oneself a method. It has traced the history through which that ambition was given institutional form, philosophical justification, and critical examination, from the Reformed classrooms where Keckermann first reorganized doctrine as a *systema* to Kant's transformation of the systematicity ideal and the counter-voice that accompanied it from Cavendish

through Hegel to Williams. Within a decade of the first *Critique*, Fichte and Schelling would treat the imperative to systematize as obvious and urgent, pursuing some the most ambitious system-building in the history of philosophy. The genealogy has been asking how the very idea that knowledge could form a system came to seem so compelling, and at what cost.

The litter that James observed philosophers tidying up turns out to be the texture of thought before it has been made machine-like: the wavering, the polyphony, the shifting plurality of voices and values that characterize cognition in its unsystematized condition. Onto this polyphonic, dynamic environment, the *systema* imposed a structure that was static, explicit, and reproducible. What was gained was genuine and, in many domains, indispensable. What was lost was harder to name, because it consisted in what only the unsystematized process could have cultivated.

An LLM is, quite literally, a method without character: a system that produces judgment-shaped outputs through an impersonal procedure, with no situated perception, no cultivated disposition, and no formative history behind them at all. What Keckermann's *systema* began, the relocation of cognitive authority from persons and institutions into articulated structure, the LLM completes at a scale its inventor could not have imagined. The question whether understanding can be compressed without remainder into its products is as old as the *systema* itself. It has lost none of its force. The genealogy does not resolve the tension between system and character. It equips us to navigate it, by specifying what the compression gains, what it costs, and when the cost is worth paying. In an age when machines that systematize thought are reshaping how knowledge is produced, transmitted, and used to govern, the counter-voice that has accompanied the machine emulation thesis from the beginning deserves to be heard with renewed attention. Williams heard in the demand for system the "rattle of machinery" that signals the point at which the pursuit of systematicity has outrun what the practical rationales warrant (2006, 120). Philosophy, he insisted, needs to be on the lookout for that sound. The rattle is louder now than it has ever been.

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